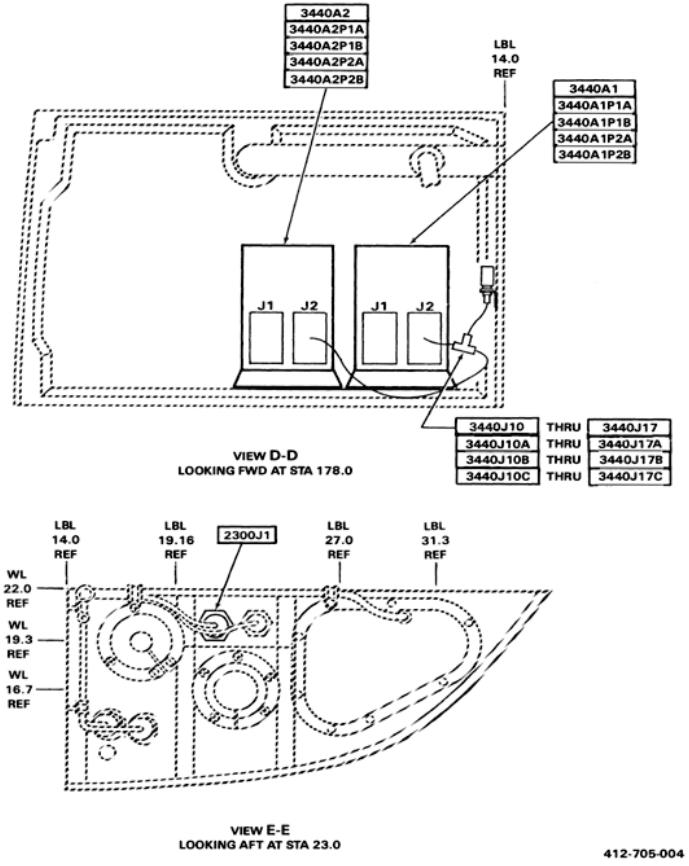
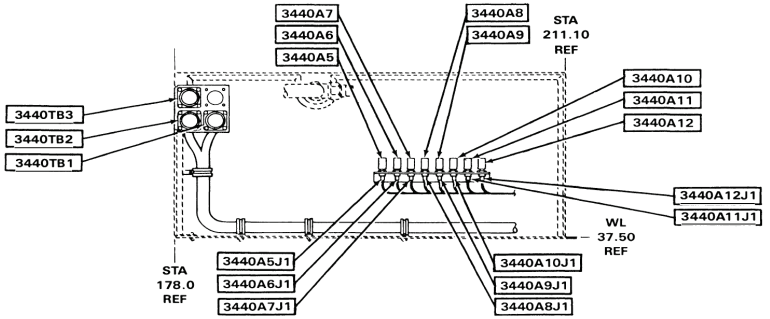
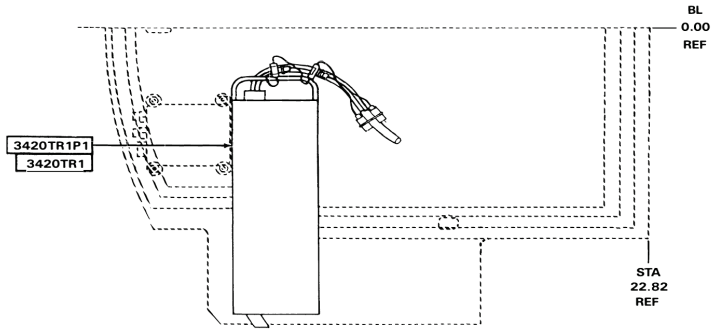


ICN-412-A-960001-B-97499-00009-A-001-01

Figure 4. EFIS Related Component Location (Sheet 5 of 8)



VIEW F
LOOKING INBD LH SIDE
SYMBOL GENEATORS REMOVED FOR CLARITY

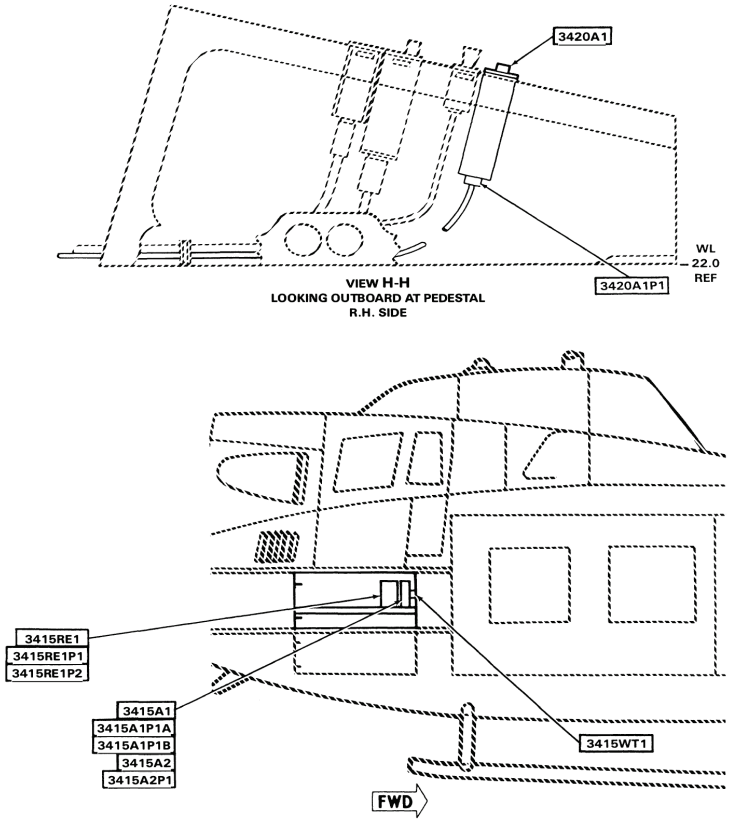


VIEW G-G
LOOKING DOWN AT LOWER NOSE
SHELF WL 22.00

412-570-002
412-705-004

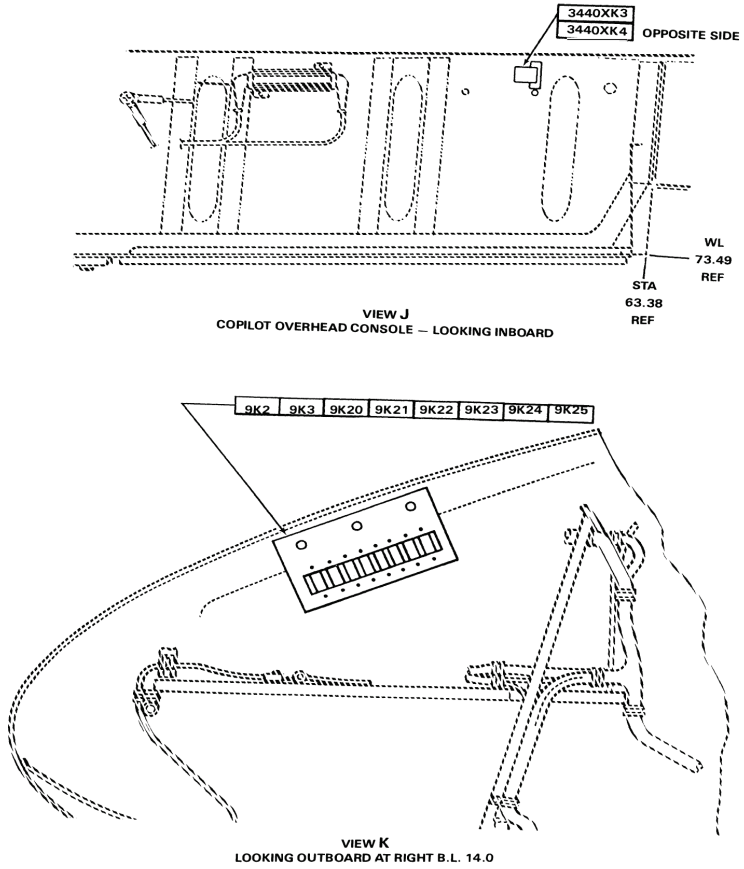
ICN-412-A-960001-B-97499-00010-A-001-01

Figure 4. EFIS Related Component Location (Sheet 6 of 8)



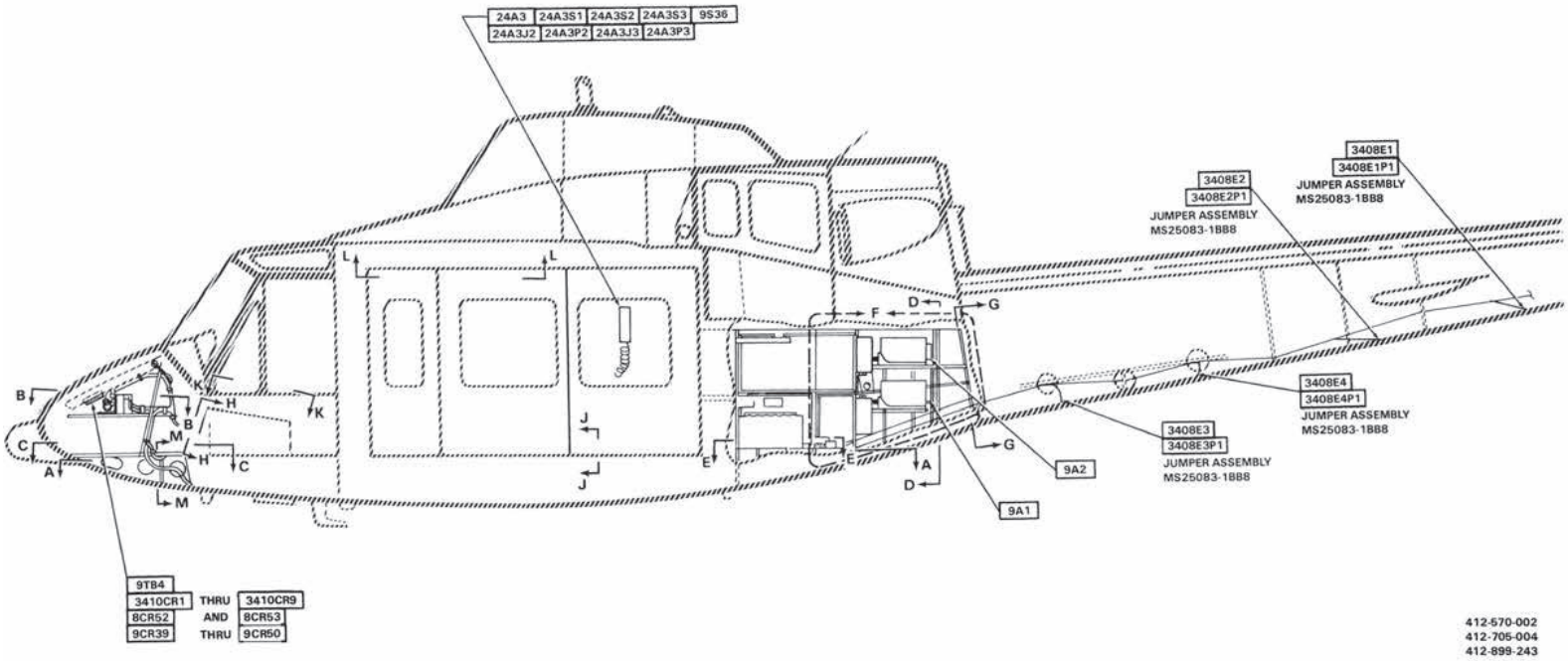
ICN-412-A-960001-B-97499-00011-A-001-01

Figure 4. EFIS Related Component Location (Sheet 7 of 8)



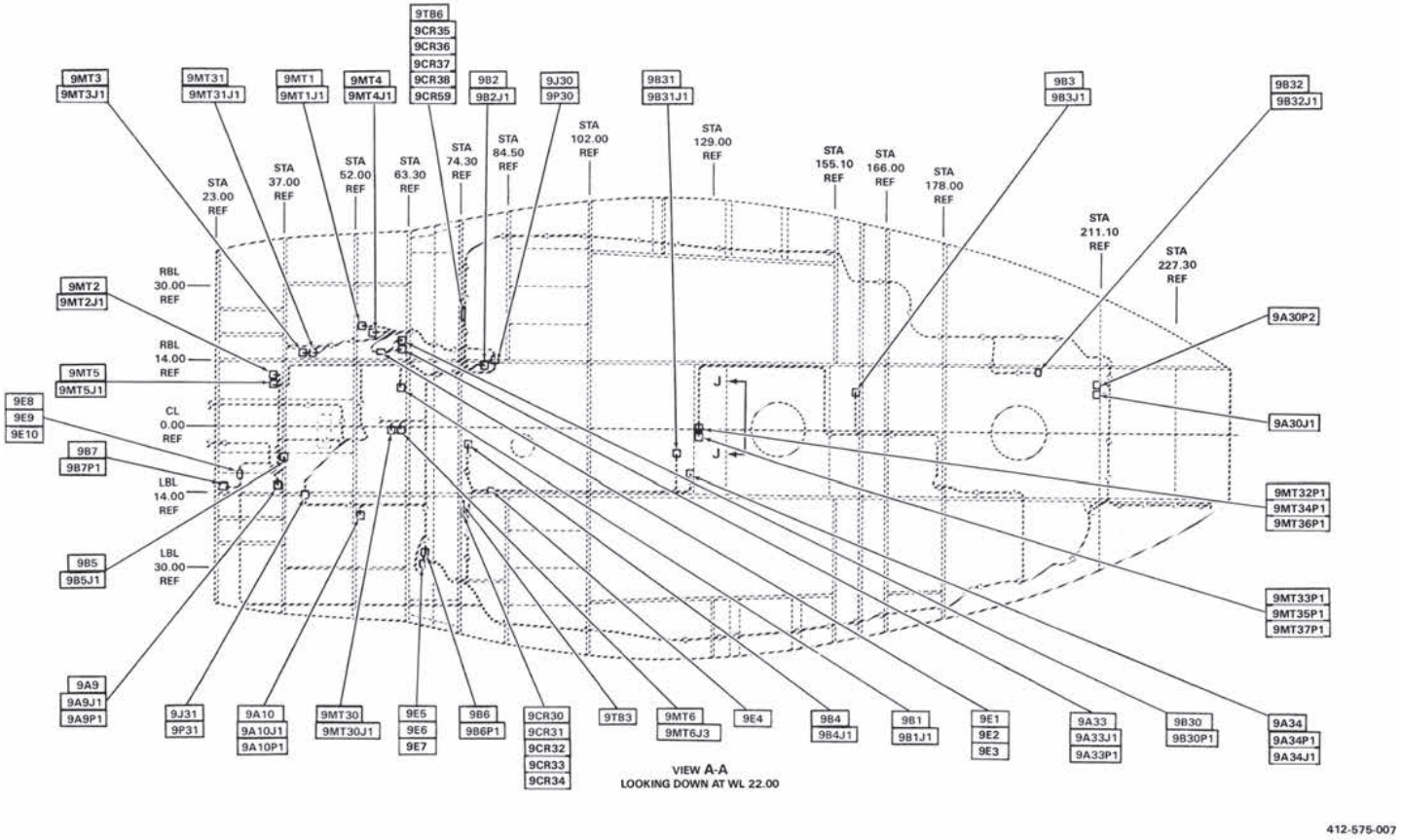
ICN-412-A-960001-B-97499-00012-A-001-01

Figure 4. EFIS Related Component Location (Sheet 8 of 8)



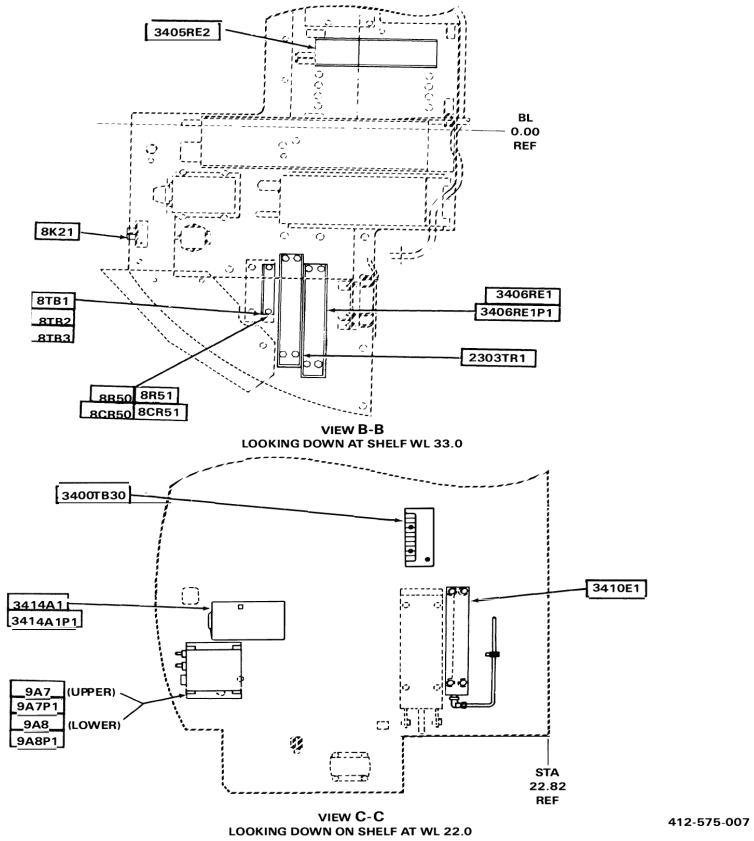
ICN-412-A-960001-B-97499-00013-A-001-01

Figure 5. DAFCS Related Component Location (Sheet 1 of 11)



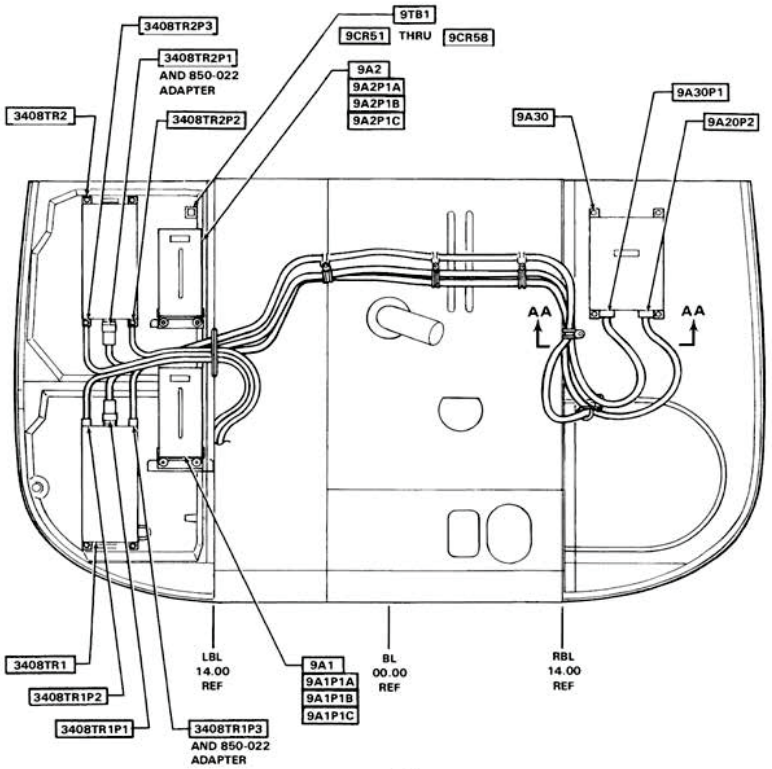
ICN-412-A-960001-B-97499-00014-A-001-01

Figure 5. DAFCS Related Component Location (Sheet 2 of 11)



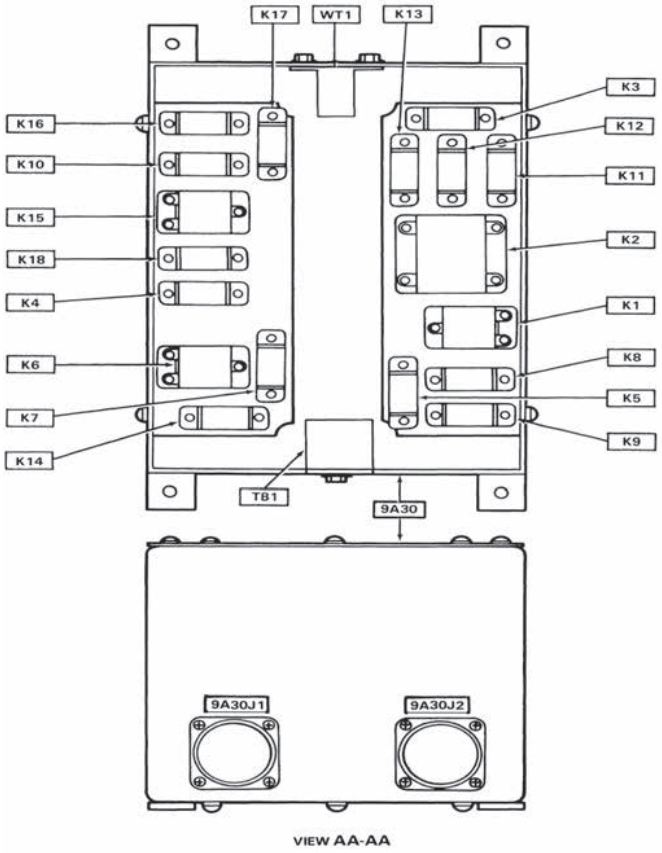
ICN-412-A-960001-B-97499-00015-A-001-01

Figure 5. DAFCS Related Component Location (Sheet 3 of 11)



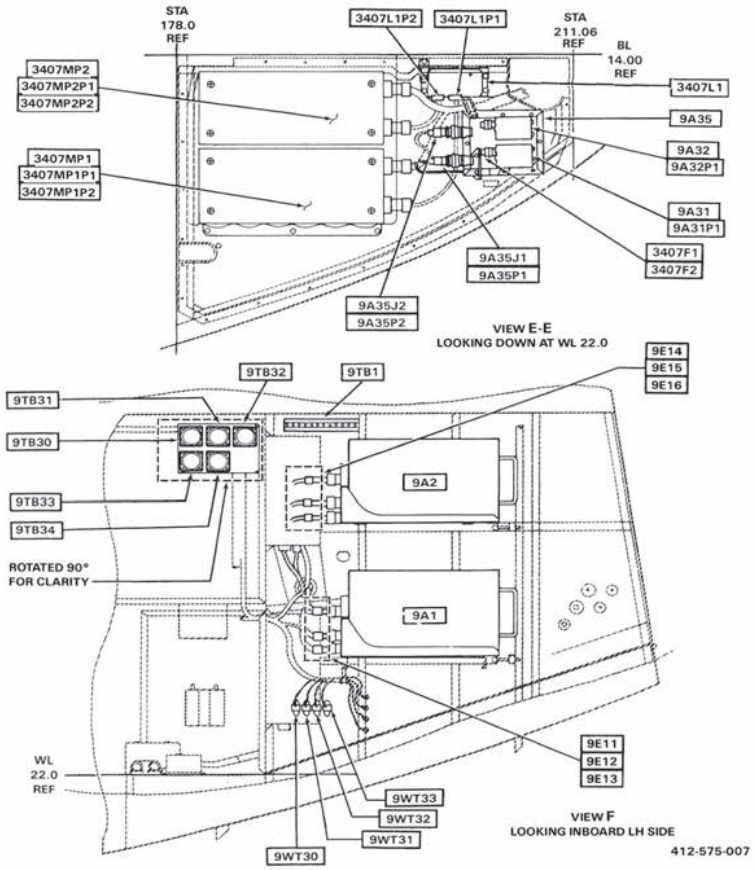
ICN-412-A-960001-B-97499-00016-A-001-01

Figure 5. DAFCS Related Component Location (Sheet 4 of 11)



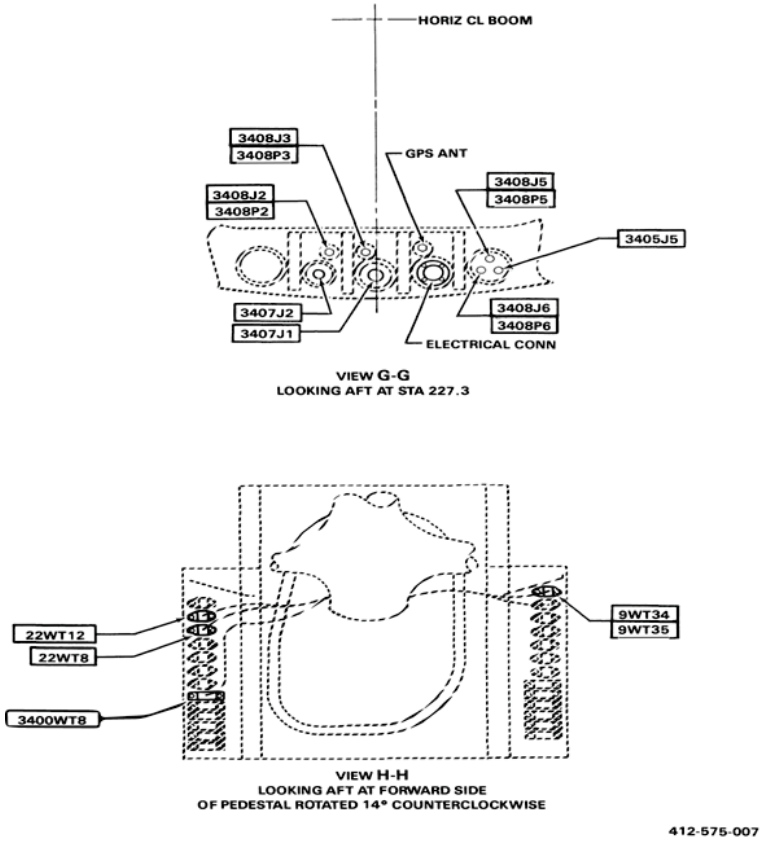
ICN-412-A-960001-B-97499-00017-A-001-01

Figure 5. DAFCS Related Component Location (Sheet 5 of 11)



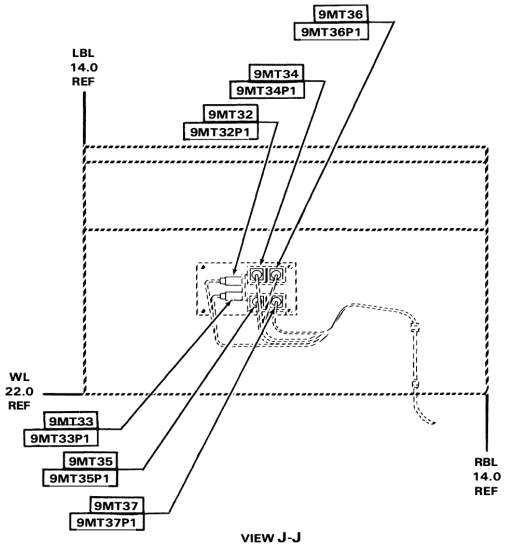
ICN-412-A-960001-B-97499-00018-A-001-01

Figure 5. DAFCS Related Component Location (Sheet 6 of 11)



ICN-412-A-960001-B-97499-00019-A-001-01

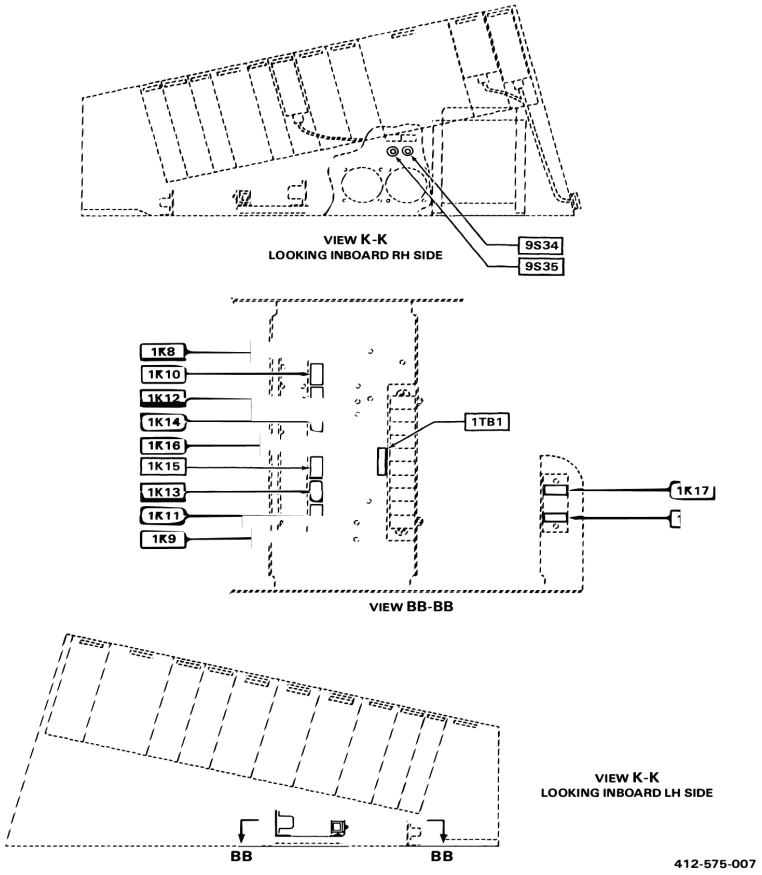
Figure 5. DAFCS Related Component Location (Sheet 7 of 11)



412-575-007

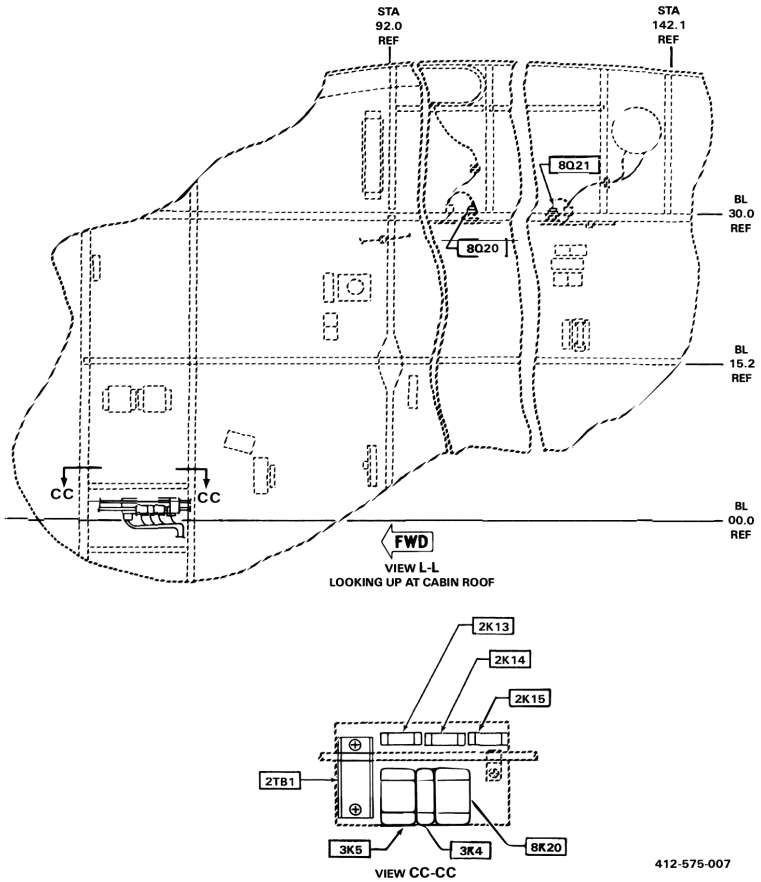
ICN-412-A-960001-B-97499-00020-A-001-01

Figure 5. DAFCS Related Component Location (Sheet 8 of 11)



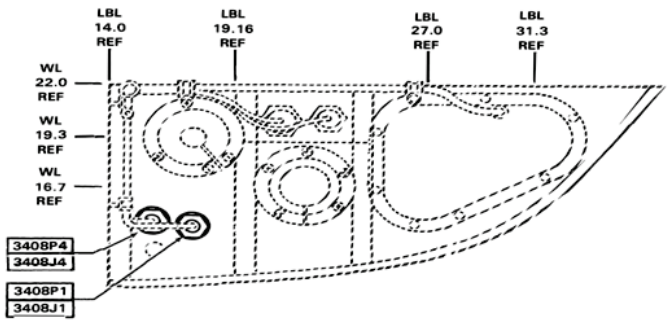
ICN-412-A-960001-B-97499-00021-A-001-01

Figure 5. DAFCS Related Component Location (Sheet 9 of 11)



ICN-412-A-960001-B-97499-00022-A-001-01

Figure 5. DAFCS Related Component Location (Sheet 10 of 11)



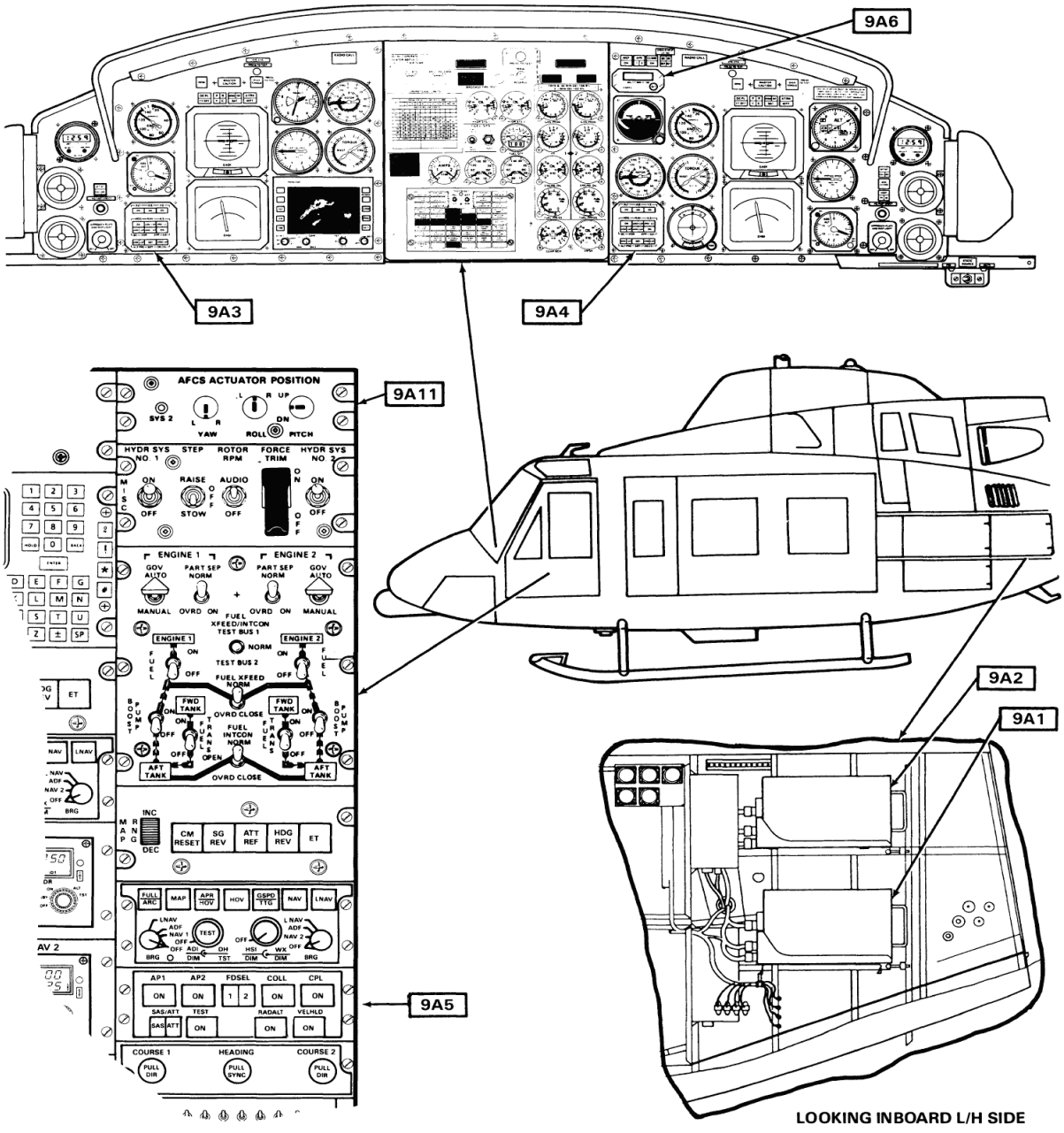
VIEW M-M
LOOKING AFT AT STA 23.0
ROTATED 90° COUNTERCLOCKWISE

412-575-007

ICN-412-A-960001-B-97499-00023-A-001-01

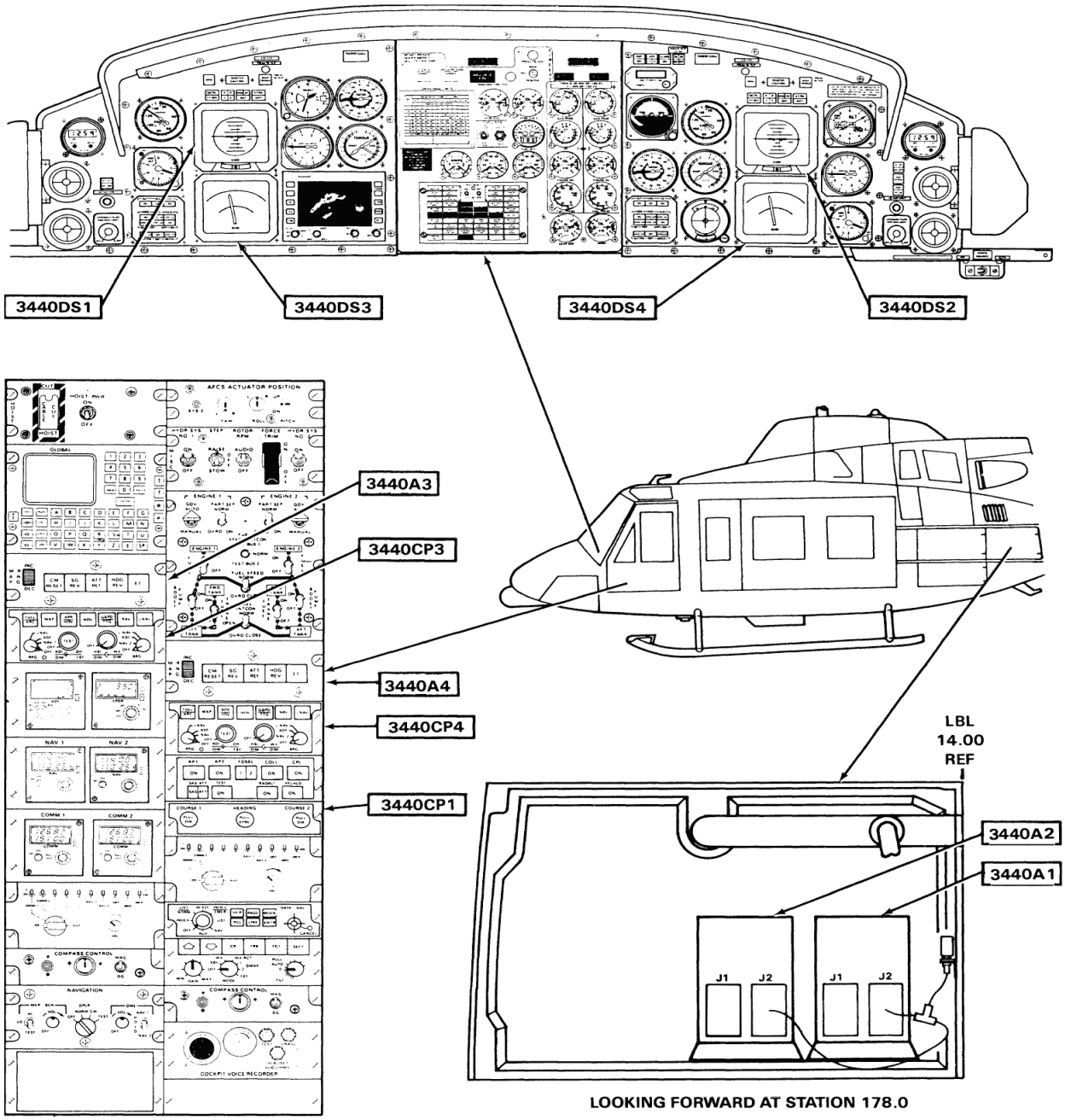
Figure 5. DAFCS Related Component Location (Sheet 11 of 11)

Intentionally blank page



ICN-412-A-960001-B-97499-00024-A-001-01

Figure 6. Digital Automatic Flight Control System - Major Component Location (Sheet 1 of 1)



ICN-412-A-960001-B-97499-00025-A-001-01

Figure 7. Electronic Flight Instrument System - Component Location (Sheet 1 of 1)

Miscellaneous Systems - Description

The miscellaneous system peculiar to SAR configured helicopters is the EFIS cooling system. Refer to 412-A-98-01-01-01A-051A-A / 02975, 412-A-98-01-01-02A-051A-A / 02976, 412-A-98-01-01-03A-051A-A / 02977, 412-A-98-01-01-04A-051A-A / 02978 for EFIS wiring diagrams.

Intentionally blank page

EFIS Cooling System - Description

One external cooling source is provided for each EADI/EHSI pair of electronic displays to provide adequate cooling and proper display operation. Each fan supplies a minimum of 120 cubic feet per minute free air delivery at sea level.

Intentionally blank page

EFIS Cooling System - Operational Check

Procedure

1. The EFIS cooling system operates when the EFIS system is operating. Refer to Honeywell, SPZ-7600 Integrated Flight Control System Bell 412HP System Maintenance Manual, Publication Number A09-1169-024.

Intentionally blank page

EFIS Cooling System - EFIS Fan - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Disconnect electrical connector.
2. Remove and retain mounting hardware.
3. Remove fan.

Intentionally blank page

EFIS Cooling System - EFIS Fan - Installation

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Position fan in mounting bracket.
2. Install mounting hardware.
3. Connect electrical connector.

Intentionally blank page

Integrated Flight Control System - Description

The Integrated Flight Control System consists of the Digital Automatic Flight Control System (DAFCS) and the Electronic Flight Instrument System (EFIS) .

Refer to Honeywell, SPZ-7600 Integrated Flight Control System Bell 412HP System Maintenance Manual, Publication Number A09-1169-024.

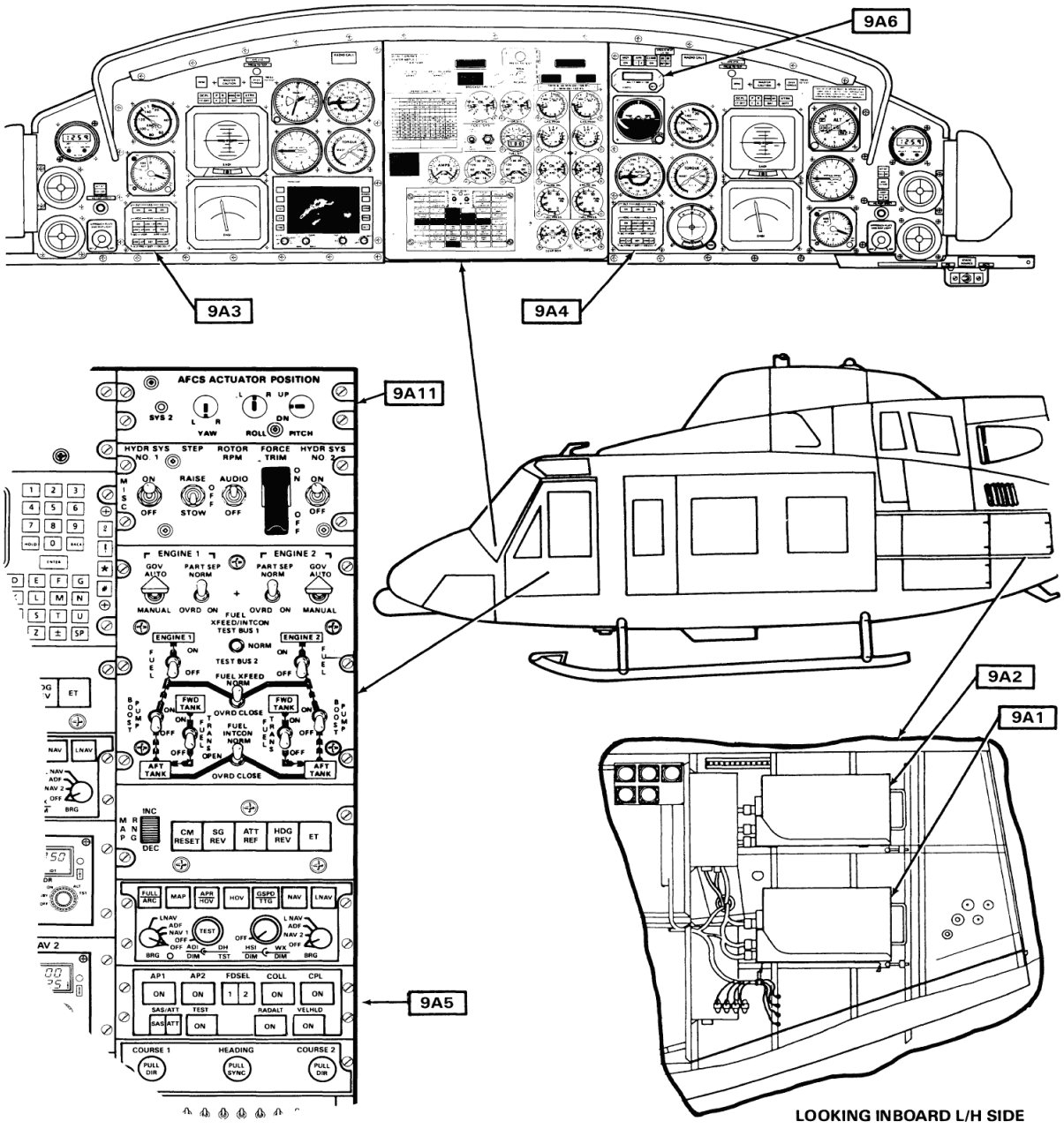
Intentionally blank page

Digital Automatic Flight Control System (DAFCS) - Description

The DAFCS is comprised of the following major components: copilot and pilot Digital Flight Control (DFC) computers (9A1 and 9A2), copilot and pilot flight director mode selectors (9A3 and 9A4), autopilot controller (9A5), attitude preselect indicator (9A6), and AFCS position indicator panel (9A11) (Figure 1).

The DAFCS provides fully coupled autopilot and flight director functions in the pitch, roll, yaw, and collective axis. The autopilot and flight director functions are performed by two identical, interchangeable digital flight control computers.

The DFC computer utilizes a combination of digital and analog computational techniques to implement various automatic flight control system functions. The computer processes information about actual aircraft attitude versus a desired attitude as a function of selected flight mode to produce autopilot pitch, roll, yaw, and collective control outputs and flight director pitch and roll steering commands.



ICN-412-A-960001-B-97499-00024-A-001-01

Figure 1. Digital Automatic Flight Control System - Major Component Location (Sheet 1 of 1)

SAR DAFCS - Operational Check

Procedure

1. Visually inspect equipment and correct any faulty mechanical mountings or electrical connections. Connect external power, adjusted to 27.5 Vdc, to helicopter power receptacle.

2. Apply external hydraulic pressure, refer to 412-A-29-00-00-00A-320A-A / 00616.

3. Energize power buses per 412-A-97-01-00-03A-340A-A / 02886.

4. Close EFIS No. 1 circuit breakers ADI (3440CB1), HSI (3440CB3), SG/DL (3440CB5), EFIS 26V NO. 1 (3440CB7), BLWR (3440CB9), and EFIS NO. 2 circuit breakers ADI (3440CB2), HSI (3440CB4), SG/DL (3440CB6), EFIS 26V NO. 2 (3440CB8), and BLWR (3440CB10).

5. Close circuit breakers AFCS 26V NO. 1 (9CB4), RATE GYRO NO. 1 (9CB31), AFCS NO. 1 PWR (9CB31), AFCS NO. 1 ADD/CAUT (9CB32), and FORCE TRIM (9CB3).

6. Close circuit breakers AFCS 26V NO. 2 (9CB7), RATE GYRO NO. 2 (9CB30), AFCS NO. 2 PWR (9CB2), AFCS NO. 2 CAUT (9CB33), and ENC ALT (3410CB1).

7. Set EFIS MASTER switch to ON.

8. Perform operational checkout according to Honeywell, SPZ-7600 Integrated Flight Control System Bell 412HP System Maintenance Manual, Publication Number A09-1169-024.

Intentionally blank page

Digital Automatic Flight Control System (DAFCS) - Troubleshooting

Procedure

1. It is assumed that all appropriate circuit breakers are closed, interconnecting cabling is in normal operational condition, and power is applied to the system. (See appropriate wiring diagram.)

Table 1. Troubleshooting - DAFCS

INDICATION OF TROUBLE	PROBABLE CAUSE	CORRECTIVE ACTION
	NOTE Error codes are displayed on the attitude preselect indicator (9A6). Refer to Honeywell, SPZ-7600 Integrated Flight Control System Bell 412HP System Maintenance Manual, Publication Number A09-1169-024 for an explaination of error codes.	

Intentionally blank page

Accelerometer Assembly - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.



ACCELEROMETERS ARE VERY DELICATE; HANDLE VERY CAREFULLY.

1. Disconnect electrical connectors.
2. Remove and retain mounting screws and other hardware.
3. Remove accelerometer assembly.

Intentionally blank page

AFCS Actuator Indicator Panel - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Refer to 412-A-96-00-01-04A-520A-A / 01882 for removal of pedestal mounted component.

Intentionally blank page

AFCS Relay Control Box - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Disconnect electrical connectors.
2. Remove and retain mounting screws and other hardware.
3. Remove relay control box.

Intentionally blank page

Air Data Sensor - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Disconnect pitot and static lines.
2. Disconnect electrical connector.
3. Remove and retain mounting screws and other hardware.
4. Remove air data sensor.
5. Protect all openings with caps or tape.

Intentionally blank page

Autopilot Controller - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Refer to 412-A-96-00-01-04A-520A-A / 01882 for removal of pedestal mounted component.

Intentionally blank page

DFC Computer - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Loosen knurled nuts, at front of computer mounting rack, until holddown clamps can be slipped clear of the lug at the front of computer.

2. Slide computer out of its mounting rack.

Intentionally blank page

Hoist Pendant Control Unit - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Disconnect electrical connector.
2. Remove pendant control unit.

Intentionally blank page

Miscellaneous Control Panel - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Refer to 412-A-96-00-01-04A-520A-A / 01882 for removal of pedestal mounted component.

Intentionally blank page

Mode Selector - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Refer to 412-A-95-00-01-00A-520A-A / 01791 for removal of panel mounted instrument.

Intentionally blank page

Yaw Rate Gyro - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.



A GYRO MAY BE DAMAGED BY MOVING IT AFTER ELECTRICAL POWER IS REMOVED BUT BEFORE THE GYRO ROTOR STOPS. THE GYRO ROTOR WILL NOT COME TO A STOP FOR APPROXIMATELY 15 MINUTES AFTER ELECTRICAL POWER IS REMOVED. WHEN REMOVING OR INSTALLING A GYRO, DO SO GENTLY WITH NO QUICK MOTION.

1. Remove and retain screws and washers securing lid on yaw rate gyro assembly.
2. Remove lid.
3. Disconnect electrical connector.
4. Remove screws and mounting hardware.
5. Remove yaw rate gyro from assembly.

Intentionally blank page

Accelerometer Assembly - Installation

Procedure



ENSURE ELECTRICAL POWER IS OFF.



ACCELEROMETERS ARE VERY DELICATE; HANDLE VERY CAREFULLY.

1. Position accelerometer assembly and secure with mounting screws and other hardware.
2. Connect electrical connectors.

Intentionally blank page

AFCS Actuator Indicator Panel - Installation

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Refer to 412-A-95-00-01-00A-720A-A / 01792 for installation of pedestal mounted component.

Intentionally blank page

AFCS Relay Control Box - Installation

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Position control box and secure with mounting screws and other hardware.
2. Connect electrical connectors.

Intentionally blank page

Air Data Sensor - Installation

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Position air data sensor and secure with mounting screws and other hardware.
2. Connect pitot and static lines.
3. Connect electrical connector.

Intentionally blank page

Autopilot Controller - Installation

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Refer to 412-A-96-00-01-04A-720A-A / 01886 for installation of pedestal mounted component.

Intentionally blank page

DFC Computer - Installation

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Slide computer into its mounting rack.
2. Lift holddown clamps, at front of mount, over the lug at front of computer.
3. Secure computer by tightening knurled retainer nuts.

Intentionally blank page

Hoist Pendant Control Unit - Installation

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Position control unit in helicopter.
2. Connect electrical connector.

Intentionally blank page

Miscellaneous Control Panel - Installation

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Refer to 412-A-96-00-01-04A-720A-A / 01886 for installation of pedestal mounted component.

Intentionally blank page

Mode Selector - Installation

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Refer to 412-A-95-00-01-00A-720A-A / 01792 for installation of panel mounted instrument.

Intentionally blank page

Yaw Rate Gyro - Installation

Procedure



ENSURE ELECTRICAL POWER IS OFF.



WHEN REMOVING OR INSTALLING A GYRO, DO SO GENTLY WITH NO QUICK MOTION.

1. Position yaw rate gyro in assembly box and secure with mounting hardware.
2. Connect electrical connector.
3. Position and secure lid on assembly box.

Intentionally blank page

Digital Automatic Flight Control System (DAFCS) - Inspection

Procedure

1. Inspect DAFCS for completeness, proper installation, cleanliness, and secure mounting of each component.
2. Inspect switches and controls on DAFCS system components for proper mechanical action.
3. Inspect components to ensure panel lights illuminate with electrical power on.
4. Inspect electrical connections for corroded or bent pins and proper mating, interconnecting cable for frayed or broken insulation, and circuit breakers for proper operation.

Intentionally blank page

Digital Automatic Flight Control System (DAFCS) - Cleaning

Procedure

1. Remove moisture and loose dirt with a soft, clean cloth.
2. Remove dust and dirt from system components with a soft, clean cloth.



CLEANING COMPOUND FUMES ARE FLAMMABLE AND TOXIC.

3. Remove grease, fungus, and ground-in dirt with a soft, clean cloth dampened with approved cleaning compound.
4. Remove dirt from connectors with a soft bristle brush; remove moisture with a clean, dry cloth.

Intentionally blank page

Digital Automatic Flight Control System (DAFCS) - Repair or Replacement Procedure

1. Tighten or replace loose or cracked control knobs or system components.
2. Replace defective panel light bulbs.
3. Replace defective or inoperative units.
4. Further repairs, other than removal or replacement, must be conducted at an authorized repair facility.

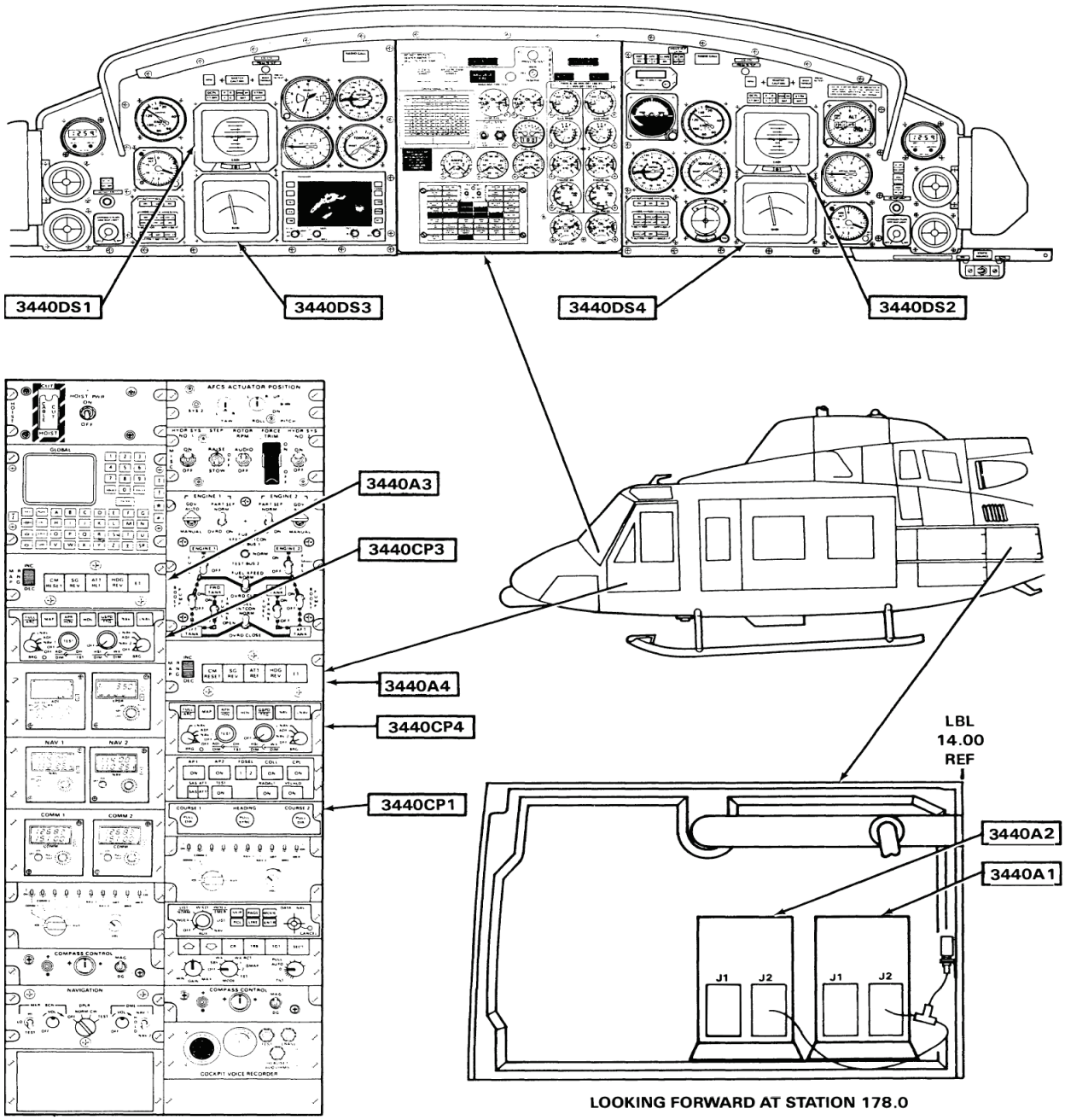
Intentionally blank page

Electronic Flight Instrument System (EFIS) - Description

The Electronic Flight Instrument System (EFIS) is comprised of the following components: copilot and pilot display controllers (3440CP3 and 3440CP4), copilot and pilot Electronic Attitude Director Indicators (EADI) (3440DS1 and 3440DS2), copilot and pilot Electronic Horizontal Situation Indicators (EHSI) (3440DS3 and 3440DS4), copilot and pilot symbol generators (3440A1 and 3440A2), and remote instrument controller (3440CP1) (Figure 1).

The EFIS is a dual system that provides attitude and heading information to the pilot and copilot. The components of each system are identical and interchangeable. The EFIS displays pitch and roll attitude, heading, course orientation, weather presentations, source annunciations and doppler velocities.

The major component of the EFIS is the symbol generator. The symbol generator receives heading, attitude, radio altitude, rate of turn, doppler/velocities, short and long range navigation, and weather inputs. All inputs are processed and transmitted to the electronics displays as a function of the selections made on the display controller and the remote instrument controller. If one symbol generator fails, backup is provided by the other symbol generator. The symbol generators communicate over the Avionics Standard Communications Bus (ASCB) to perform comparison monitoring.



ICN-412-A-960001-B-97499-00025-A-001-01

Figure 1. Electronic Flight Instrument System - Component Location (Sheet 1 of 1)

SAR EFIS - Operational Check

Procedure

1. Visually inspect equipment and correct any faulty mechanical mountings or electrical connections. Connect external power, adjusted to 27.5 Vdc, to helicopter power receptacle.

2. Energize power buses per 412-A-97-01-00-03A-340A-A / 02886.

3. Apply external hydraulic pressure, refer to 412-A-29-00-00-00A-320A-A / 00616.

4. Close EFIS No. 1 circuit breakers ADI (3440CB1), HSI (3440CB3) , SG/DL (3440CB5), EFIS 26V NO. 1 (3440CB7), BLWR (3440CB9), and EFIS No. 2 circuit breakers ADI (3440CB2), HSI (3440CB4), SG/DL (3440CB6), EFIS 26V NO. 2 (3440CB8), and BLWR (3440CB10).

5. Close circuit breakers AFCS 26V NO. 1 (9CB4), RATE GYRO NO. 1 (9CB31), AFCS NO. 1 PWR (9CB31), AFCS NO. 1 ADD/CAUT (9CB32), and FORCE TRIM (9CB3).

6. Close circuit breakers AFCS 26V NO. 2 (9CB7), RATE GYRO NO. 2 (9CB30), AFCS NO. 2 PWR (9CB2), AFCS NO. 2 CAUT (9CB33), and ENC ALT (3410CB1).

7. Set EFIS MASTER switch to ON.

8. Perform operational checkout according to Honeywell, SPZ-7600 Integrated Flight Control System Bell 412HP System Maintenance Manual, Publication Number A09-1169-024.

Intentionally blank page

Electronic Flight Instrument System (EFIS) - Troubleshooting

Procedure

1. It is assumed that all appropriate circuit breakers are closed, interconnecting cabling is in normal operational condition, and power is applied to the system. (See appropriate wiring diagram.) Refer to Honeywell, SPZ-7600 Integrated Flight Control System Bell 412HP System Maintenance Manual, Publication Number A09-1169-024. for troubleshooting.

Intentionally blank page

Auxiliary Switch Panel - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Refer to 412-A-96-00-01-04A-520A-A / 01882 for removal of pedestal mounted component.

Intentionally blank page

Display Controller - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Refer to 412-A-96-00-01-04A-520A-A / 01882 for removal of pedestal mounted component.

Intentionally blank page

Instrument Controller - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Refer to 412-A-96-00-01-04A-520A-A / 01882 for removal of pedestal mounted component.

Intentionally blank page

Symbol Generator - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Loosen mounting tray knob, at front of symbol generator.
2. Slide symbol generator out of its mounting rack.

Intentionally blank page

Auxiliary Switch Panel - Installation

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Refer to 412-A-96-00-01-04A-720A-A / 01886 for installation of pedestal mounted component.

Intentionally blank page

Display Controller - Installation

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Refer to 412-A-96-00-01-04A-720A-A / 01886 for installation of pedestal mounted component.

Intentionally blank page

Instrument Controller - Installation

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Refer to 412-A-96-00-01-04A-720A-A / 01886 for installation of pedestal mounted component.

Intentionally blank page

Symbol Generator - Installation

Procedure



ENSURE ELECTRICAL POWER IS OFF.



DO NOT FORCE FIT. IF MATING IS DIFFICULT, REMOVE THE UNIT AND CHECK FOR CONNECTOR PINS THAT MAY BE BENT OR OUT OF ALIGNMENT. ALSO CHECK THE ALIGNMENT OF THE RECEPTACLE IN THE MOUNTING TRAY.

1. Slide symbol generator into its mounting rack.
2. Tighten mounting tray knob, at front of symbol generator.

Intentionally blank page

Electronic Flight Instrument System (EFIS) - Inspection

Procedure

1. Inspect electronic flight instrument system for completeness, proper installation, cleanliness, and secure mounting of each component.
2. Inspect switches and controls on electronic flight instrument system components for proper mechanical action.
3. Inspect components to ensure panel lights illuminate with electrical power on.
4. Inspect electrical connections for corroded or bent pins and proper mating, interconnecting cable for frayed or broken insulation, and circuit breakers for proper operation.

Intentionally blank page

Electronic Flight Instrument System (EFIS) - Cleaning

Procedure

1. Remove moisture and loose dirt with a soft, clean cloth.
2. Remove dust and dirt from system components with a soft, clean cloth.



CLEANING COMPOUND FUMES ARE FLAMMABLE AND TOXIC.

3. Remove grease, fungus, and ground-in dirt with a soft, clean cloth dampened with approved cleaning compound.
4. Remove dirt from connectors with a soft bristle brush; remove moisture with a clean, dry cloth.

Intentionally blank page

Electronic Flight Instrument System (EFIS) - Repair or Replacement Procedure

1. Tighten or replace loose or cracked control knobs or system components.
2. Replace defective panel light bulbs.
3. Replace defective or inoperative units.
4. Further repairs, other than removal or replacement, must be conducted at an authorized repair facility.

Intentionally blank page

Avionics Systems - Description

The supplemental avionic equipment installed in the Search and Rescue (SAR) configured model 412 helicopter consists of the radio altimeter system, and the weather radar system. Other customizing avionic equipment installed includes Doppler, DATA NAV III, and navigation system (GNS-X).

Intentionally blank page

Airframe Components - Description

Components that are a part of the avionics configuration and are termed airframe components, include the following avionics/electrical items:

ac and dc power busses	panel lights
circuit board assemblies	plugs
circuit breakers	receptacles
connectors	relays
diode assemblies	resistors
interconnecting wiring	switches
electrical leads	terminal boards

Intentionally blank page

Energizing Power Buses - Functional Tests

Procedure

1. The following is a general procedure for energizing ac and dc buses when using an external power source during functional testing of the avionics systems and sets.

1.1. Set BATTERY BUS 1 switch and BATTERY BUS 2 switch to ON.

1.2. Close the following circuit breakers:

- MASTER CAUTION
- AC FEEDERS BUS 1 115V
- AC FEEDERS BUS 2 115V
- AC FEEDERS BUS 3 115V
- AC FEEDERS EMERG BUS 115V
- AC FEEDERS BUS 1 26V
- AC FEEDERS BUS 2 26V
- AC FEEDERS BUS 3 26V
- AC FEEDERS EMERG BUS 26V
- NO. 1 ESNTL BUS FEEDERS (4)
- NO. 2 ESNTL BUS FEEDERS (4)
- GEN 1 FIELD
- GEN 2 FIELD
- INV 1 PWR
- INV 2 PWR
- EMERG BUS INTCT
- BUS 1 VM
- BUS 2 VM

1.3. Set INV 1 switch and INV 2 switch to ON.

1.4. Close all remaining ac circuit breakers to load the inverters.

1.5. Set NON-ESNTL BUS switch to MANUAL.

2. To energize avionics systems or sets, refer to functional test procedures for individual systems or sets.

Intentionally blank page

Avionics Systems - Illustrations - Description

For information/location of supplemental avionics systems or customizing components installed in search and rescue configuration helicopter, refer to Figure 1 through Figure 7. For code designation correlation, refer to Avionics Equipment List (Table 1) and Figure 8 through Figure 13 for supplemental avionic systems wiring diagrams installed in search and rescue configured helicopters.

For maintenance information of airframe components refer to 412-A-96-00-01-04A-042A-A / 01881.

Table 1. Avionics Equipment List

REF DES	NOMENCLATURE
2301CP2J1A	CONN, PILOT ICS
2301CP2P1A	CONN, PILOT ICS
2303CP2J1A	CONN, COM 2
2303CP2P1A	CONN, COM 2
24A3S2	SWITCH, ICS
3405CP2J1A	CONN, NAV 2
3405CP2P1A	CONN, NAV 2
3405J5	CONN, TAILBOOM DISCONNECT
3405RE2	RECEIVER, NAV
3405RE2P2	CONN, NAV RECEIVER
3406CB1	CIRCUIT BREAKER, ADF
3406RE1	RECEIVER, ADF
3406RE1P1	CONN, ADF RECEIVER
3408CB1	CIRCUIT BREAKER, RAD ALT 1
3408CB2	CIRCUIT BREAKER, RAD ALT 2
3408DS1	INDICATOR, COPILOT RAD ALT
3408DS1P1	CONN, COPILOT RAD ALT INDICATOR
3408DS2	INDICATOR, PILOT RAD ALT
3408DS2P1	CONN, PILOT RAD ALT INDICATOR
3408E1	ANTENNA
3408E1P1	CONN, RADIO ALTIMETER ANTENNA
3408E2	ANTENNA
3408E2P1	CONN, RADIO ALTIMETER ANTENNA
3408E3	ANTENNA
3408E3P1	CONN, RADIO ALTIMETER ANTENNA
3408E4	ANTENNA
3408E4P1	CONN, RADIO ALTIMETER ANTENNA
3408J1	CONN
3408J2	CONN, TAILBOOM DISCONNECT
3408J3	CONN, TAILBOOM DISCONNECT
3408J4	CONN
3408J5	CONN, TAILBOOM DISCONNECT
3408J6	CONN, TAILBOOM DISCONNECT
3408P1	CONN
3408P2	CONN
3408P3	CONN
3408P4	CONN

Table 1. Avionics Equipment List (continued)

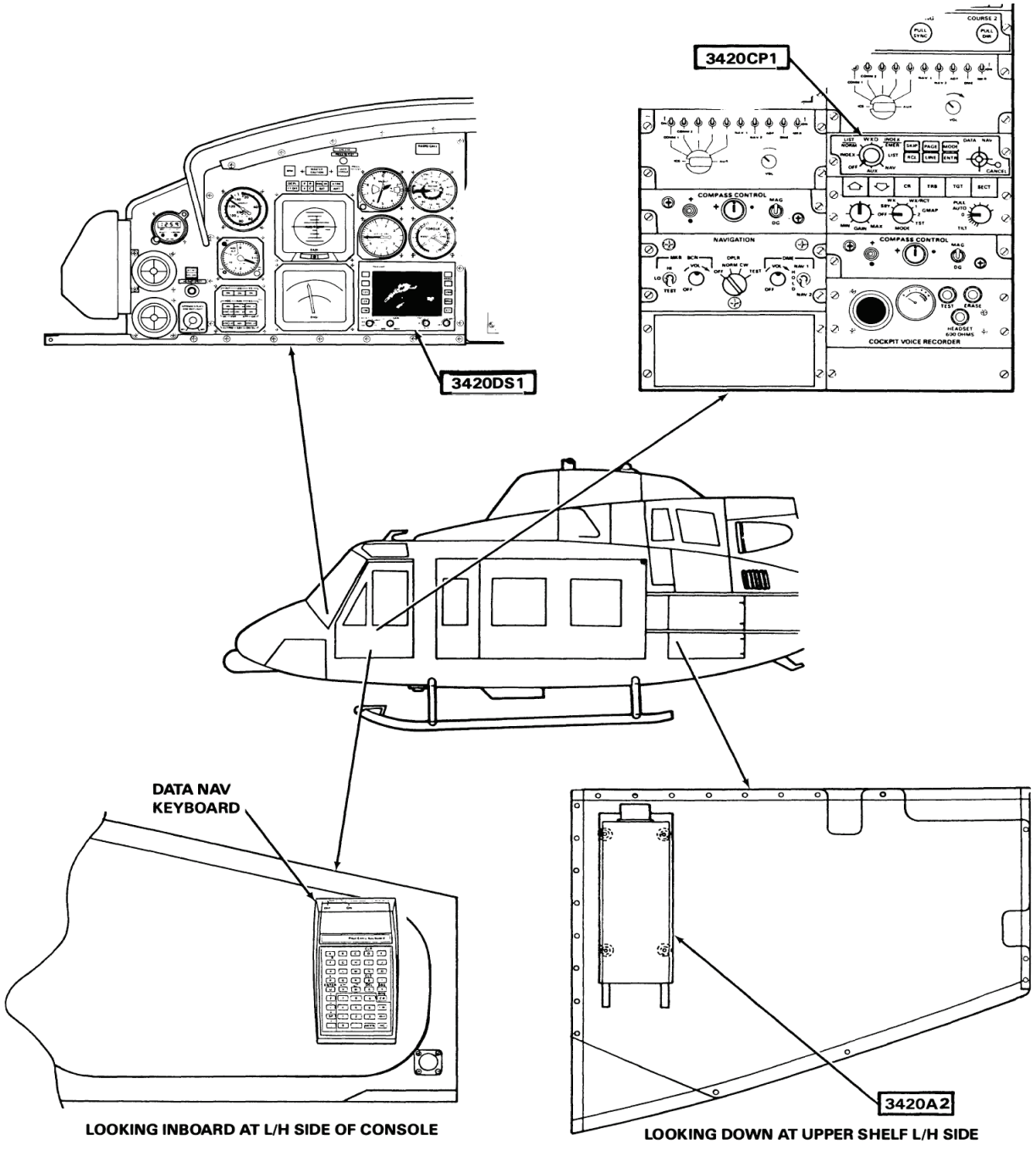
REF DES	NOMENCLATURE
3408P5	CONN
3408P6	CONN
3408TR1	ALTIMETER, RADIO
3408TR1P1	CONN, RADIO ALTIMETER
3408TR1P2	CONN, RADIO ALTIMETER
3408TR1P3	CONN, RADIO ALTIMETER
3408TR2	ALTIMETER, RADIO
3408TR2P1	CONN, RADIO ALTIMETER
3408TR2P2	CONN, RADIO ALTIMETER
3408TR2P3	CONN, RADIO ALTIMETER
3410CB1	CIRCUIT BREAKER, ENC ALTM
3410CP1J1A	CONN, TRANSPONDER
3410CP1P1A	CONN, TRANSPONDER
3410DS1	ALTIMETER, ENCODING
3410DS1P1	CONN, ENCODING ALTIMETER
3414A1	ADAPTER, KING CHANNELING
3414A1P1	CONN, KING CHANNELING ADAPTER
3414K2	RELAY, ANNUNCIATOR TEST
3415A1	NAVIGATION MANAGEMENT UNIT (NMU)
3415A1P1A	CONN, GNS-X NMU
3415A1P1B	CONN, GNS-X NMU
3415A2	CONFIGURATION MODULE
3415A2P1	CONN, CONFIGURATION MODULE
3415A3	PORTABLE DATA TRANSFER UNIT
3415A3P1	CONN, PORTABLE DATA TRANSFER UNIT
3415CB1	CIRCUIT BREAKER, NMU
3415CB2	CIRCUIT BREAKER, RPU
3415CP1	CONTROL DISPLAY UNIT (CDU)
3415CP1P1	CONN,CDU
3415CR1	DIODE
3415CR2	DIODE
3415CR3	DIODE
3415CR4	DIODE
3415CR5	DIODE
3415DS1	LIGHT, COPILOT DISPLAY
3415DS2	LIGHT, PILOT DISPLAY
3415DS3	LIGHT, COPILOT DISPLAY
3415DS4	LIGHT, PILOT DISPLAY
3415E1	ANTENNA,VLF
3415E1P1	CONN, VLF ANTENNA
3415E3	ANTENNA, LORAN
3415E3P1	CONN, LORAN ANTENNA
3415E4	ANTENNA,GPS

Table 1. Avionics Equipment List (continued)

REF DES	NOMENCLATURE
3415E4P1	CONN, GPS ANTENNA
3415G2	DOWN CONVERTER
3415G2P1	CONN, DOWN CONVERTER
3415G2P2	CONN, DOWN CONVERTER
3415J2	CONN
3415P2	CONN
3415RE1	RECEIVER PROCESSOR UNIT
3415RE1P1	CONN, RECEIVER PROCESSOR UNIT
3415RE1P2	CONN, RECEIVER PROCESSOR UNIT
3415WT1	TERMINAL BOARD, GROUND
3420A1	CONTROL, WEATHER RADAR
3420A2	COMPUTER, DATA NAV
3420A2P1	CONN, DATA NAV COMPUTER
3420CB1	CIRCUIT BREAKER, WXRAD STB
3420CB3	CIRCUIT BREAKER, DATA NAV
3420CP1	DATA NAV CONTROL UNIT
3420CP1P1	CONN, DATA NAV CONTROL UNIT
3420DS1	INDICATOR, RADAR
3420DS1P1	CONN, RADAR INDICATOR
3420DS1P2	CONN, COLOR INDICATOR
3420E1	ANTENNA
3420E1P1	CONN,ANTENNA
3420J1	CONN,TEE
3420J1C	CONN
3420J2	CONN,TEE
3420J2C	CONN
3420J3	CONN, TEE
3420J3C	CONN
3420J4	CONN,TEE
3420J4C	CONN
3420J5	CONN,TEE
3420J5C	CONN
3420P1A	CONN
3420P1B	CONN
3420P1P1	CONN, WEATHER RADAR CONTROL
3420P2A	CONN
3420P2B	CONN
3420P3A	CONN
3420P3B	CONN
3420P4A	CONN
3420P4B	CONN
3420P5A	CONN
3420P5B	CONN

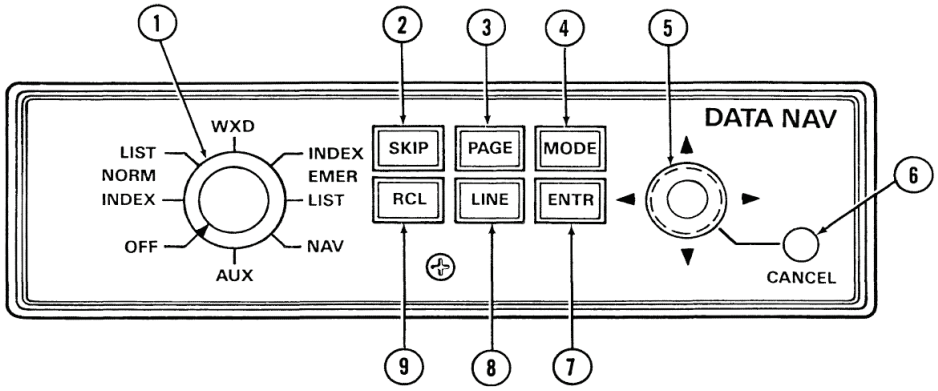
Table 1. Avionics Equipment List (continued)

REF DES	NOMENCLATURE
3420TR1	RECEIVER/TRANSMITTER
3420TR1P1	CONN, RECEIVER/TRANSMITTER
3425CB1	CIRCUIT BREAKER, DPLR
3425TR1	DOPPLER
3425TR1PLA	CONN, DOPPLER



ICN-412-A-970100-B-97499-00001-A-001-01

Figure 1. DATA NAV III System - Major Component Location (Sheet 1 of 1)



1 **OFF/NORM INDEX, NORM LIST/WDX/EMER INDEX, EMER LIST/NAV/AUX***
Rotary switch used to turn DATA NAV III OFF or to turn it on by selection of checklist, weather (WXD), long-range NAVigation, or AUXiliary operation.

2 **SKIP**
Momentary pushbutton switch used to advance from one LINE to next during checklist operation.

3 **PAGE**
Momentary pushbutton switch used to advance from one PAGE to next during checklist operation.

4 **MODE**
Momentary pushbutton switch used in navigation operation to change nav modes in fixed rotation.

5 **Joystick**
Wobble-type control used to move designator forward, backward, left, or right in navigation operation. Initial activation steps designator one step (dot width) in requested direction. If joystick is held in given direction for approximately one second, designator will slew in that direction. When designator comes to rest, latitude/longitude position of designator appears in upper left corner of display.

6 **CANCEL**
Momentary pushbutton switch used in navigation operation to return waypoint designator to home position. With same long-range nav systems, CANCEL will return original waypoint to screen and cancel offset waypoint.

7 **ENTR**
Momentary pushbutton switch used in navigation operation to send latitude/longitude position of designator to long-range nav system. Not all nav systems can use this feature.

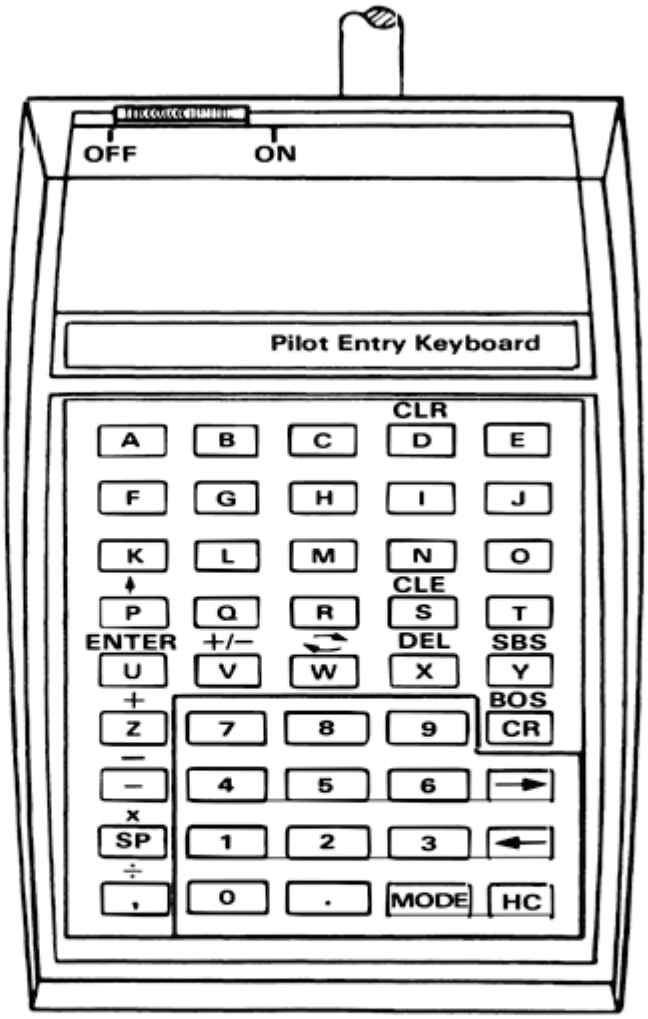
8 **LINE**
Momentary pushbutton switch used to advance from one LINE to next during checklist operation.

9 **RCL**
Momentary pushbutton switch used to RECALL first item skipped during checklist operation. Not enabled in INDEXes.

*If the pilot interrupts a nav display by switching to WXD or AUX and then back again to NAV, the nav display mode will be the same as that which was interrupted. Since the checklists and nav map displays are supported by individual memories, the pilot can switch between these functions without disturbing the display as last used.

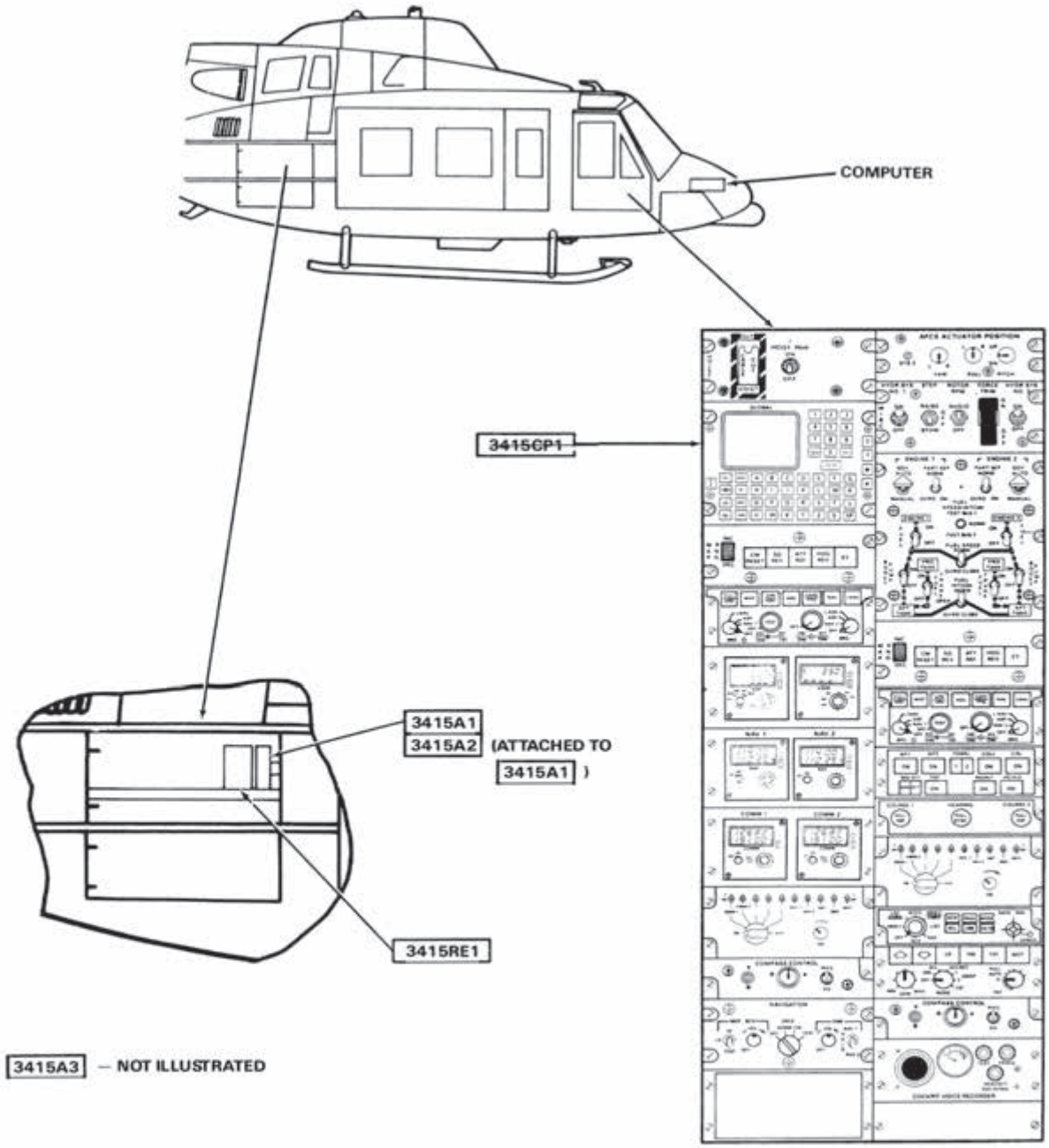
ICN-412-A-970100-B-97499-00002-A-001-01

Figure 2. DATA NAV Control Panel Assembly (Sheet 1 of 1)



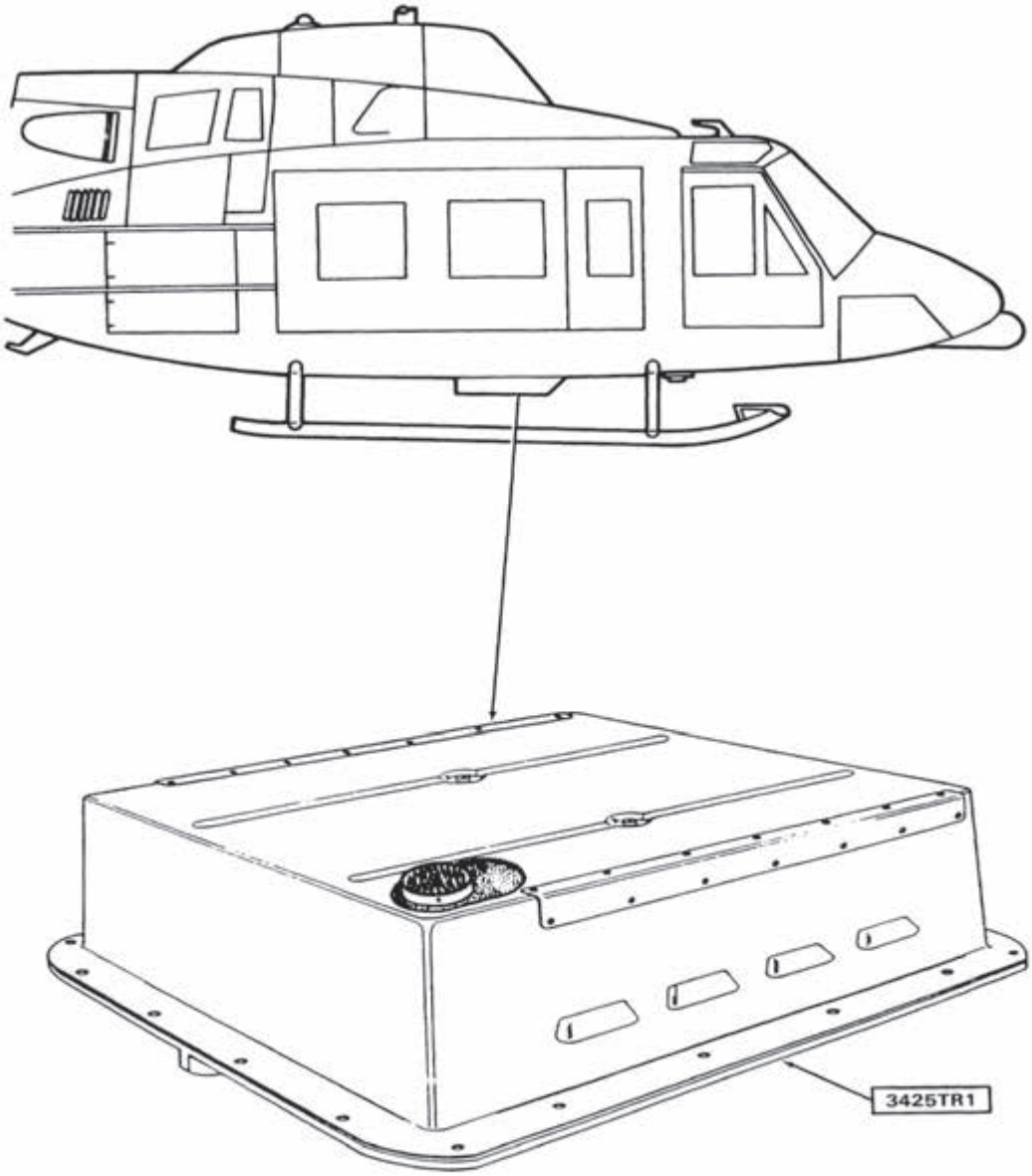
ICN-412-A-970100-B-97499-00003-A-001-01

Figure 3. Pilot Entry Keyboard (Sheet 1 of 1)



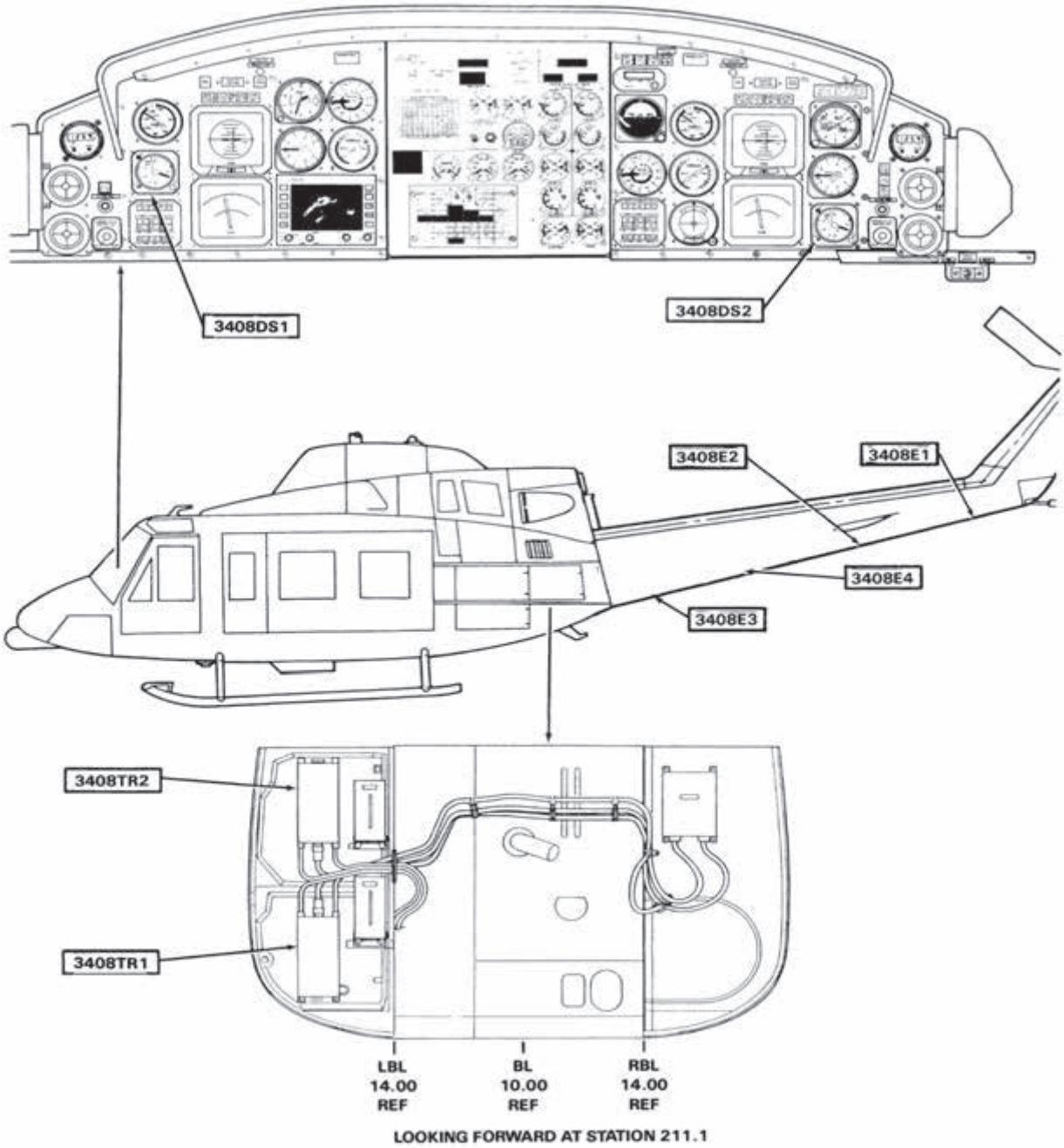
ICN-412-A-970100-B-97499-00004-A-001-01

Figure 4. Navigation System - Major Component Location (Sheet 1 of 1)



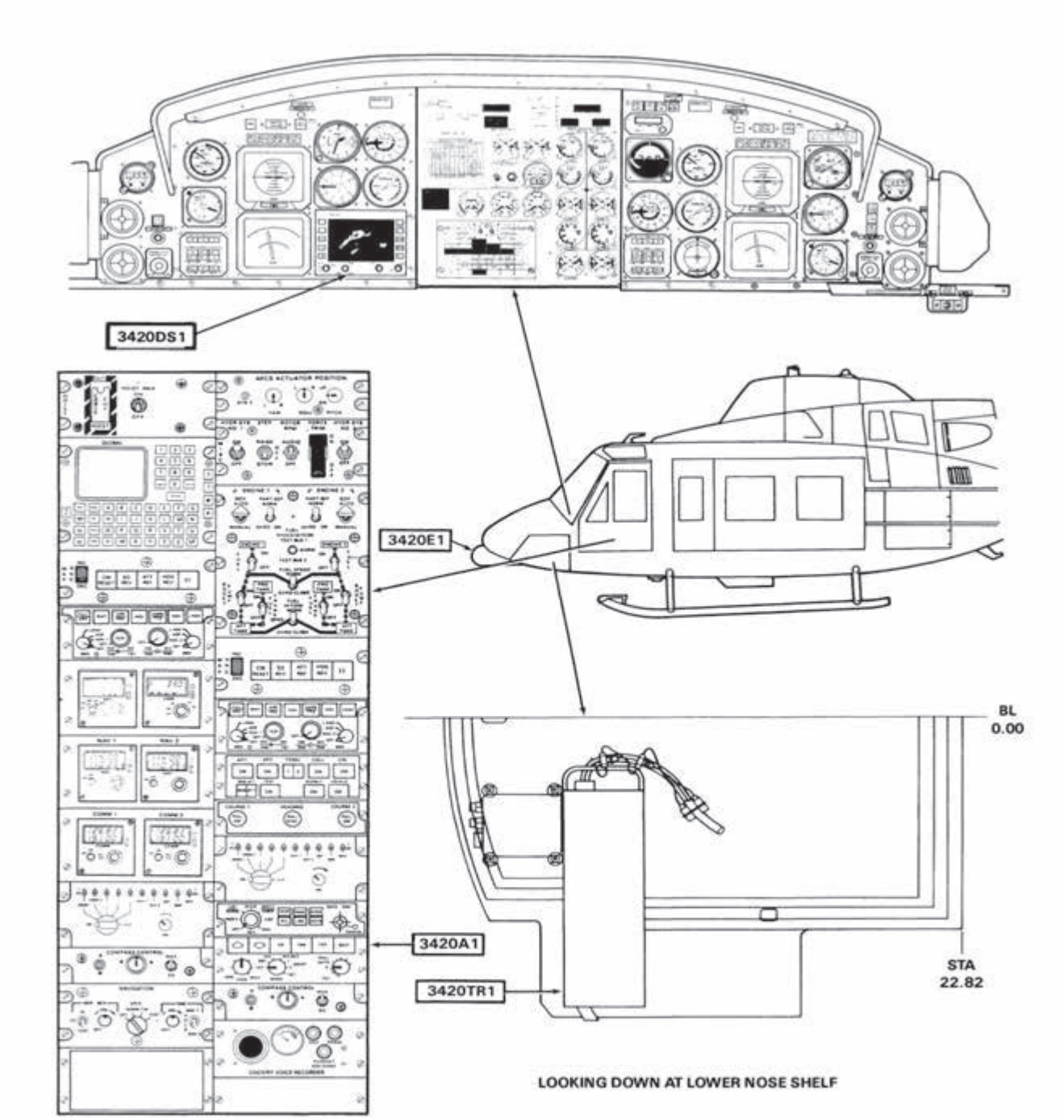
ICN-412-A-970100-B-97499-00005-A-001-01

Figure 5. Doppler - Major Component Location (Sheet 1 of 1)



ICN-412-A-970100-B-97499-00006-A-001-01

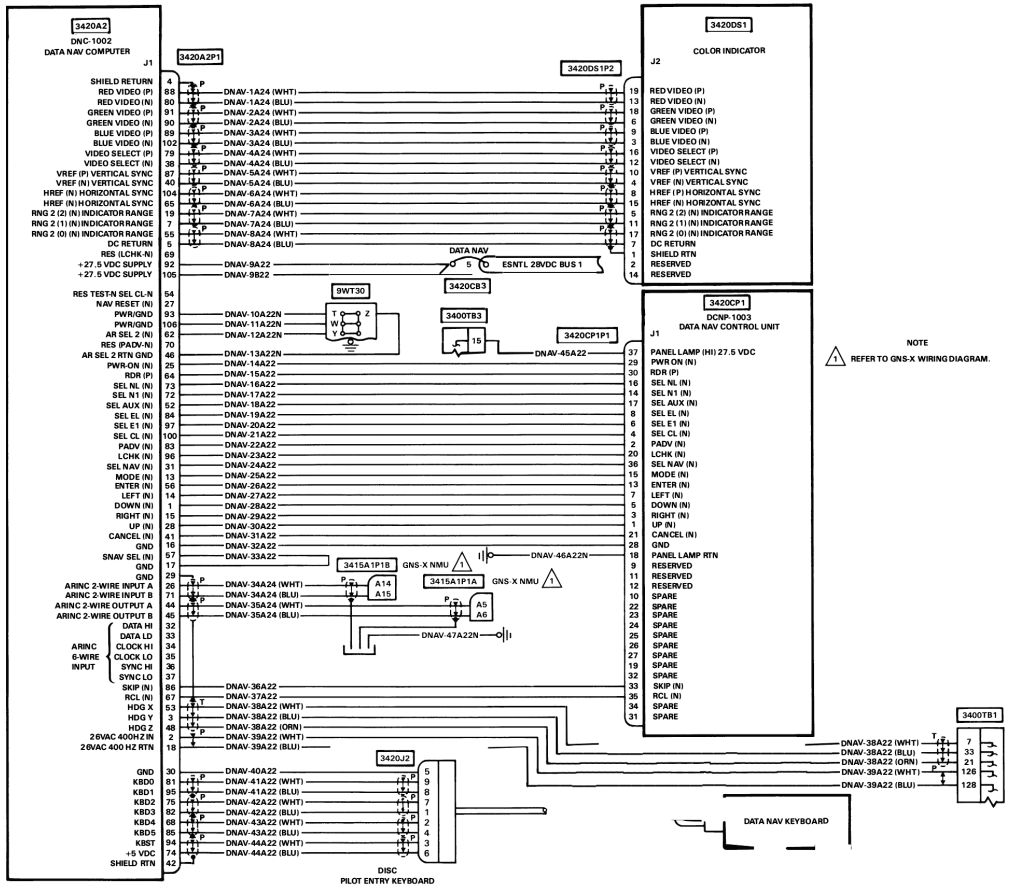
Figure 6. Radio Altimeter System - Major Component Location (Sheet 1 of 1)



ICN-412-A-970100-B-97499-00007-A-001-01

Figure 7. Weather Radar System - Major Component Location (Sheet 1 of 1)

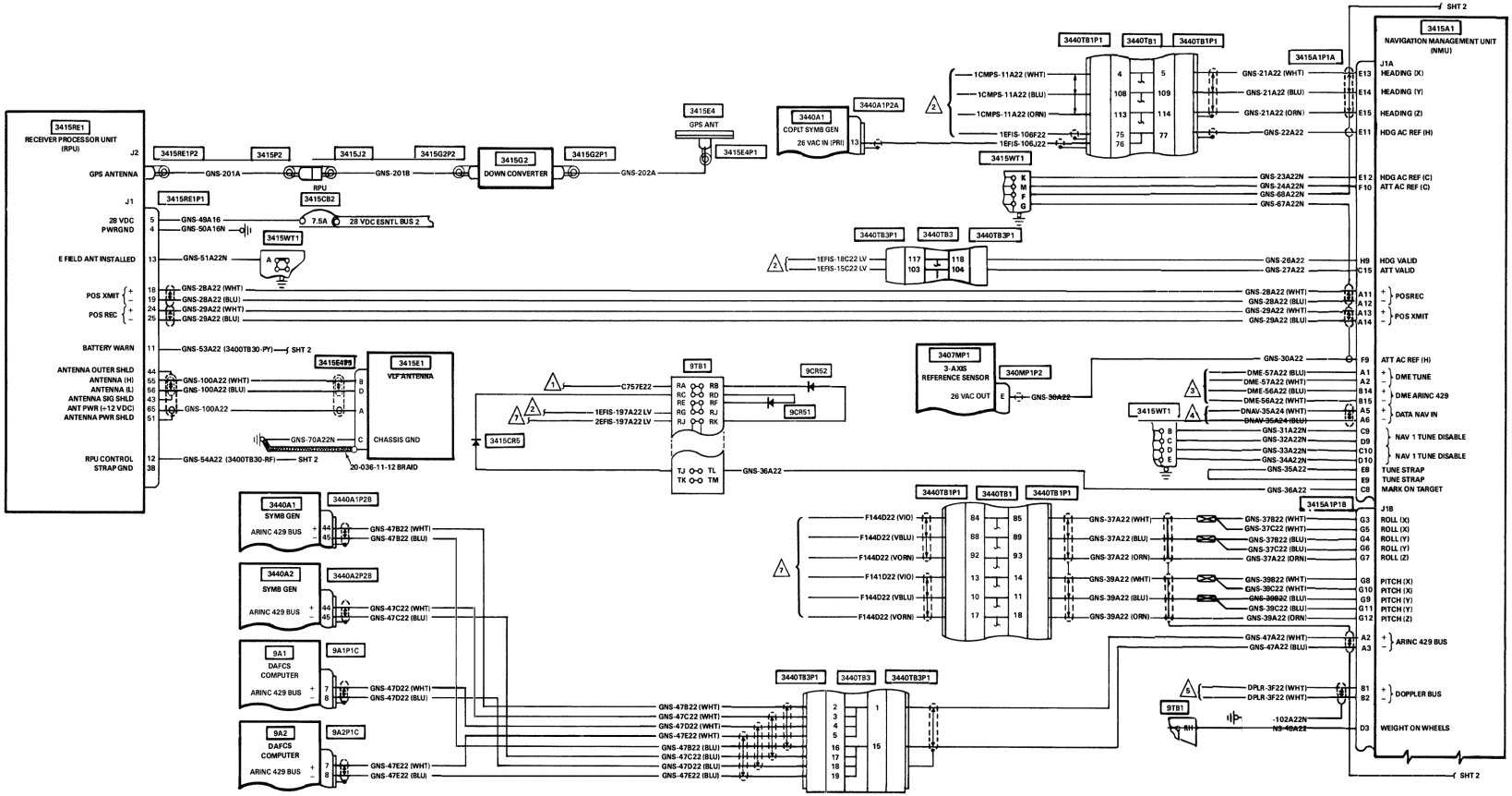
Intentionally blank page



ICN-412-A-970100-B-97499-00008-A-001-01

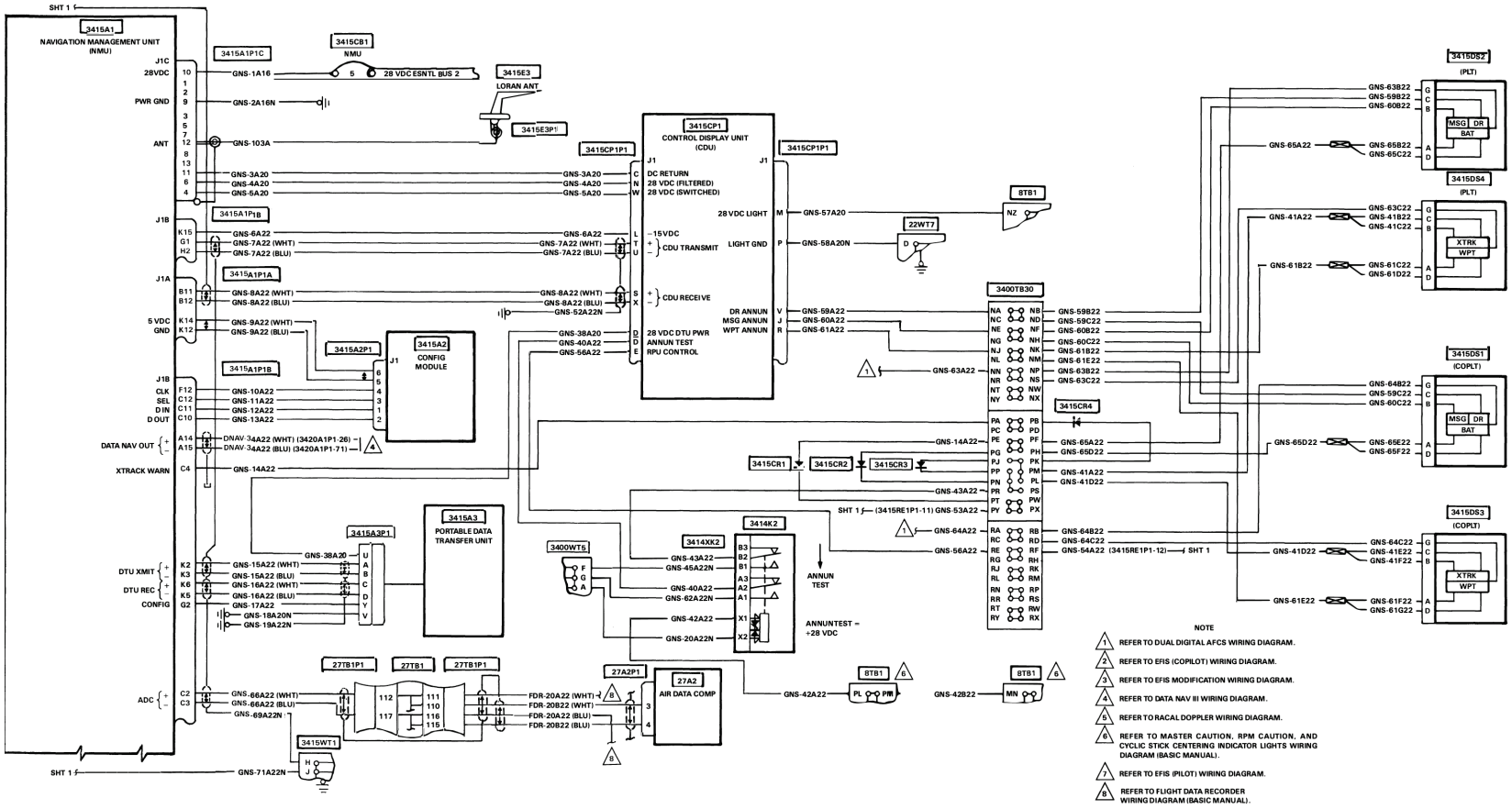
Figure 8. DATA NAV III System Wiring Diagram (Sheet 1 of 1)

Intentionally blank page



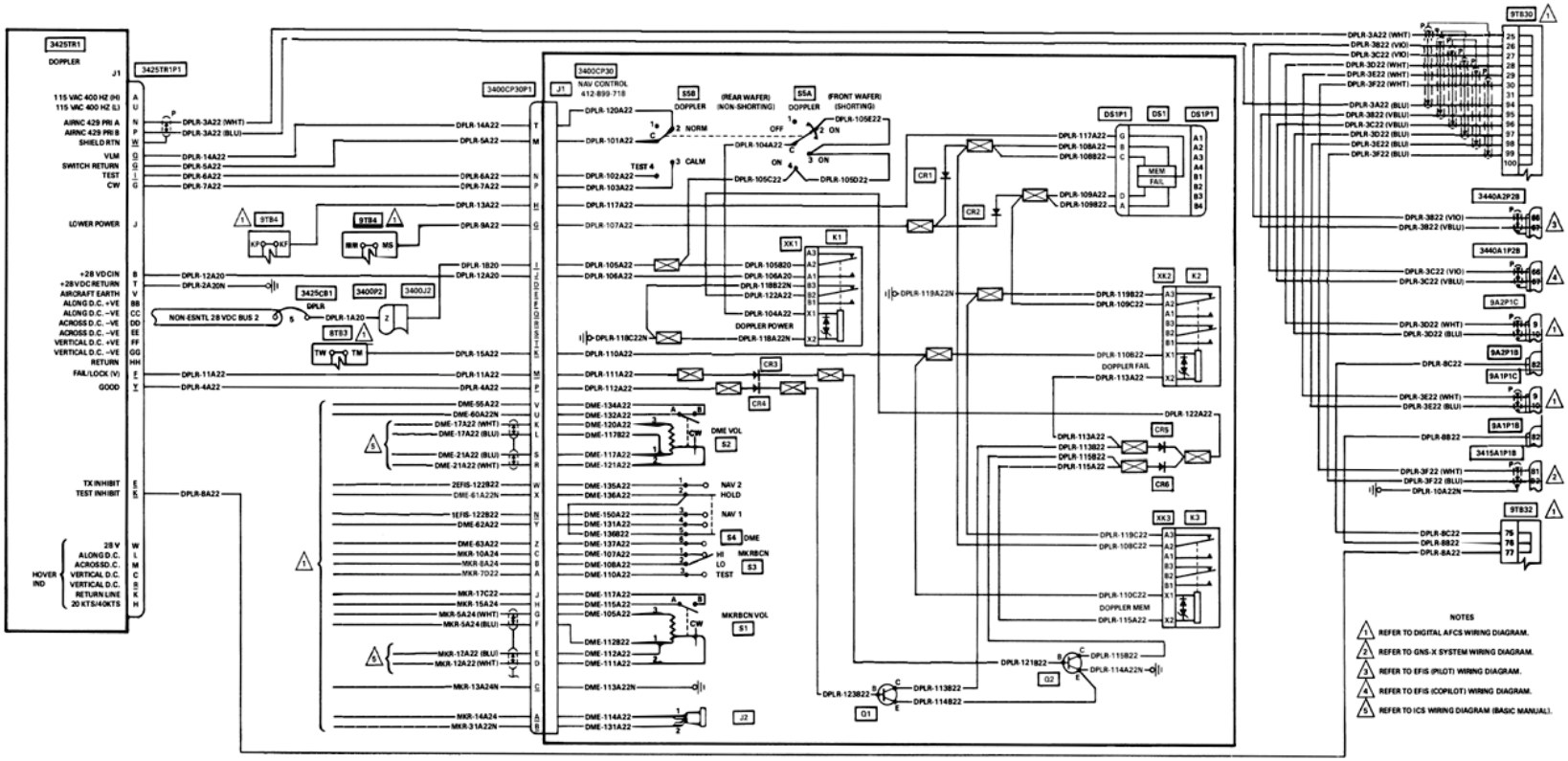
ICN-412-A-970100-B-97499-00009-A-001-01

Figure 9. GNS-X System Wiring Diagram (Sheet 1 of 2)



ICN-412-A-970100-B-97499-00010-A-001-01

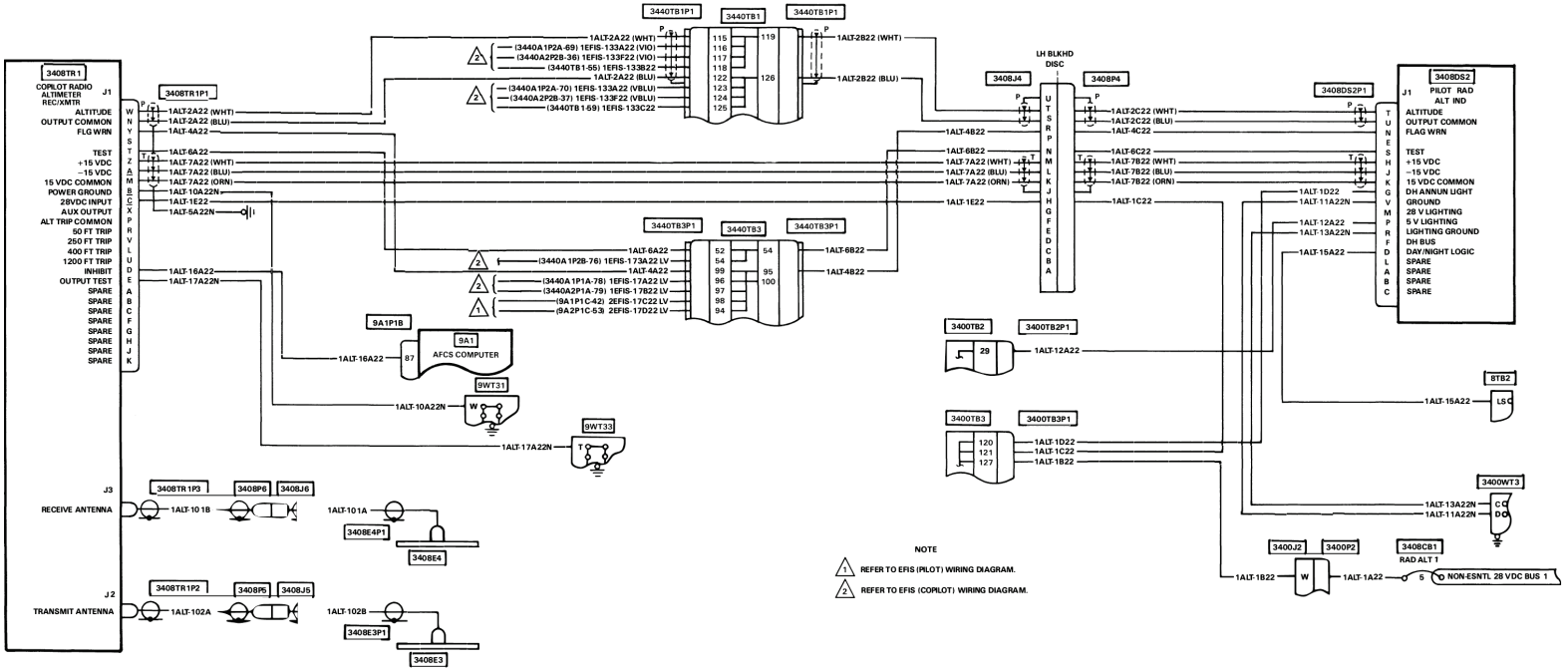
Figure 9. GNS-X System Wiring Diagram (Sheet 2 of 2)



ICN-412-A-970100-B-97499-00011-A-001-01

Figure 10. Doppler Velocity Sensor Wiring Diagram (Sheet 1 of 1)

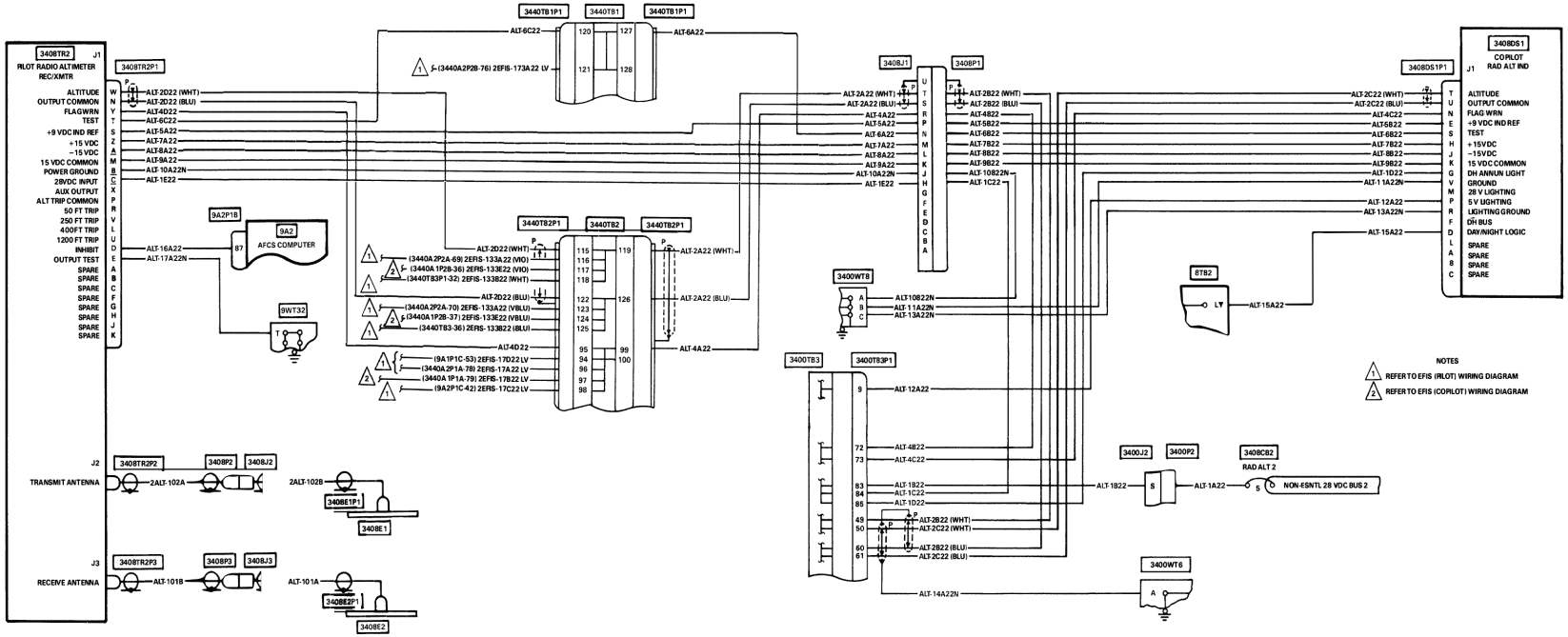
Intentionally blank page



ICN-412-A-970100-B-97499-00012-A-001-01

Figure 11. Radio Altimeter System (Copilot) Wiring Diagram (Sheet 1 of 1)

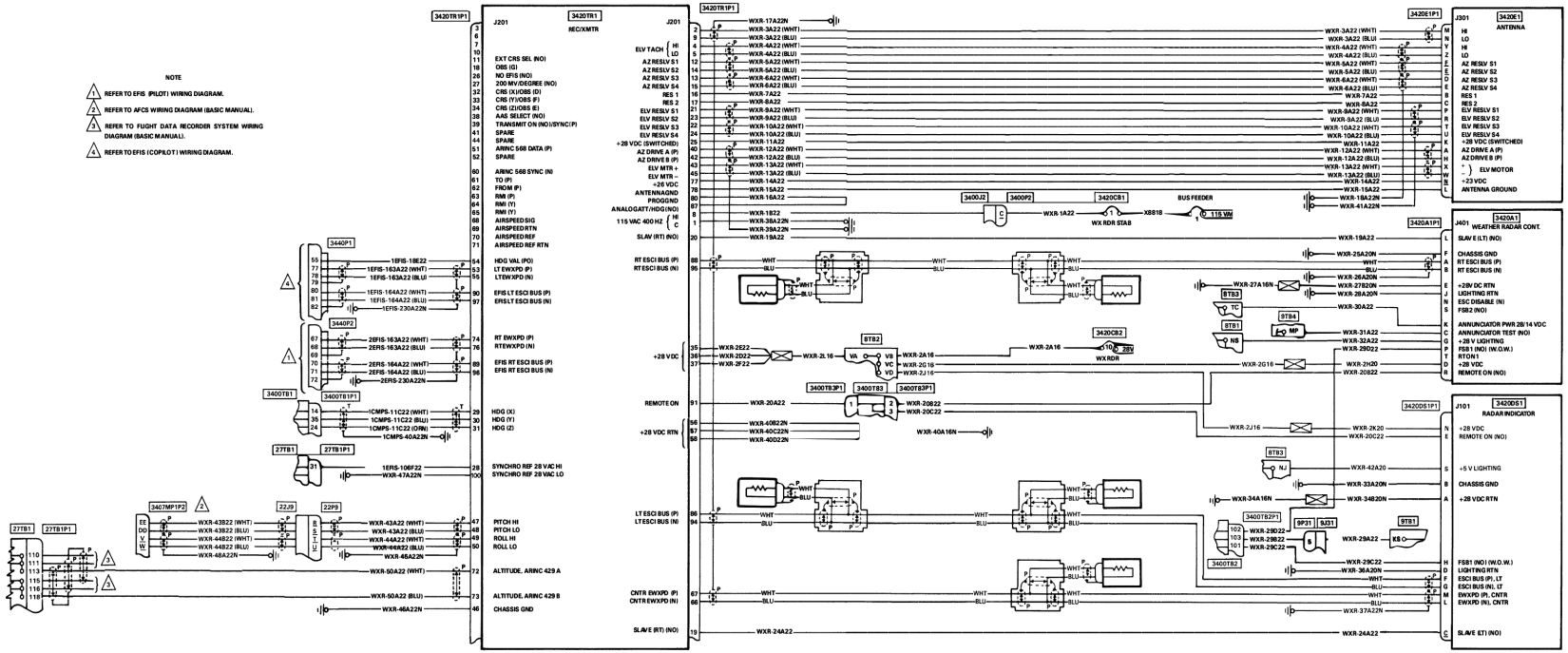
Intentionally blank page



ICN-412-A-970100-B-97499-00013-A-001-01

Figure 12. Radio Altimeter System (Pilot) Wiring Diagram (Sheet 1 of 1)

Intentionally blank page



ICN-412-A-970100-B-97499-00014-A-001-01

Figure 13. Weather Radar System Wiring Diagram (Sheet 1 of 1)

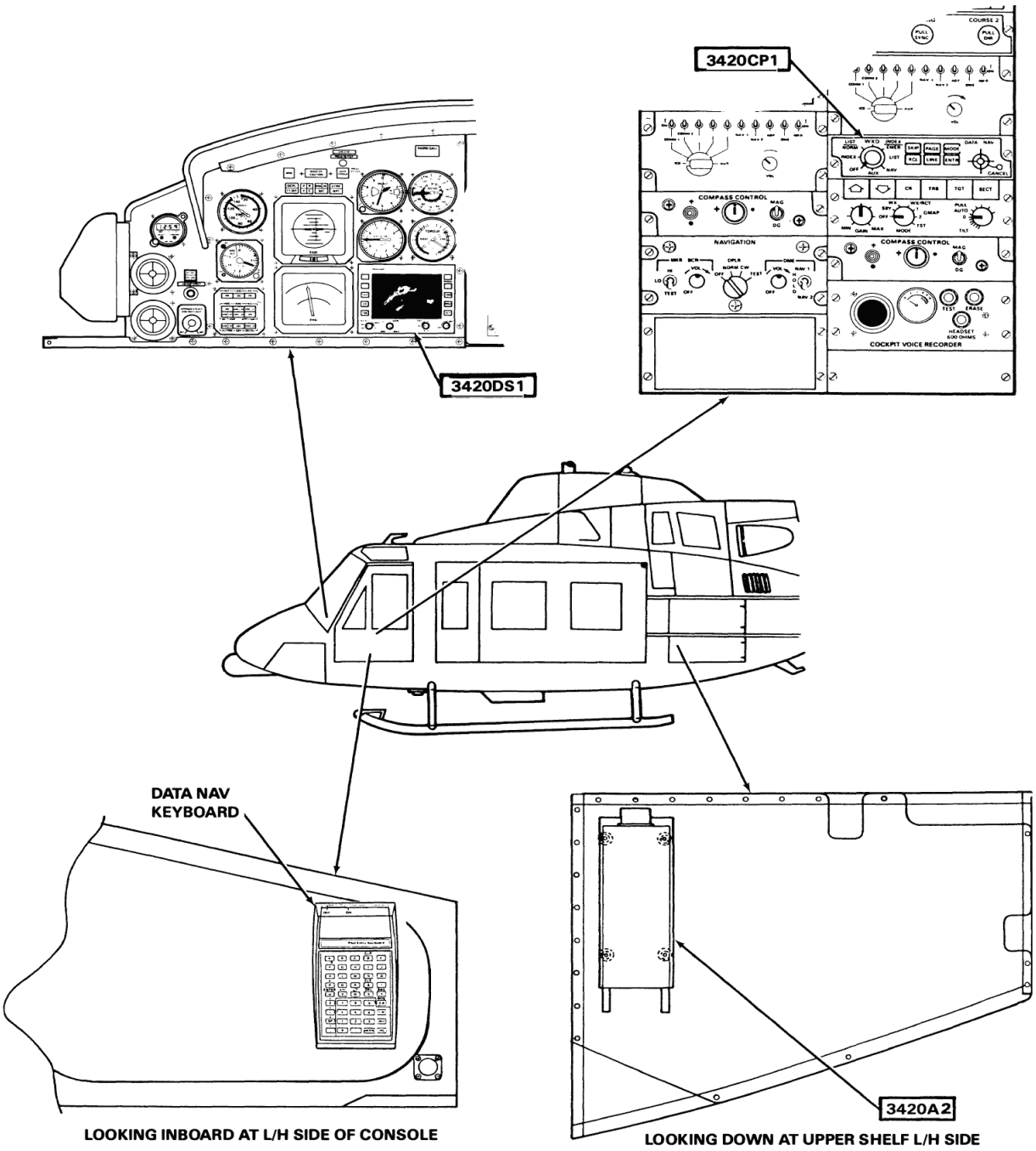
Intentionally blank page

DATA NAV III System - Description

The DATA NAV III system is comprised of the following major components: DATA NAV computer (3420A2), DATA NAV control unit (3420CP1), color indicator (3420DS1), and the DATA NAV keyboard (Figure 1).

The DATA NAV III equipment monitors the navigation system. Navigation information is received by the input/output interface and identified by the input/output processor. Data for direct display is transferred to the display processor. The navigation processor computes range and bearing information from the latitude and longitude figures of the aircraft present position and the navigation system entered waypoints. After computation, range and bearing information plus instructions for waypoint identification is passed to the display processor.

The display processor monitors the radar indicator display parameters (vertical/horizontal reference and range). The display processor presents graphical symbology generated in the DATA NAV III equipment for display in correct position to range and azimuth marks of the radar indicator display.



ICN-412-A-970100-B-97499-00001-A-001-01

Figure 1. DATA NAV III System - Major Component Location (Sheet 1 of 1)

DATA NAV III System - Test Procedure

Procedure

1. Visually inspect equipment and correct any faulty mechanical mountings or electrical connections. Connect external power, adjusted to 27.5 Vdc, to helicopter power receptacle.

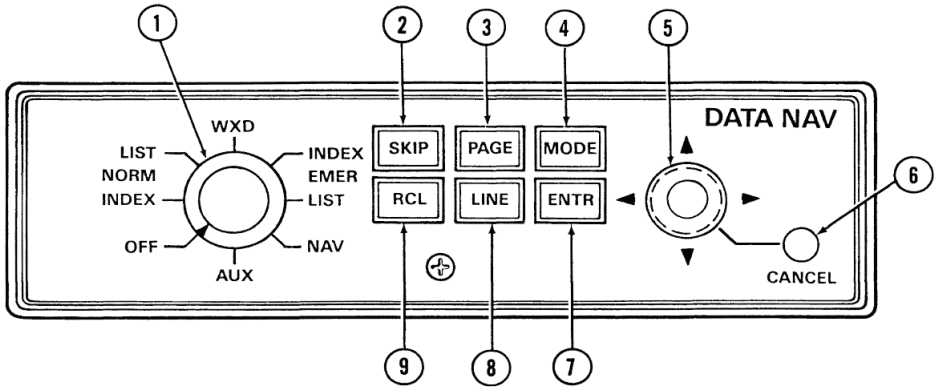
2. Energize power buses per 412-A-97-01-00-03A-340A-A / 02886.

3. Close circuit breaker DATA NAV (3420CB2).



DO NOT OPERATE RADAR SYSTEM, WITHOUT 115VAC 400HZ SUPPLY TO SYSTEM TURNED ON. WITHOUT RECEIVER-TRANSMITTER COOLING BLOWER OPERATING CIRCUIT BREAKER DAMAGE AND/OR AUTOMATIC SHUT DOWN COULD OCCUR.

4. See Figure 1 for DATA NAV control panel assembly switches and features.



1 **OFF/NORM INDEX, NORM LIST/WDX/EMER INDEX, EMER LIST/NAV/AUX***
Rotary switch used to turn DATA NAV III OFF or to turn it on by selection of checklist, weather (WXD), long-range NAVigation, or AUXiliary operation.

2 **SKIP**
Momentary pushbutton switch used to advance from one LINE to next during checklist operation.

3 **PAGE**
Momentary pushbutton switch used to advance from one PAGE to next during checklist operation.

4 **MODE**
Momentary pushbutton switch used in navigation operation to change nav modes in fixed rotation.

5 **Joystick**
Wobble-type control used to move designator forward, backward, left, or right in navigation operation. Initial activation steps designator one step (dot width) in requested direction. If joystick is held in given direction for approximately one second, designator will slew in that direction. When designator comes to rest, latitude/longitude position of designator appears in upper left corner of display.

6 **CANCEL**
Momentary pushbutton switch used in navigation operation to return waypoint designator to home position. With same long-range nav systems, CANCEL will return original waypoint to screen and cancel offset waypoint.

7 **ENTR**
Momentary pushbutton switch used in navigation operation to send latitude/longitude position of designator to long-range nav system. Not all nav systems can use this feature.

8 **LINE**
Momentary pushbutton switch used to advance from one LINE to next during checklist operation.

9 **RCL**
Momentary pushbutton switch used to RECALL first item skipped during checklist operation. Not enabled in INDEXes.

*If the pilot interrupts a nav display by switching to WXD or AUX and then back again to NAV, the nav display mode will be the same as that which was interrupted. Since the checklists and nav map displays are supported by individual memories, the pilot can switch between these functions without disturbing the display as last used.

ICN-412-A-970100-B-97499-00002-A-001-01

Figure 1. DATA NAV Control Panel Assembly (Sheet 1 of 1)

DATA NAV III System - Checklist Operation

Procedure

1. The checklist mode allows the technician to display normal or emergency procedures and to check off specific actions as the technician sequences through the procedure.

2. For typical check lists displays, refer to Honeywell, DATA NAV III manual.

Intentionally blank page

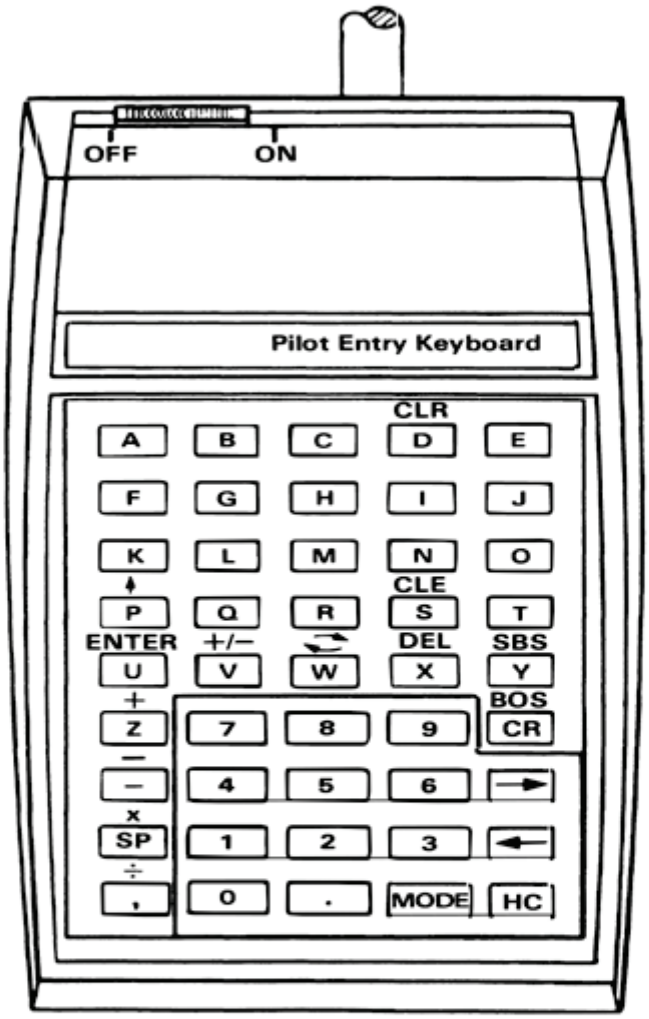
DATA NAV III System - Pilot Entry Keyboard Operation

Procedure

1. Keyboard Controls (see Figure 1).

1.1. The keyboard unit has 45 keys consisting of 26 alphabetics (A through Z); 10 numerics (0 through 9); comma; period; dash (-); backspace (←); forward space (→); carriage return (CR); space (SP); home cursor (HC); and mode. The MODE key selects either the data entry mode or the calculator mode.

1.2. Alternate functions are available for 13 keys: clear (CLR), roll up (↑); clear entry (CLE); enter (ENTER); sign change (+/-); exchange (); delete line (DEL); set beginning of stack (SBS); beginning of stack (BOS); and the arithmetic operations (+, -, x, ÷). These alternate functions are operational only during calculator mode, appear above the affected key.



ICN-412-A-970100-B-97499-00003-A-001-01

Figure 1. Pilot Entry Keyboard (Sheet 1 of 1)

DATA NAV III System - Troubleshooting

Procedure

1. It is assumed that all appropriate circuit breakers are closed, interconnecting cabling is in normal operational condition, and power is applied to the system. (See appropriate wiring diagram.)

Table 1. Troubleshooting - DATA NAV III

INDICATION OF TROUBLE	PROBABLE CAUSE	CORRECTIVE ACTION
	NOTE Refer to Honeywell, DATA NAV III System Description and Installation Manual, Publication Number IB8023133 for system troubleshooting.	

Intentionally blank page

Color Indicator - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Disconnect electrical connector.
2. Loosen captive screws securing color indicator to mount.
3. Remove color indicator.

Intentionally blank page

DATA NAV Computer - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Release fasteners that secure computer to mount.
2. Lift DATA NAV computer free of mount.

Intentionally blank page

DATA NAV Control Unit - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.



THE ON/OFF SWITCH MUST BE IN OFF POSITION BEFORE CONNECTING/DISCONNECTING THE KEYBOARD UNIT. FAILURE TO DO SO WILL BE DETRIMENTAL TO THE LIFE OF THE INTEGRATED CIRCUITS IN THE KEYBOARD UNIT.

1. Loosen spring lock fasteners that secure control unit to pedestal assembly.
2. Lift control unit far enough out of pedestal to reach and disconnect electrical connector.
3. Lift DATA NAV control unit free of pedestal assembly.

Intentionally blank page

Color Indicator - Installation

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Position color indicator in mounting bracket.
2. Tighten captive screws to secure unit.
3. Connect electrical connector.

Intentionally blank page

DATA NAV Computer - Installation

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Position DATA NAV computer in mount.
2. Secure DATA NAV computer to mount by engaging fasteners.

Intentionally blank page

DATA NAV Control Unit - Installation

Procedure



ENSURE ELECTRICAL POWER IS OFF.



THE ON/OFF SWITCH MUST BE IN OFF POSITION BEFORE CONNECTING/DISCONNECTING THE KEYBOARD UNIT. FAILURE TO DO SO WILL BE DETRIMENTAL TO THE LIFE OF THE INTEGRATED CIRCUITS IN THE KEYBOARD UNIT.

1. Connect electrical connector to back of DATA NAV control unit.
2. Position DATA NAV control unit in place on pedestal assembly.
3. Secure DATA NAV control unit to pedestal assembly with spring lock fasteners.

Intentionally blank page

DATA NAV III System - Inspection

Procedure

1. Inspect DATA NAV III system for completeness, proper installation, cleanliness, and secure mounting of each component.
2. Inspect switches and controls on DATA NAV III system components for proper mechanical action.
3. Inspect components to ensure panel lights illuminate with electrical power on.
4. Inspect electrical connections for corroded or bent pins and proper mating, interconnecting cable for frayed or broken insulation, and circuit breakers for proper operation.

Intentionally blank page

DATA NAV III System - Cleaning

Procedure

1. Remove moisture and loose dirt with a soft, clean cloth.
2. Remove dust and dirt from system components with a soft, clean cloth.



CLEANING COMPOUND FUMES ARE FLAMMABLE AND TOXIC.

3. Remove grease, fungus, and ground-in dirt with a soft, clean cloth dampened with approved cleaning compound.
4. Remove dirt from connectors with a soft bristle brush; remove moisture with a clean, dry cloth.

Intentionally blank page

DATA NAV III System - Repair or Replacement

Procedure

1. Tighten or replace loose or cracked control knobs or system components.
2. Replace defective panel light bulbs.
3. Replace defective or inoperative units.
4. Further repairs, other than removal or replacement, must be conducted at an authorized repair facility.

Intentionally blank page

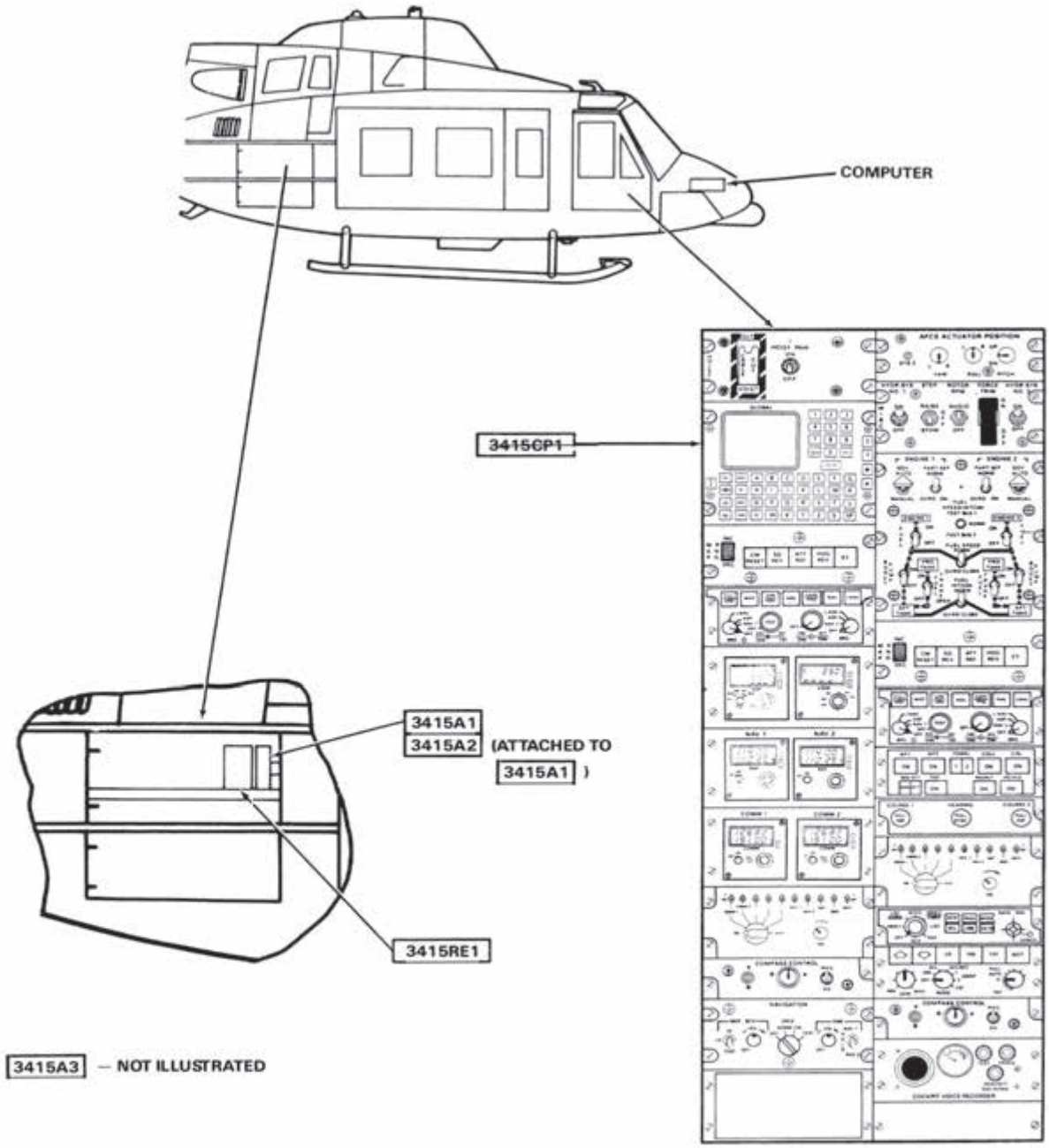
Navigation System (GNS-X) - Description

The GNS-X flight management system is comprised of the following major components: GNS-X navigation management unit (3415A1), GNS-X control display unit (3415CP1), GNS-X receiver processor unit (3415RE1), GNS-X configuration module (3415A2), and a portable data transfer unit (3415A3) (Figure 1).

The GNS-X interfaces with DAFCS, EFIS, DATA NAV III System, and Doppler velocity sensor.

In addition to standard navigation functions, the GNS-X navigation management system when used for search and rescue operations has the following capabilities added:

- Special Mission Programing (Search Patterns).
- Conversion (and compensation) of Doppler body axis velocities to inertial. Input of pitch and roll for the conversion.
- Use of Doppler velocities in composite navigation filter.
- Changes to support Mark-On-Target mode of the DAFCS autopilot.
- Storage of 10 Mark-on-Target points.
- Operations to allow renaming of Mark-on-Target points.
- Independent NAV computation based only on doppler velocities to support the Mark-on-Target and EFIS display of range/bearing to the Mark-on-Target location.
- Changes to enhance low speed operation of the system.



ICN-412-A-970100-B-97499-00004-A-001-01

Figure 1. Navigation System - Major Component Location (Sheet 1 of 1)

Navigation System (GNS-X) - Test Procedure

Procedure

1. Visually inspect equipment and correct any faulty mechanical mountings or electrical connections. Connect external power, adjusted to 27.5 Vdc, to helicopter power receptacle.

2. Energize power buses per 412-A-97-01-00-03A-340A-A / 02886.

3. Perform test procedure per Global-Wulfsburg Systems GNS-X Operator's Manual.

Intentionally blank page

Navigation System (GNS-X) - Troubleshooting

Procedure

1. It is assumed that all appropriate circuit breakers are closed, interconnecting cabling is in normal operational condition, and power is applied to the system. (See appropriate wiring diagram.)

Table 1. Troubleshooting - GNS - X

INDICATION OF TROUBLE	PROBABLE CAUSE	CORRECTIVE ACTION
	NOTE Refer to Global-Wulfsburg System GNS-X manual for troubleshooting.	

Intentionally blank page

Air Data Computer - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Disconnect electrical connectors.
2. Remove and retain screws and other mounting hardware.
3. Remove air data computer.

Intentionally blank page

Navigation Management Unit (NMU) - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Disengage NMU retaining handle.
2. Remove NMU from mount.

Intentionally blank page

Configuration Module - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.

NOTE

Configuration module is attached to rear of navigation management unit.

1. Remove navigation management unit.
2. Remove and retain screws and other mounting hardware securing configuration module to the NMU mounting tray.
3. Remove configuration module.

Intentionally blank page

Control Display Unit - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Loosen spring lock fasteners that secure control display unit to pedestal assembly.
2. Lift control display unit far enough out of pedestal assembly to reach and disconnect electrical connector from back of unit.
3. Lift control display unit free of pedestal assembly.

Intentionally blank page

GPS Downconverter - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Disconnect electrical connectors.
2. Remove and retain screws and other mounting hardware.
3. Remove down converter.

Intentionally blank page

Receiver Processor Unit (RPU) - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Disengage RPU retaining handle.
2. Remove RPU from mount.

Intentionally blank page

VLF Antenna - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Remove attaching screws from antenna mount.
2. Cut and remove sealant between airfoil and antenna mount.
3. Disconnect antenna coaxial cable connector from antenna base and protect open end with cap and electrical tape.

Intentionally blank page

GPS Antenna - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Remove attaching screws from antenna mount.
2. Cut and remove sealant between airfoil and antenna mount.
3. Disconnect antenna coaxial cable connector from antenna base and protect open end with cap and electrical tape.

Intentionally blank page

Loran-C Antenna - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Remove attaching screws from antenna mount.
2. Cut and remove sealant between airfoil and antenna mount.
3. Disconnect antenna coaxial cable connector from antenna base and protect open end with cap and electrical tape.

Intentionally blank page

Air Data Computer - Installation

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Position and secure air data computer with attaching screws and mounting hardware.
2. Connect electrical connectors.

Intentionally blank page

Configuration Module - Installation

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Position and secure configuration module with attaching screws and mounting hardware to rear of NMU mounting tray.
2. Install navigation management unit.

Intentionally blank page

Navigation Management Unit (NMU) - Installation

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Position NMU in mount.
2. Secure unit by engaging NMU retaining handle.

Intentionally blank page

Control Display Unit - Installation

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Connect electrical connector to back of control display unit.
2. Position control display unit in place on pedestal assembly.
3. Secure control display unit to pedestal assembly with spring lock fasteners.

Intentionally blank page

GPS Downconverter - Installation

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Position and secure down converter with attaching screws and mounting hardware.
2. Connect electrical connectors.

Intentionally blank page

Receiver Processor Unit (RPU) - Installation

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Position RPU in mount.
2. Secure unit by engaging RPU retaining handle.

Intentionally blank page

VLF Antenna - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealant	C-308	AR	

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Remove protective cap from antenna coaxial cable connector.
2. Connect antenna cable to connector receptacle on base of antenna.
3. Position antenna on its mounting surface and secure with antenna base mounting screws.
4. Apply sealant (C-308) bead to mounting surface at antenna base periphery.

Intentionally blank page

GPS Antenna - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealant	C-308	AR	

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Remove protective cap from antenna coaxial cable connector.
2. Connect antenna cable to connector receptacle on base of antenna.
3. Position antenna on its mounting surface and secure with antenna base mounting screws.
4. Apply sealant (C-308) bead to mounting surface at antenna base periphery.

Intentionally blank page

Loran-C Antenna - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealant	C-308	AR	

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Remove protective cap from antenna coaxial cable connector.
2. Connect antenna cable to connector receptacle on base of antenna.
3. Position antenna on its mounting surface and secure with antenna base mounting screws.
4. Apply sealant (C-308) bead to mounting surface at antenna base periphery.

Intentionally blank page

Navigation System (GNS-X) - Inspection

Procedure

1. Inspect navigation system for completeness, proper installation, cleanliness, and secure mounting of each component.
2. Inspect switches and controls on navigation system components for proper mechanical action.
3. Inspect components to ensure panel lights illuminate with electrical power on.
4. Inspect electrical connections for corroded or bent pins and proper mating, interconnecting cable for frayed or broken insulation, and circuit breakers for proper operation.

Intentionally blank page

Navigation System (GNS-X) - Cleaning

Procedure

1. Remove moisture and loose dirt with a soft, clean cloth.
2. Remove dust and dirt from system components with a soft, clean cloth.



CLEANING COMPOUND FUMES ARE FLAMMABLE AND TOXIC.

3. Remove grease, fungus, and ground-in dirt with a soft, clean cloth dampened with approved cleaning compound.
4. Remove dirt from connectors with a soft bristle brush; remove moisture with a clean, dry cloth.

Intentionally blank page

Navigation System (GNS-X) - Repair or Replacement

Procedure

1. Tighten or replace loose or cracked control knobs or system components.
2. Replace defective panel light bulbs.
3. Replace defective or inoperative units.
4. Further repairs, other than removal or replacement, must be conducted at an authorized repair facility.

Intentionally blank page

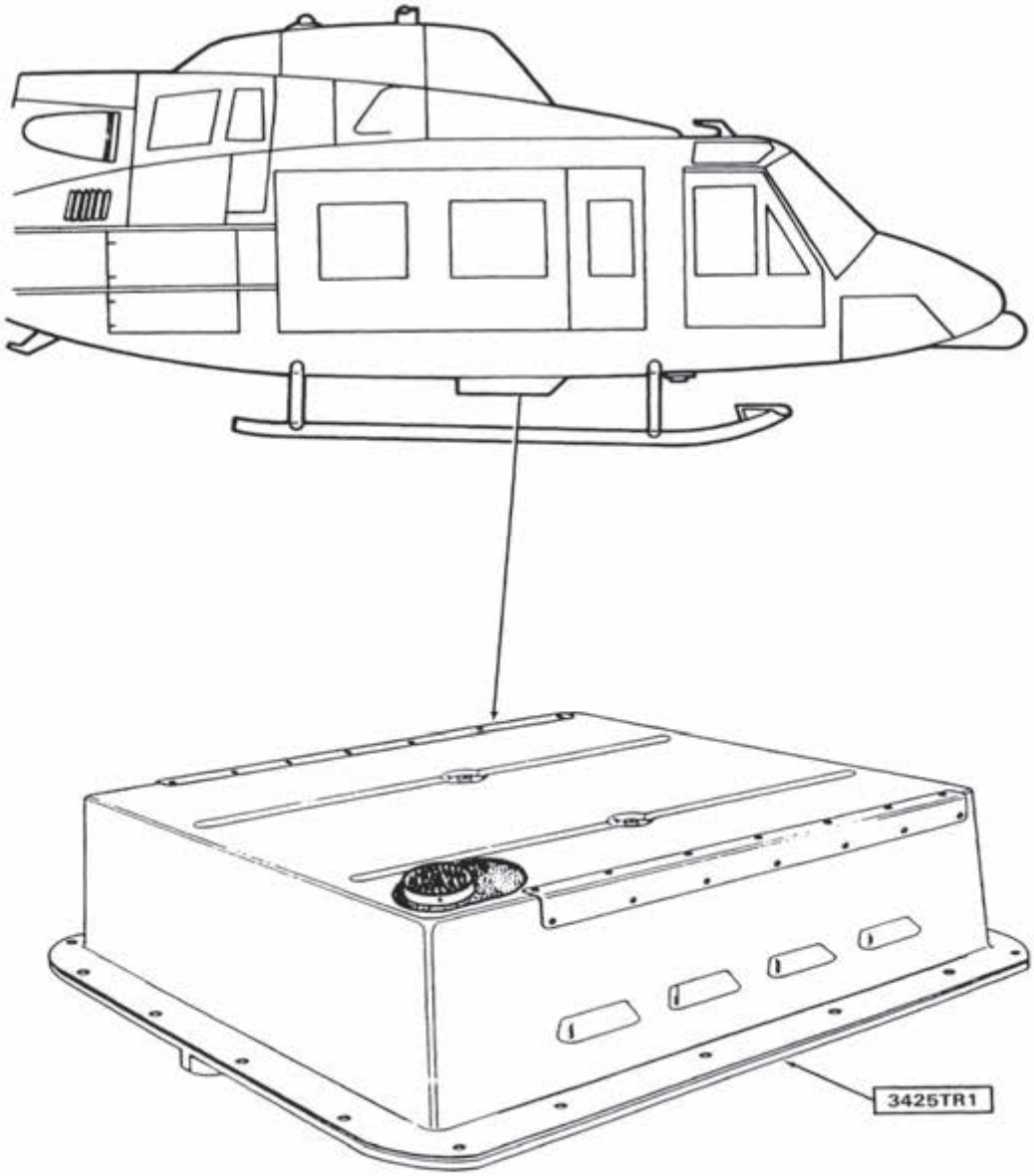
Doppler Velocity Sensor - Description

The Doppler sensor unit (3425TR1), contains transmitting and receiving antennae, a solid state transmitter and receiver, and the electronic circuits required to compute and output aircraft velocities, in the three axes of aircraft movement, to the computer of an aircraft navigation system (Figure 1).

The Doppler sensor unit provides velocity data to the computer of the helicopters navigation system.

The Doppler sensor unit uses two modes of transmission: interrupted continuous wave, and continuous wave. During normal operation in the interrupted continuous wave mode a significant reduction in signals from rain, when flying in or above the rain, is achieved.

The Doppler sensor unit transmits four beam frequencies that are continuously checked for consistency with each other and to detect the tracking of a false signal in any beam. During approach to and pull-away from a hover, the Doppler sensor unit checks the signal strengths of the four beams and the axis velocities in order to detect false tracking. This is known as Valid Lock Monitoring.



ICN-412-A-970100-B-97499-00005-A-001-01

Figure 1. Doppler - Major Component Location (Sheet 1 of 1)

Doppler Velocity Sensor - Test Procedure

Procedure

1. Visually inspect equipment and correct any faulty mechanical mountings or electrical connections. Connect external electrical power, adjusted to 27.5 Vdc, to helicopter power receptacle.

2. Energize power buses per 412-A-97-01-00-03A-340A-A / 02886.

3. Close circuit breakers DPLR (3425CB1), NMU (3415CB1), and RPU (3415CB2).

4. Close EFIS No. 1 circuit breakers ADI (3440CB1), HSI (3440CB3), SG/DL (3440CB5), EFIS 26V NO. 1 (3440CB7), BLWR (3440CB9), and EFIS No. 2 circuit breakers ADI (3440CB2), HSI (3440CB4), SG/DL (3440CB6), EFIS 26V NO. 2 (3440CB8), and BLWR (3440CB10).

5. Close circuit breakers AFCS 26V NO. 1 (9CB4), RATE GYRO NO.1 (9CB31), AFCS NO.1 PWR (9CB31), AFCS NO. 1 ADD/CAUT (9CB32), and FORCE TRIM (9CB3).

6. Close circuit breakers AFCS 26V NO. 2 (9CB7), RATE GYRO NO. 2 (9CB30), AFCS NO . 2 PWR (9CB2), AFCS NO. 2 CAUT (9CB33), and ENC ALT (3410CB1).

7. Set EFIS MASTER switch to ON.

8. Set NAV control panel, DPLR switch to NORM.

9. After 30 seconds check operation of Doppler by pressing HOV switch on EFIS display controller. Verify Doppler is valid.

NOTE

Amber Doppler display and no velocity bars - use light lateral cyclic inputs to gently rock helicopter. Velocity bars are displayed and amber Doppler extiguished, validates Doppler.

10. Failure of the Doppler is indicated by 'FAIL' being displayed on the NAV control panel.

Intentionally blank page

Doppler Velocity Sensor - Troubleshooting

Procedure

1. It is assumed that all appropriate circuit breakers are closed, interconnecting cabling is in normal operational condition, and power is applied to the system. (See appropriate wiring diagram.)

Table 1. Troubleshooting - Doppler System

INDICATION OF TROUBLE	PROBABLE CAUSE	CORRECTIVE ACTION
Operational checkout indicates a 'FAIL' condition.	Doppler unit failure.	Replace Doppler unit.
	Defective nav control unit.	Replace nav control unit.

Intentionally blank page

Navigation Control Unit - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Loosen spring lock fasteners that secure mounting panel to pedestal assembly.
2. Lift navigation control unit, far enough out of pedestal assembly to reach, and disconnect electrical connector from back of control unit.
3. Lift navigation control unit free of pedestal assembly.

Intentionally blank page

Doppler Velocity Sensor - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Disconnect electrical connector.
2. Support Doppler velocity sensor and remove attaching screws.
3. Remove Doppler velocity sensor from helicopter.

Intentionally blank page

Navigation Control Unit - Installation

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Connect electrical connector to back of navigation control unit.
2. Position navigation control unit in place on pedestal assembly.
3. Secure navigation control unit to pedestal assembly with spring lock fasteners.

Intentionally blank page

Doppler Velocity Sensor - Installation

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Support Doppler velocity sensor and secure with attaching screws.
2. Connect electrical connectors.

Intentionally blank page

Doppler Velocity Sensor - Inspection

Procedure

1. Inspect Doppler system for completeness, proper installation, cleanliness, and secure mounting of each component.
2. Inspect switches and controls on Doppler system components for proper mechanical action.
3. Inspect components to ensure panel lights illuminate with electrical power on.
4. Inspect electrical connections for corroded or bent pins and proper mating, interconnecting cable for frayed or broken insulation, and circuit breakers for proper operation.

Intentionally blank page

Doppler Velocity Sensor - Cleaning

Procedure

1. Remove moisture and loose dirt with a soft, clean cloth.
2. Remove dust and dirt from system components with a soft, clean cloth.



CLEANING COMPOUND FUMES ARE FLAMMABLE AND TOXIC.

3. Remove grease, fungus, and ground-in dirt with a soft, clean cloth dampened with approved cleaning compound.
4. Remove dirt from connectors with a soft bristle brush; remove moisture with a clean, dry cloth.

Intentionally blank page

Doppler Velocity Sensor - Repair or Replacement

Procedure

1. Tighten or replace loose or cracked control knobs or system components.
2. Replace defective panel light bulbs.
3. Replace defective or inoperative units.
4. Further repairs, other than removal or replacement, must be conducted at an authorized repair facility.

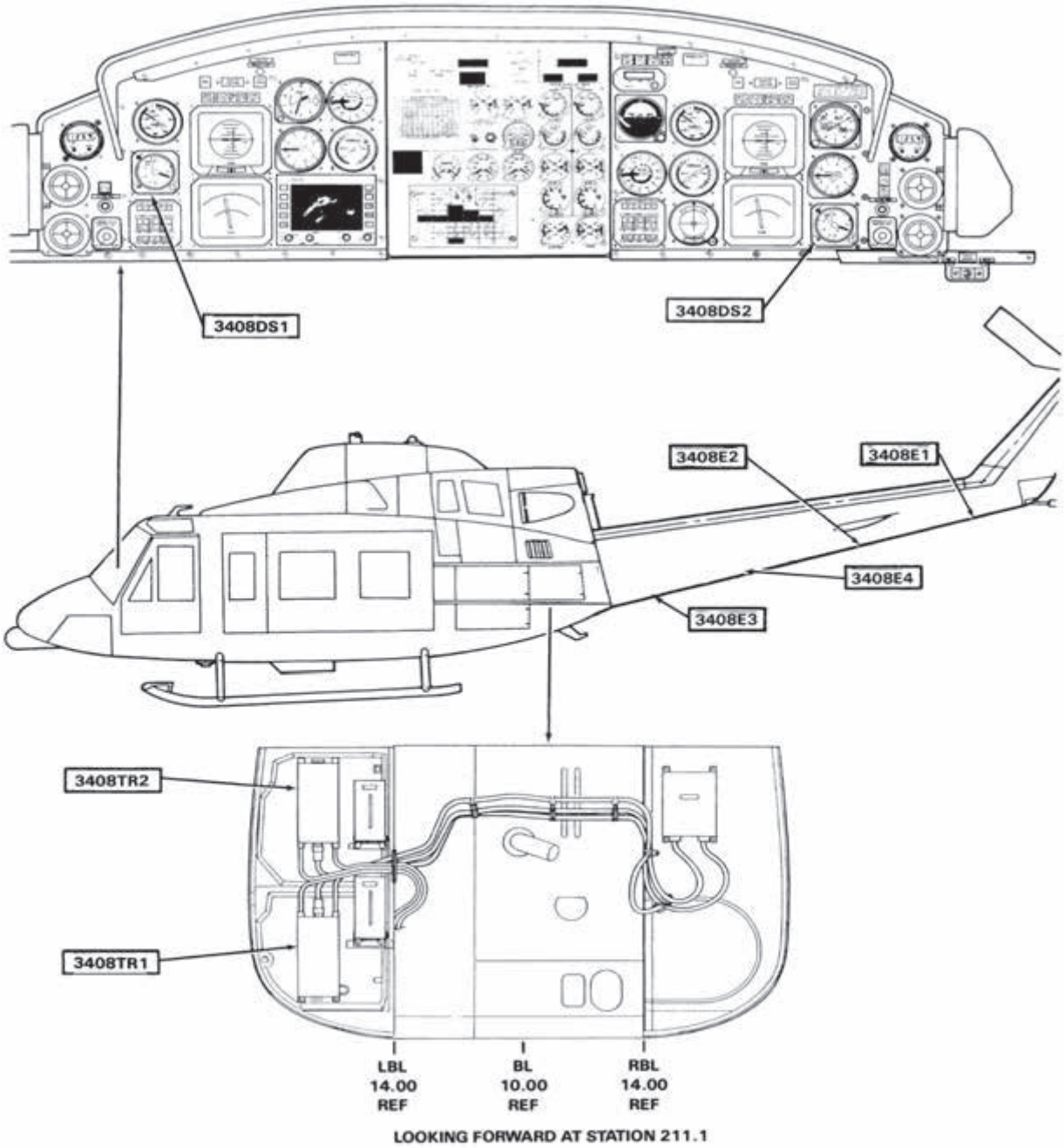
Intentionally blank page

Radio Altimeter System - Description

The radio altimeter system is comprised of the radio altimeter receiver/transmitters (3408TR1 and 3408TR2) and the radio altimeter indicator (3408DS1 and 3408DS2) (Figure 1).

The primary display for the No. 1 RAD ALT is on the copilot EADI and EHSI. The secondary display is the pilot electromechanical indicator. The primary display for the No. 2 RAD ALT is the pilot EADI and EHSI. The secondary display is the copilot electromechanical indicator.

The radio altimeter system provides altitude display of 0 to 1500 feet. The indicator has, in addition to the altitude display, a red power OFF warning flag, a TEST switch (button) for confidence testing, a Decision Height (DH) knob for setting a DH index, and a DH warning annunciator. Two antennas are used to allow simultaneous reception and transmission.



ICN-412-A-970100-B-97499-00006-A-001-01

Figure 1. Radio Altimeter System - Major Component Location (Sheet 1 of 1)

Radio Altimeter System - Test Procedure

Procedure

1. Visually inspect equipment and correct any faulty mechanical mountings or electrical connections. Connect external electrical power, adjusted to 27.5 Vdc, to helicopter power receptacle.

2. Energize power buses per 412-A-97-01-00-03A-340A-A / 02886.

3. Close RAD ALT 1 and RAD ALT 2 circuit breakers (3408CB1 and 3408CB2).



DO NOT ENERGIZE THE SYSTEM WITH TRANSMIT ANTENNA DISCONNECTED.

NOTE

Positioning the DH SET knob to 'X' deactivates the DH function and the DH warning annunciator.

4. Rotate the DH SET knob on the radio altimeter indicator (3808DS1 and 3408DS2) to set the decision height index to "2" (200 feet). The indicator pointer should immediately mask. After approximately 35 seconds warmup, the red power OFF flag on the indicator should mask, the DH annunciator should illuminate, and the indicator pointer should indicate 0 ± 5 feet. If this accuracy is not indicated, allow a 2 minute stabilization period then adjust the zero height control, located below the 3408TR1J1 and 3408TR2J2 connector of the radio altimeter receiver-transmitter (3408TR1 and 3408TR2). Remove hole plug and adjust zero height control for a pointer indication of 0 ± 5 feet.

5. Press and hold the TEST button on the indicator. The pointer should indicate 100 ± 20 feet, the DH annunciator should remain lighted, and the power OFF flag should illuminate.

6. Release the TEST button. The pointer should return to 0 ± 5 feet, and the DH annunciator should remain lighted; the power OFF flag should mask.

NOTE

The DH index may be positioned to below zero feet to deactivate the DH function and annunciator.

7. Position the DH index below zero and observe that the DH annunciator extinguishes.

8. Turn off all equipment, open all circuit breakers and disconnect the external power source.

Intentionally blank page

Radio Altimeter System - Troubleshooting

Procedure

1. It is assumed that all appropriate circuit breakers are closed, interconnecting cabling is in normal operational condition, and power is applied to the system. (See appropriate wiring diagram.)

Table 1. Troubleshooting - Radio Altimeter System

INDICATION OF TROUBLE	PROBABLE CAUSE	CORRECTIVE ACTION
Power OFF flag remains in view with pointer remaining in zero position after warmup.	Defective radio altimeter receiver-transmitter.	Replace radio altimeter receiver-transmitter.
	Defective radio altimeter indicator.	Replace radio altimeter indicator.
	Defective power wiring.	Repair or replace wiring.
Power OFF flag in view with pointer behind the mask after warmup.	Defective radio altimeter receiver-transmitter.	Replace radio altimeter receiver-transmitter.
	Defective radio altimeter antenna.	Replace radio altimeter antenna(s) (one or both).
	Defective coaxial cables or connectors.	Repair or replace coaxial cables or connectors.
Pointer remains behind mask but responds correctly to pressed TEST condition, after power OFF flag is masked from view.	Defective radio altimeter receiver-transmitter.	Replace radio altimeter receiver-transmitter.
	Defective coaxial cables or antenna.	Replace radio altimeter antenna(s) (one or both) or cable(s).
	NOTE TEST button tolerance is 100 ± 20 feet.	
Pointer fails to respond, or responds incorrectly, to the TEST condition, after power OFF flag is masked from view.	Defective radio altimeter receiver-transmitter.	Replace radio altimeter receiver-transmitter.
	Defective radio altimeter indicator.	Replace radio altimeter indicator.
	Defective power wiring.	Repair or replace wiring.
Indicator shows TEST altitude at all times.	Defective radio altimeter indicator.	Replace radio altimeter indicator.
	Defective radio altimeter receiver-transmitter.	Replace radio altimeter receiver-transmitter.
	NOTE Ground test tolerance is plus or minus 5 feet. (This may be adjusted with zero height control located on top of receiver-transmitter.)	
Altimeter pointer reading out of tolerance. All other indications normal.	Defective radio altimeter receiver-transmitter.	Replace radio altimeter receiver-transmitter.
	Defective radio altimeter indicator.	Replace radio altimeter indicator.
Low warning lamp inoperative. (DH SET knob not positioned to “X”.)	Defective radio altimeter indicator or lamp.	Replace radio altimeter indicator or lamp.
	Defective wiring.	Repair or replace wiring.

Intentionally blank page

Radio Altimeter Receiver/Transmitter - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Disconnect electrical connectors.
2. Remove and retain mounting screws and other hardware.
3. Remove radio altimeter receiver/transmitter.

Intentionally blank page

Radio Altimeter Indicator - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Disconnect cable connector from receptacle on rear of indicator.
2. Loosen mounting clamp and remove indicator from instrument panel.

Intentionally blank page

Radio Altimeter Antenna(s) - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Remove screws attaching tailboom access panel.
2. Lower access panel and disconnect antenna connector.
3. Remove jumper assembly.
4. Remove screws attaching antenna to its panel and remove antenna.

Intentionally blank page

Radio Altimeter Receiver/Transmitter - Installation

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Position and secure radio altimeter with mounting screws and other hardware.
2. Connect electrical connectors.

Intentionally blank page

Radio Altimeter Indicator - Installation

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Carefully position indicator in instrument panel and tighten screw of mounting clamp.



EXCESSIVE CLAMP TENSION MAY DEFORM INDICATOR AND CAUSE ERRONEOUS READING.

2. Connect cable connector to receptacle on rear of indicator.

Intentionally blank page

Radio Altimeter Antenna(s) - Installation

Procedure



ENSURE ELECTRICAL POWER IS OFF.

NOTE

Correct antenna orientation is required for proper system operation. Assure forward two antennas point to each other. Assure aft two antennas also point to each other.

1. Position antenna in place on access panel and install mounting screws.
2. Lift panel into approximate position and connect antenna connector.
3. Install jumper assembly.
4. Position access panel in place and install mounting screws.

Intentionally blank page

Radio Altimeter System - Inspection

Procedure

1. Inspect radio altimeter system for completeness, proper installation, cleanliness, and secure mounting of each component.
2. Inspect switches and controls on radio altimeter system components for proper mechanical action.
3. Inspect components to ensure panel lights illuminate with electrical power on.
4. Inspect electrical connections for corroded or bent pins and proper mating, interconnecting cable for frayed or broken insulation, and circuit breakers for proper operation.

Intentionally blank page

Radio Altimeter System - Cleaning

Procedure

1. Remove moisture and loose dirt with a soft, clean cloth.
2. Remove dust and dirt from system components with a soft, clean cloth.



CLEANING COMPOUND FUMES ARE FLAMMABLE AND TOXIC.

3. Remove grease, fungus, and ground-in dirt with a soft, clean cloth dampened with approved cleaning compound.
4. Remove dirt from connectors with a soft bristle brush; remove moisture with a clean, dry cloth.

Intentionally blank page

Radio Altimeter System - Repair or Replacement

Procedure

1. Tighten or replace loose or cracked control knobs or system components.
2. Replace defective panel light bulbs.
3. Replace defective or inoperative units.
4. Further repairs, other than removal or replacement, must be conducted at an authorized repair facility.

Intentionally blank page

Weather Radar System - Description

The weather radar system consists of the following major components: receiver/transmitter unit (3420TR1), antenna (3420E1), indicator (3420DS1), and radar controller (3420A1) (Figure 1).

The weather radar system is a high-resolution, high-power airborne ground/sea system optimized for over-water search and rescue missions.

Dual operation is allowed within any combination of two control sources (two radar controllers, two indicators or a mix) along with a left and right display.

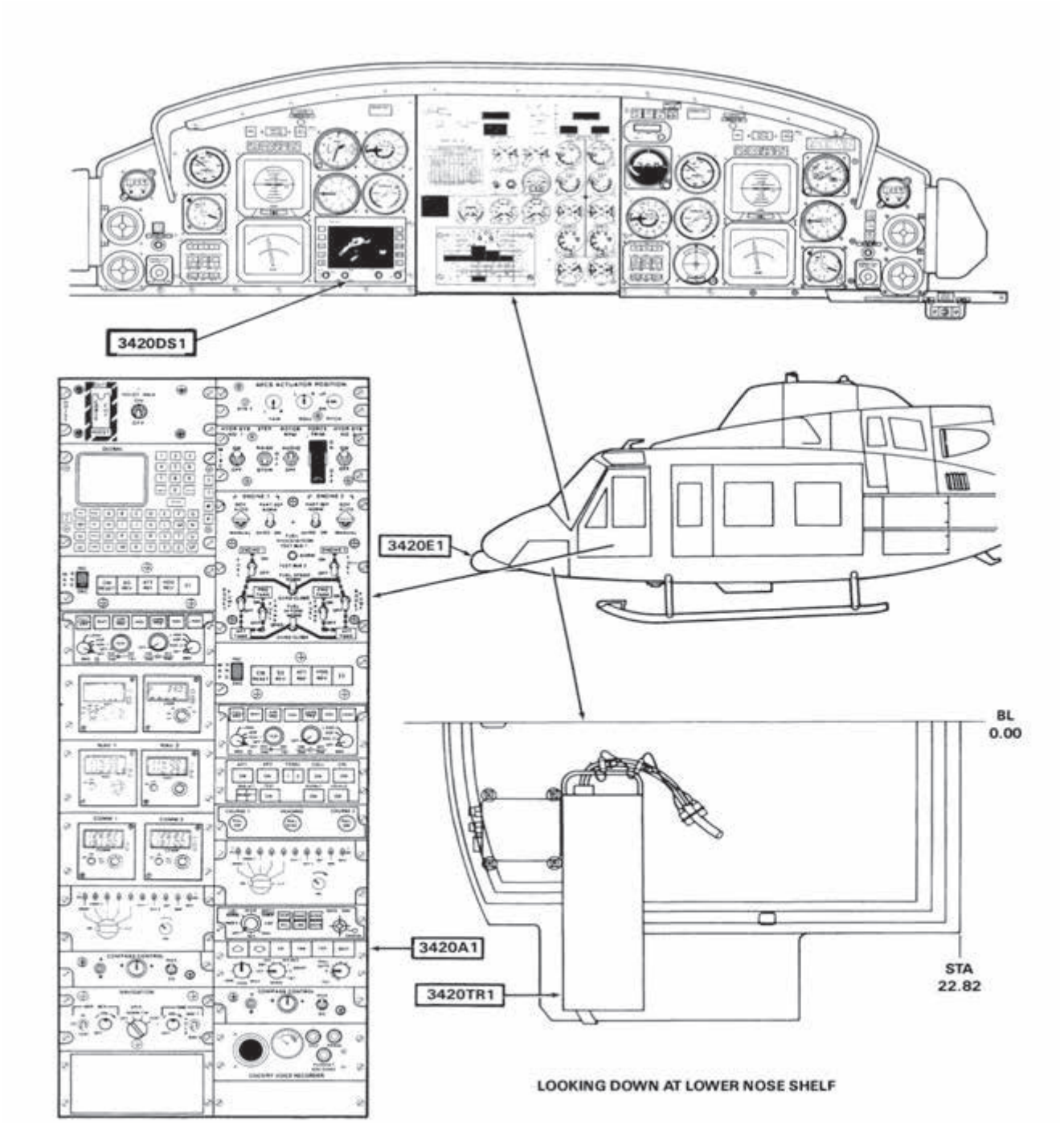
The dual mode allows the pilot and copilot to each select independent radar mode, range, tilt, and gain settings. This is accomplished by time sharing the antenna. On the right-to-left antenna scan, the system switches to the mode, range, tilt, and gain selected by the left control and updates the left display. On the reverse antenna scan, the system switches to the mode, range, tilt, and gain setting selected by the right control and updates the right display. Also, either control can be slaved to the other control, resulting in identical displays on both displays. Dual mode gives the appearance of having two radar systems on the aircraft.

The receiver/transmitter is a remote-mounted, line replaceable unit which contains the microwave transmitter, receiver and all data processing circuitry.

The antenna is also a remote-mounted, line replaceable unit which contains the flat plate radiator, drive motors and associated electronics.

The indicator is an instrument-panel-mounted unit which contains a color display tube, memory and logic circuitry, and power supply circuitry. The indicator contains all primary controls required to operate the weather radar system.

The radar controller, used for dual mode operation, contains secondary controls to operate the weather radar system.



ICN-412-A-970100-B-97499-00007-A-001-01

Figure 1. Weather Radar System - Major Component Location (Sheet 1 of 1)

Weather Radar System - Test Procedure

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Inclinometer	- Commercial	1	

Procedure



THIS SYSTEM GENERATES MICROWAVE RADIATION THAT COULD BE DANGEROUS TO YOUR HEALTH. DO NOT OPERATE THIS EQUIPMENT UNTIL YOU HAVE READ FAA ADVISORY CIRCULAR AC 20-68B DATED 8-8-80.

1. Check antenna for security of mount and that it can be moved in tilt and azimuth axis with no binding action.
2. Check waveguide, cable, and black box installation to determine they are mechanically and electrically secure.
3. Inspect equipment installation to detect faulty mechanical mountings or electrical connections. If no damage or defects are apparent, proceed with testing.
4. Verify helicopter is pointed clear of buildings, personnel, and other obstacles in a zone ± 90 degrees dead ahead, for at least 200 feet.
5. Connect external electrical power, adjusted to 27.5 Vdc, to helicopter power receptacle.
6. Energize power buses per 412-A-97-01-00-03A-340A-A / 02886.
7. Close WX RDR circuit breaker and WX RDR STAB circuit breaker (3420CB2 and 3420CB1).

Intentionally blank page

Weather Radar System - Test Mode Check

Procedure

WARNING

TESTS INVOLVING RADIATION OF RF ENERGY BY THE RADAR ANTENNA MUST NOT BE MADE IN THE VICINITY OF FUELING OPERATIONS, OR WHEN PERSONNEL ARE STANDING DIRECTLY IN FRONT OF THE RADAR ANTENNA. RF ENERGY IS GENERATED IN ALL MODES EXCEPT STANDBY AND TEST. DO NOT ALLOW PERSONNEL WITHIN 15 FEET OF AREA BEING SCANNED BY ANTENNA WHEN SYSTEM IS TRANSMITTING.

1. Set indicator controls as follows:

   BRT - ROTATE FULL CW

   MODE - SBY

   GAIN - MAX (Push for preset)

   TILT - +15 DEG.

2. Set RAD controller as follows:

   MODE - OFF

   GAIN - MAX (Push for preset)

   TILT - +15 DEG.

3. Set RAD controller MODE select knob to SBY.

4. Adjust BRT control on indicator for a comfortable level. Verify standby mode is indicated.

WARNING

R/T TRANSMITS IN TEST MODE. ENSURE AREA AHEAD OF HELICOPTER IS CLEAR OF PERSONNEL AND OBSTRUCTIONS.

5. Select TST mode on indicator. Verify TEST pattern is displayed.

6. Adjust TILT knob on indicator between +15 and -15 degrees. Verify tilt angle is displayed.

7. Verify Red/Green sense indication block is Red at the left.

8. Press VRM button on indicator. Verify that variable range mark is displayed.

9. Slew range mark in/out on indicator. Verify correct range is displayed.

10. Press (← →) buttons on indicator. Verify azimuth mark is displayed dead ahead.

11. Slew azimuth mark left and right with (← →) buttons. Verify correct azimuth is displayed.

NOTE

HDG source must be on and valid for next step.

12. Select HDG display (via menu). Verify correct heading is displayed at top center of indicator.

13. Select BRG feature. Verify variable bearing cursor is displayed.

NOTE

The displayed bearing is the compass heading of the cursor. When the cursor is ahead, the bearing will match the heading.

14. Slew variable bearing cursor left and right. Verify correct bearing is displayed.

15. Select WX mode, set range to 10 NM, and adjust tilt for ground or weather targets.

15.1. Select 300 NM range and MAX variable gain.

15.2. Verify receiver noise is visible in display.

16. Slew ranges through all selections. Verify proper operation.

Weather Radar System - Stabilization Check

Procedure

NOTE

Use the following procedures to check and calibrate antenna stabilization whenever a new installation is made, whenever a stabilization circuit in either the R/T or antenna has been serviced, or whenever faulty operation of the stabilization system is suspected.

The following checks require accurate positioning of the vertical gyro in both pitch and roll. If a tilt table is not available, the gyro may be positioned by reference to the flight director or artificial horizon, or some other method of accurately tilting the gyro in the helicopter. This procedure should be performed after a period of gyro or antenna maintenance.

1. Preliminary Setup.

1.1. Position helicopter on a level area, with vertical reference mounting surface horizontal ± 2 degrees in pitch and roll.

1.2. Set radar controls as follows:

XMIT - OFF (on R/T unit)

SCAN - OFF (on antenna).

1.3. Switch radar on and select the following:

TILT - ZERO

GAIN - OFF (Use menu or simultaneous TGT and *CR on the controller)

MODE - SBY.

1.4. Verify only one control source is active.

1.5. Position flat plate dead ahead by hand.

1.6. Use an inclinometer to measure and adjust flat plate tilt angle so that it is vertical ± 0.1 degree.

1.7. Remove vertical reference sensor from its mount and install it on a tilt table or similar device.

2. Pitch Adjustment.

2.1. Set the following:

GAIN - MANUAL

STAB - ON.

NOTE

Take care not to roll the vertical reference sensor to avoid gimbal error.

2.2. Tilt the vertical reference sensor to an attitude of approximately 20 degrees nose down. Measure and record exact pitch angle.

2.3. Measure flat plate angle with inclinometer.

2.4. Flat plate angle should be 20 degrees nose up ±1 degree. If necessary, adjust pitch gain potentiometer on R/T unit so that flat plate looks up by the amount recorded in Step 2.2..

NOTE

Take care not to roll the vertical reference sensor.

2.5. Tilt vertical reference sensor to an attitude of approximately 20 degrees noseup. Measure and record the exact pitch angle.

2.6. Measure flat plate angle and determine difference between vertical reference sensor plate recorded in Step 2.5. and the look down angle of the flat plate. Difference should not exceed 1 degree.

2.7. if necessary, adjust pitch gain potentiometer to reduce error noted in Step 2.6. by 50 percent.

3. Roll Adjustment.

3.1. Set the following:

GAIN - PRESET

STAB - ON.

NOTE

Take care not to pitch the vertical reference sensor.

3.2. Tilt the vertical reference sensor to an attitude of approximately 20 degrees left wing down. Measure and record exact roll angle.

3.3. Measure flat plate angle with inclinometer.

3.4. Flat plate angle should be 20 degrees nose up ±1 degree. If necessary, adjust roll gain potentiometer on R/T unit so that flat plate looks down by the amount recorded in Step 3.2..

3.5. Tilt vertical reference sensor to an attitude of approximately 20 degrees right wing down. Measure and record the exact roll angle.

3.6. Measure flat plate angle and determine difference between vertical reference roll sensor plate recorded in Step 3.5. and the look-up angle of the flat plate. Difference should not exceed 1 degree.

3.7. if necessary, adjust roll gain potentiometer to reduce error noted in Step 3.6. by 50 percent.

4. Return system to operation.

MODE SBY

STAB - ON

SCAN - ON

XMIT - ON.

Weather Radar System - Troubleshooting

Procedure

1. It is assumed that all appropriate circuit breakers are closed, interconnecting cabling is in normal operational condition, and power is applied to the system. (See appropriate wiring diagram.)

Table 1. Troubleshooting - Weather Radar System

INDICATION OF TROUBLE	PROBABLE CAUSE	CORRECTIVE ACTION
Noise band broken.	Defective R/T unit.	Replace R/T unit.
	Broken/flooded waveguide.	Replace waveguide.
Noise band missing or weak or too strong.	Defective R/T unit.	Replace R/T unit.
No TRB noise in noise band.	Defective R/T unit.	Replace R/T unit.
No range marks, pattern ok.	Defective indicator.	Replace indicator.
Wrong colors.	Defective indicator.	Replace indicator.
No azimuth/bearing/ track graphics.	Defective R/T unit.	Replace R/T unit.
	Defective indicator.	Replace indicator.
ANT error.	Defective antenna.	Replace antenna.
R/T error.	Defective R/T unit.	Replace R/T unit.
IND error.	Defective indicator.	Replace indicator.
SENS error.	Sensor (gyro) failure.	Replace sensor.

Intentionally blank page

Weather Radar System - On-Line Error Monitors

Procedure

1.

NOTE

Critical functions in the R/T and indicator are continuously monitored. The indicator displays both indicator and R/T errors on a “hidden” display page that may be viewed by selecting the “ERRORS ON” function from MENU 2 on the indicator. All errors are stored in a nonvolatile memory. Exit from the “hidden” page is accomplished by using the menu feature.

Table 1. Indicator Error Codes and Code Translations.

Error Code	Source	Name	Description
02	R/T	Transmitter Error	Indicates that the transmitter has failed.
03	R/T	AFC Error	Indicates that the AFC is unlocked while the transmitter is on.
04	R/T	AGC Error	Indicates that the receiver AGC is operating outside its permitted control range.
05	R/T	Mixer Error	Indicates that one or both mixer diode currents are out of tolerance.
06	R/T	IOP Error	Error in input/output processor circuit.
07	R/T	DSP Fault (INT)	Error in the digital signal processor circuit which processes radar reflectivity data.
08	R/T	DSP Fault (PPP)	Error in the digital signal processor circuit which processes radar turbulence data.
09	R/T	Antenna Drive	Future function.
0A	R/T	IOP RAM	RAM data error.
0B	R/T	Modulator	Modulator errors for four seconds.
0D	R/T	Boot Checksum	Processor detects a corrupt checksum in the boot program.
0F	R/T	HS 429	Indicates errors in the hardware circuit which filters high speed ARINC 429 data.
10	R/T	LS 429	Indicates errors in the hardware circuit which filters low speed ARINC 429 data.
11	R/T	Power Supply	Indicates that the +5 V, +15 V, or -15 V power supplies are out of tolerance.
12	R/T	Right ESCI	Indicates errors in the hardware circuit which filters the control bus data.
13	R/T	Left ESCI	Indicates errors in the hardware circuit which filter the control bus data.

Table 1. Indicator Error Codes and Code Translations. (continued)

Error Code	Source	Name	Description
14	R/T	Magnetron Drive	Indicates that the magnetron drive voltage is out of tolerance.
17	R/T	Overheat	Indicates that the R/T is overheated, probably due to fan malfunction.
18	R/T	DSP	Indicates errors in the digital signal processor circuit.
31	Indicator	Horiz Def	Malfunction in horizontal deflection circuitry.
32	Indicator	Vert Def	Malfunction in vertical deflection circuitry.
33	Indicator	Computer	Error in indicator's control processor system.
34	Indicator	Graphics	Error in indicator's graphics processor system.
35	Indicator	HVPS	Malfunction in the indicator's high voltage power supply.
36	Indicator	LVPS	Malfunction in the indicator's low voltage power supply.
37	Indicator	EWXPD	Error in the indicator's EWXPD bus input circuitry or data.
38	Indicator	ESCI	Error in the indicator's ESCI bus input circuitry or data.
39	Indicator	UDI	Error in the indicator's UDI input interface circuitry or data.
61	Antenna	Scan	Azimuth errors greater than 5 degrees are detected.
62	Antenna	Elevation	Elevation errors greater than 5 degrees are detected.
69	External	LS 429	No digital altitude data. Only annunciated if the R/T is programmed for digital altitude data.
6A	External	HS 429	Either pitch or roll data on the HS 429 bus is absent or invalid. ---OR--- Digital heading is required but not available.
6B	Transmit	Switch off	Only annunciated if the R/T is programmed for digital data.
6C	External	Analog ALT	Indicates that autotilt selected, and analog altitude signal is invalid. Only annunciated if R/T is programmed for analog altitude.

Table 1. Indicator Error Codes and Code Translations. (continued)

Error Code	Source	Name	Description
6E	External	Analog HDG	Indicates that analog (synchro) heading is invalid, and that it is required for NAV functions or HDG display. Only annunciated if the R/T is programmed for analog heading.
6F	External	Selected Course	Indicates that the synchro or course OBS selected course input is invalid. Only annunciated if the R/T is programmed for external course input.
70	External	115 V ac	Indicates that the 115 V ac 400 Hz input is below 90 or above 135 volts.
71	External	ESCI	Indicates that the R/T is not receiving control bus signals from the indicator or controller.

Intentionally blank page

Antenna - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Disconnect electrical connectors.
2. Disconnect waveguide.
3. Remove and retain screws and other mounting hardware securing antenna and antenna pedestal to airframe.

Intentionally blank page

Radar Indicator - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Disconnect electrical connector from rear of radar indicator.
2. Loosen fastener that secure radar indicator to instrument panel.
3. Remove radar indicator from instrument panel.

Intentionally blank page

Receiver/Transmitter - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Disconnect electrical connectors.
2. Remove and retain screws and other mounting hardware.
3. Remove receiver/transmitter.

Intentionally blank page

Weather Radar Control - Removal

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Loosen spring lock fasteners that secure weather radar control to pedestal assembly.
2. Lift weather radar control far enough out of pedestal assembly to reach and disconnect electrical connector at back of unit.
3. Lift weather radar control free of pedestal assembly.

Intentionally blank page

Antenna - Installation

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Position and secure antenna and antenna pedestal with attaching screws and other mounting hardware.
2. Connect waveguide.
3. Connect electrical connectors.

Intentionally blank page

Radar Indicator - Installation

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Position radar indicator in instrument panel.
2. Secure radar indicator to instrument panel with fasteners.
3. Connect electrical connector to back of radar indicator.

Intentionally blank page

Receiver/Transmitter - Installation

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Position and secure receiver/transmitter with attaching screws and other mounting hardware.
2. Disconnect electrical connectors.

Intentionally blank page

Weather Radar Control - Installation

Procedure



ENSURE ELECTRICAL POWER IS OFF.

1. Connect electrical connector to back of weather radar control.
2. Position weather radar control in place on pedestal assembly.
3. Secure weather radar control with spring lock fasteners.

Intentionally blank page

Weather Radar System - Inspection

Procedure

1. Inspect weather radar system for completeness, proper installation, cleanliness, and secure mounting of each component.
2. Inspect switches and controls on weather radar system components for proper mechanical action.
3. Inspect components to ensure panel lights illuminate with electrical power on.
4. Inspect electrical connections for corroded or bent pins and proper mating, interconnecting cable for frayed or broken insulation, and circuit breakers for proper operation.

Intentionally blank page

Weather Radar System - Cleaning

Procedure

1. Remove moisture and loose dirt with a soft, clean cloth.
2. Remove dust and dirt from system components with a soft, clean cloth.



CLEANING COMPOUND FUMES ARE FLAMMABLE AND TOXIC.

3. Remove grease, fungus, and ground-in dirt with a soft, clean cloth dampened with approved cleaning compound.
4. Remove dirt from connectors with a soft bristle brush; remove moisture with a clean, dry cloth.

Intentionally blank page

Weather Radar System - Repair or Replacement

Procedure

1. Tighten or replace loose or cracked control knobs or system components.
2. Replace defective panel light bulbs.
3. Replace defective or inoperative units.
4. Further repairs, other than removal or replacement, must be conducted at an authorized repair facility.

Intentionally blank page

VHF/AM Communication System and ADF Navigation System - Description

The KTR-908 factory-adjusted radio is incompatible with the NAT intercommunication system installed in the JMSA SAR helicopters. When it is necessary to replace the radio, it shall be replaced with a radio which has been readjusted for compatibility with the NAT ICS. Refer to the appropriate customizing drawing for radio settings.

Likewise, to maintain the direction finding (DF) function provided by VHF COMM 2, the KTR-908 radios shall only be replaced by King Radio P/N 064-1023-04 or 064-1023-0004.

ADF King Radio P/N 066-1077-01 shall be installed in SAR helicopters. P/N 066-1077-01 is not interchangeable with basic P/N 066-1077-00.

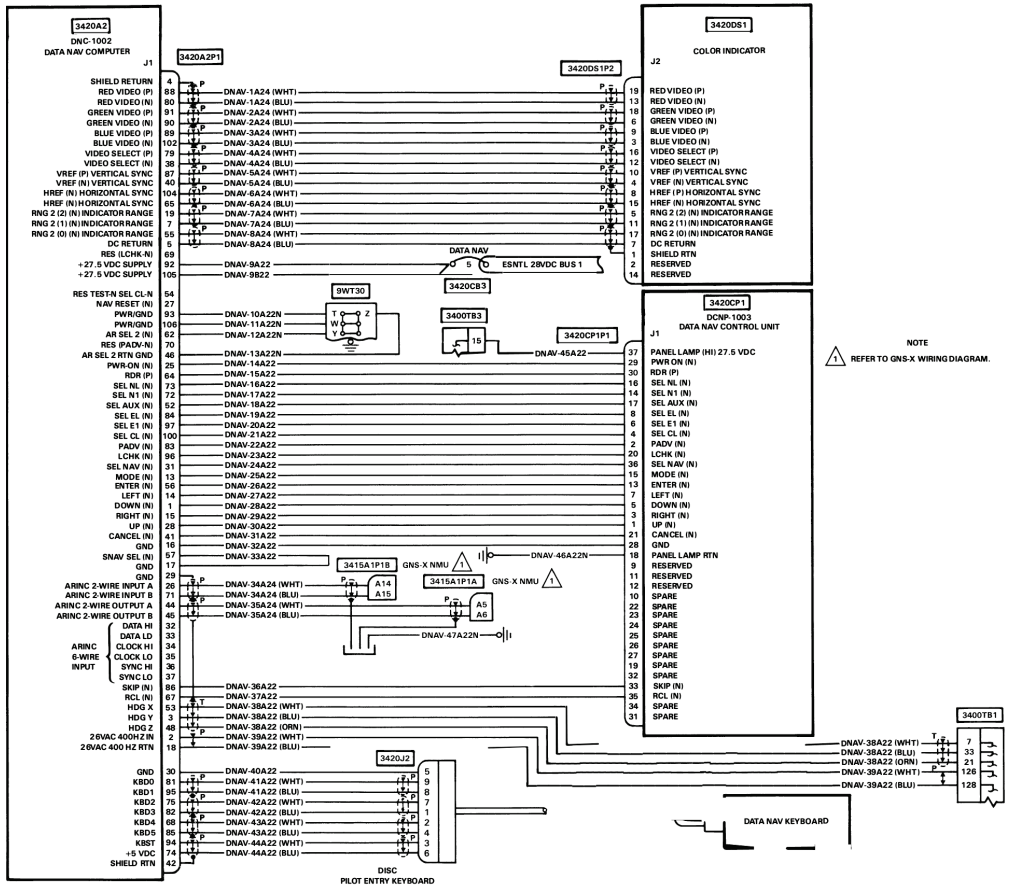
Minor modification to the VHF COMM and ADF airframe wiring is illustrated in EFIS Modification Wiring Diagram, 412-A-98-01-01-04A-051A-A / 02978.

Intentionally blank page

Avionics System Wiring Diagrams - Description

This module contains wiring diagrams for each of the SAR avionic systems that may be installed in the Bell Model 412 helicopter (Figure 1 through Figure 6).

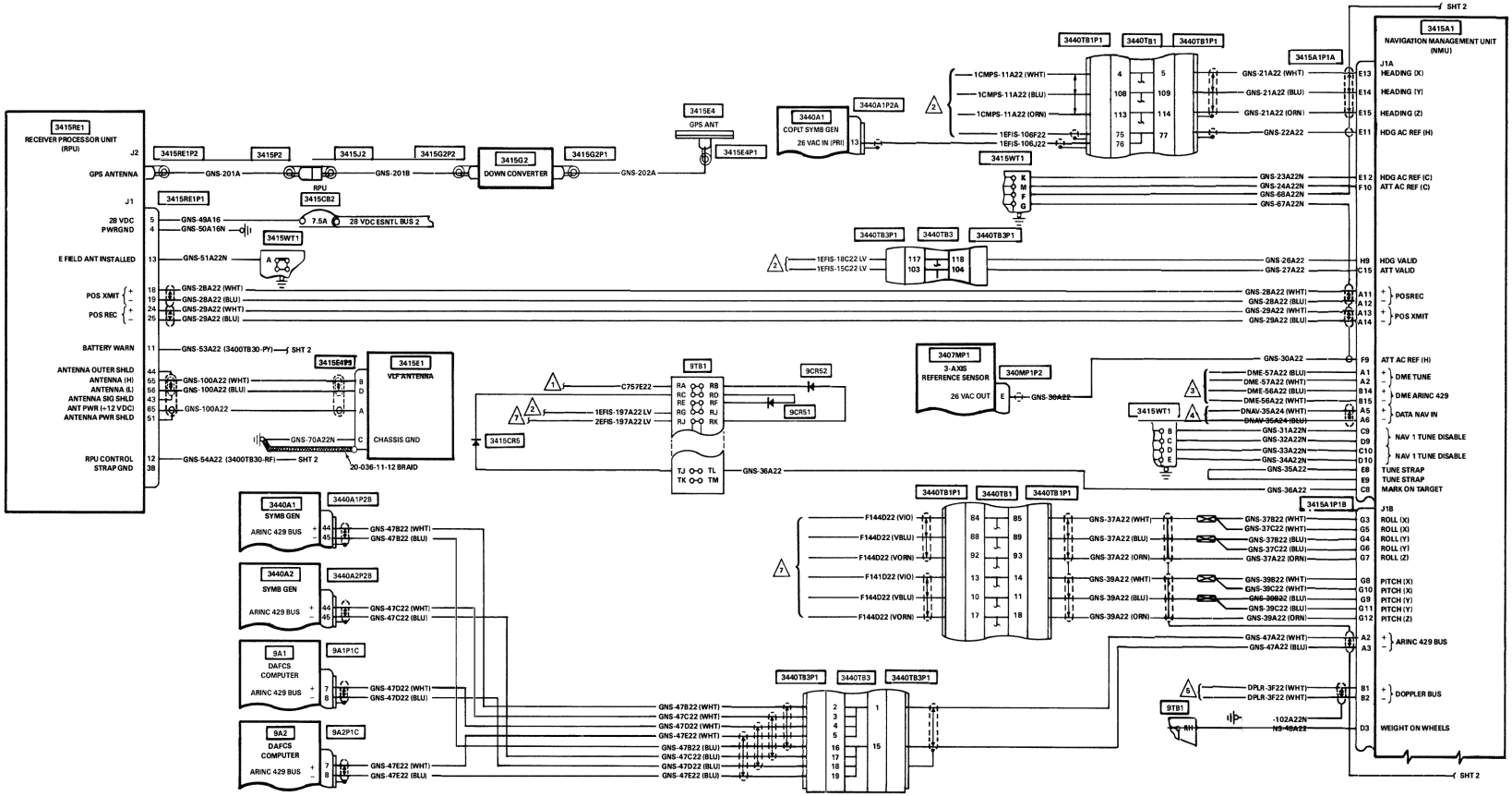
Intentionally blank page



ICN-412-A-970100-B-97499-00008-A-001-01

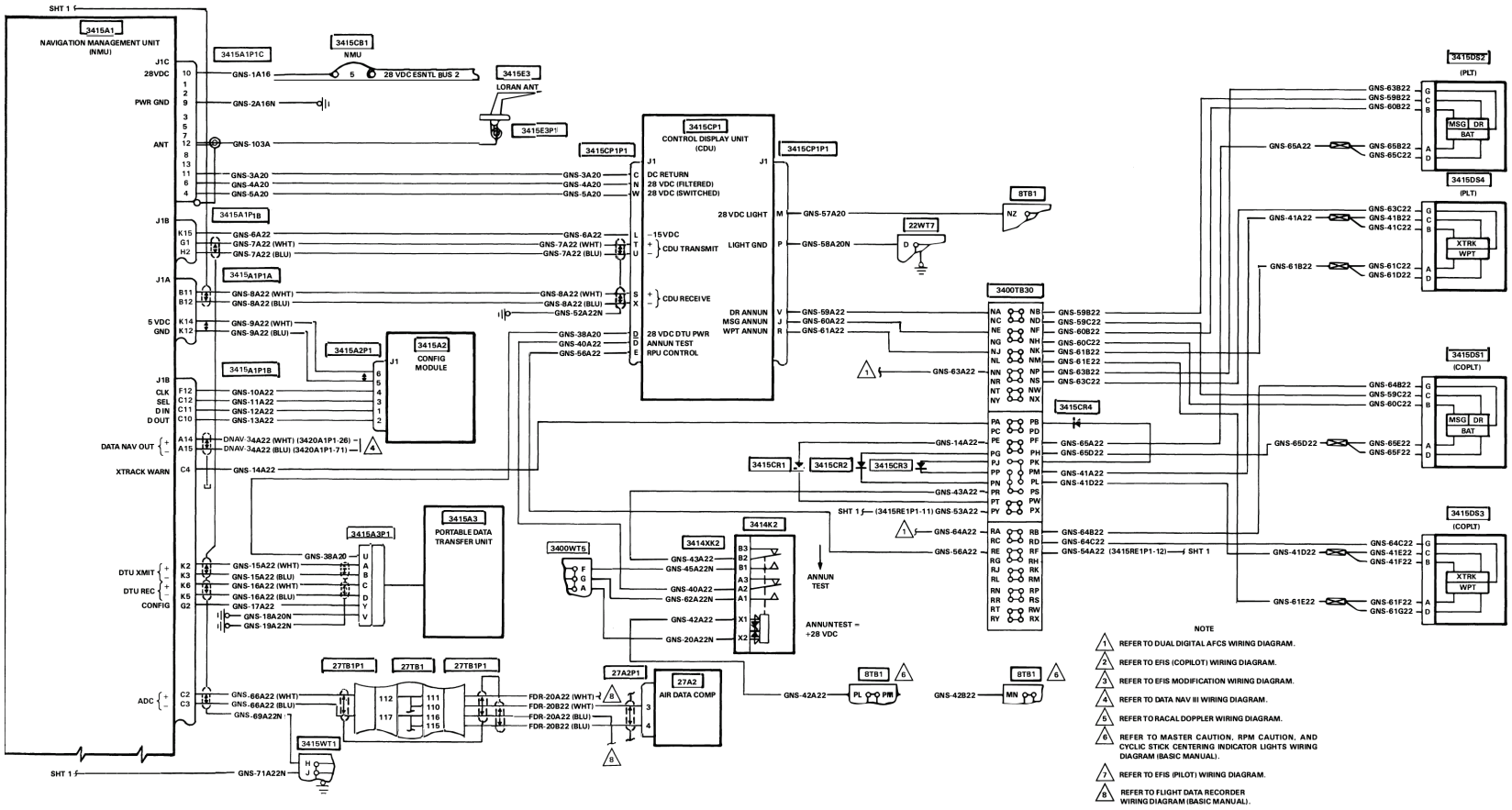
Figure 1. DATA NAV III System Wiring Diagram (Sheet 1 of 1)

Intentionally blank page



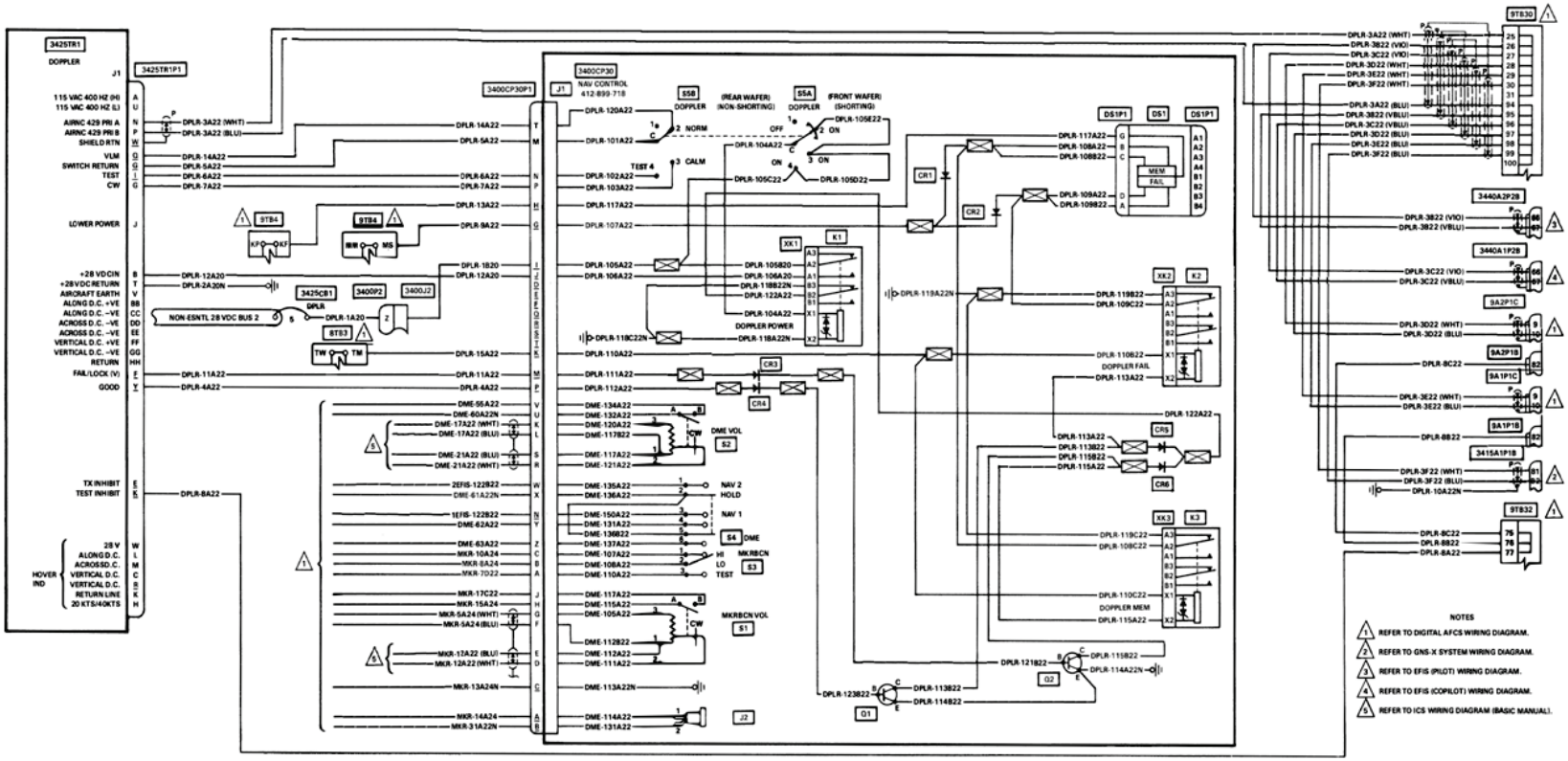
ICN-412-A-970100-B-97499-00009-A-001-01

Figure 2. GNS-X System Wiring Diagram (Sheet 1 of 2)



ICN-412-A-970100-B-97499-00010-A-001-01

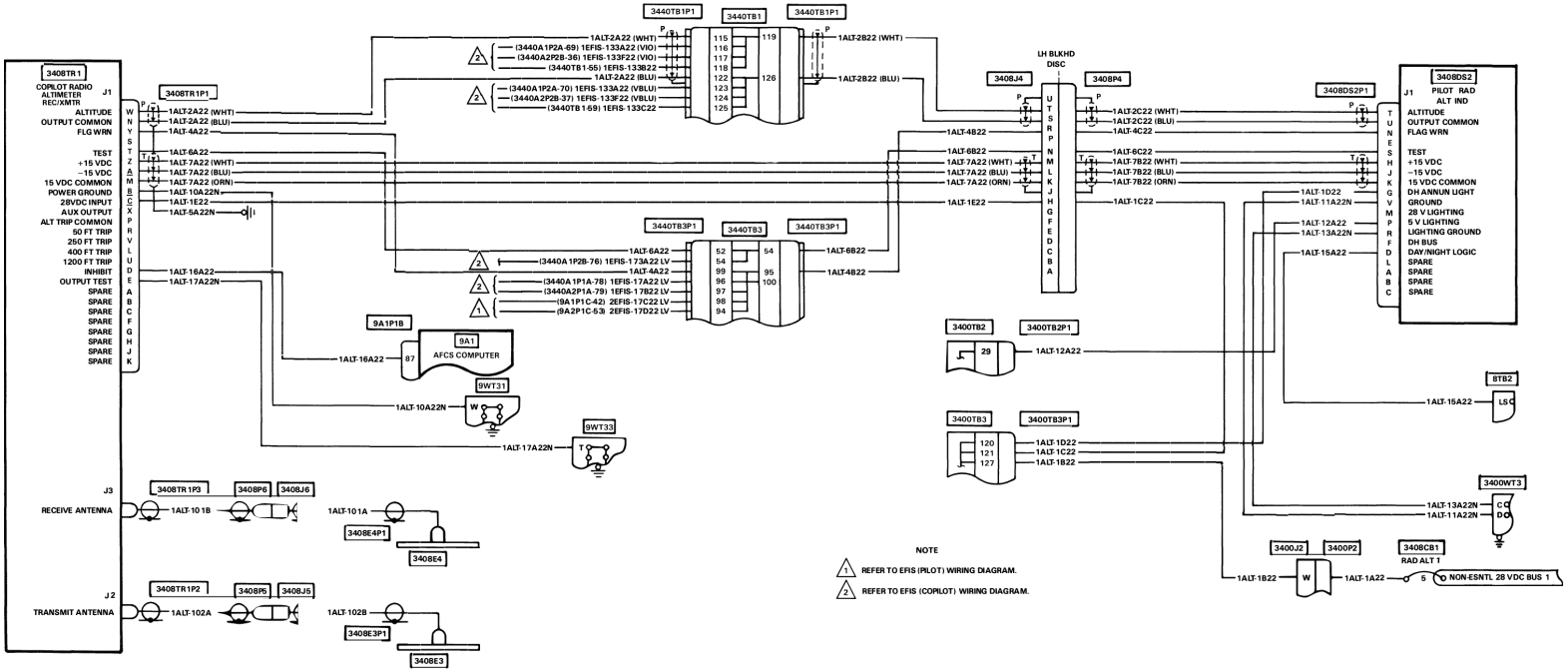
Figure 2. GNS-X System Wiring Diagram (Sheet 2 of 2)



ICN-412-A-970100-B-97499-00011-A-001-01

Figure 3. Doppler Velocity Sensor Wiring Diagram (Sheet 1 of 1)

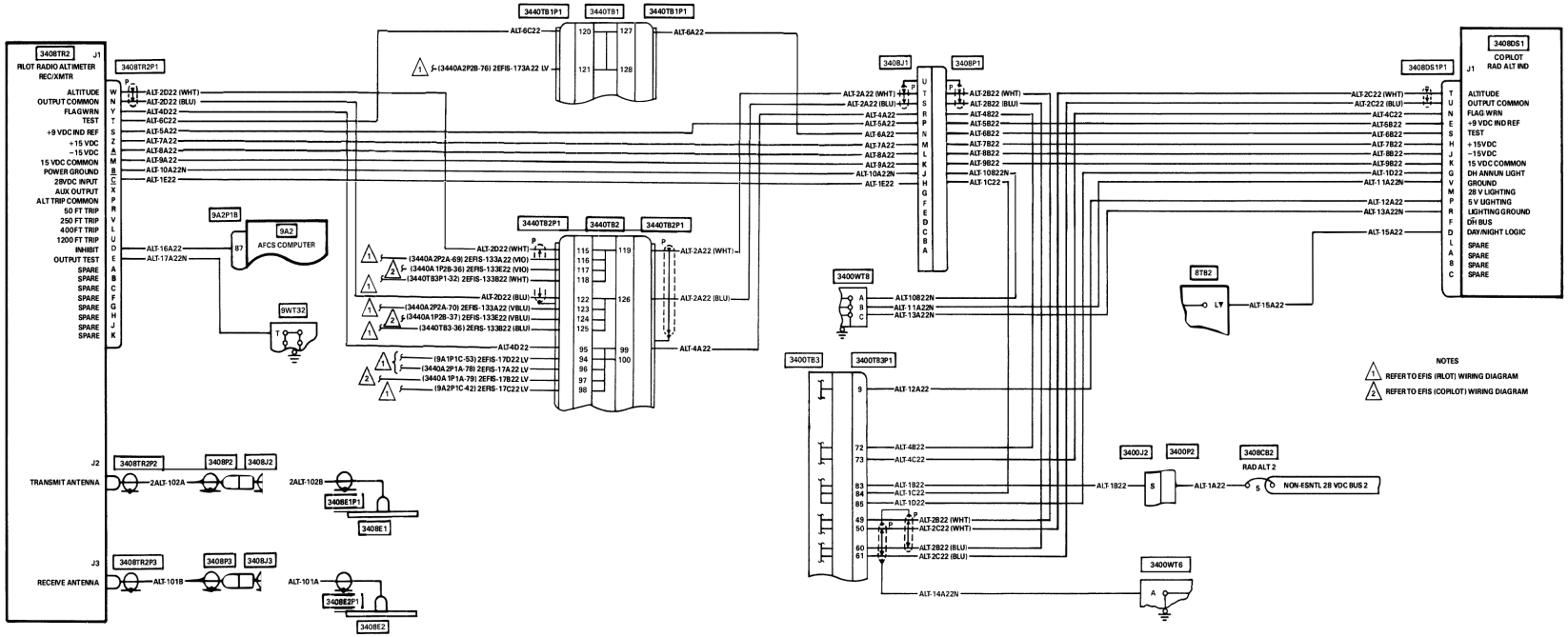
Intentionally blank page



ICN-412-A-970100-B-97499-00012-A-001-01

Figure 4. Radio Altimeter System (Copilot) Wiring Diagram (Sheet 1 of 1)

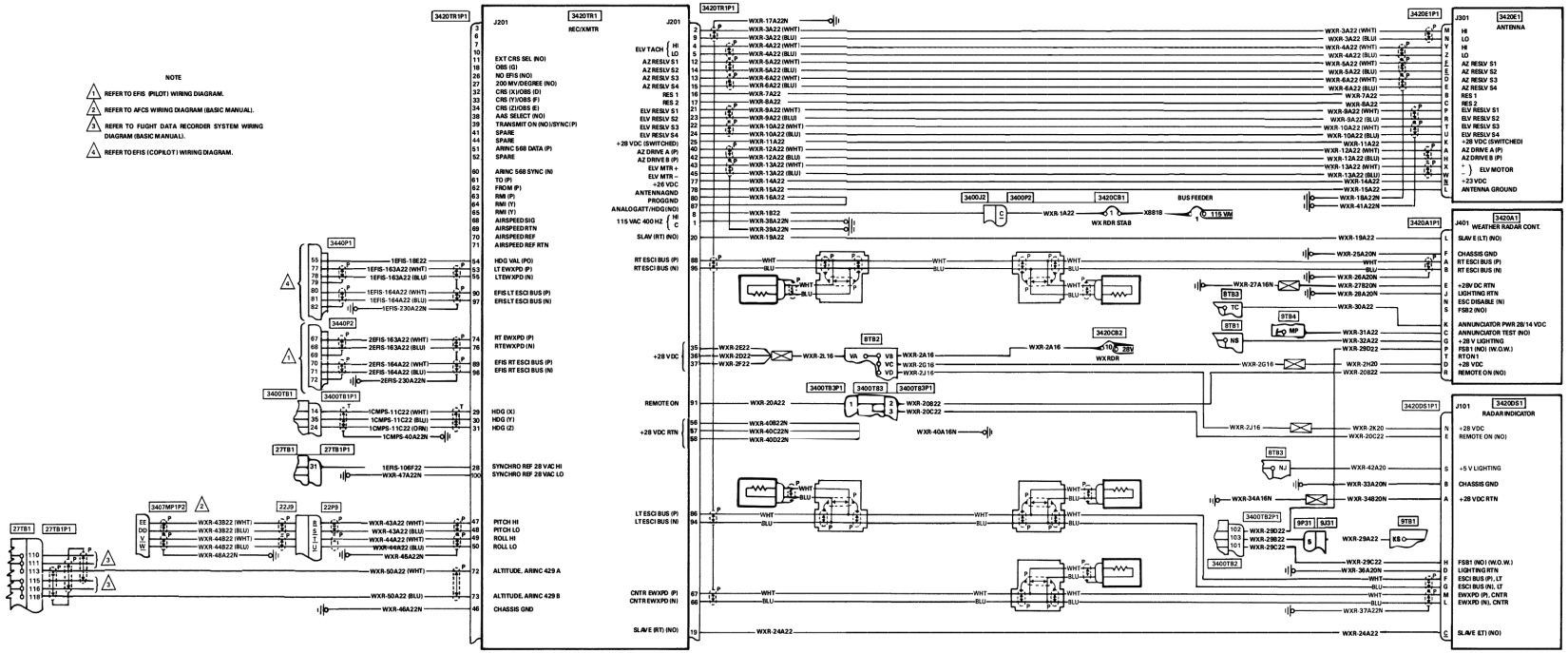
Intentionally blank page



ICN-412-A-970100-B-97499-00013-A-001-01

Figure 5. Radio Altimeter System (Pilot) Wiring Diagram (Sheet 1 of 1)

Intentionally blank page



ICN-412-A-970100-B-97499-00014-A-001-01

Figure 6. Weather Radar System Wiring Diagram (Sheet 1 of 1)

Intentionally blank page

Wiring Identification - Description

All wires on the wiring diagrams are identified by codes exactly as they are marked in the helicopter. The code or wire number indicates the circuit function, wire number, wire segment letter and the wire size of the conductor. Identification of wires may be accomplished by using the wiring identification code figure in 412-A-98-01-01-00A-042A-A / 02970.

Intentionally blank page

Use of Underlined Capital Letters - Description

Many letters have similar upper case and lower case forms. To eliminate this possible source of error in identifying connector pin designations, underlined upper case (Capital) letters (A, B, C, etc) are used to signify the lower case pin letters (a, b, c, etc).

Intentionally blank page

Wiring Diagrams - General

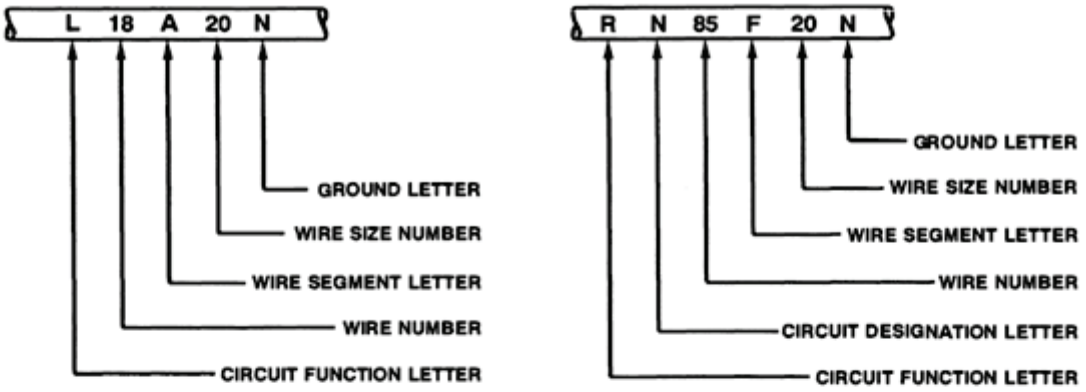
This section contains electrical wiring diagrams and component replacement charts. This information is provided to assist maintenance personnel in understanding components and circuit operations, to perform maintenance, and troubleshooting of electrical and avionics systems installed in the SAR helicopter.

Intentionally blank page

Wiring Identification - Description

All wires on the wiring diagrams are identified by codes exactly as they are marked in the helicopter. The code or wire number indicates the circuit function, wire number, wire segment letter and the wire size of the conductor. Identification of wires may be accomplished by using the wiring identification code figure in Figure 1.

WIRING IDENTIFICATION CODE



CIRCUIT FUNCTIONS

CODE	NOMENCLATURE	CODE	NOMENCLATURE
A	ARMAMENT	RC	COMMAND
C	CONTROL SURFACES	RF	VHF LIAISON
D	INSTRUMENTS (OTHER THAN FLIGHT OR ENGINE)	RL	LIAISON
E	SINGLE INSTRUMENTS	RM	MARKER BEACON
F	FLIGHT INSTRUMENTS	RN	NAVIGATION
H	HEATING, VENTILATING AND DE-ICING	RU	UHF COMMAND
J	IGNITION	RV	VHF COMMAND
K	ENGINE CONTROL	RZ	INTERPHONE AND HEADPHONE
L	LIGHTING	S	RADAR
M	MISCELLANEOUS ELECTRIC	SX	RECOGNITION (IFF)
N	GROUND	TN	TRACKING NAVIGATION
P	DC POWER	V	DC POWER AND DC CONTROL CABLES FOR AC SYSTEM
Q	FUEL AND OIL	W	WARNING AND EMERGENCY
R	RADIO (NAVIGATION AND COMMUNICATION)	X	AC POWER

COOOO

ICN-412-A-980101-B-97499-00001-A-001-01

Figure 1. Wiring Identification (Sheet 1 of 1)

Use of Underlined Capital Letters - Description

Many letters have similar upper case and lower case forms. To eliminate this possible source of error in identifying connector pin designations, underlined upper case (Capital) letters (A, B, C, etc) are used to signify the lower case pin letters (a, b, c, etc).

Intentionally blank page

Use of Zones - Description

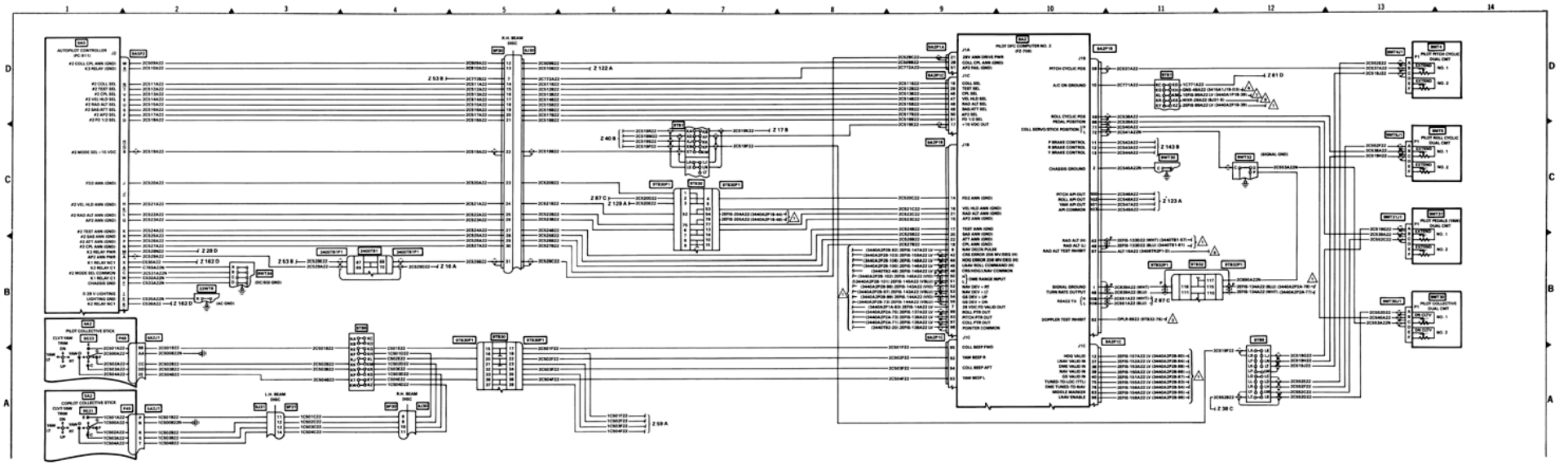
Some wiring diagrams are several sheets long. To facilitate tracking wires between the sheets, zone numbers and section letters (A, B, C, and D) have been added to the sheets. Wires going to another sheet are identified ie., Z 123B.

Intentionally blank page

Component Replacement Diagrams - Description

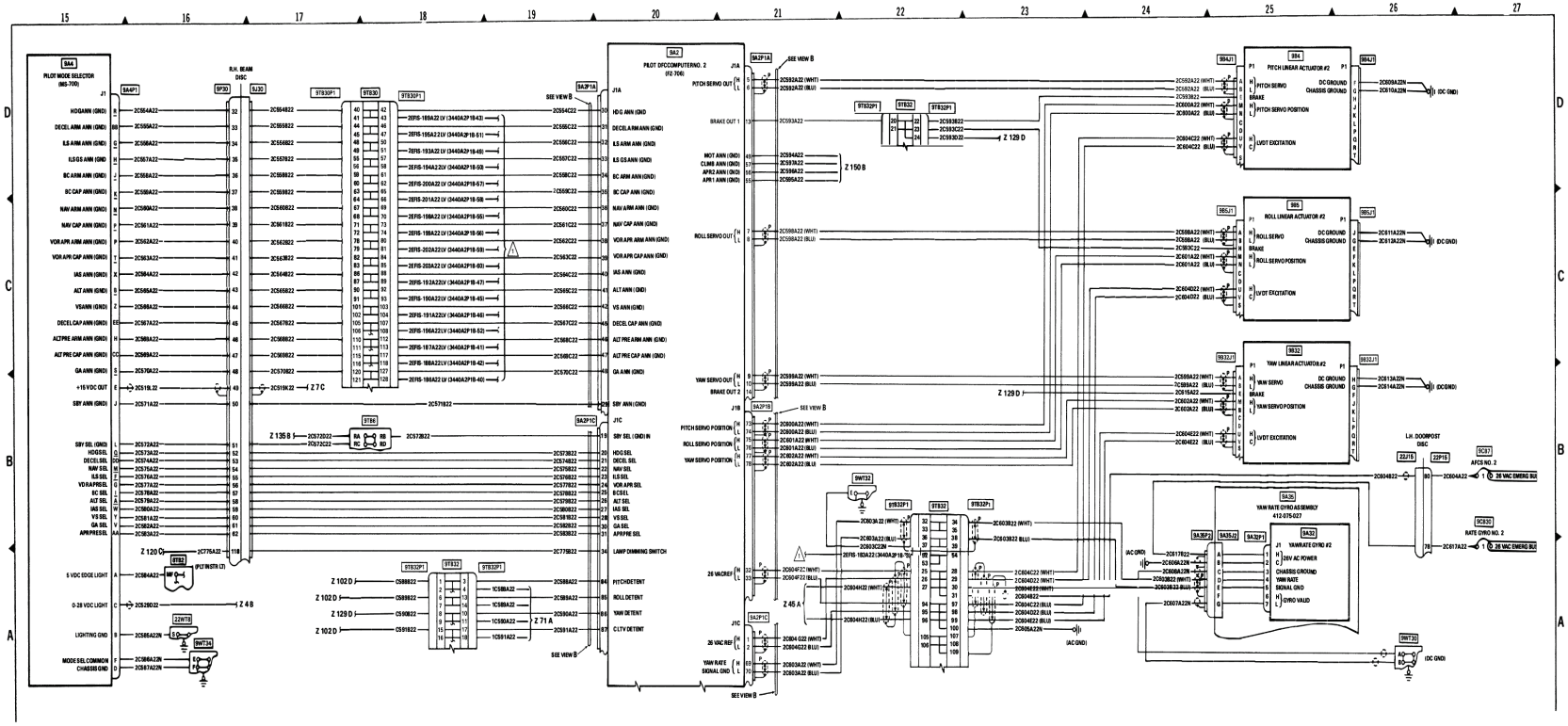
Some components are not shown complete on some figures. If a component has to be replaced, all wire numbers and connections are required. The component replacement diagram is used to show a complete component with the wire numbers and connections.

Intentionally blank page



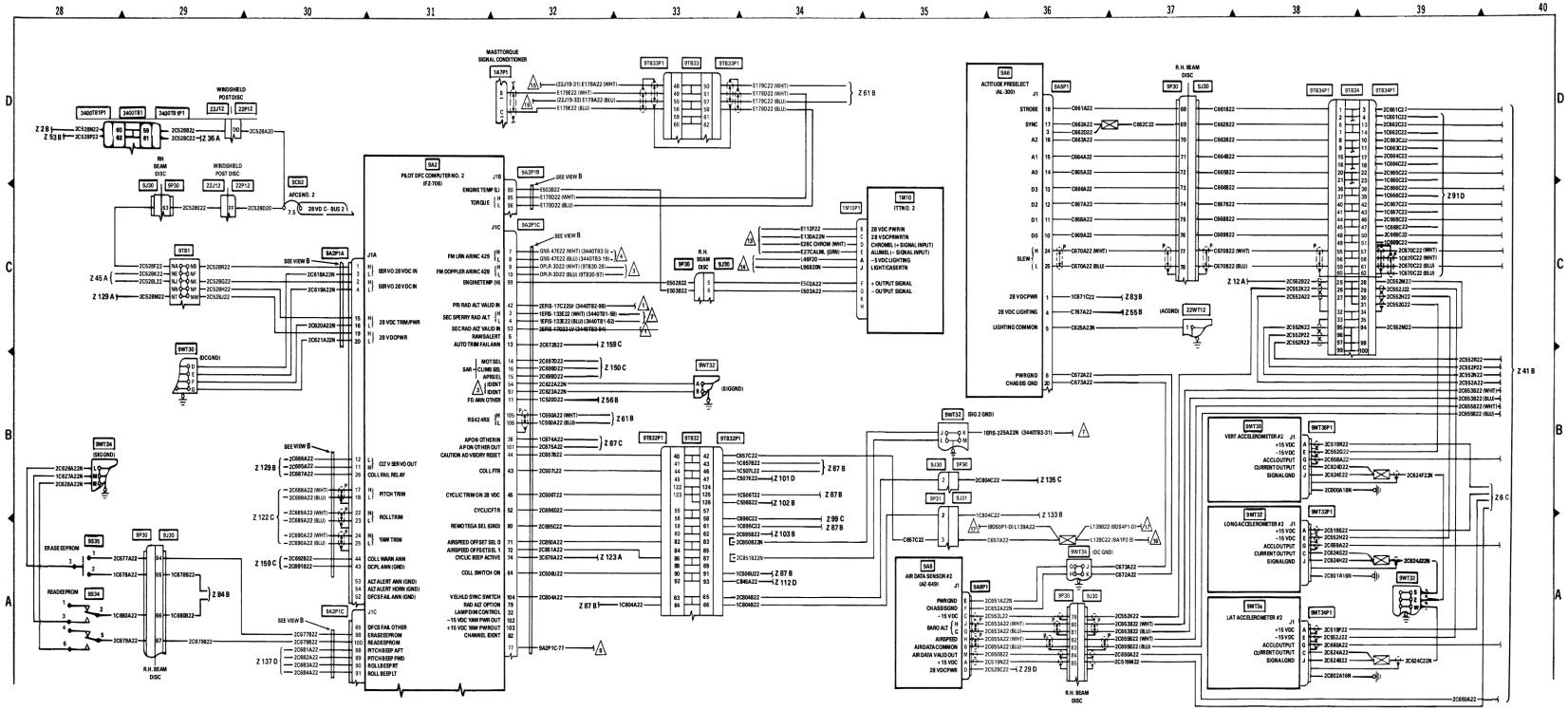
ICN-412-A-980101-B-97499-00002-A-001-01

Figure 1. Digital AFCS Wiring Diagram (Sheet 1 of 13)



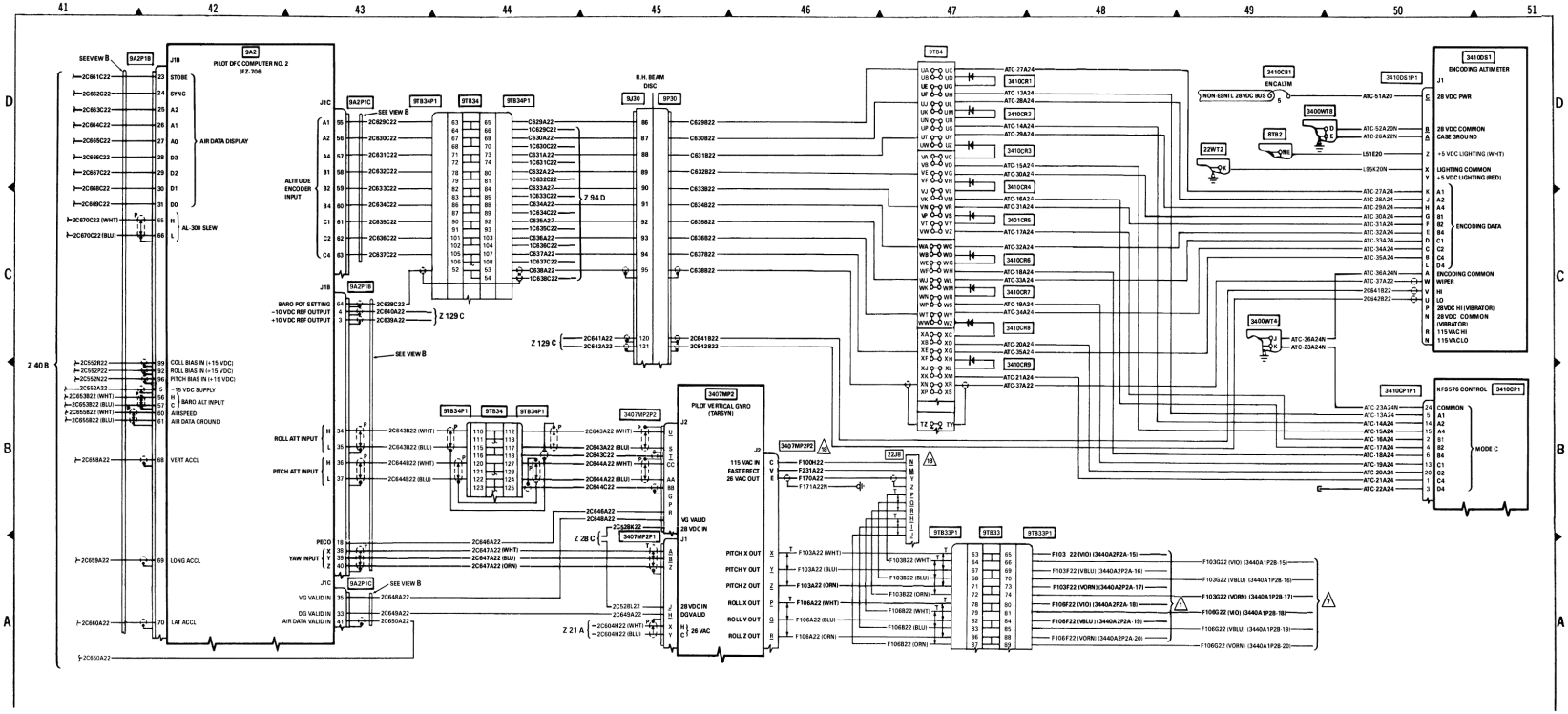
ICN-412-A-980101-B-97499-00003-A-001-01

Figure 1. Digital AFCS Wiring Diagram (Sheet 2 of 13)



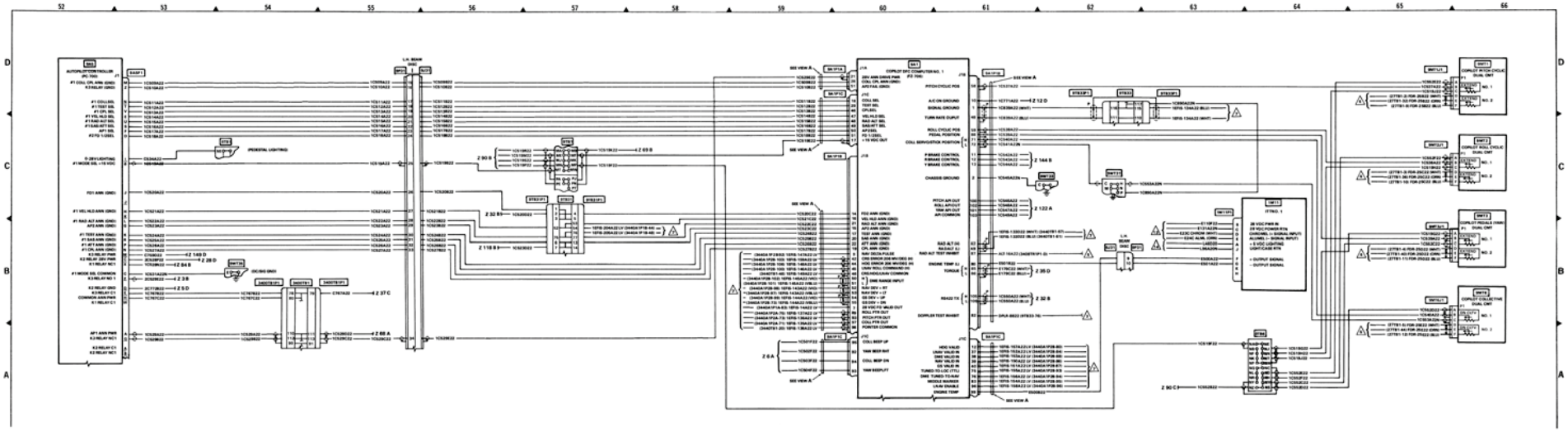
ICN-412-A-980101-B-97499-00004-A-001-01

Figure 1. Digital AFCS Wiring Diagram (Sheet 3 of 13)



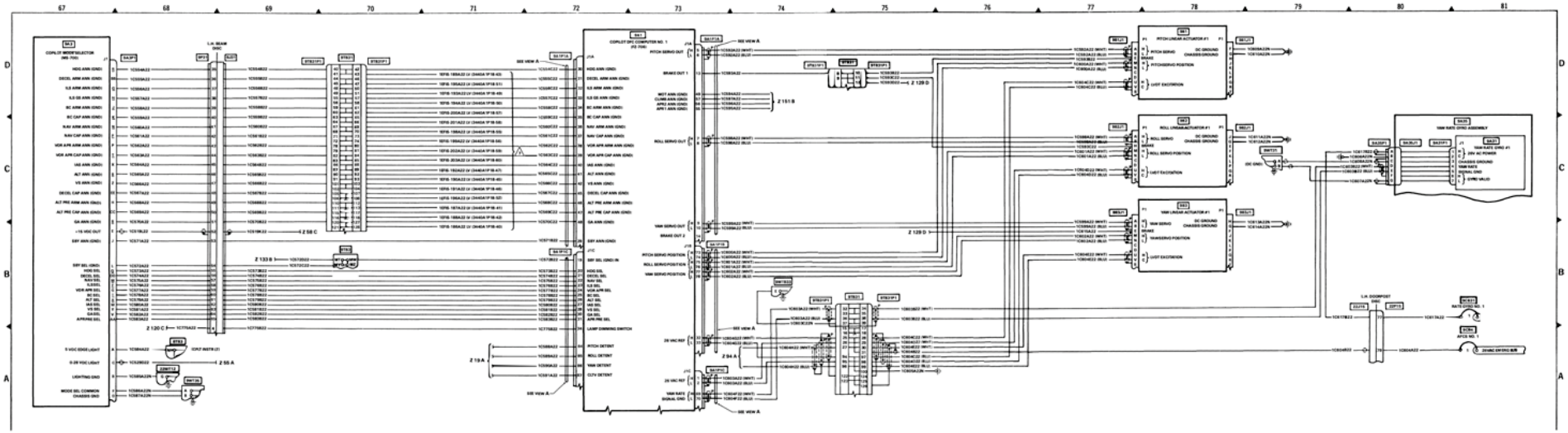
ICN-412-A-980101-B-97499-00005-A-001-01

Figure 1. Digital AFCS Wiring Diagram (Sheet 4 of 13)



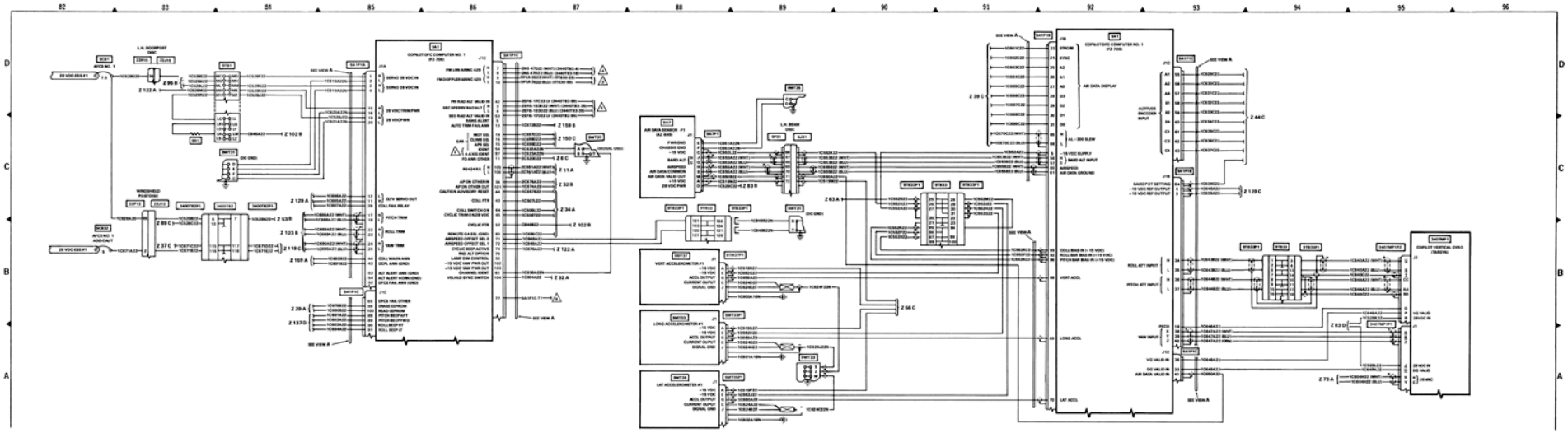
ICN-412-A-980101-B-97499-00006-A-001-01

Figure 1. Digital AFCS Wiring Diagram (Sheet 5 of 13)



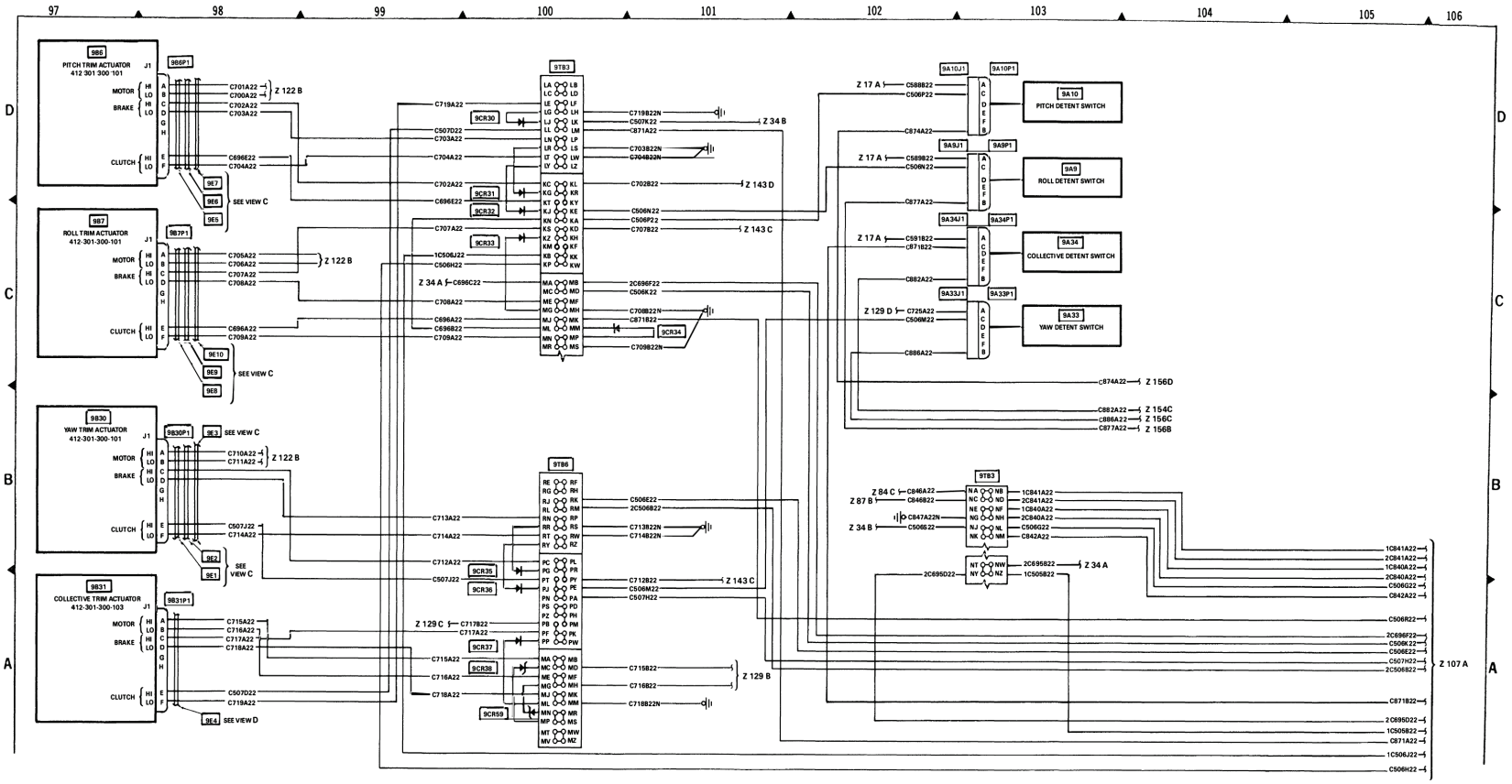
ICN-412-A-980101-B-97499-00007-A-001-01

Figure 1. Digital AFCS Wiring Diagram (Sheet 6 of 13)



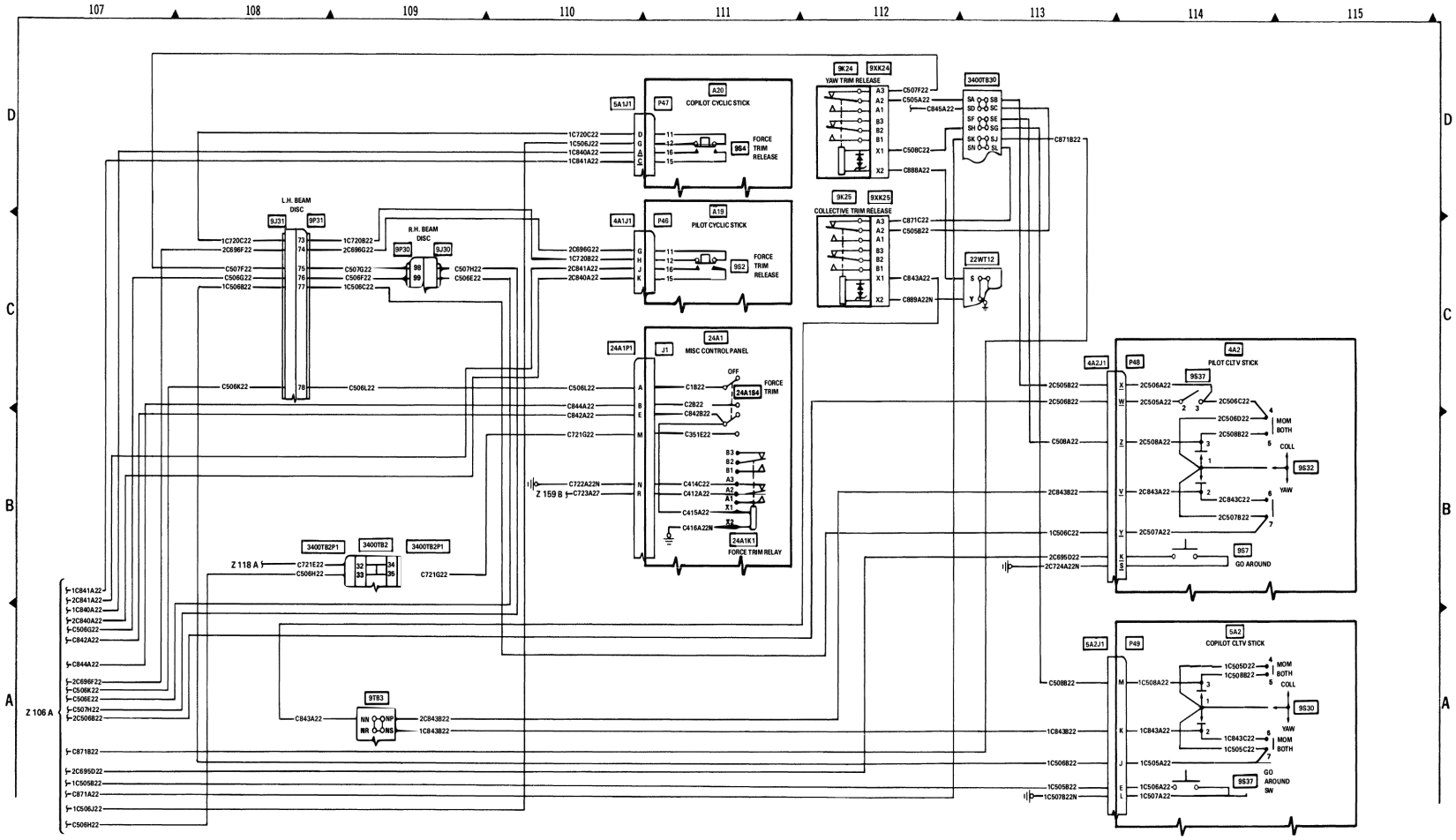
ICN-412-A-980101-B-97499-00008-A-001-01

Figure 1. Digital AFCS Wiring Diagram (Sheet 7 of 13)



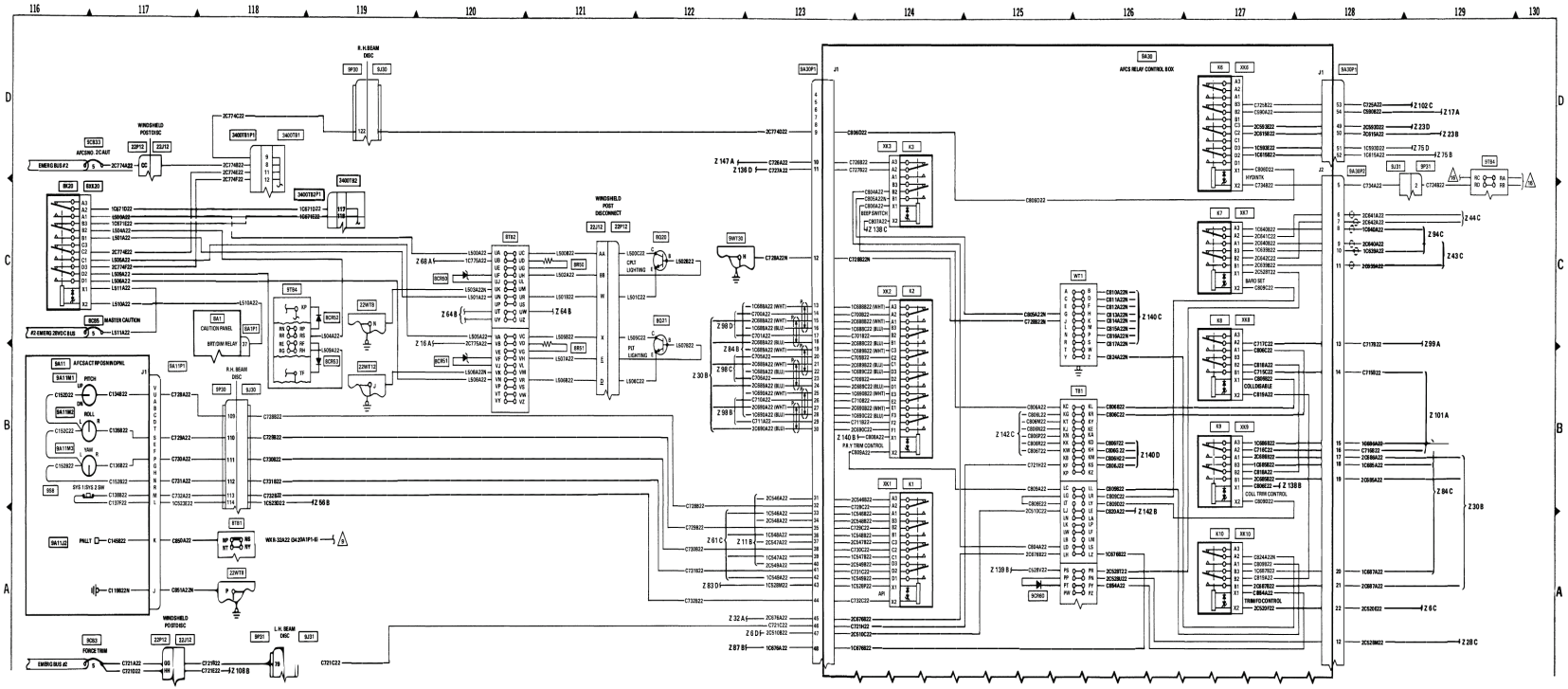
ICN-412-A-980101-B-97499-00009-A-001-01

Figure 1. Digital AFCS Wiring Diagram (Sheet 8 of 13)



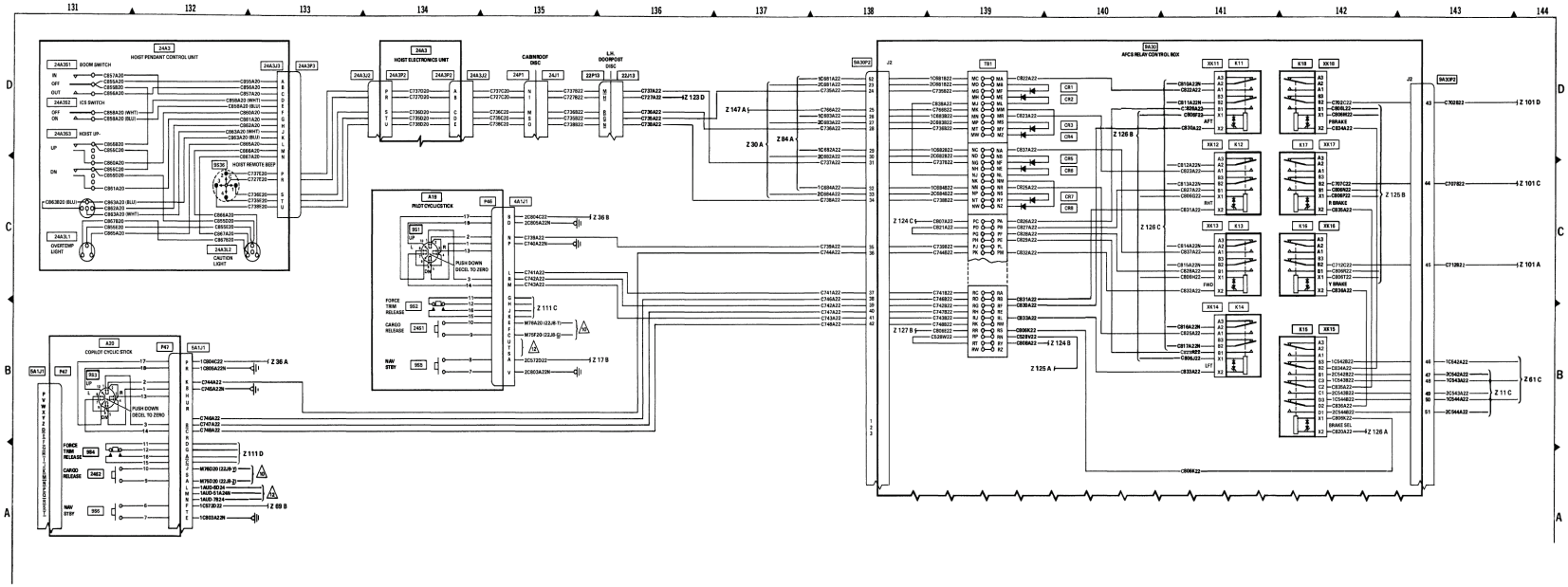
ICN-412-A-980101-B-97499-00010-A-001-01

Figure 1. Digital AFCS Wiring Diagram (Sheet 9 of 13)



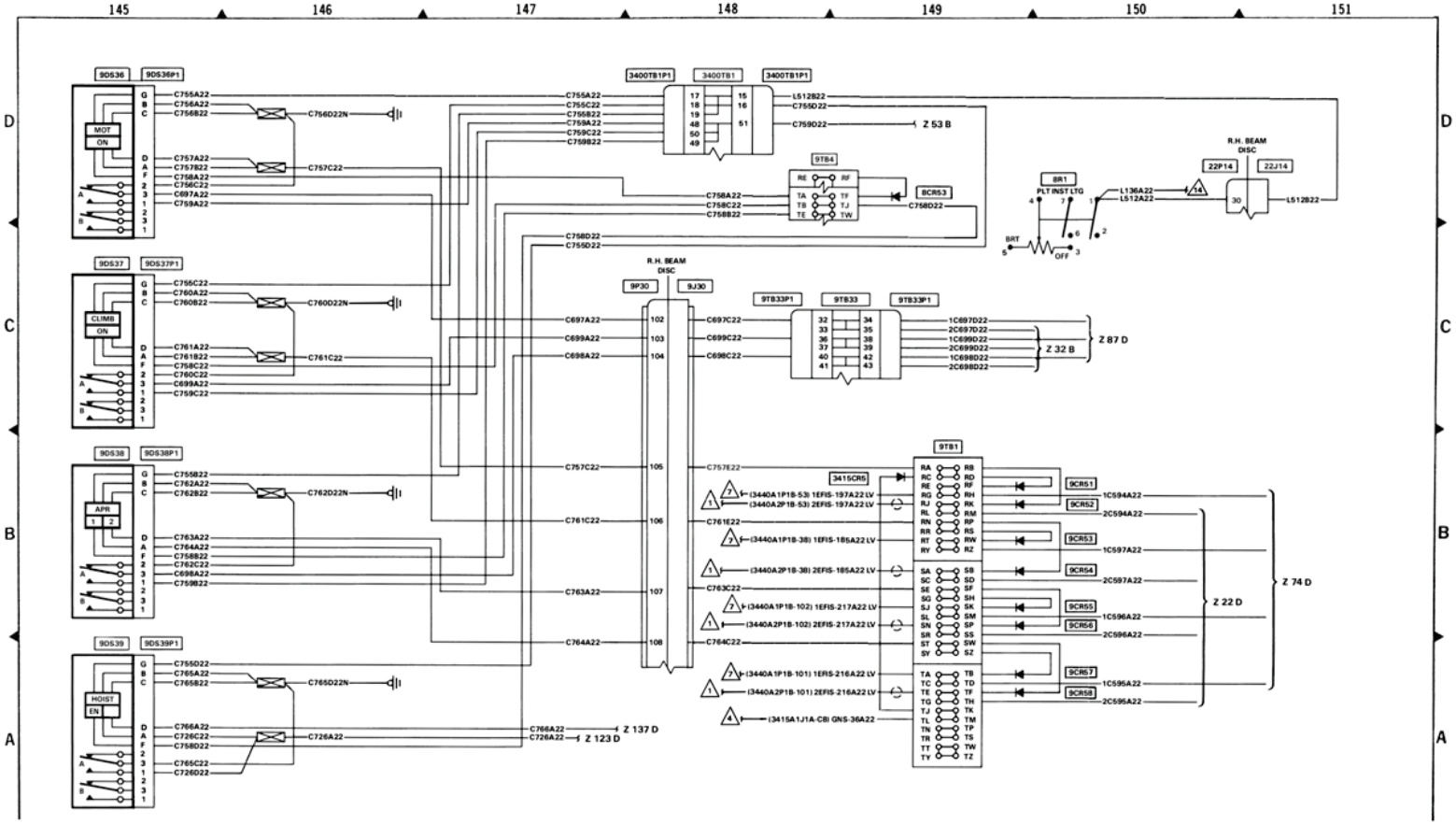
ICN-412-A-980101-B-97499-00011-A-001-01

Figure 1. Digital AFCS Wiring Diagram (Sheet 10 of 13)



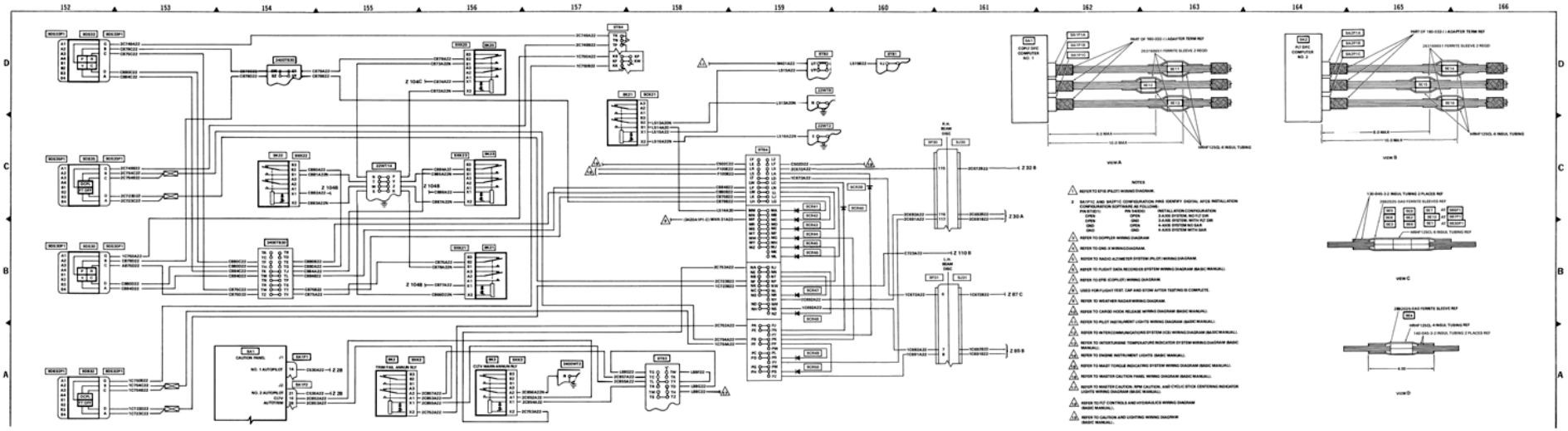
ICN-412-A-980101-B-97499-00012-A-001-01

Figure 1. Digital AFCS Wiring Diagram (Sheet 11 of 13)



ICN-412-A-980101-B-97499-00013-A-001-01

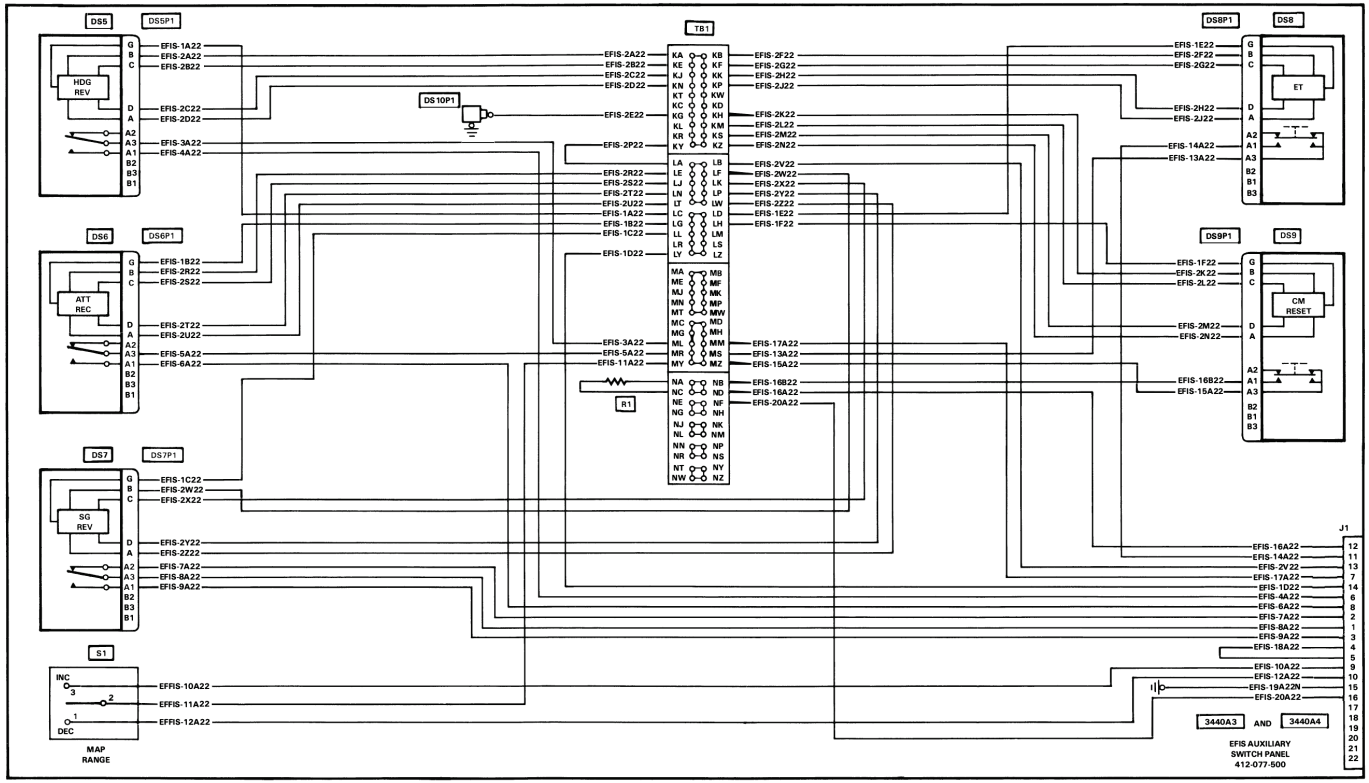
Figure 1. Digital AFCS Wiring Diagram (Sheet 12 of 13)



ICN-412-A-980101-B-97499-00014-A-001-01

Figure 1. Digital AFCS Wiring Diagram (Sheet 13 of 13)

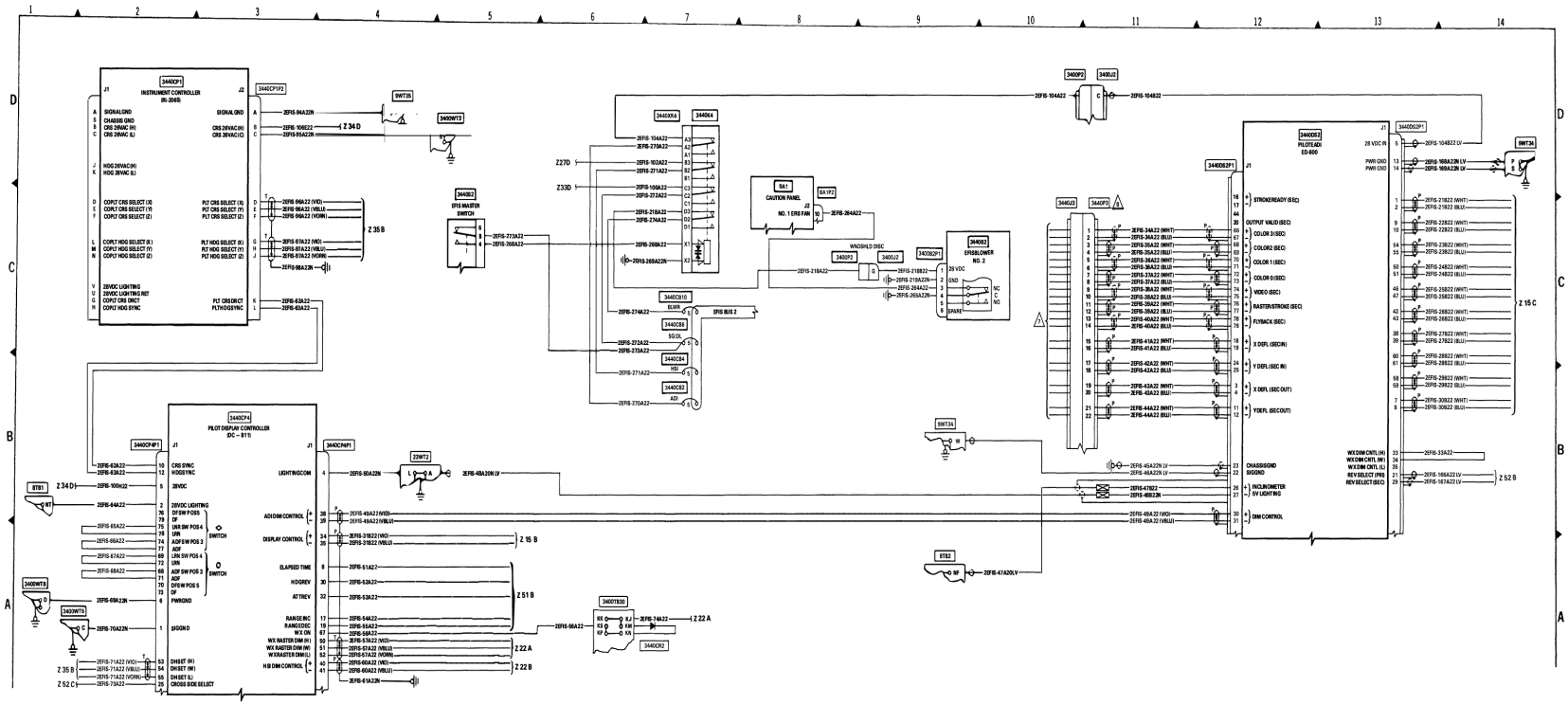
Intentionally blank page



ICN-412-A-980101-B-97499-00015-A-001-01

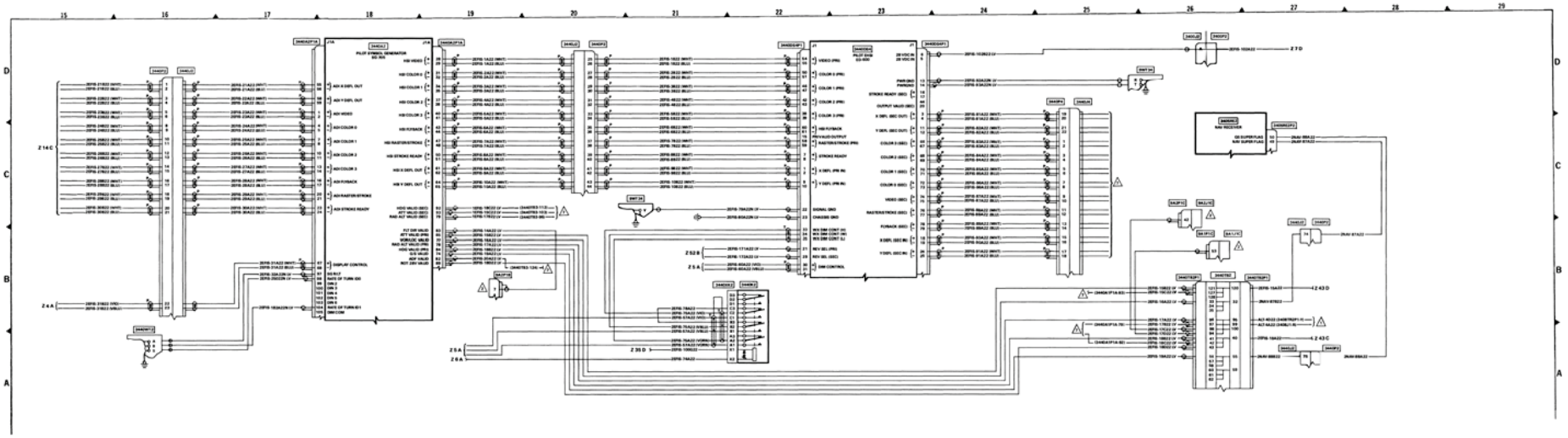
Figure 1. EFIS AUX Switch Panel Wiring Diagram (Sheet 1 of 1)

Intentionally blank page



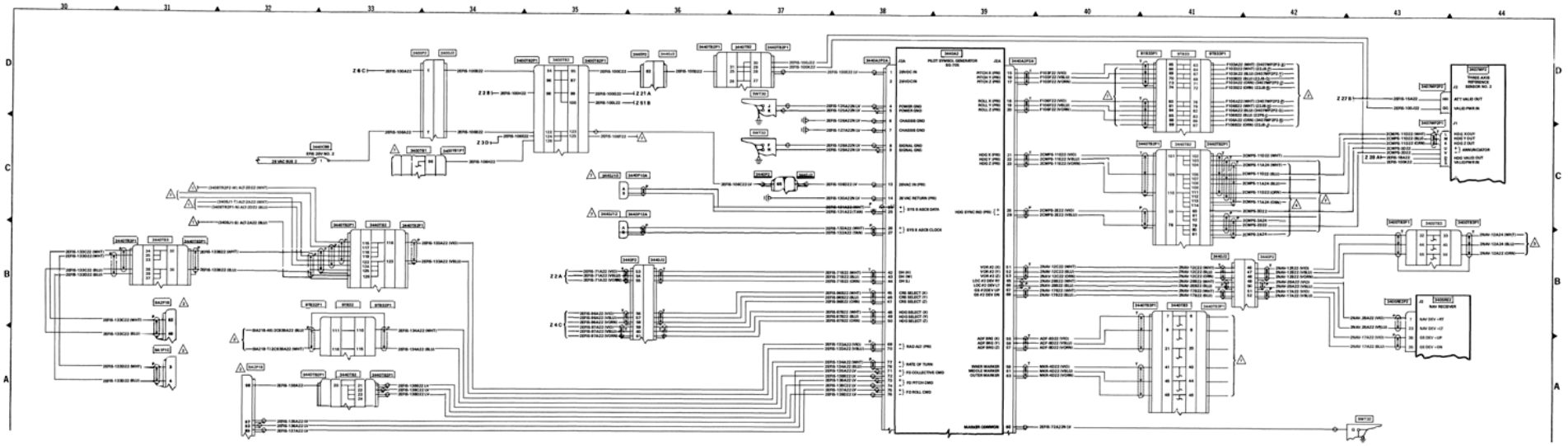
ICN-412-A-980101-B-97499-00016-A-001-01

Figure 1. EFIS (Pilot) System Wiring Diagram (Sheet 1 of 4)



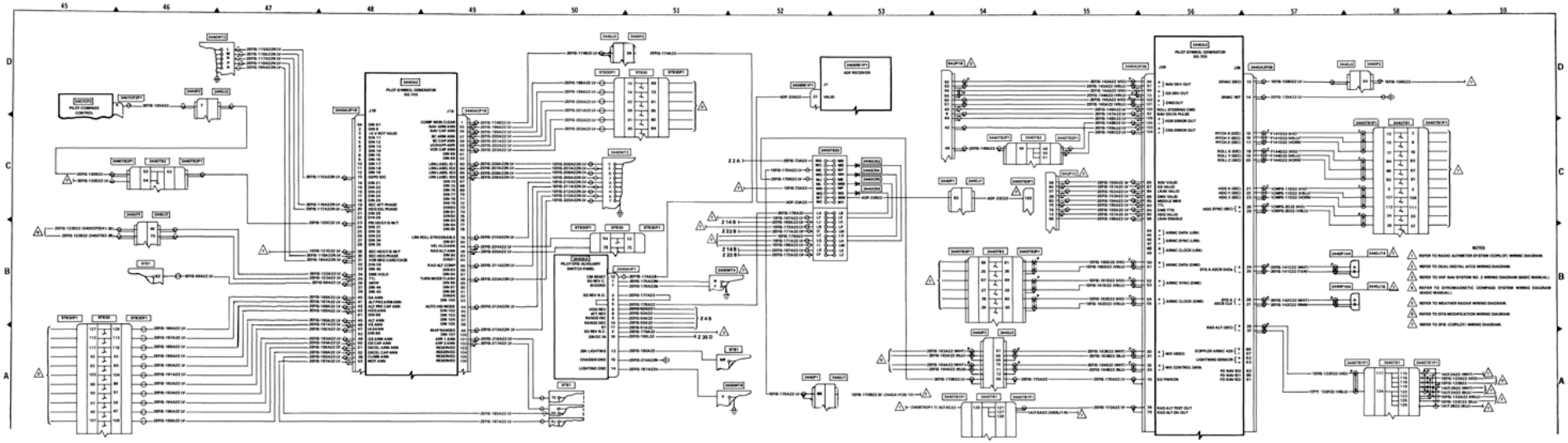
ICN-412-A-980101-B-97499-00017-A-001-01

Figure 1. EFIS (Pilot) System Wiring Diagram (Sheet 2 of 4)



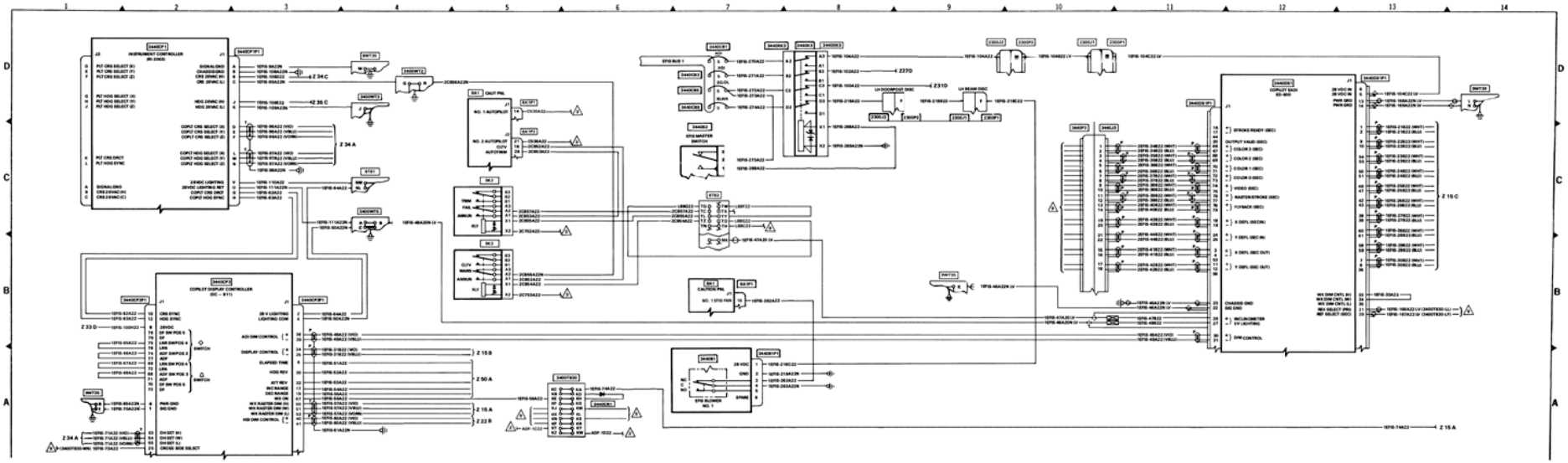
ICN-412-A-980101-B-97499-00018-A-001-01

Figure 1. EFIS (Pilot) System Wiring Diagram (Sheet 3 of 4)



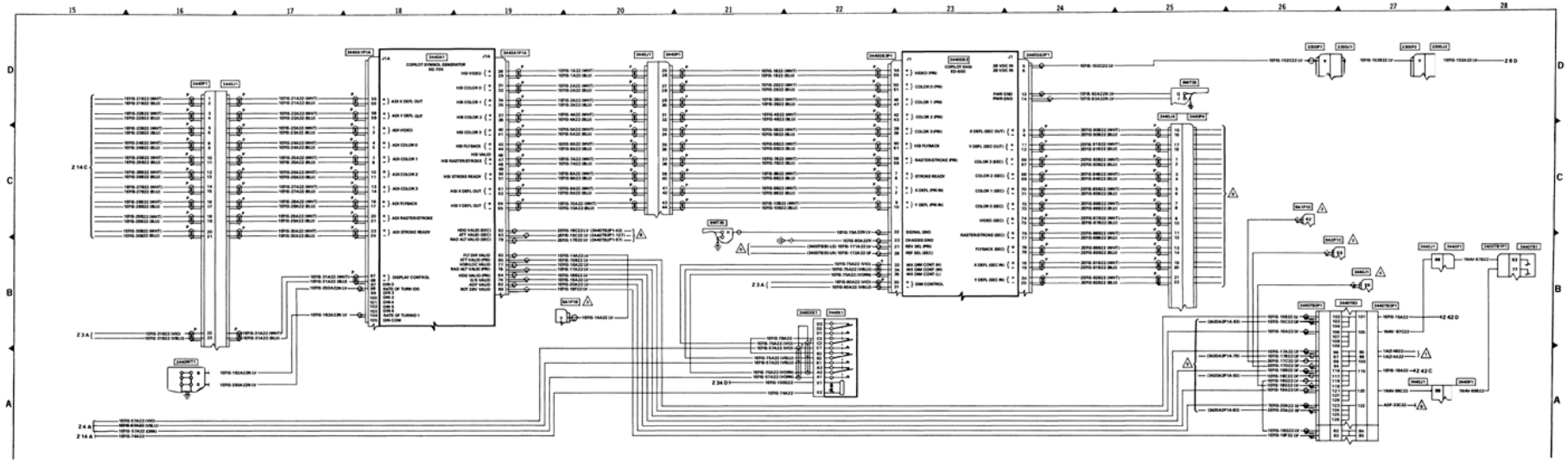
ICN-412-A-980101-B-97499-00019-A-001-01

Figure 1. EFIS (Pilot) System Wiring Diagram (Sheet 4 of 4)



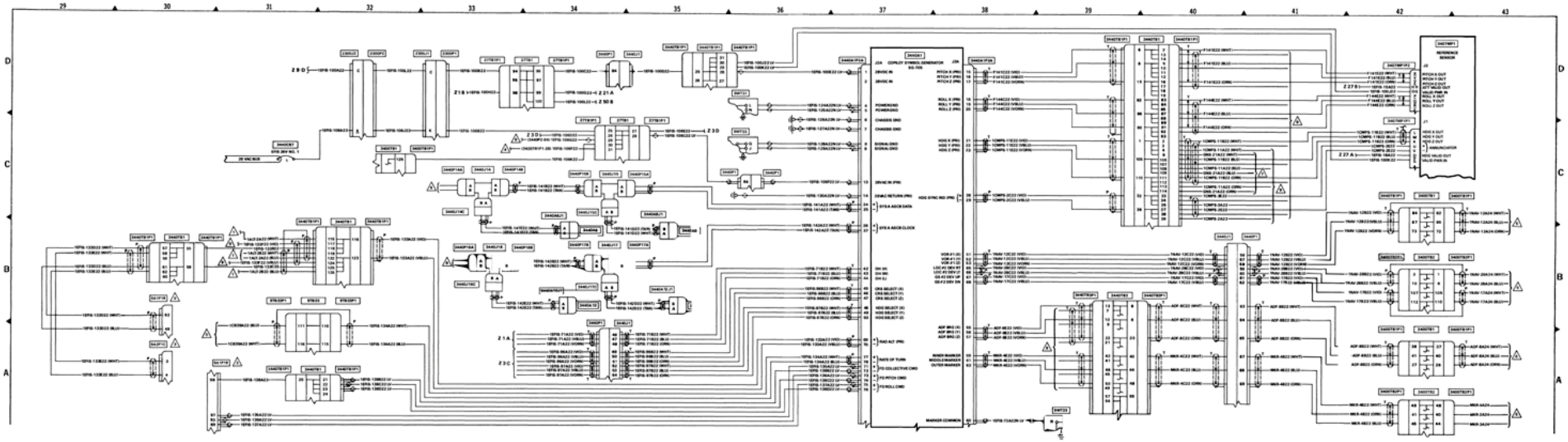
ICN-412-A-980101-B-97499-00020-A-001-01

Figure 1. EFIS (Copilot) System Wiring Diagram (Sheet 1 of 4)



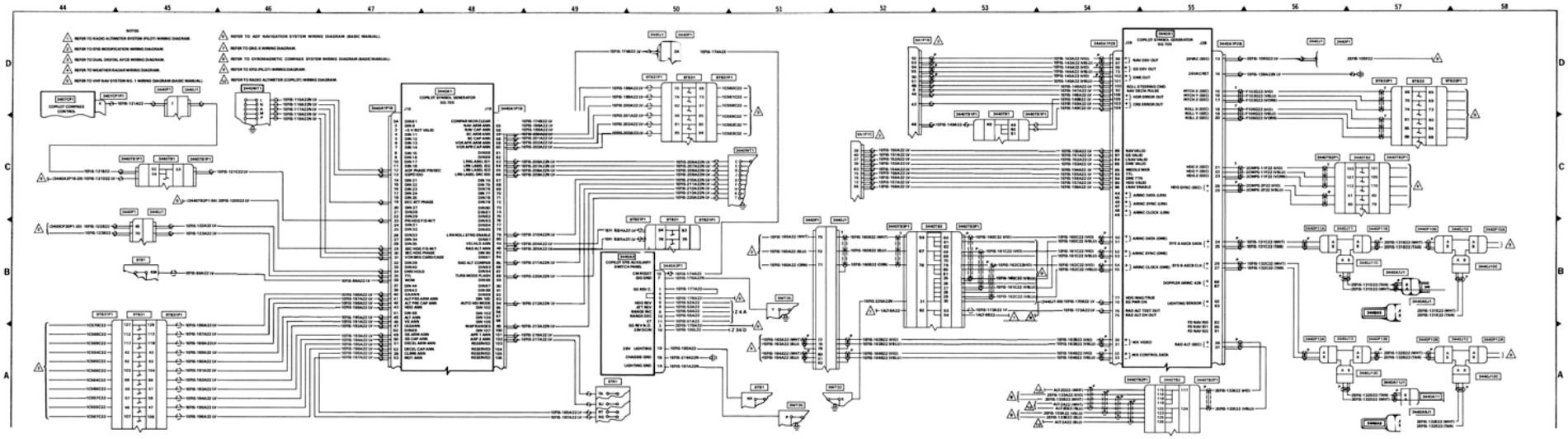
ICN-412-A-980101-B-97499-00021-A-001-01

Figure 1. EFIS (Copilot) System Wiring Diagram (Sheet 2 of 4)



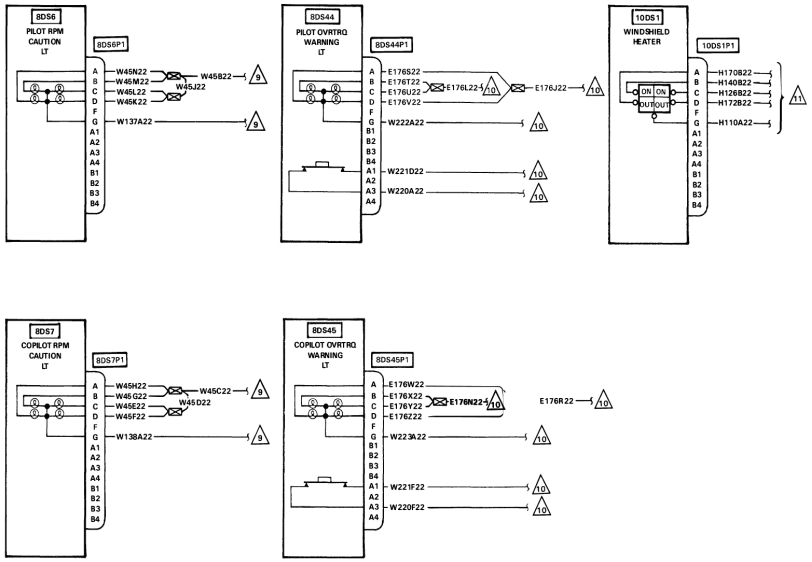
ICN-412-A-980101-B-97499-00022-A-001-01

Figure 1. EFIS (Copilot) System Wiring Diagram (Sheet 3 of 4)



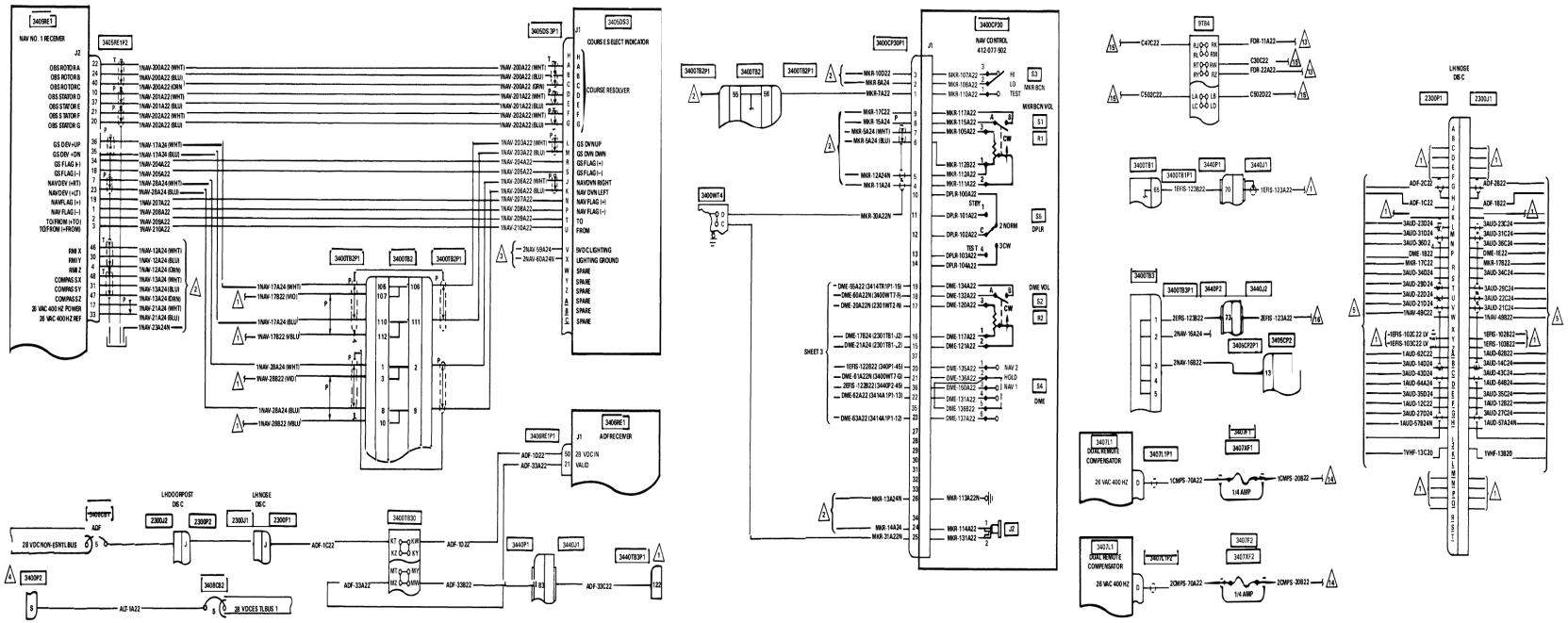
ICN-412-A-980101-B-97499-00023-A-001-01

Figure 1. EFIS (Copilot) System Wiring Diagram (Sheet 4 of 4)



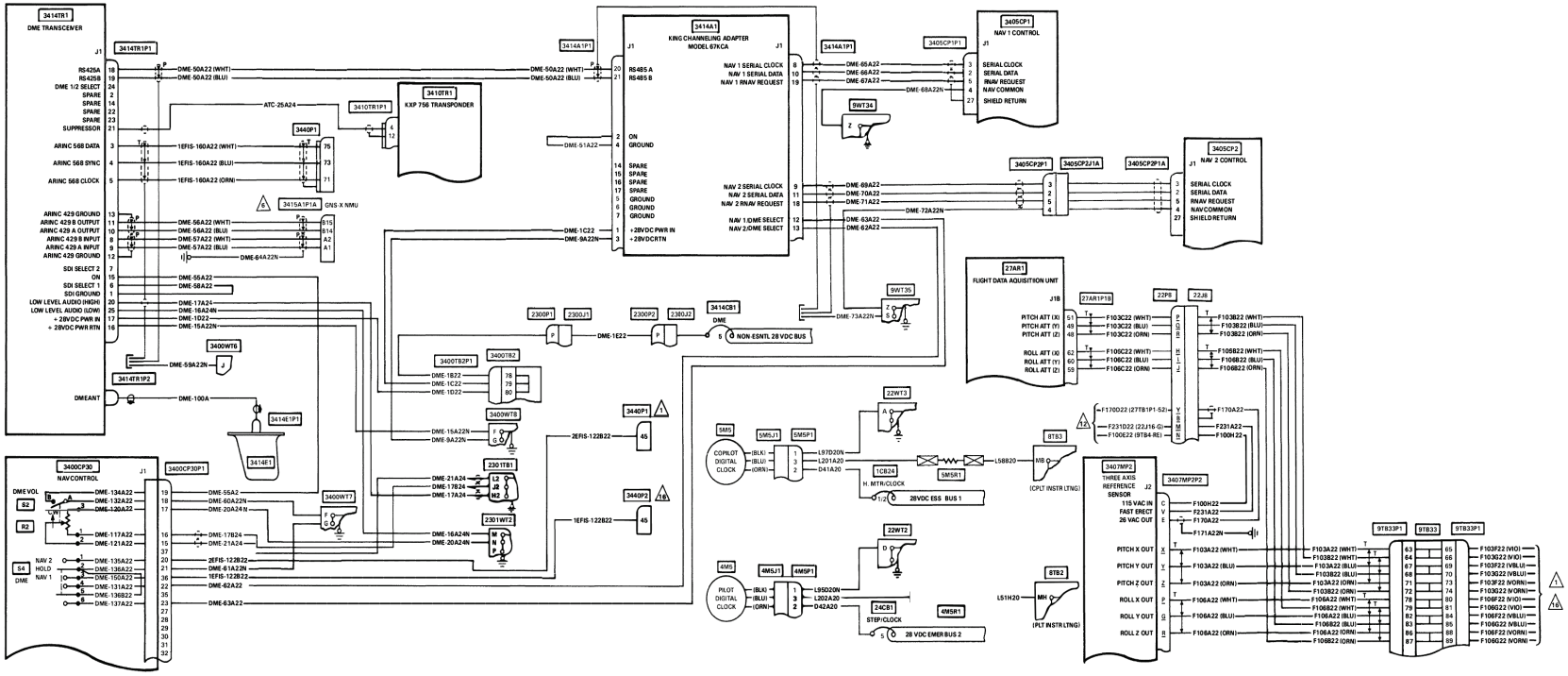
ICN-412-A-980101-B-97499-00024-A-001-01

Figure 1. EFIS (Modification) System Wiring Diagram (Sheet 1 of 5)



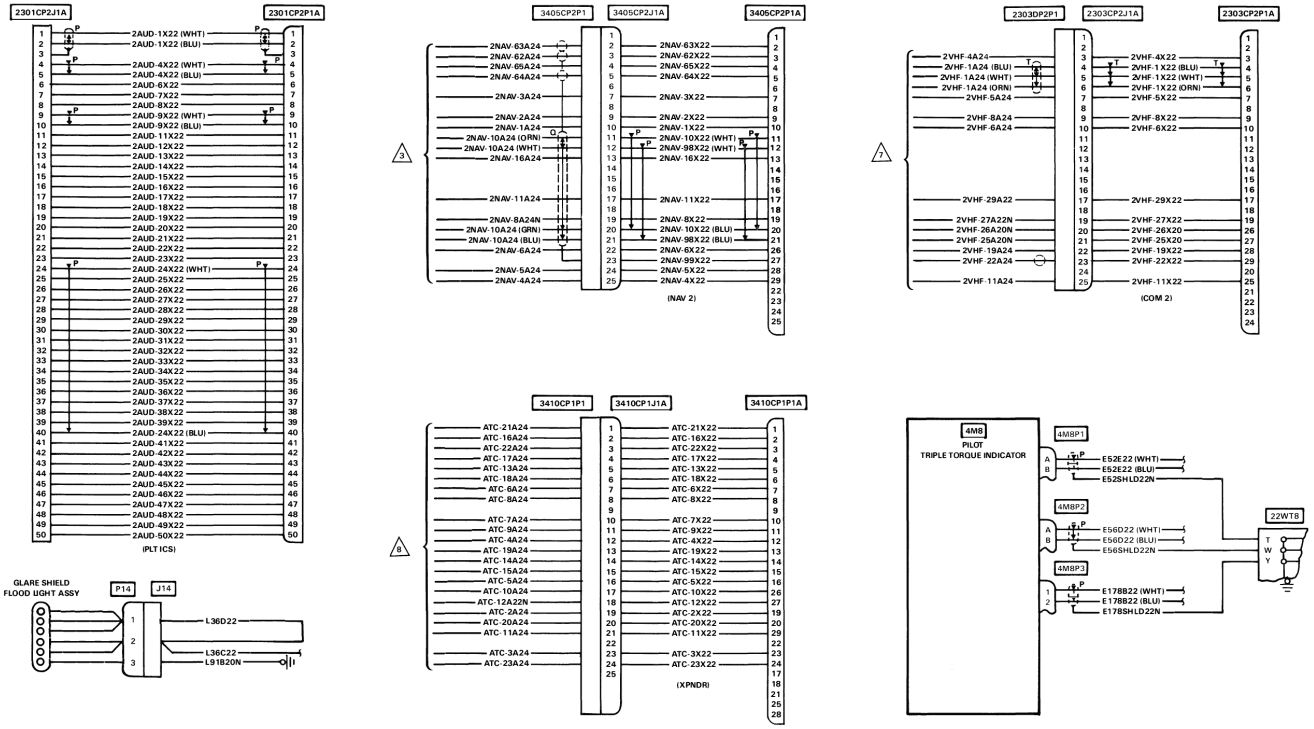
ICN-412-A-980101-B-97499-00025-A-001-01

Figure 1. EFIS (Modification) System Wiring Diagram (Sheet 2 of 5)



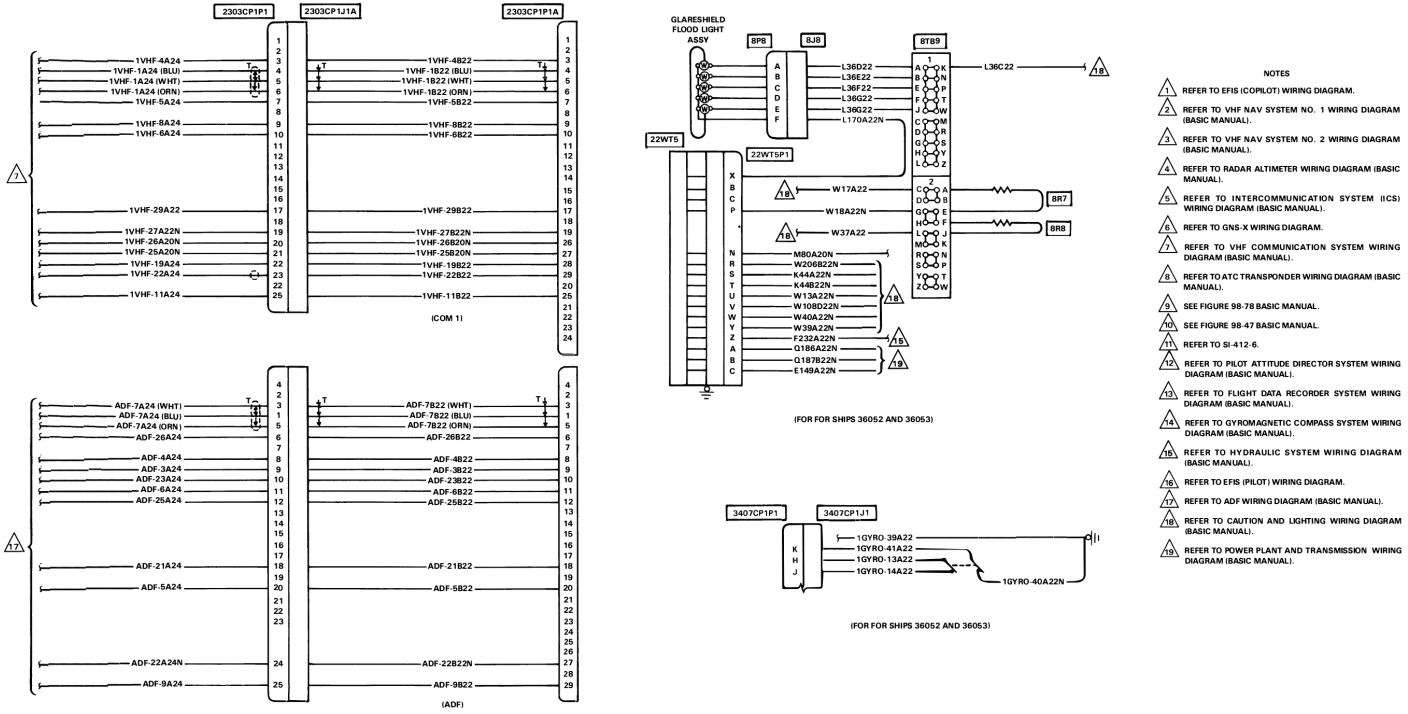
ICN-412-A-980101-B-97499-00026-A-001-01

Figure 1. EFIS (Modification) System Wiring Diagram (Sheet 3 of 5)



ICN-412-A-980101-B-97499-00027-A-001-01

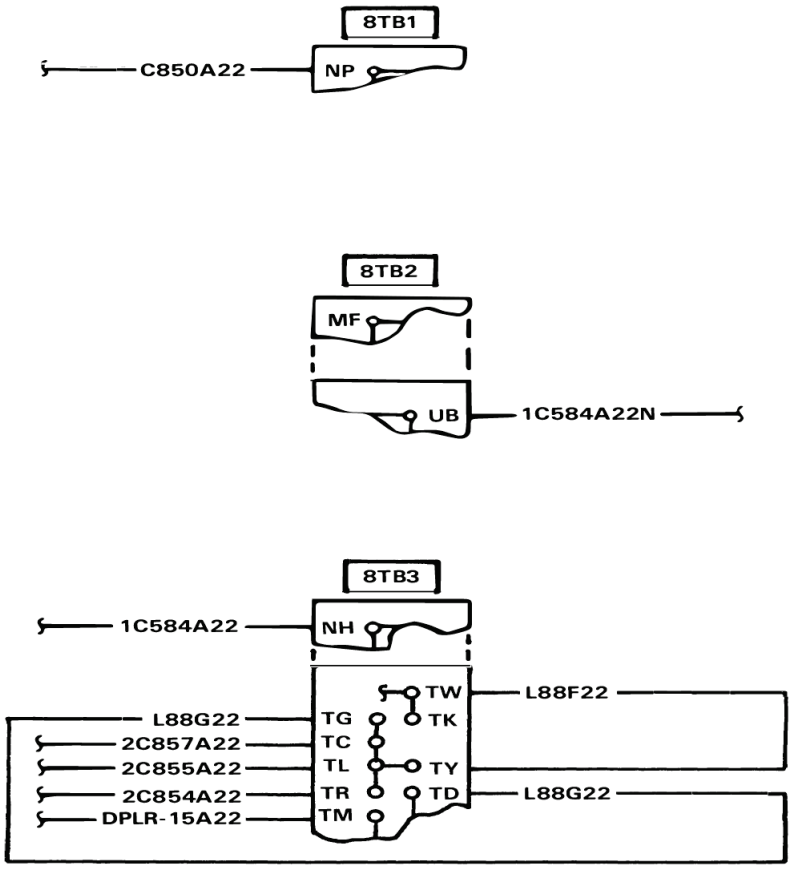
Figure 1. EFIS (Modification) System Wiring Diagram (Sheet 4 of 5)



ICN-412-A-980101-B-97499-00028-A-001-01

Figure 1. EFIS (Modification) System Wiring Diagram (Sheet 5 of 5)

Intentionally blank page

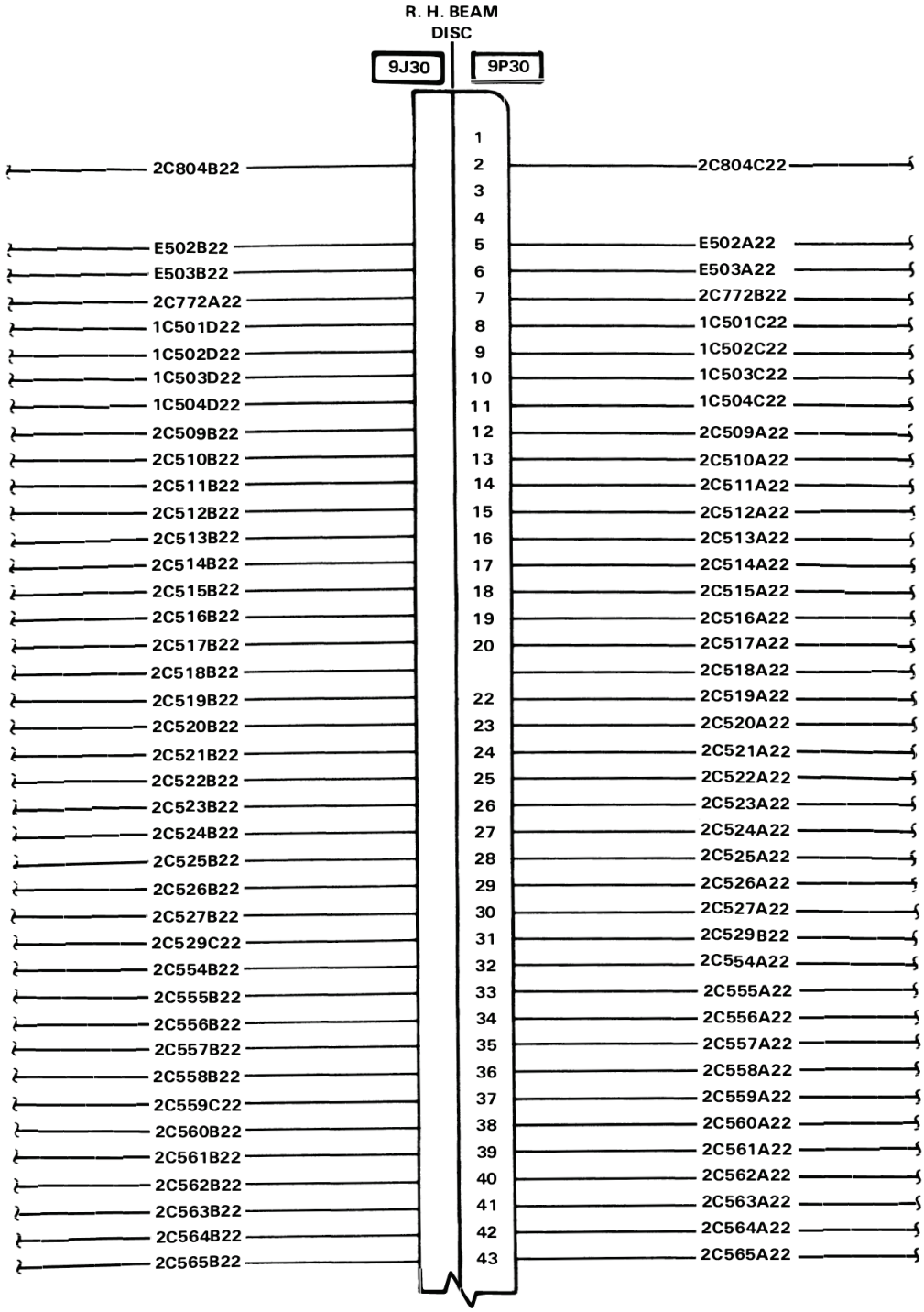


NOTE

REFER TO CHAPTER 98 OF BASIC MANUAL FOR REMAINDER OF 8TB1, 8TB2, and 8TB3.

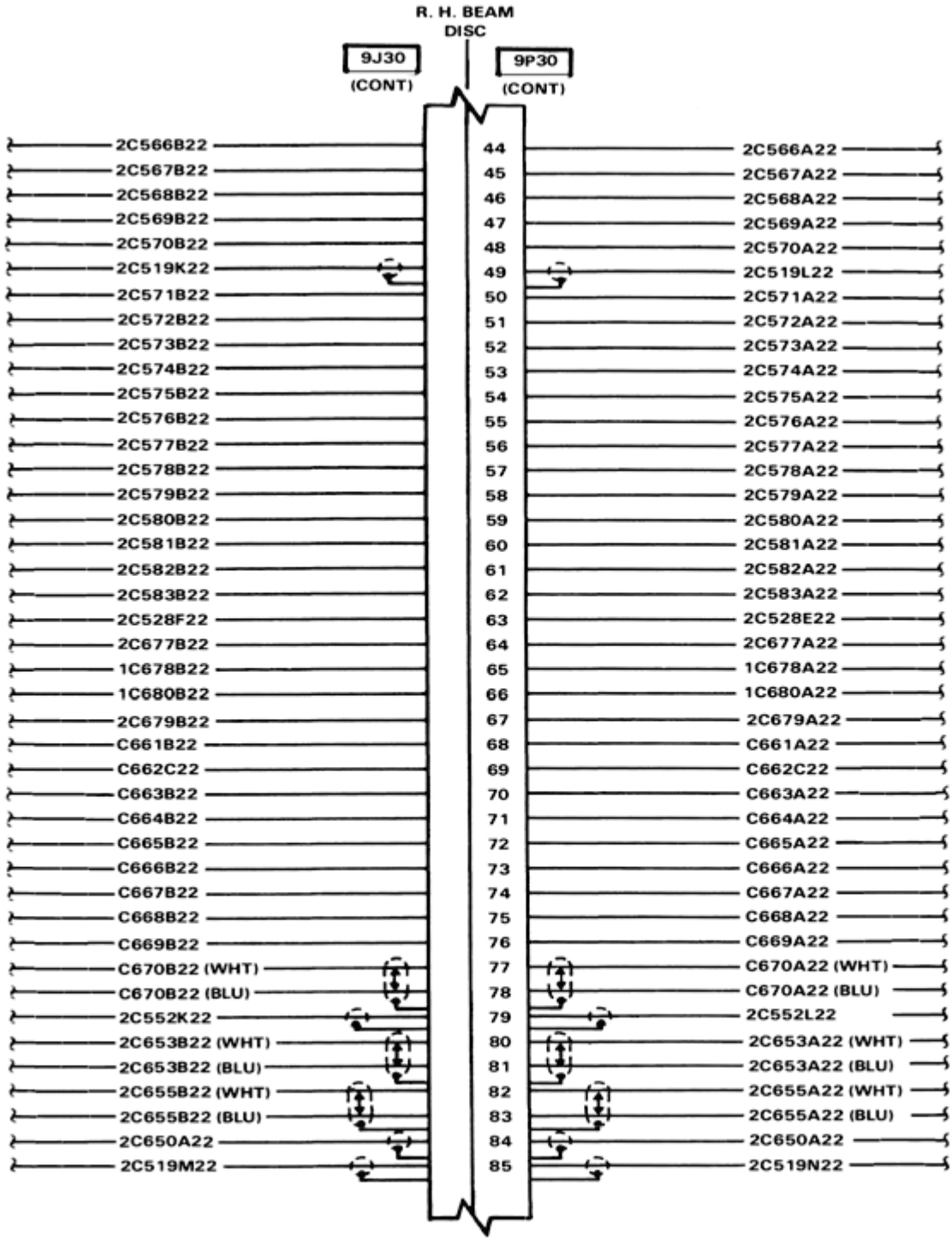
ICN-412-A-980101-B-97499-00029-A-001-01

Figure 1. Component Replacement (Sheet 1 of 59)



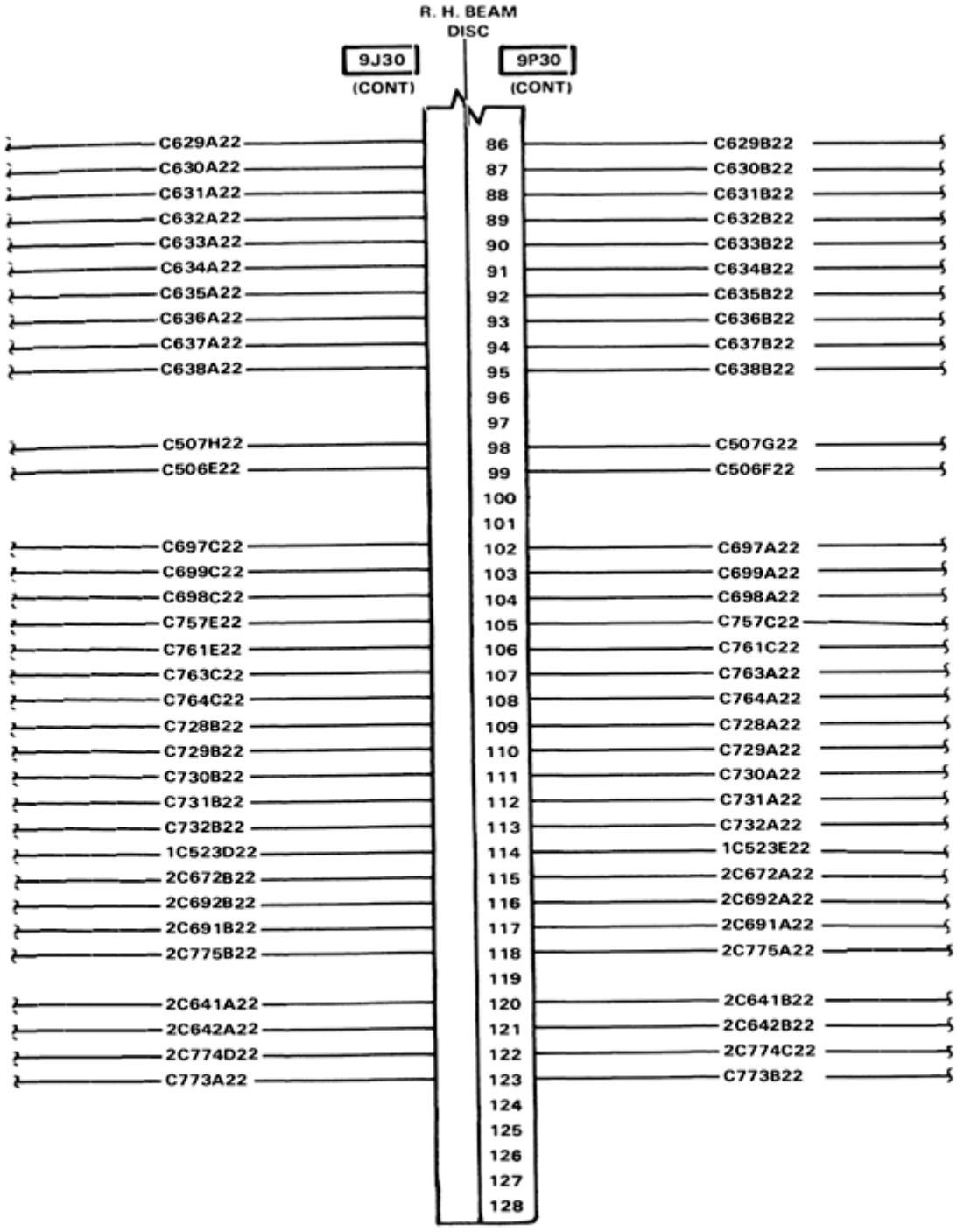
ICN-412-A-980101-B-97499-00030-A-001-01

Figure 1. Component Replacement (Sheet 2 of 59)



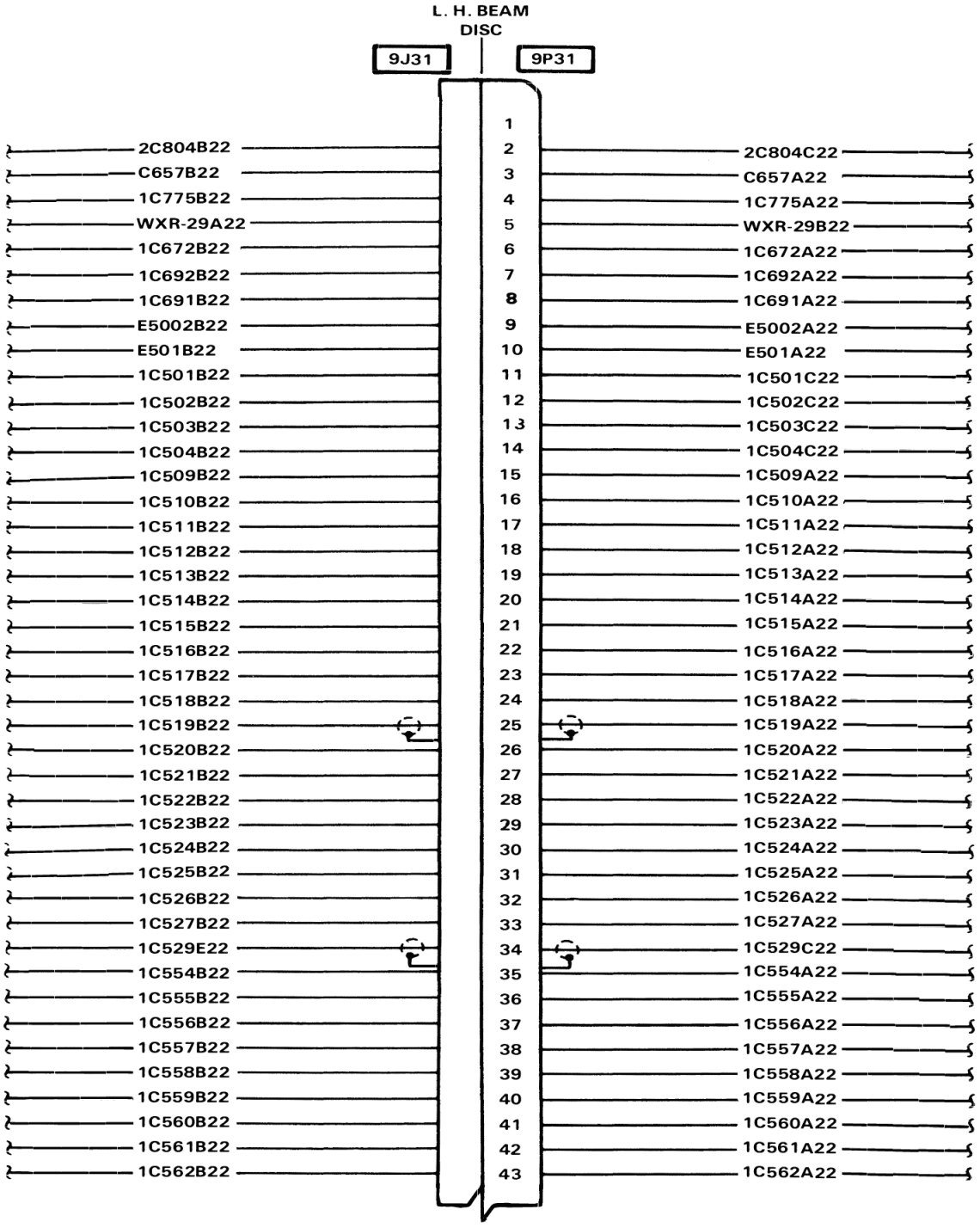
ICN-412-A-980101-B-97499-00031-A-001-01

Figure 1. Component Replacement (Sheet 3 of 59)



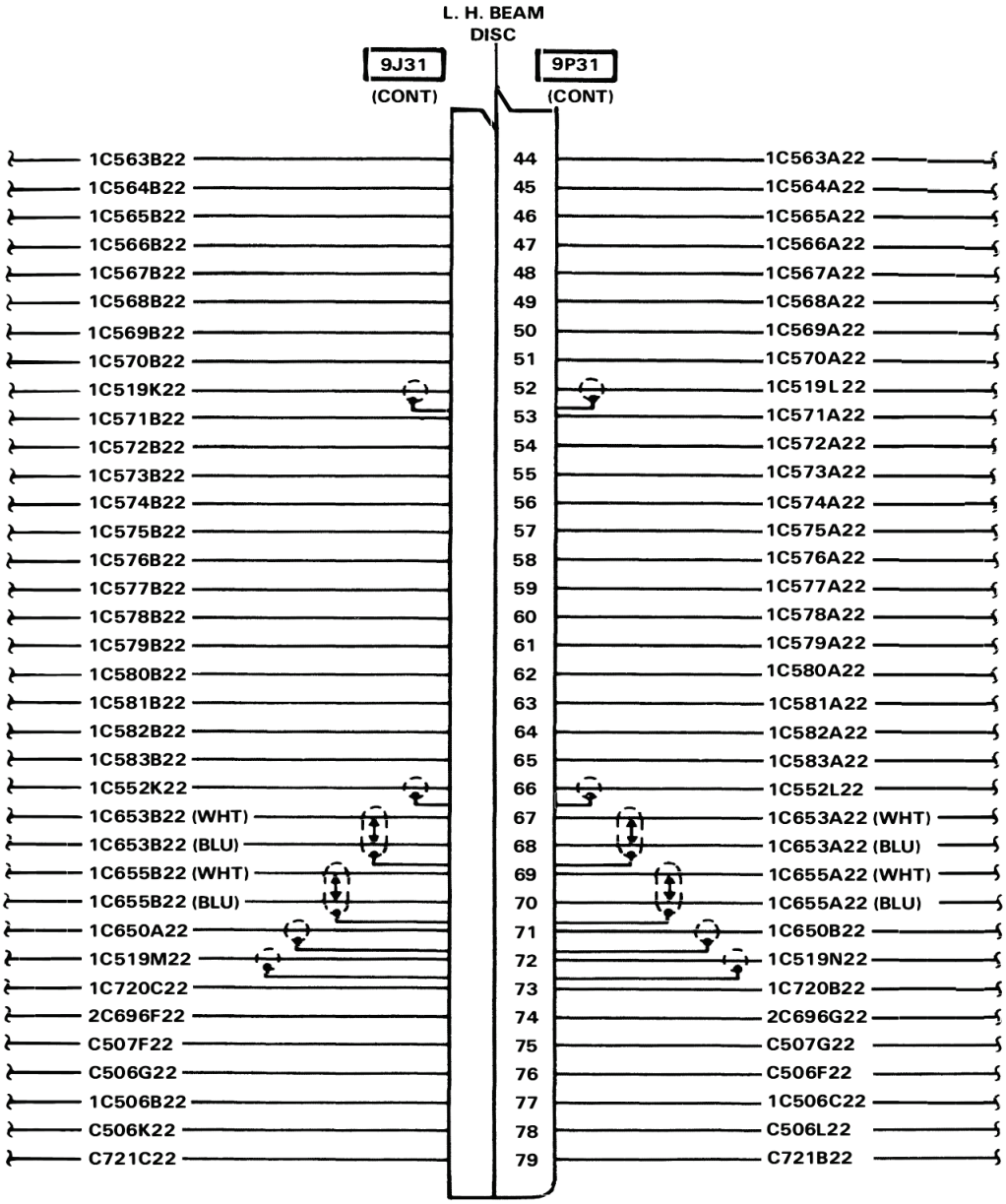
ICN-412-A-980101-B-97499-00032-A-001-01

Figure 1. Component Replacement (Sheet 4 of 59)



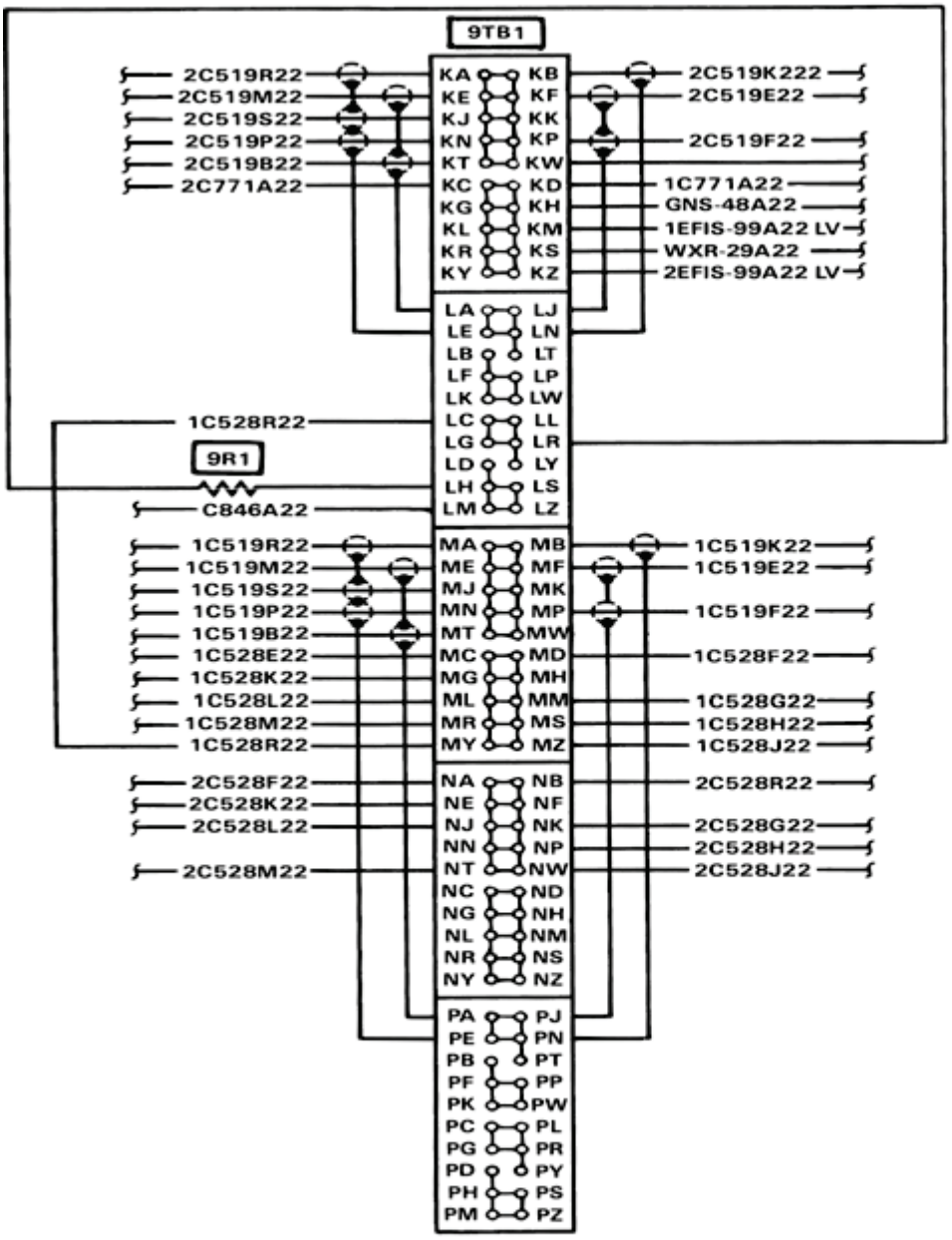
ICN-412-A-980101-B-97499-00033-A-001-01

Figure 1. Component Replacement (Sheet 5 of 59)



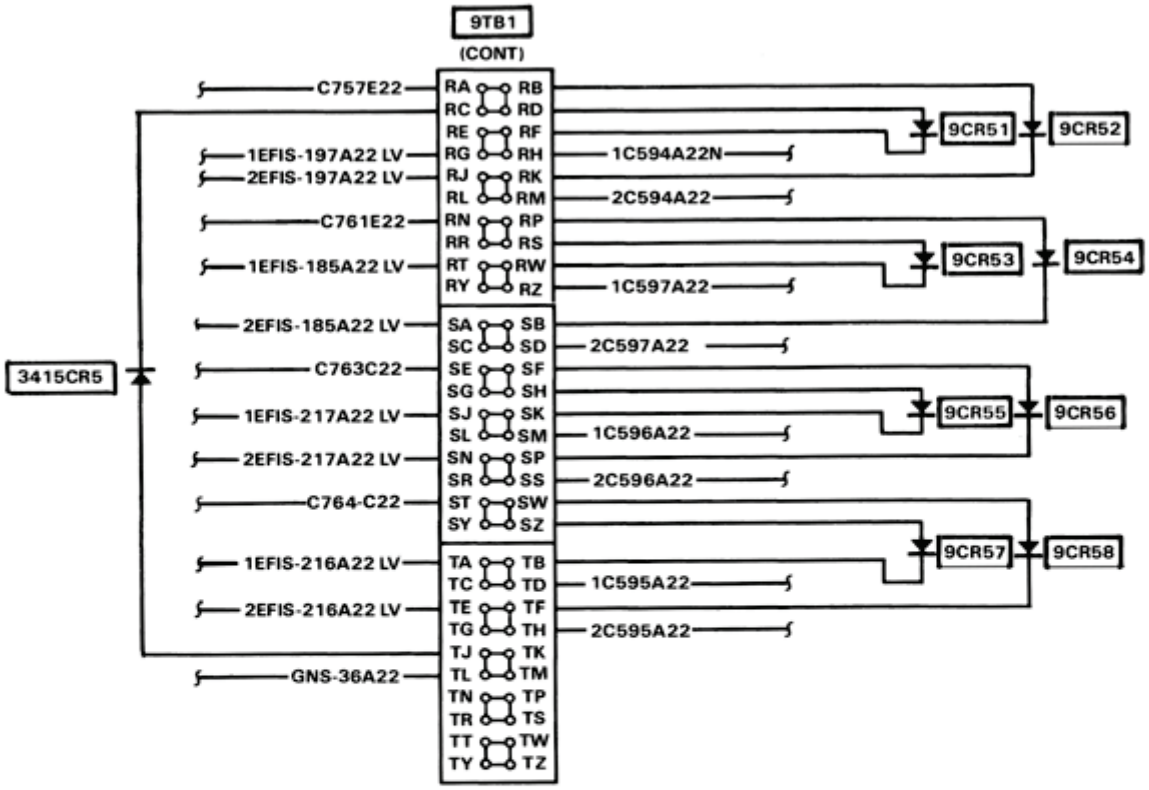
ICN-412-A-980101-B-97499-00034-A-001-01

Figure 1. Component Replacement (Sheet 6 of 59)



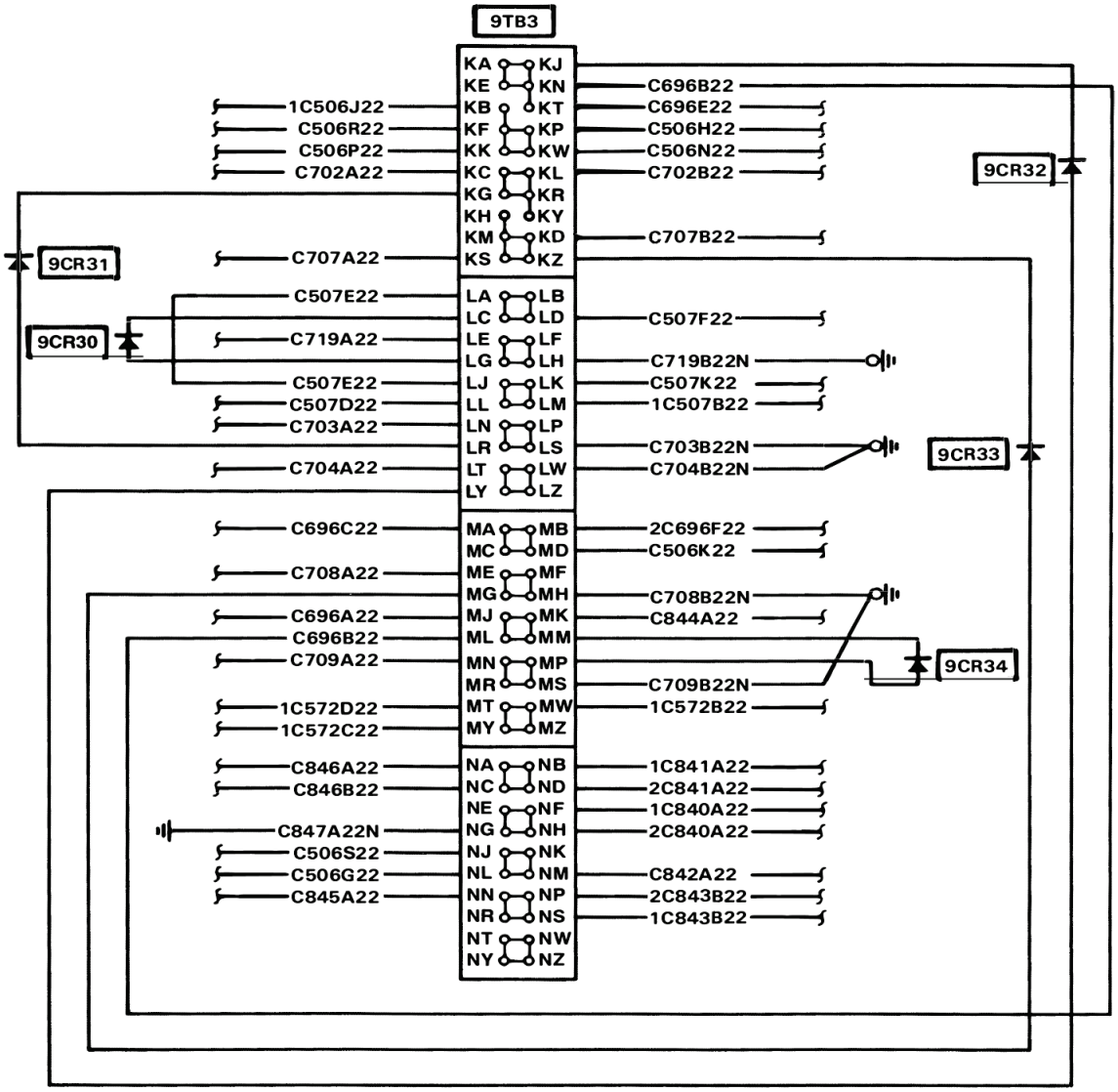
ICN-412-A-980101-B-97499-00035-A-001-01

Figure 1. Component Replacement (Sheet 7 of 59)



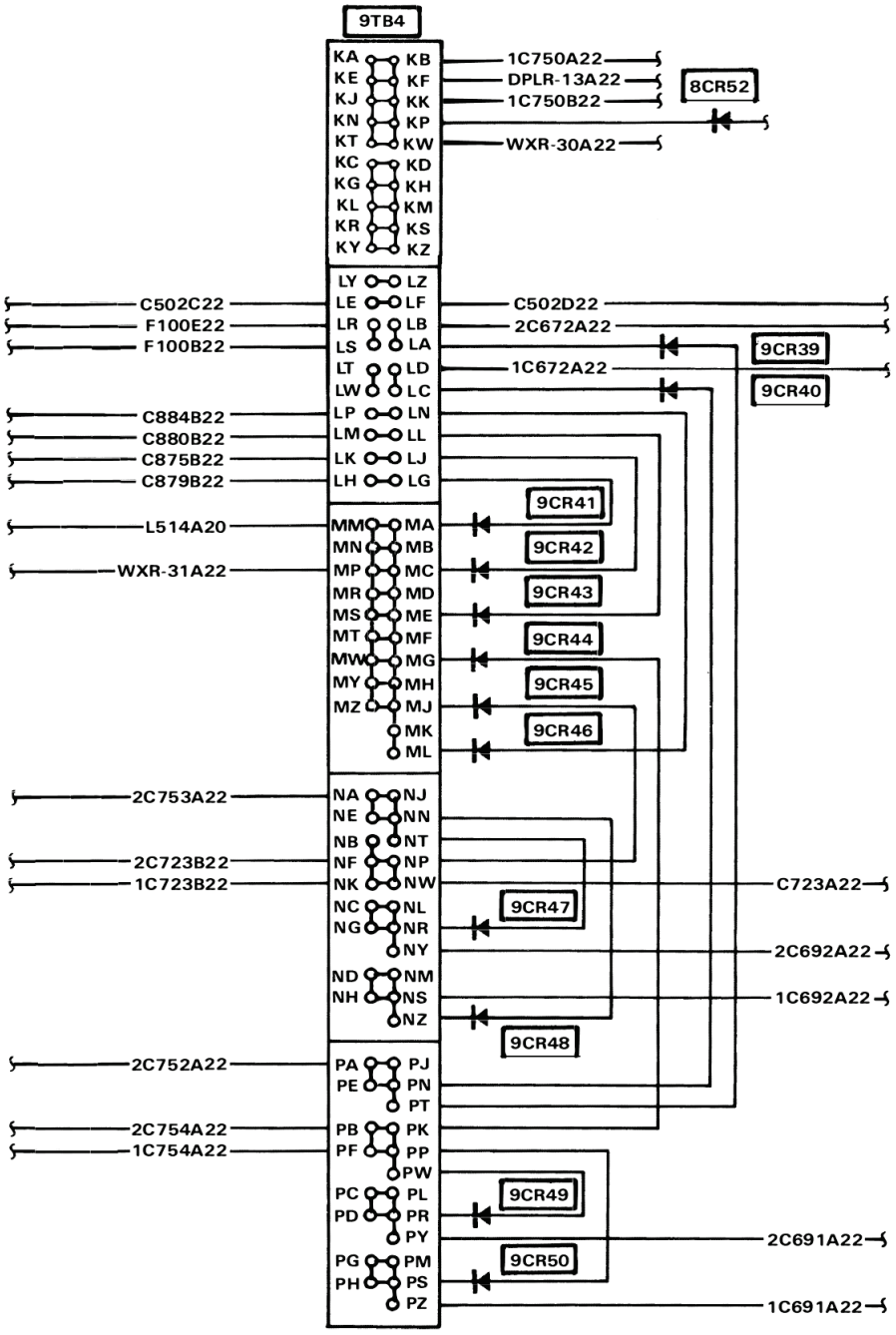
ICN-412-A-980101-B-97499-00036-A-001-01

Figure 1. Component Replacement (Sheet 8 of 59)



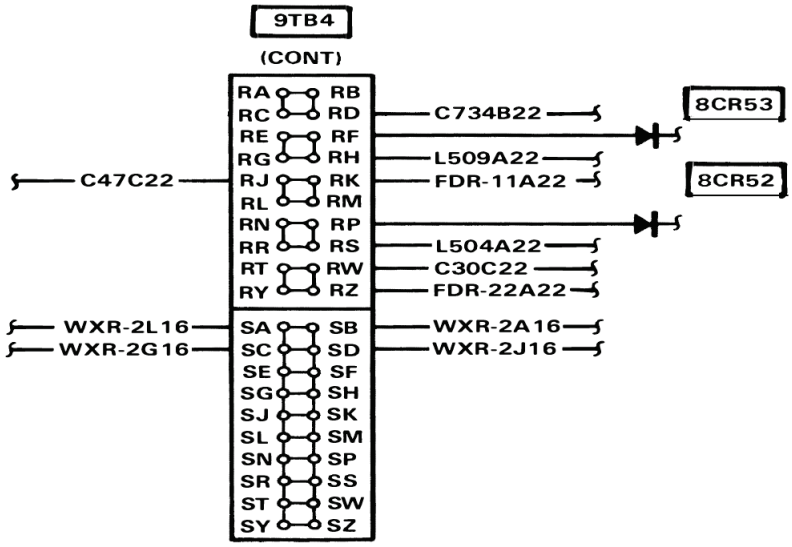
ICN-412-A-980101-B-97499-00037-A-001-01

Figure 1. Component Replacement (Sheet 9 of 59)



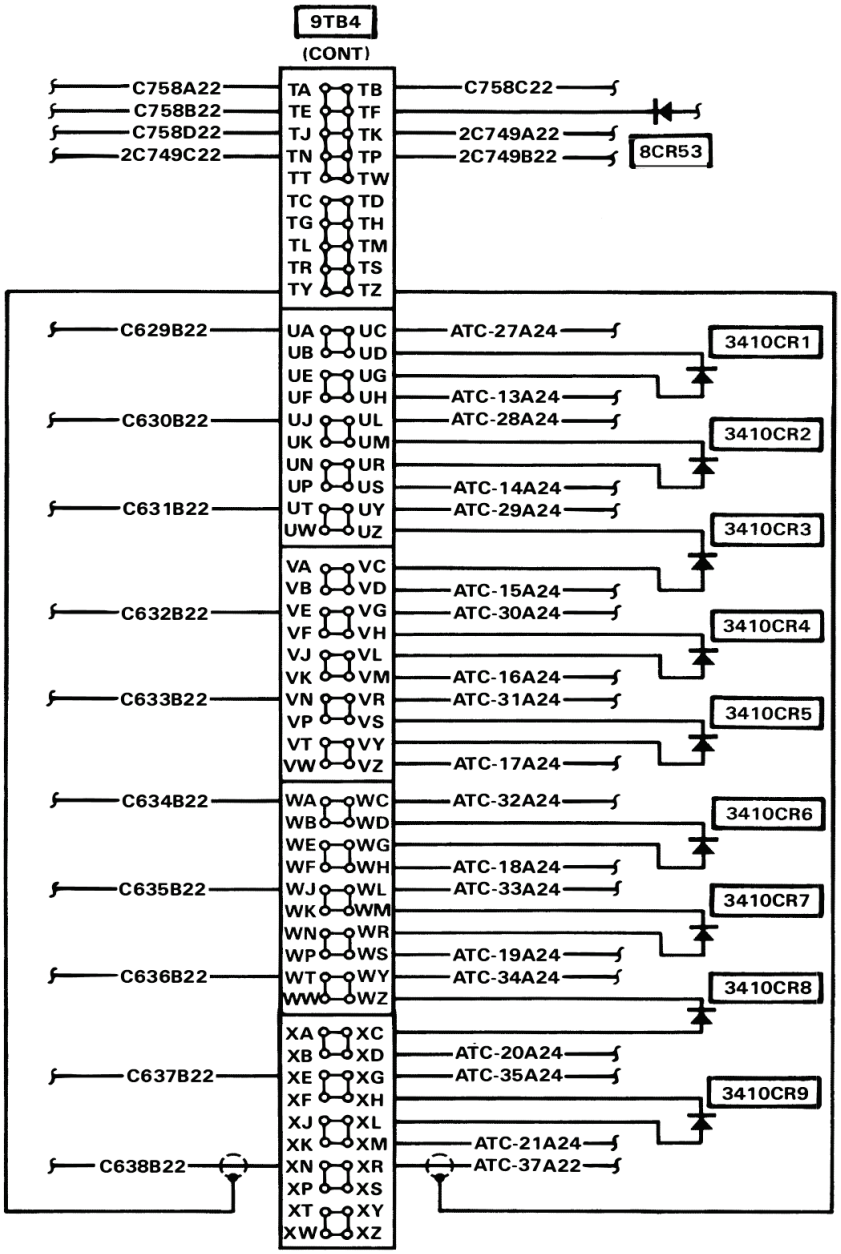
ICN-412-A-980101-B-97499-00038-A-001-01

Figure 1. Component Replacement (Sheet 10 of 59)



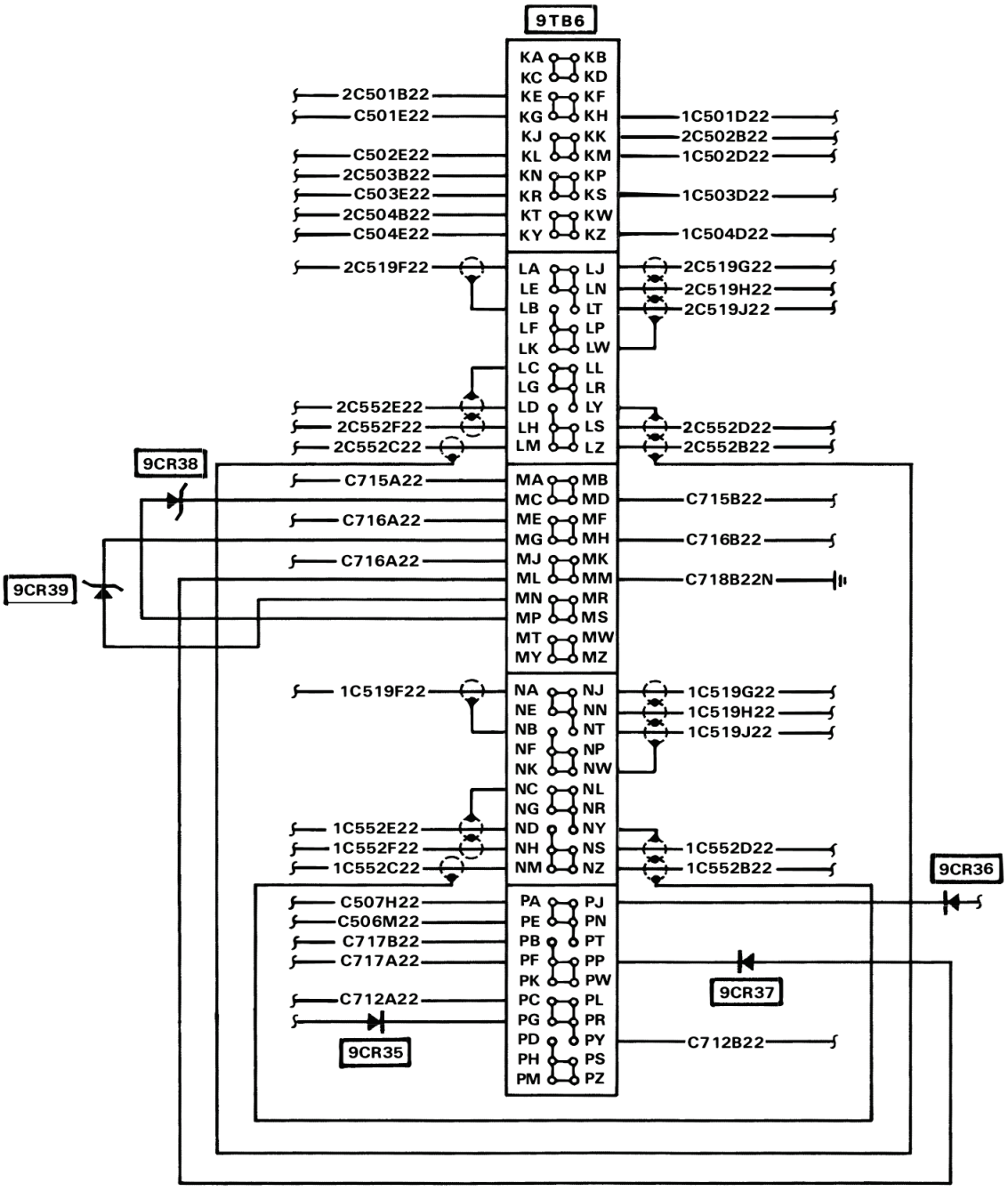
ICN-412-A-980101-B-97499-00039-A-001-01

Figure 1. Component Replacement (Sheet 11 of 59)



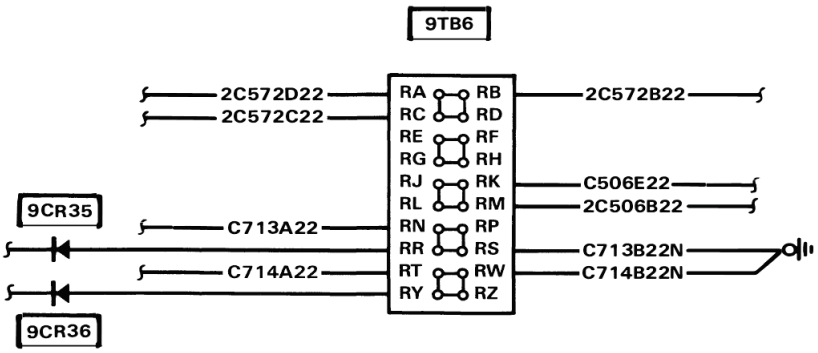
ICN-412-A-980101-B-97499-00040-A-001-01

Figure 1. Component Replacement (Sheet 12 of 59)



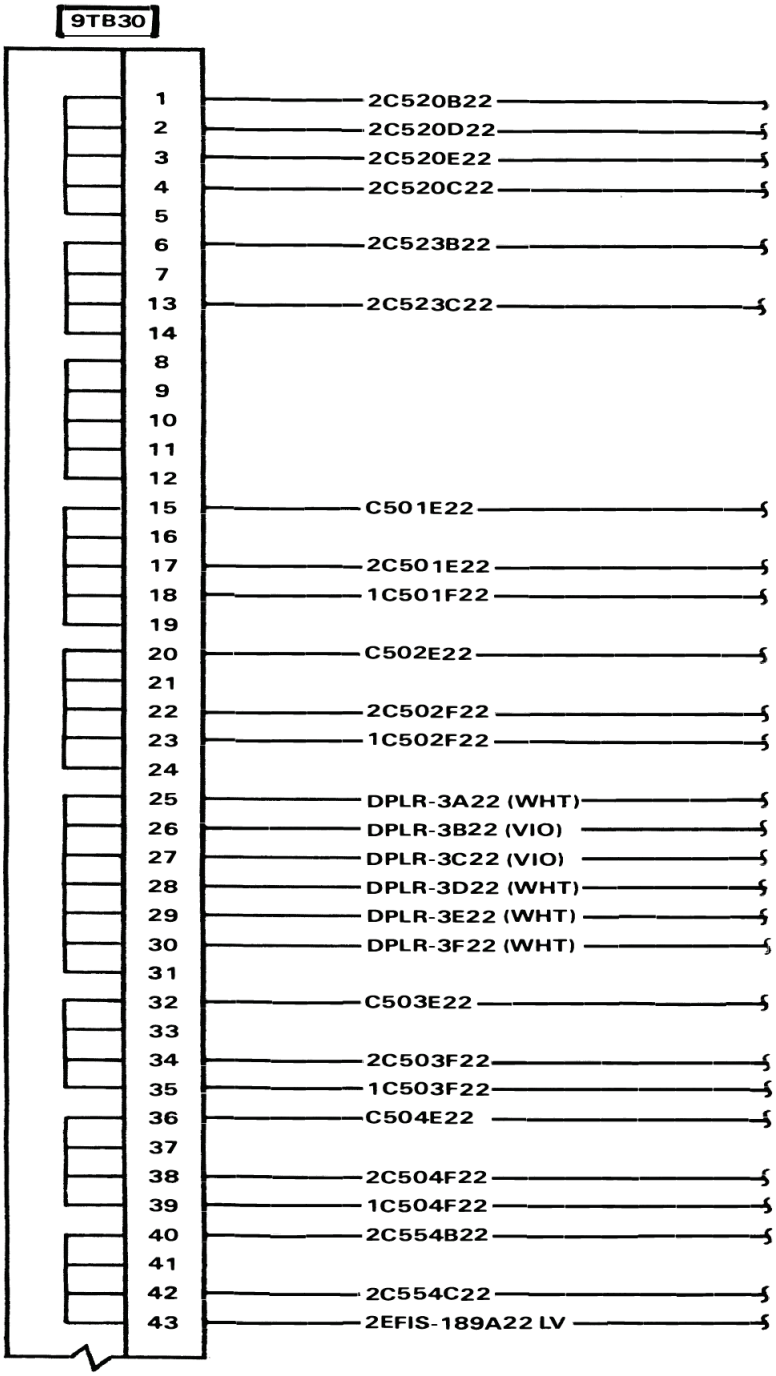
ICN-412-A-980101-B-97499-00041-A-001-01

Figure 1. Component Replacement (Sheet 13 of 59)



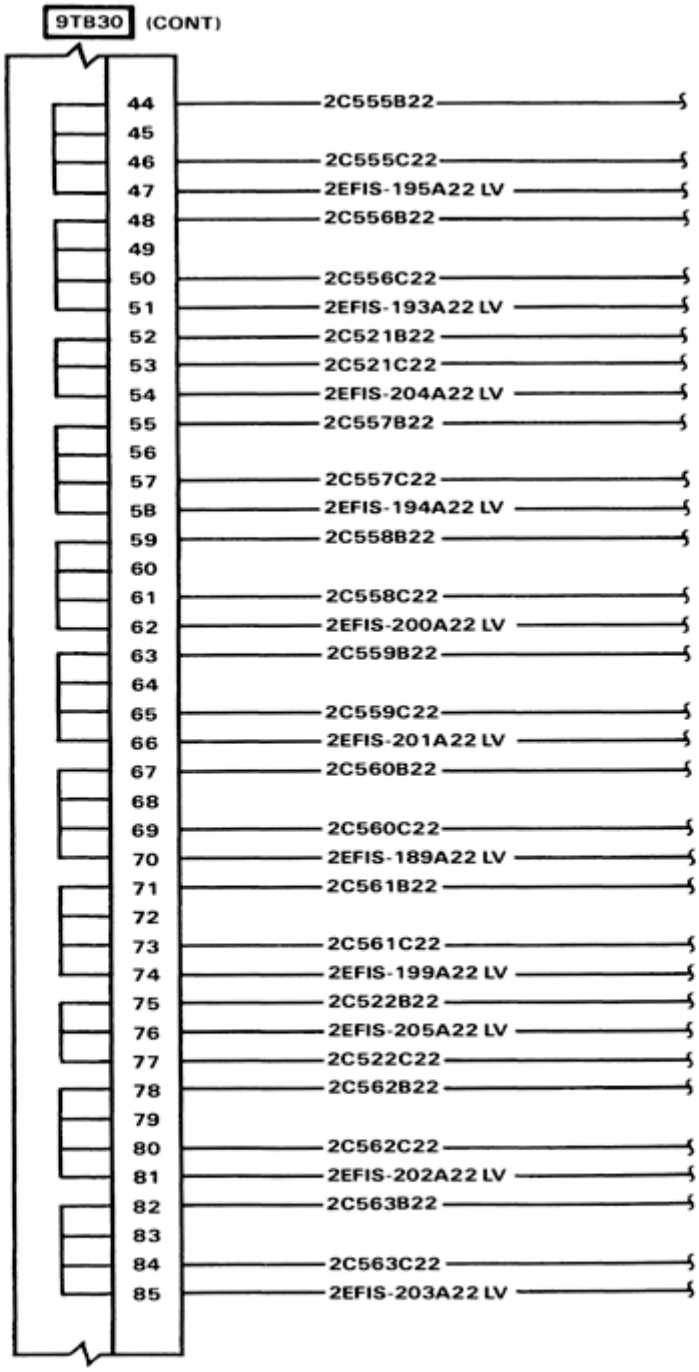
ICN-412-A-980101-B-97499-00042-A-001-01

Figure 1. Component Replacement (Sheet 14 of 59)



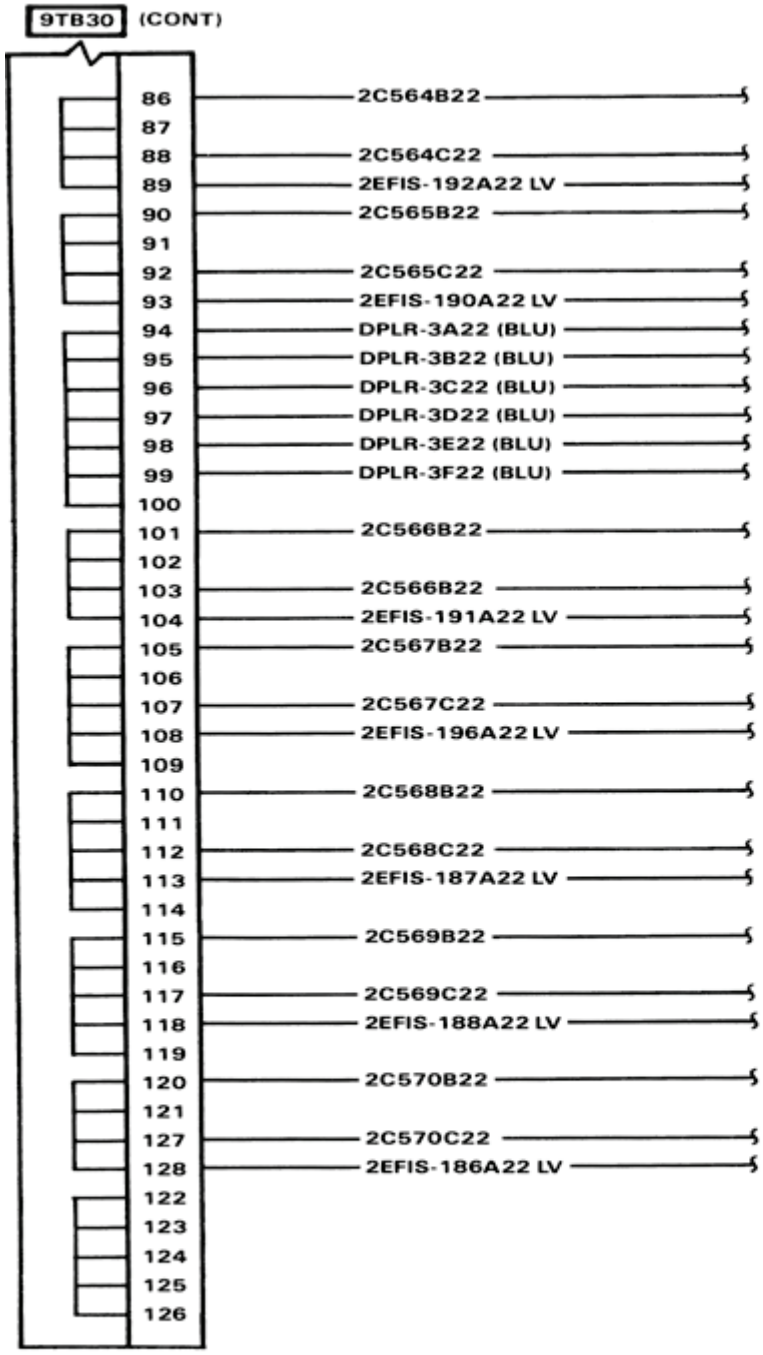
ICN-412-A-980101-B-97499-00043-A-001-01

Figure 1. Component Replacement (Sheet 15 of 59)



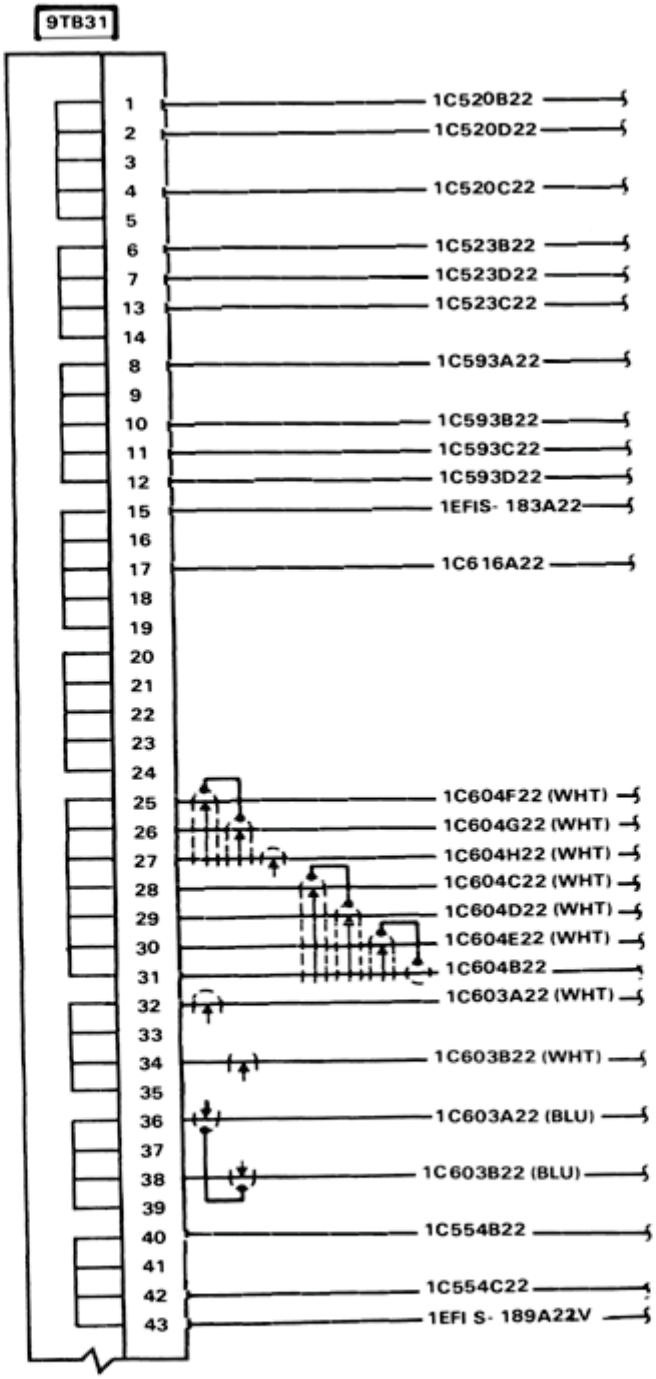
ICN-412-A-980101-B-97499-00044-A-001-01

Figure 1. Component Replacement (Sheet 16 of 59)



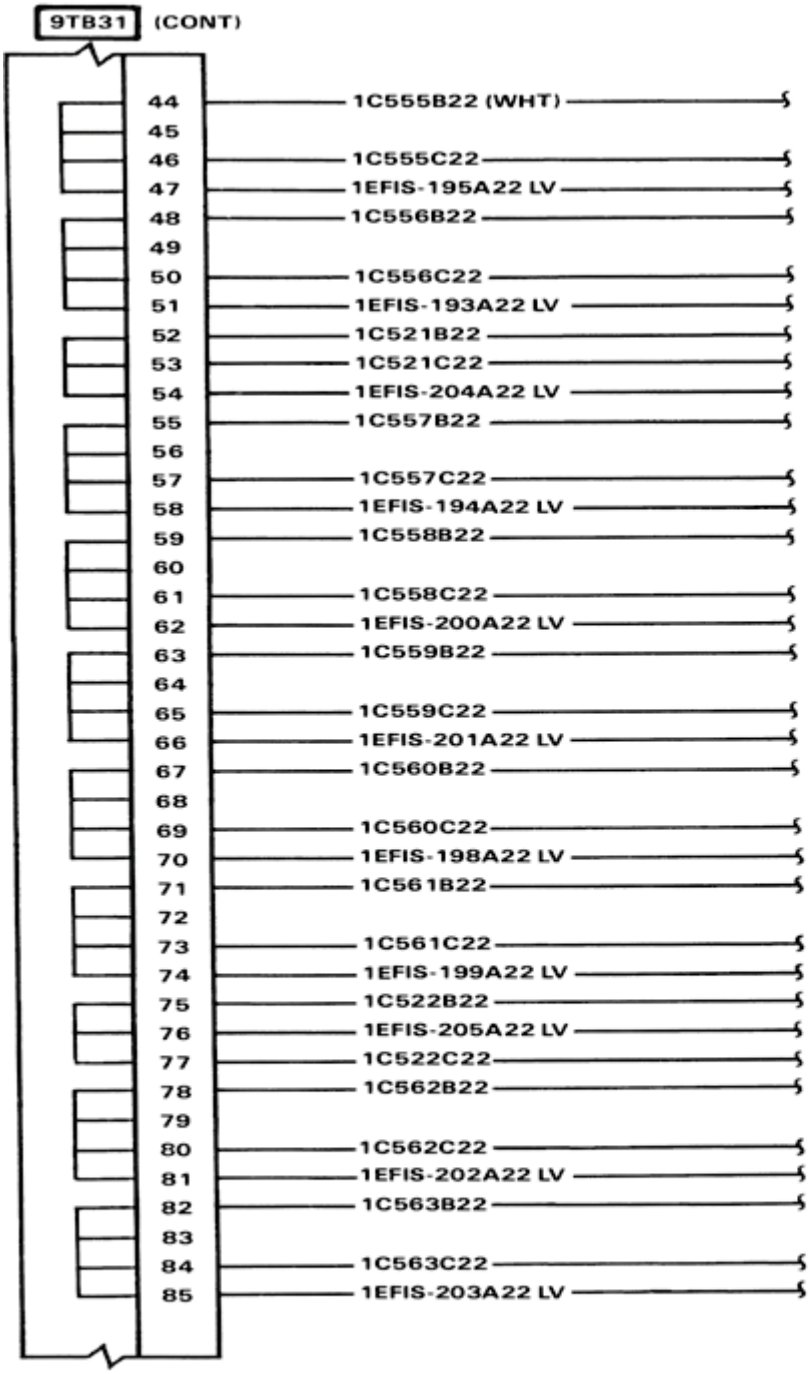
ICN-412-A-980101-B-97499-00045-A-001-01

Figure 1. Component Replacement (Sheet 17 of 59)



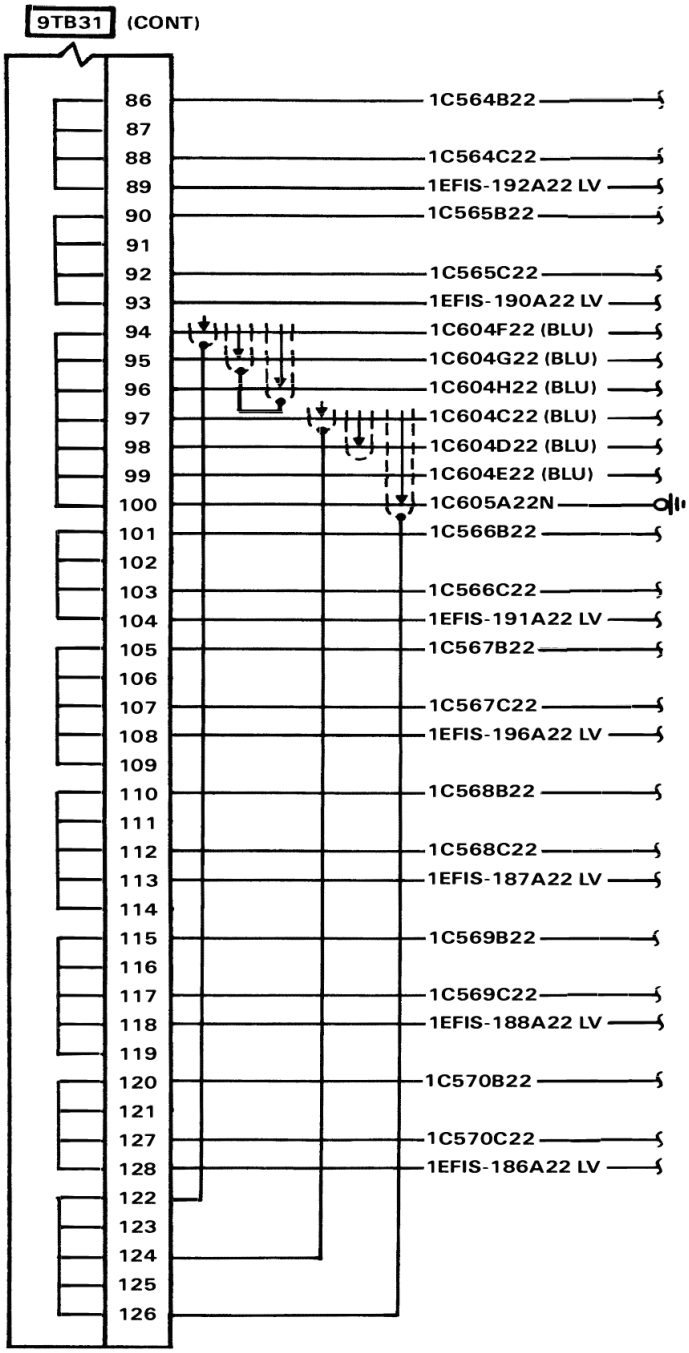
ICN-412-A-980101-B-97499-00046-A-001-01

Figure 1. Component Replacement (Sheet 18 of 59)



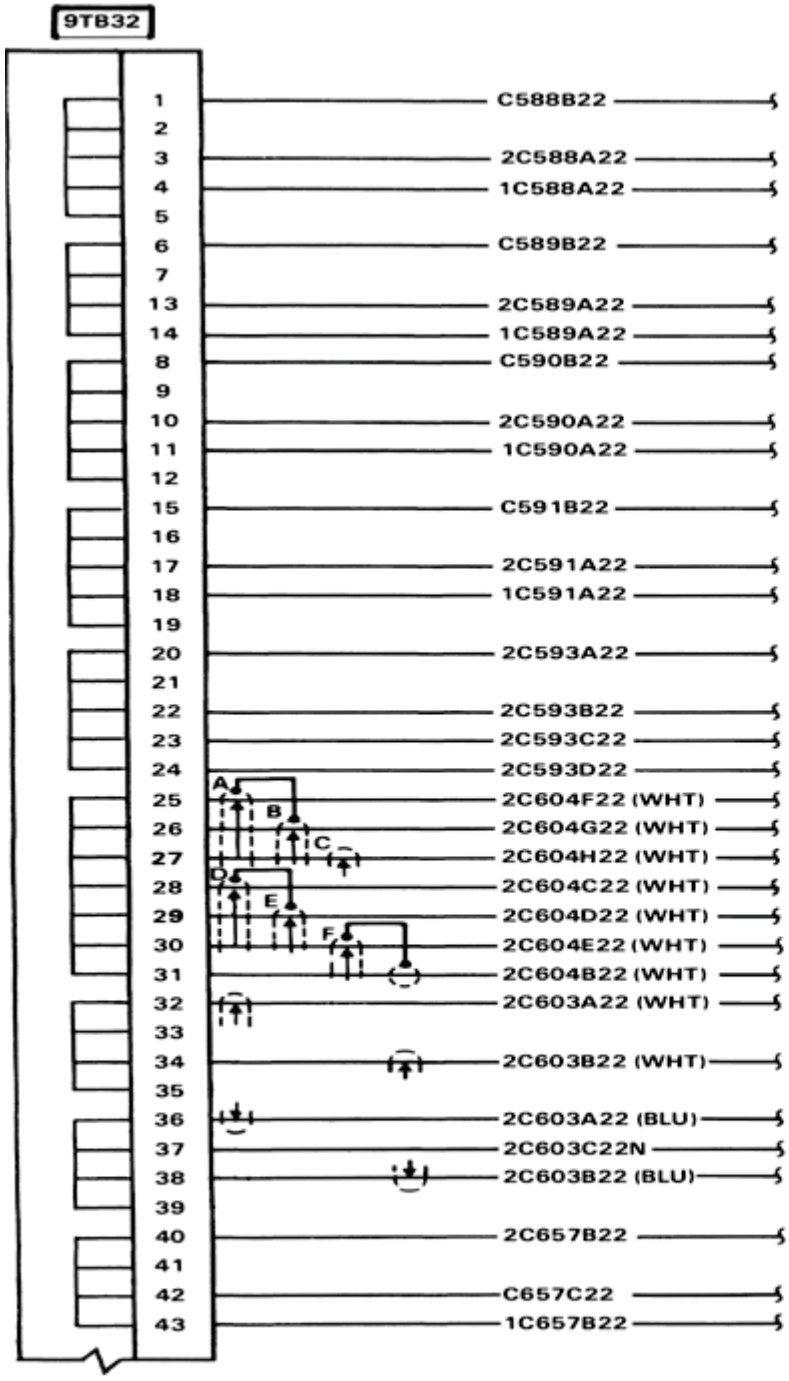
ICN-412-A-980101-B-97499-00047-A-001-01

Figure 1. Component Replacement (Sheet 19 of 59)



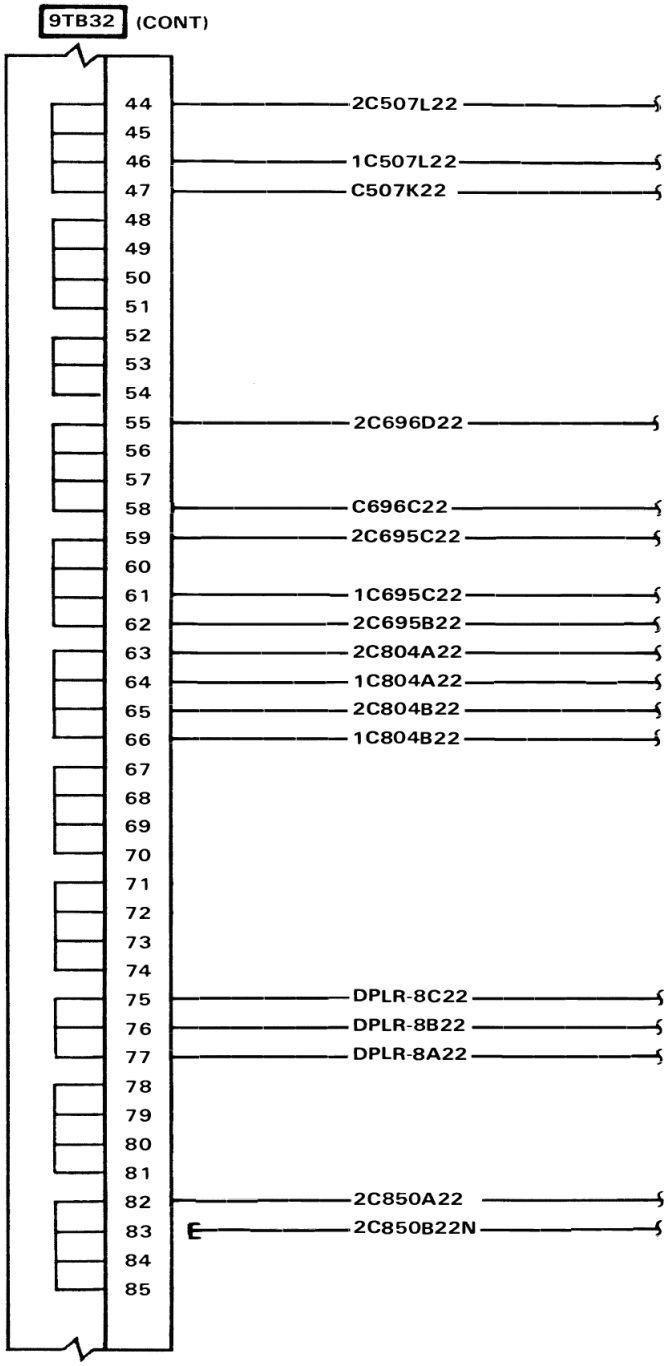
ICN-412-A-980101-B-97499-00048-A-001-01

Figure 1. Component Replacement (Sheet 20 of 59)



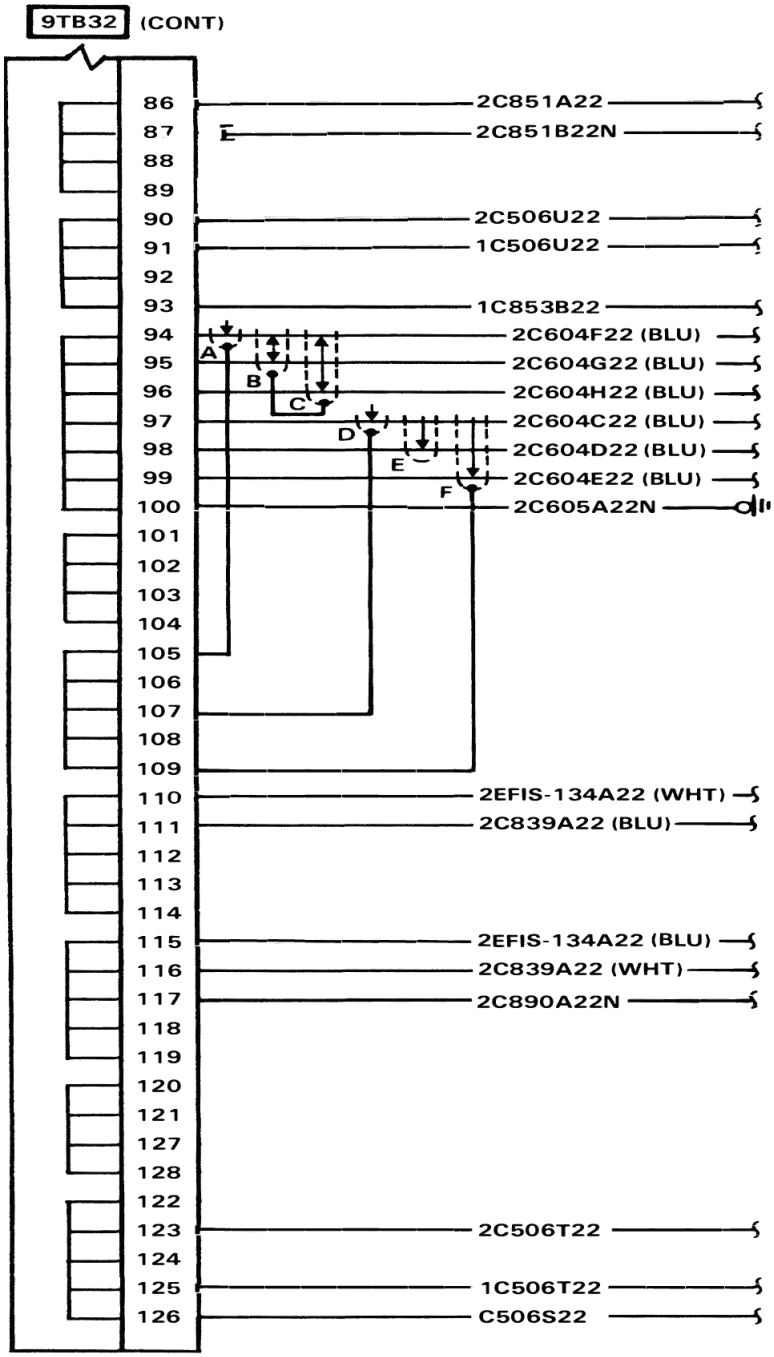
ICN-412-A-980101-B-97499-00049-A-001-01

Figure 1. Component Replacement (Sheet 21 of 59)



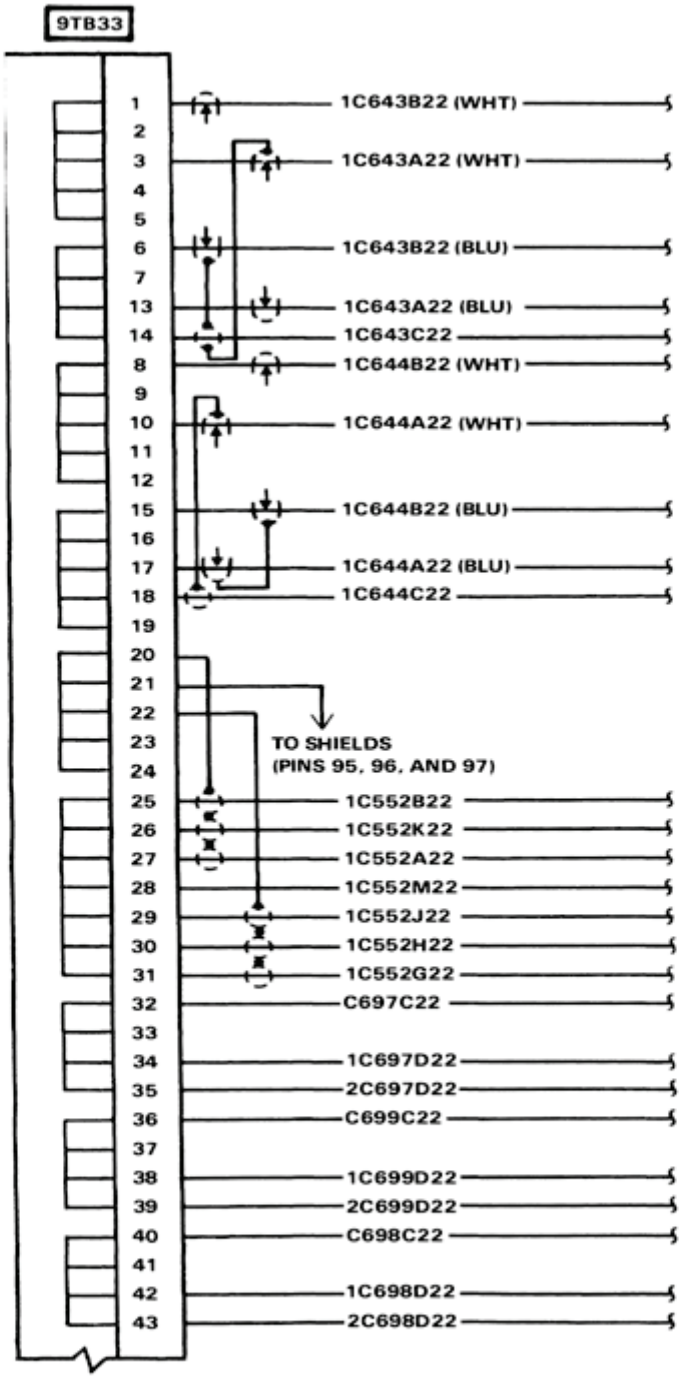
ICN-412-A-980101-B-97499-00050-A-001-01

Figure 1. Component Replacement (Sheet 22 of 59)



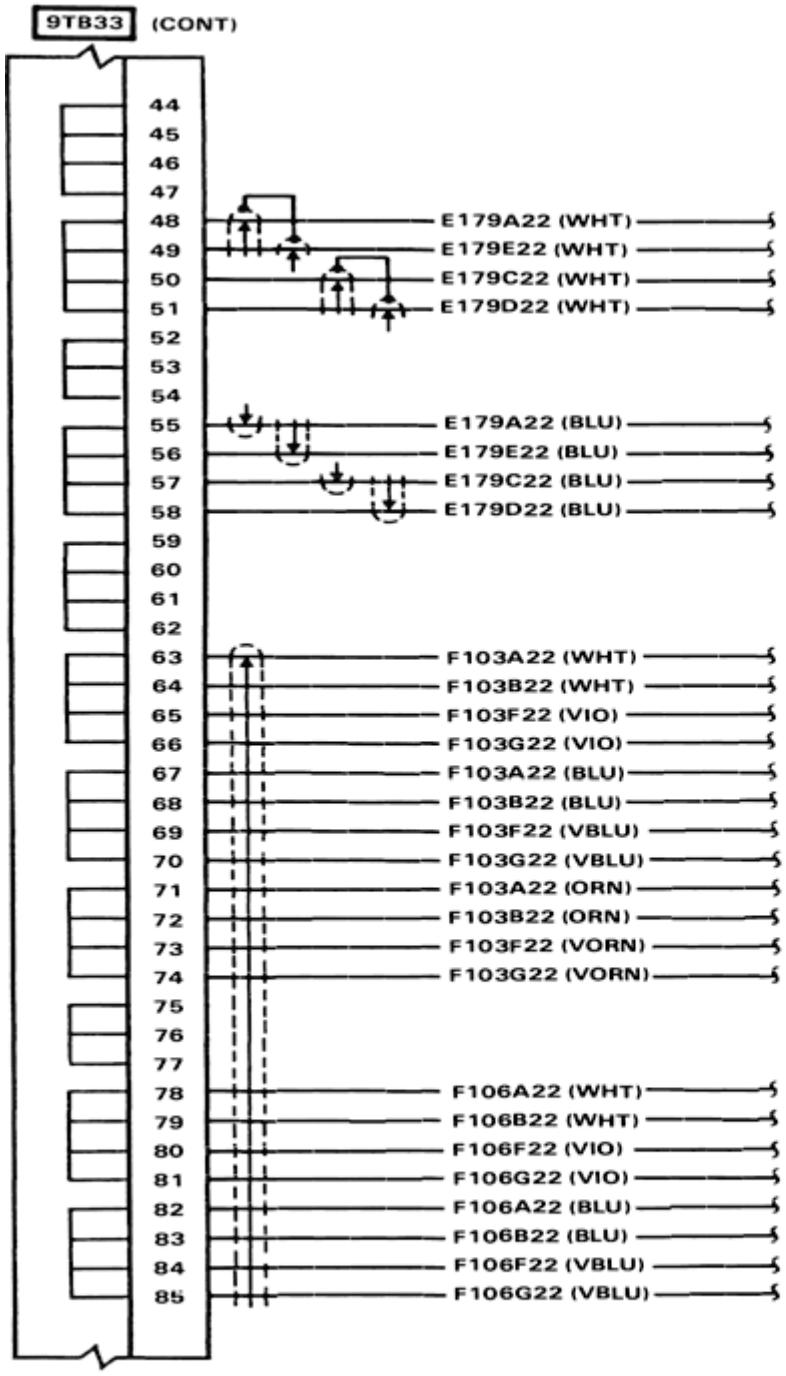
ICN-412-A-980101-B-97499-00051-A-001-01

Figure 1. Component Replacement (Sheet 23 of 59)



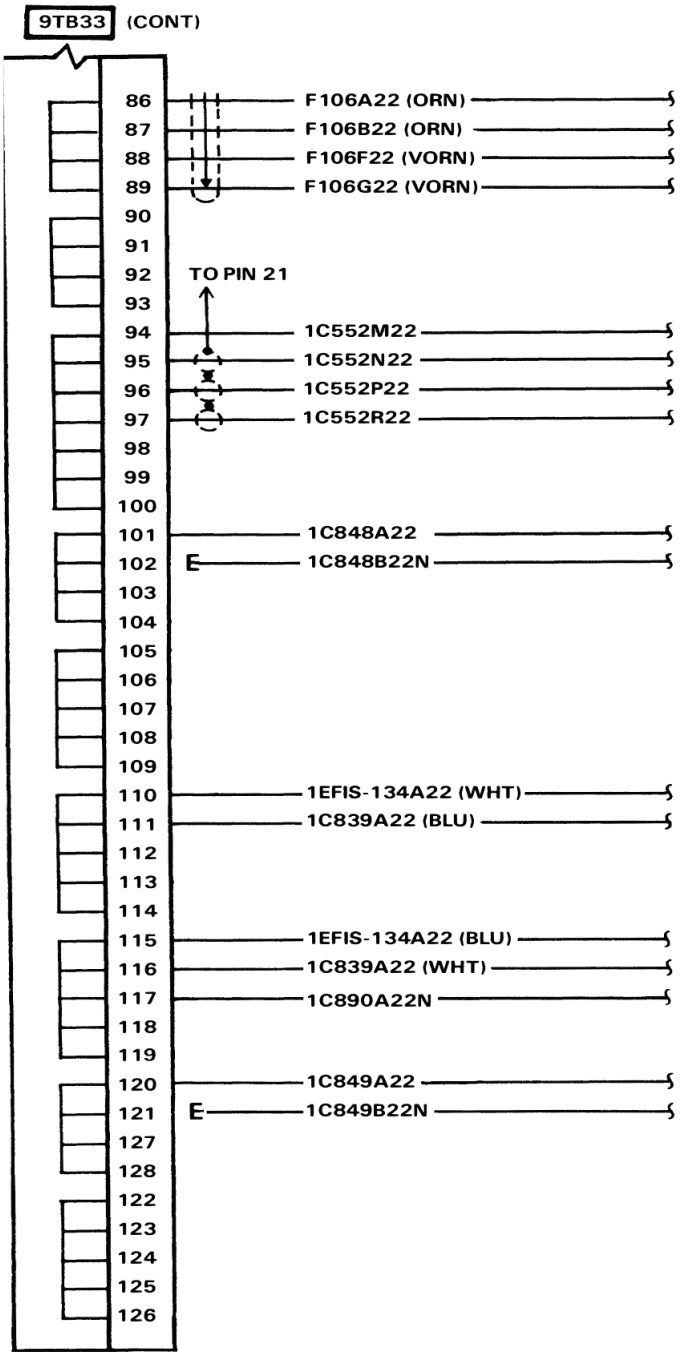
ICN-412-A-980101-B-97499-00052-A-001-01

Figure 1. Component Replacement (Sheet 24 of 59)



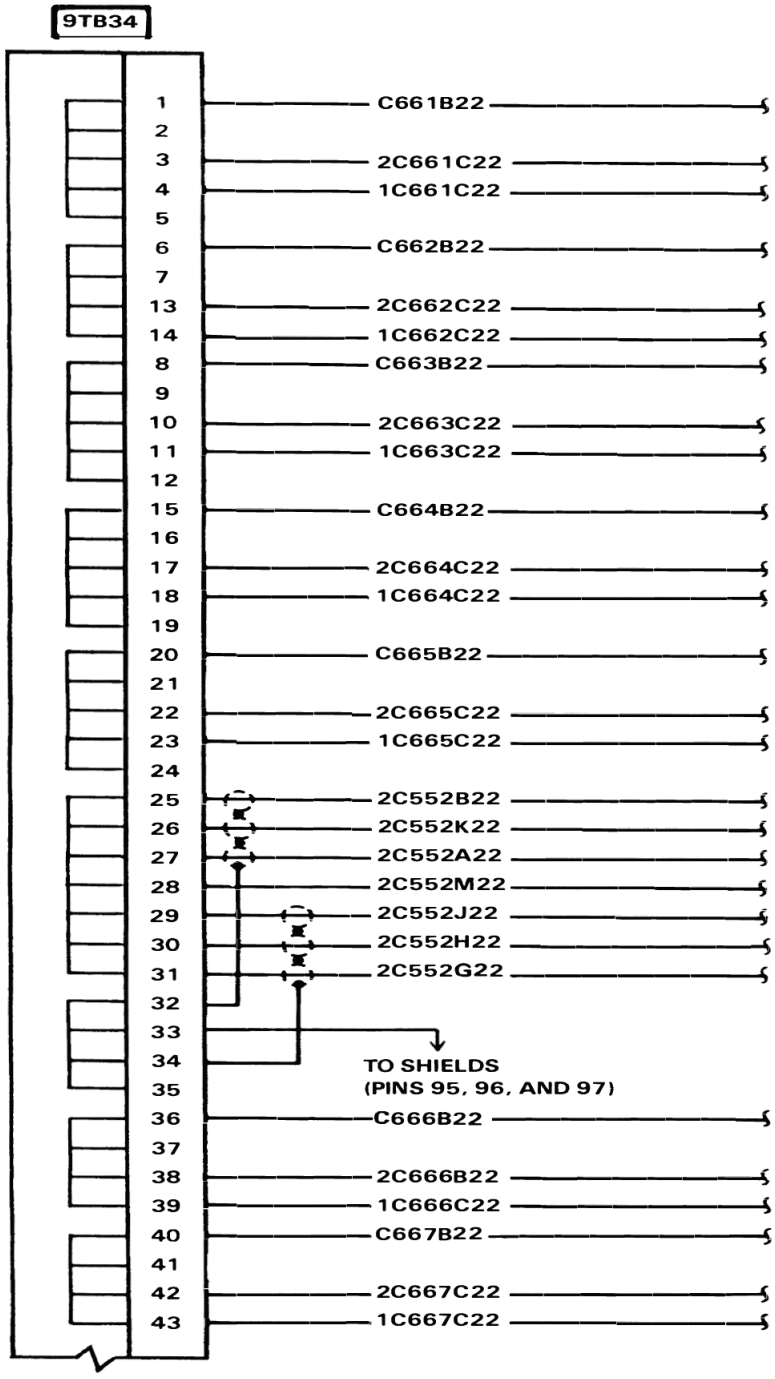
ICN-412-A-980101-B-97499-00053-A-001-01

Figure 1. Component Replacement (Sheet 25 of 59)



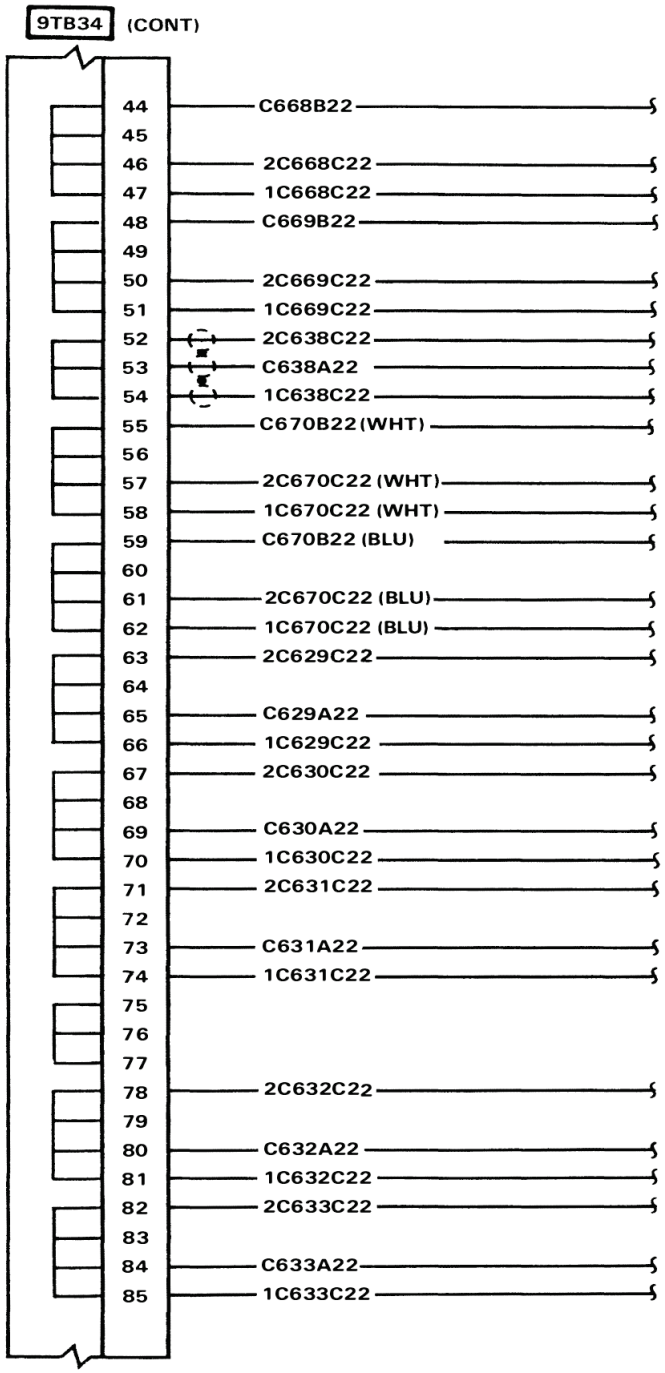
ICN-412-A-980101-B-97499-00054-A-001-01

Figure 1. Component Replacement (Sheet 26 of 59)



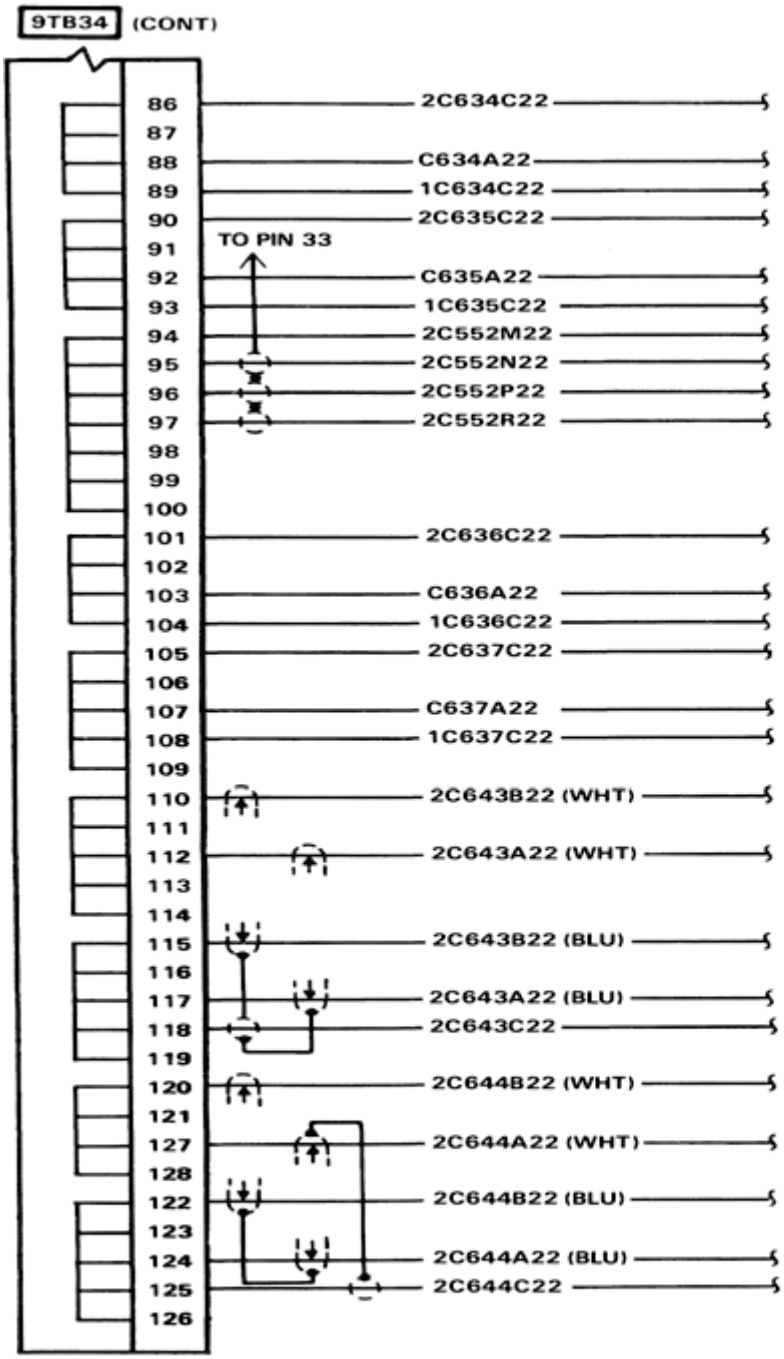
ICN-412-A-980101-B-97499-00055-A-001-01

Figure 1. Component Replacement (Sheet 27 of 59)



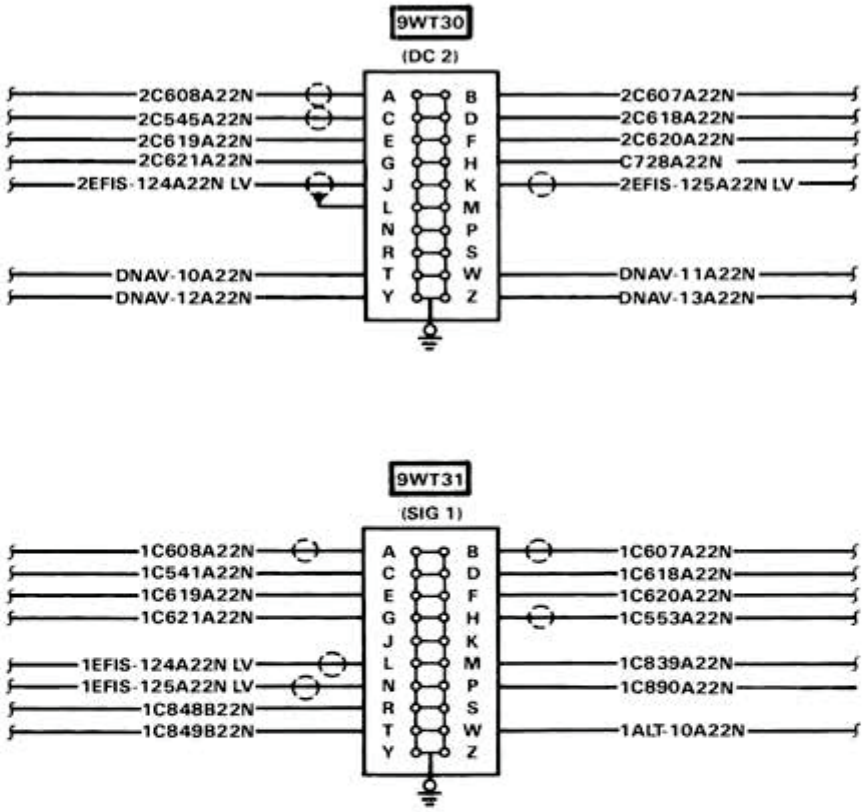
ICN-412-A-980101-B-97499-00056-A-001-01

Figure 1. Component Replacement (Sheet 28 of 59)



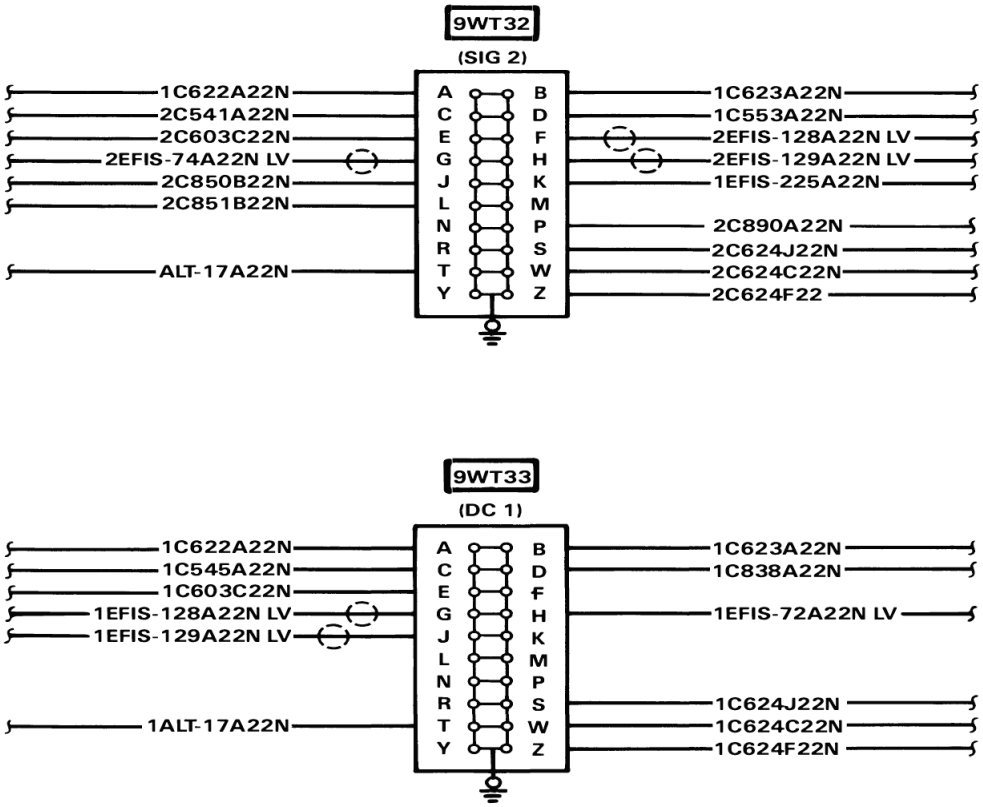
ICN-412-A-980101-B-97499-00057-A-001-01

Figure 1. Component Replacement (Sheet 29 of 59)



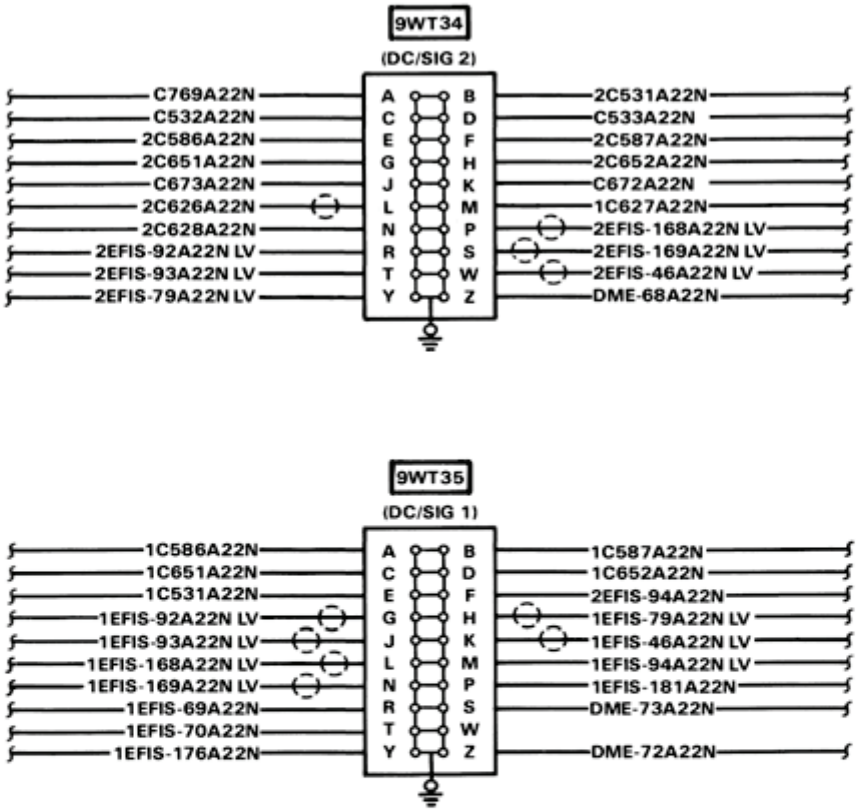
ICN-412-A-980101-B-97499-00058-A-001-01

Figure 1. Component Replacement (Sheet 30 of 59)



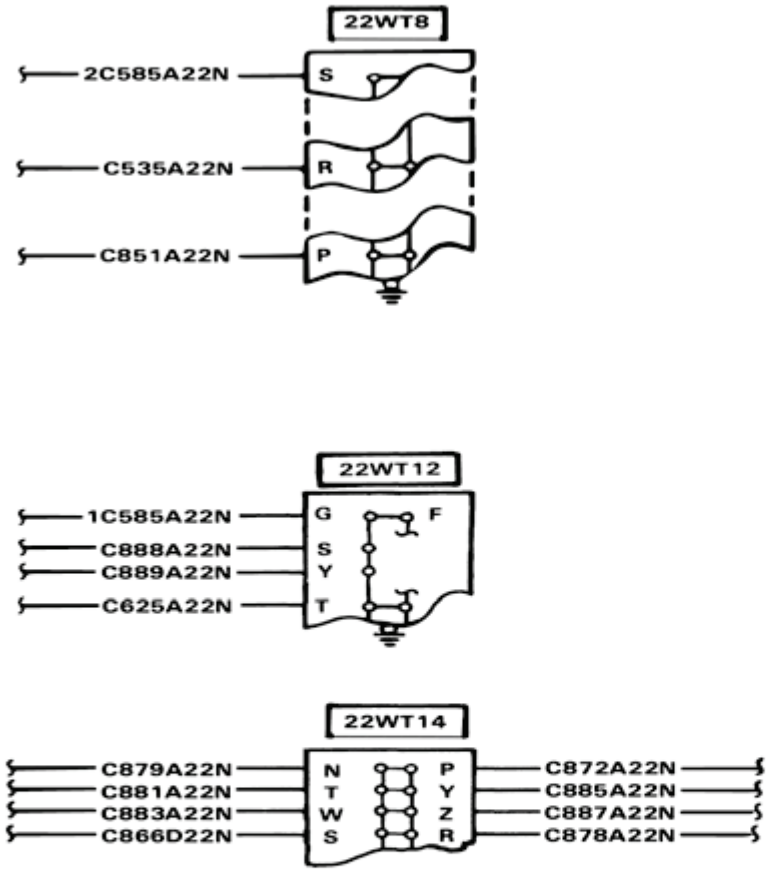
ICN-412-A-980101-B-97499-00059-A-001-01

Figure 1. Component Replacement (Sheet 31 of 59)



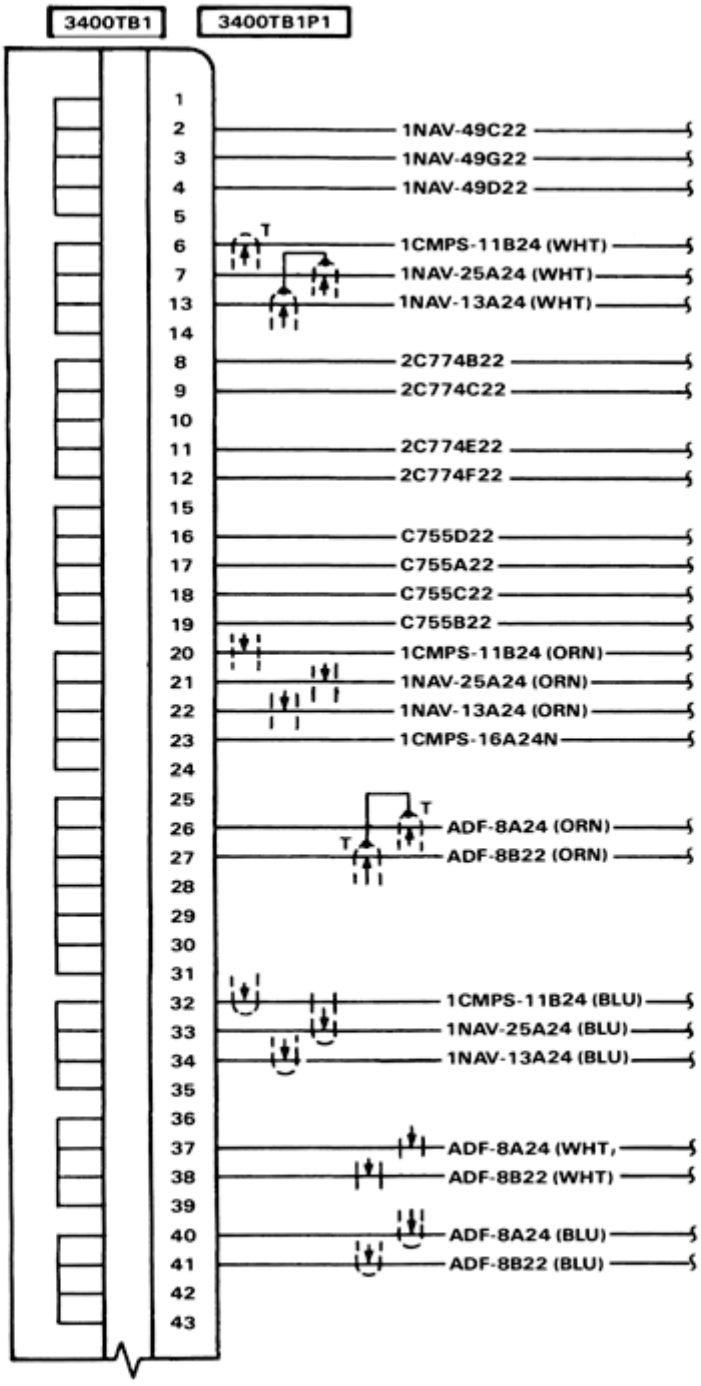
ICN-412-A-980101-B-97499-00060-A-001-01

Figure 1. Component Replacement (Sheet 32 of 59)



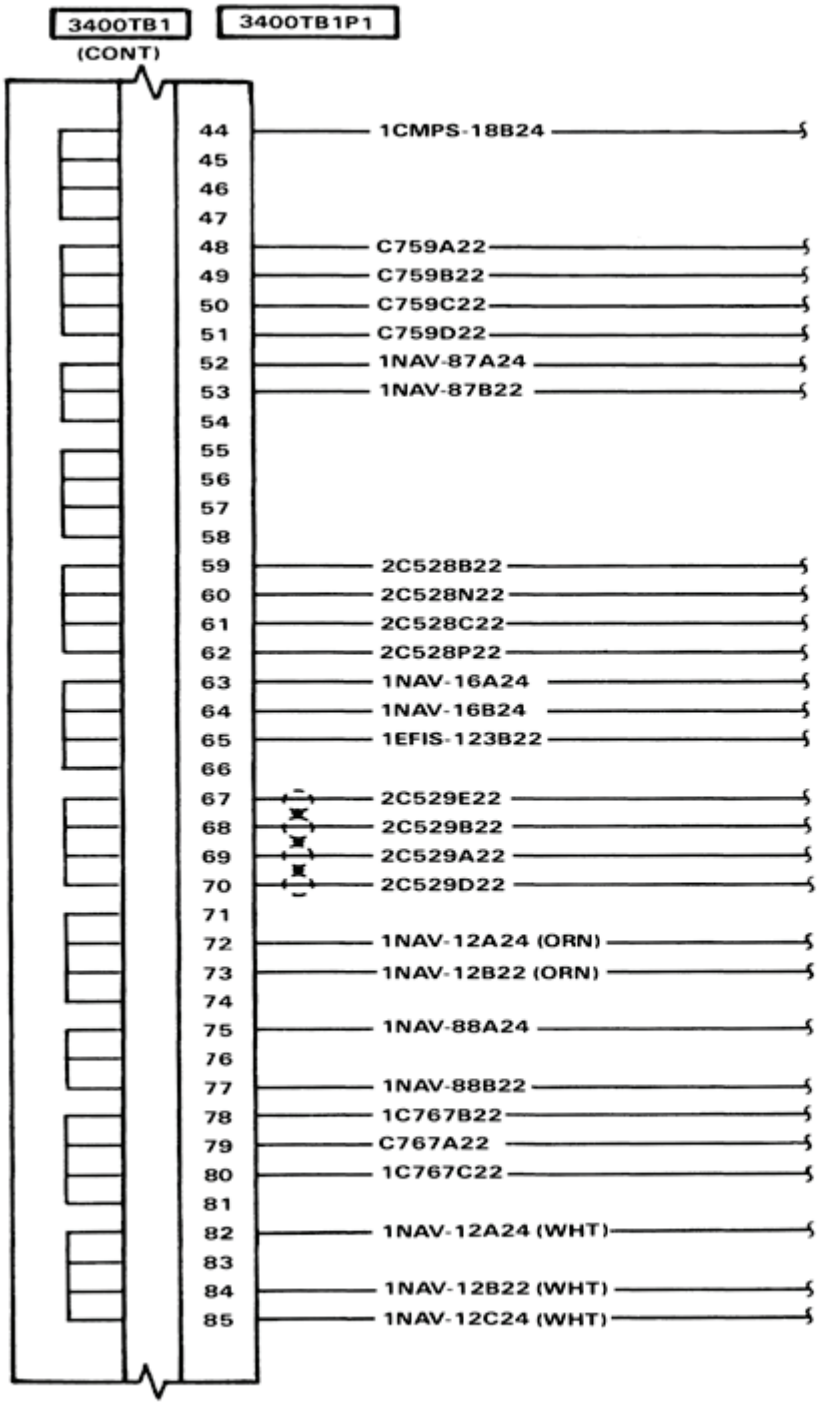
ICN-412-A-980101-B-97499-00061-A-001-01

Figure 1. Component Replacement (Sheet 33 of 59)



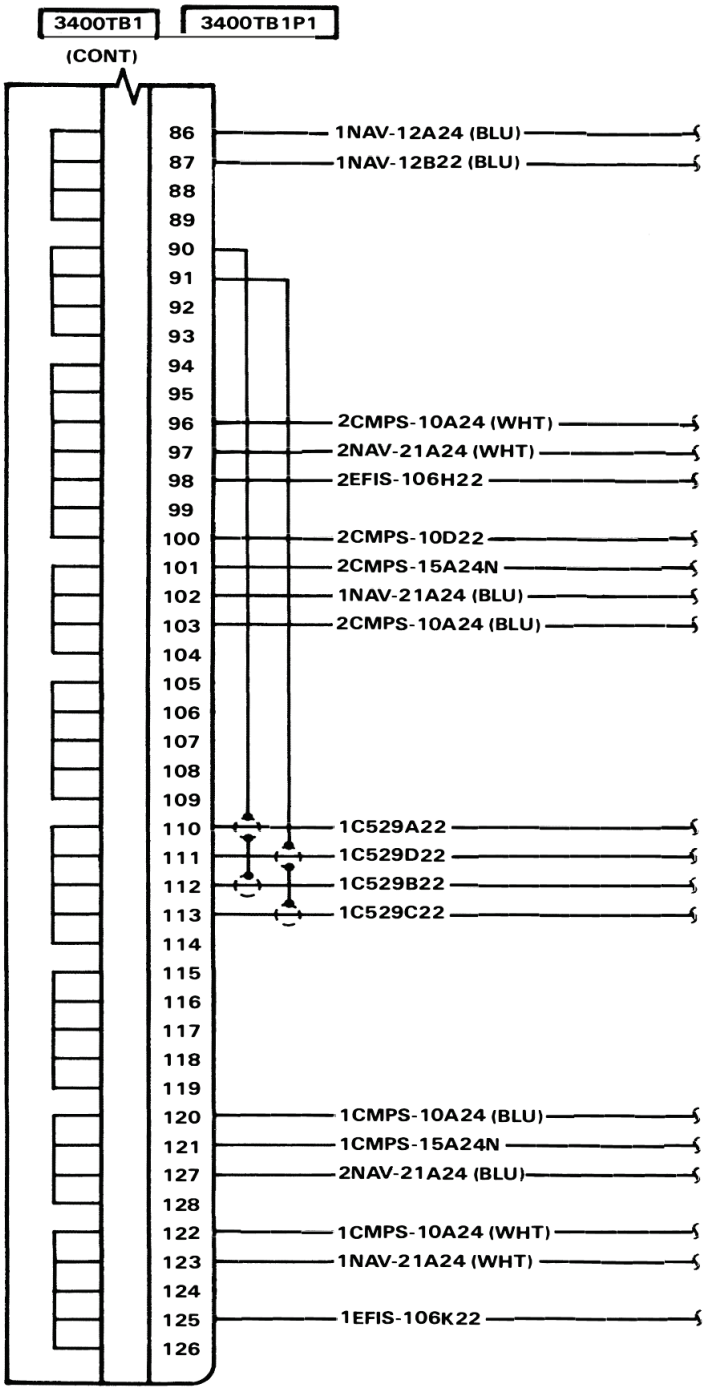
ICN-412-A-980101-B-97499-00062-A-001-01

Figure 1. Component Replacement (Sheet 34 of 59)



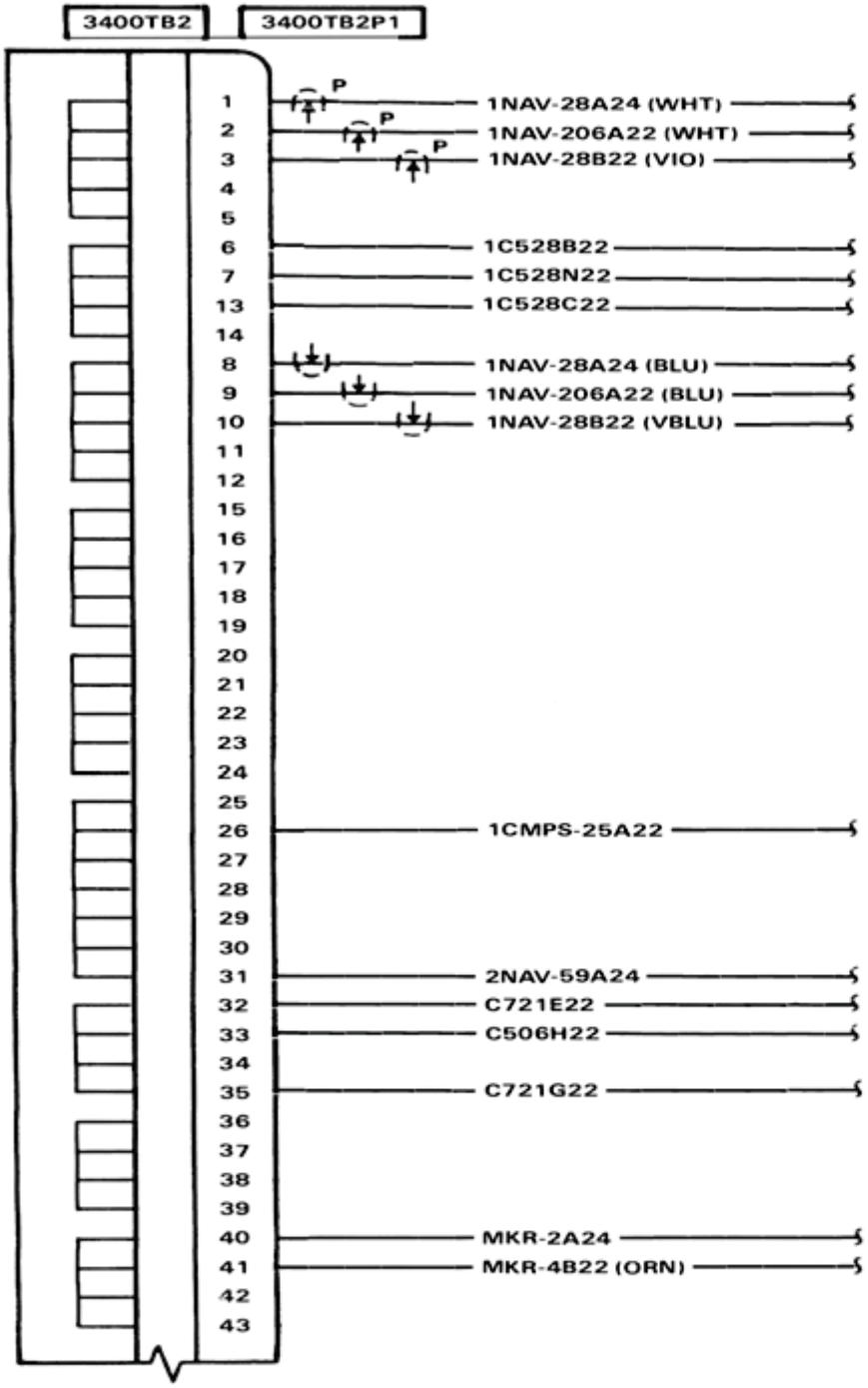
ICN-412-A-980101-B-97499-00063-A-001-01

Figure 1. Component Replacement (Sheet 35 of 59)



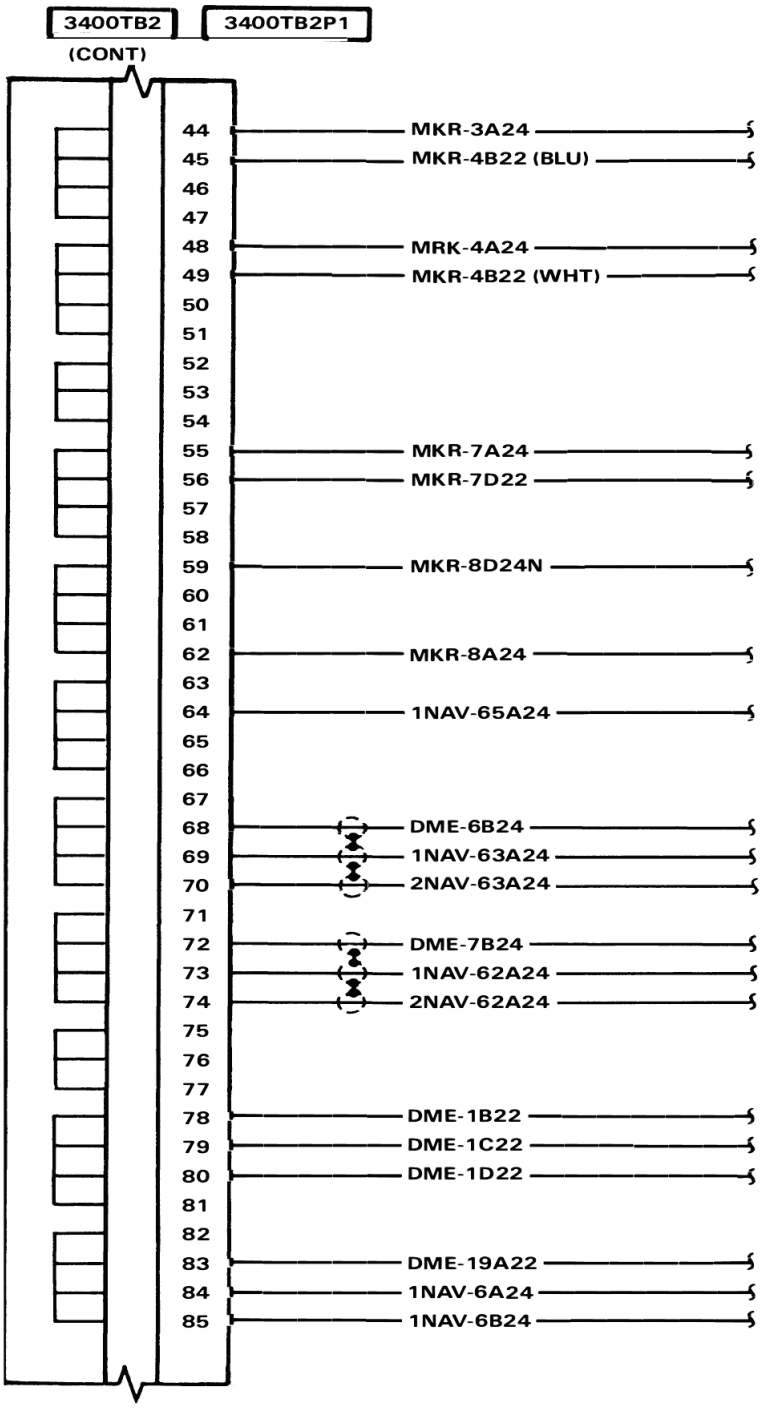
ICN-412-A-980101-B-97499-00064-A-001-01

Figure 1. Component Replacement (Sheet 36 of 59)



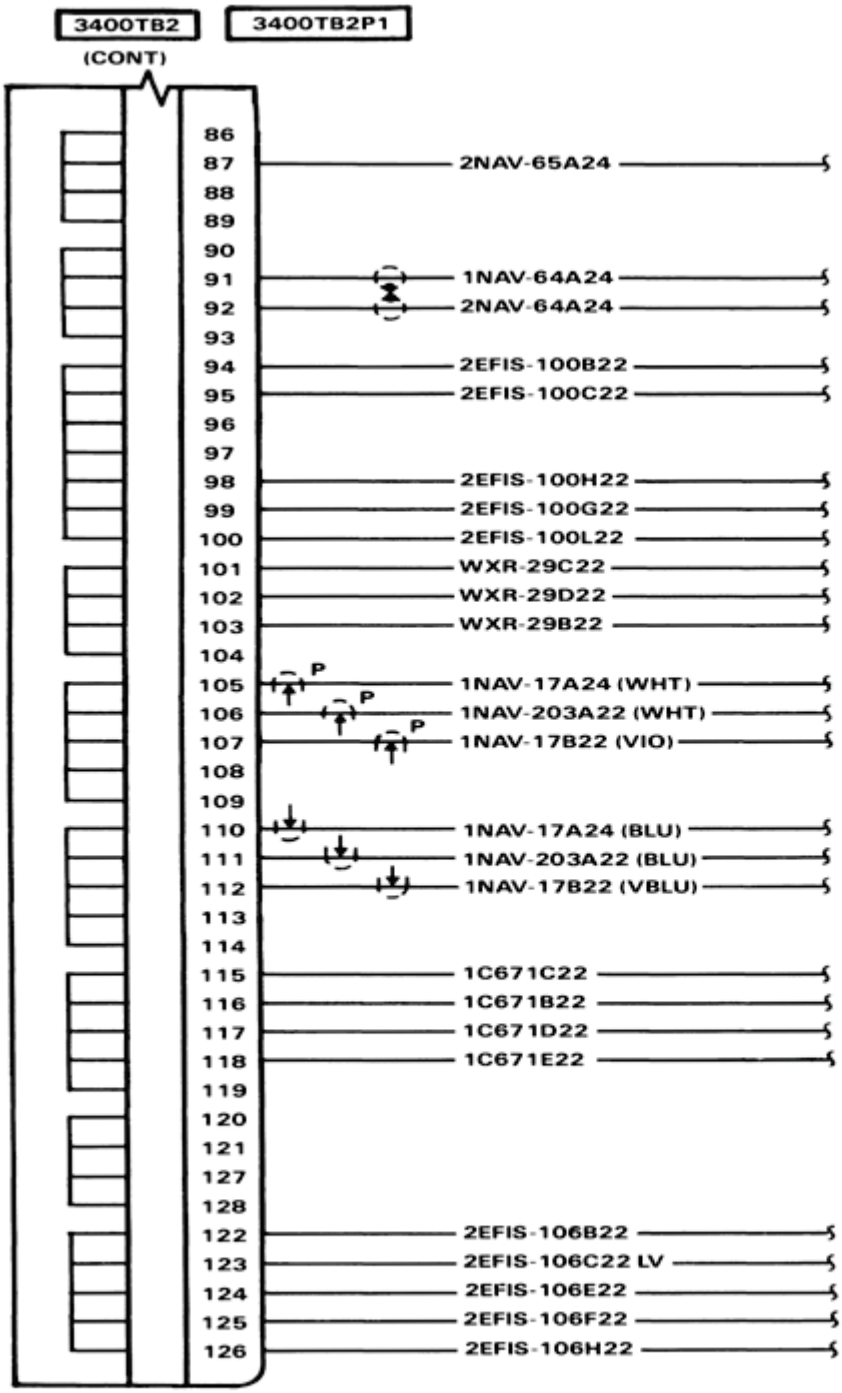
ICN-412-A-980101-B-97499-00065-A-001-01

Figure 1. Component Replacement (Sheet 37 of 59)



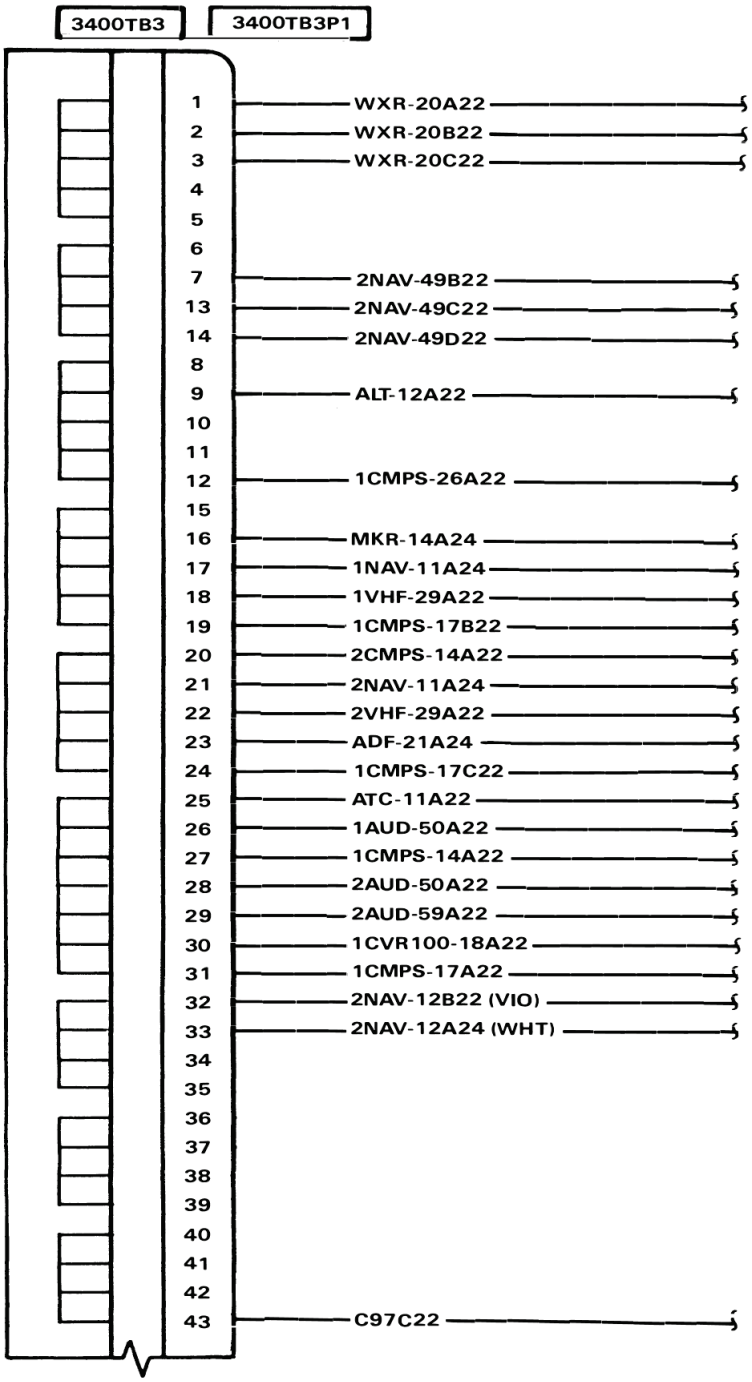
ICN-412-A-980101-B-97499-00066-A-001-01

Figure 1. Component Replacement (Sheet 38 of 59)



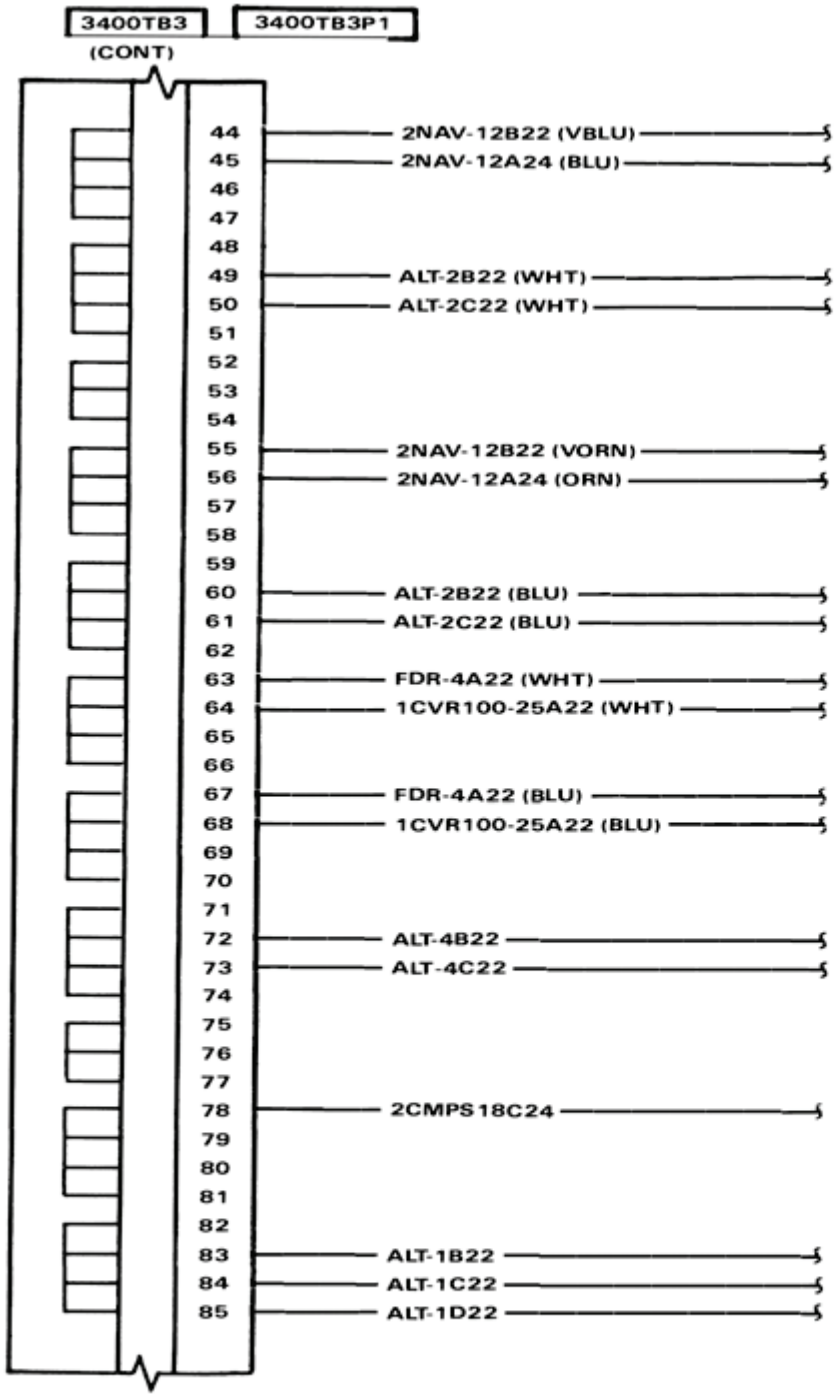
ICN-412-A-980101-B-97499-00067-A-001-01

Figure 1. Component Replacement (Sheet 39 of 59)



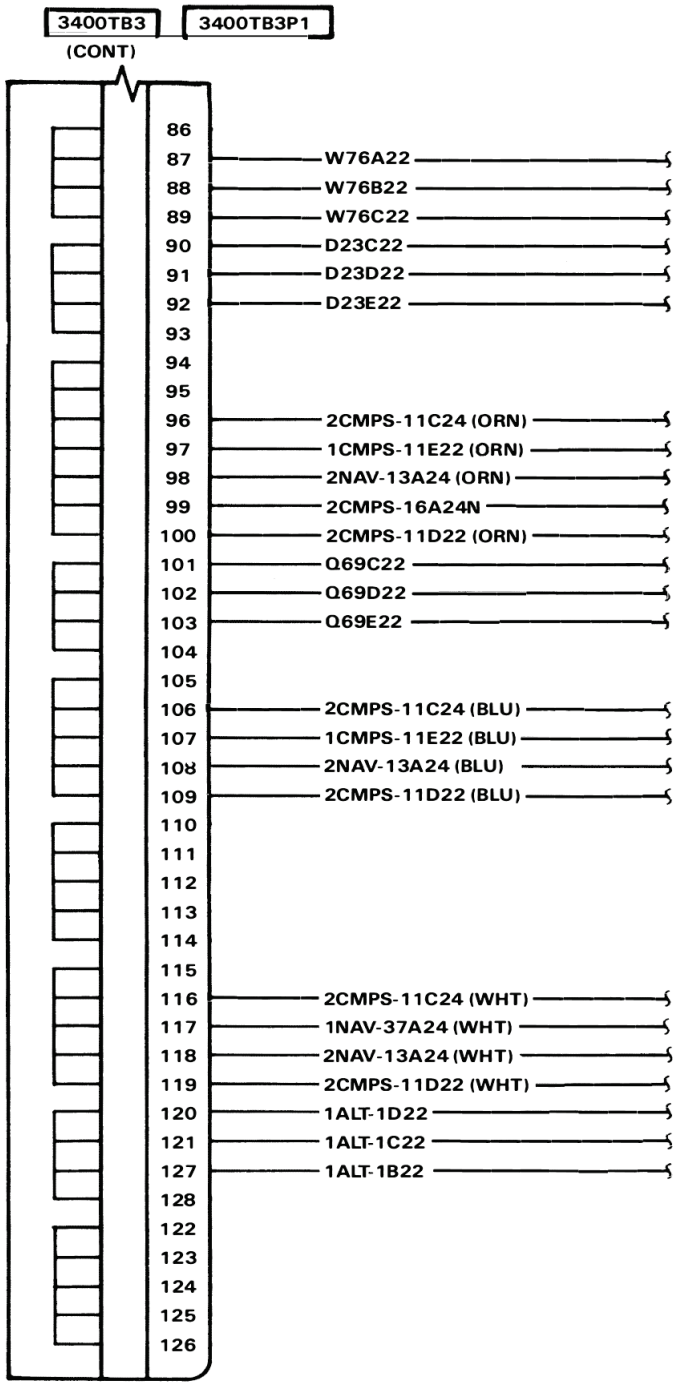
ICN-412-A-980101-B-97499-00068-A-001-01

Figure 1. Component Replacement (Sheet 40 of 59)



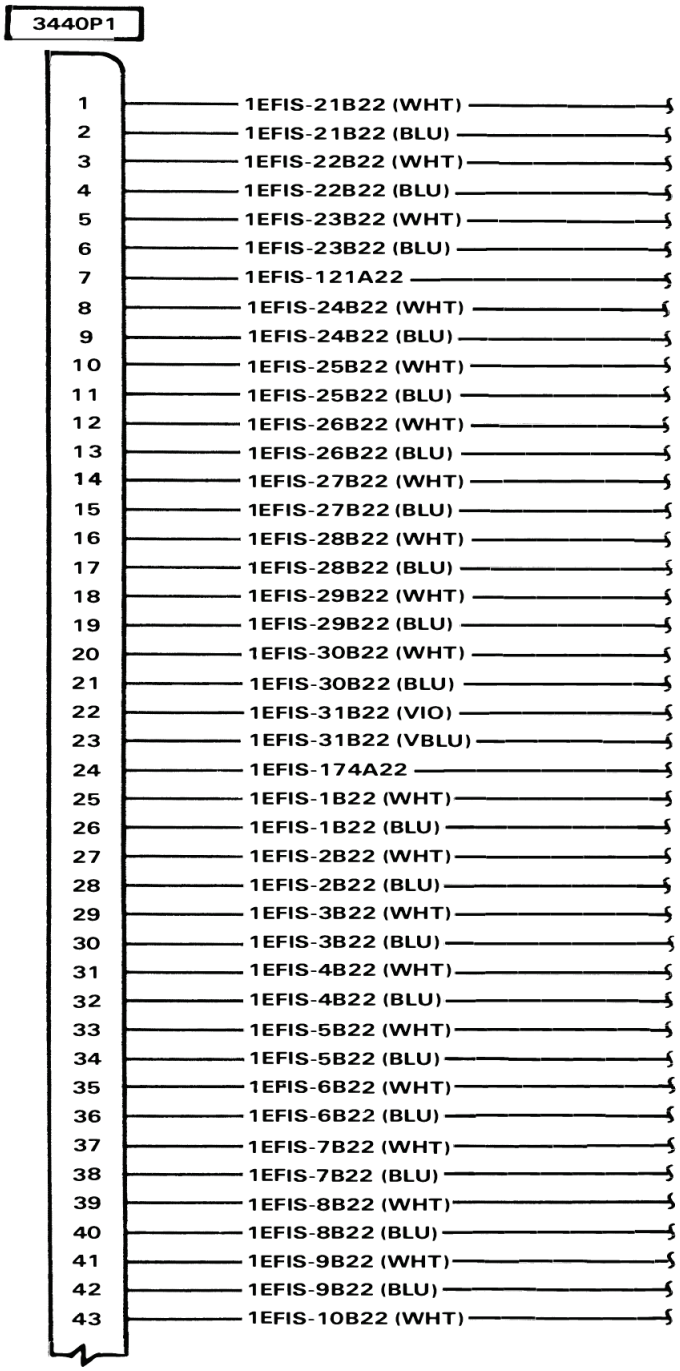
ICN-412-A-980101-B-97499-00069-A-001-01

Figure 1. Component Replacement (Sheet 41 of 59)



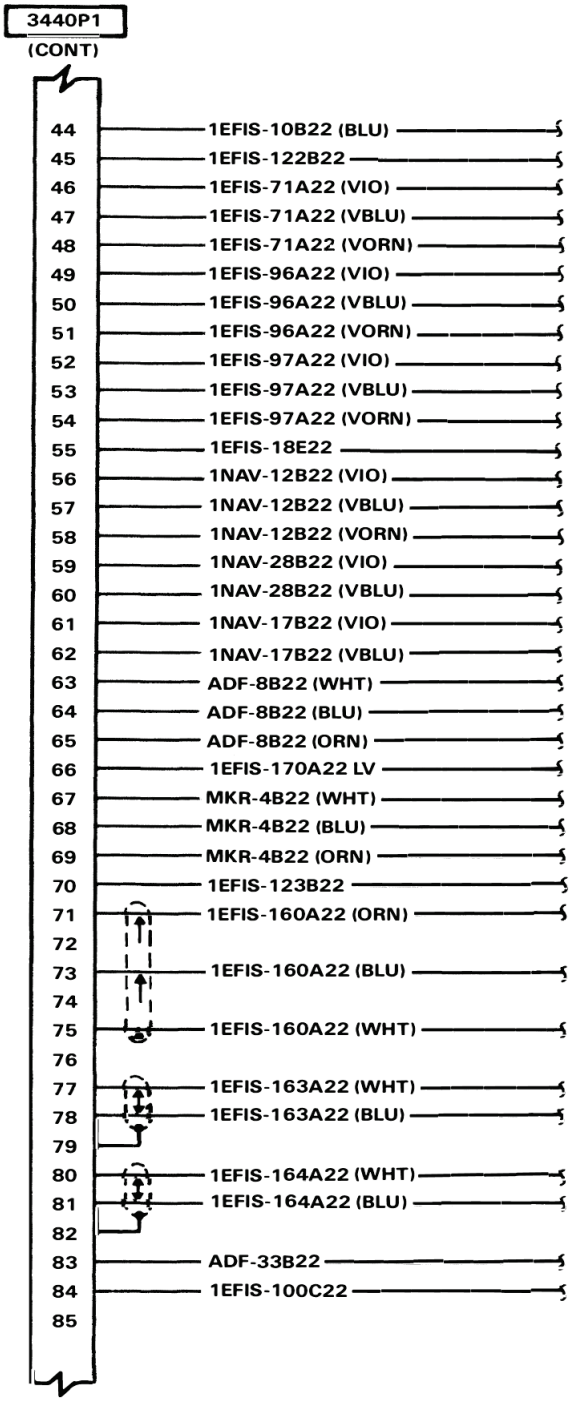
ICN-412-A-980101-B-97499-00070-A-001-01

Figure 1. Component Replacement (Sheet 42 of 59)



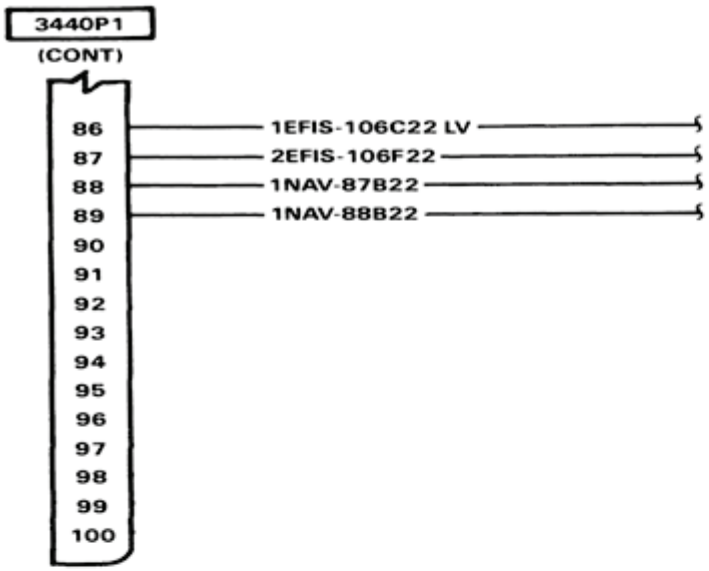
ICN-412-A-980101-B-97499-00071-A-001-01

Figure 1. Component Replacement (Sheet 43 of 59)



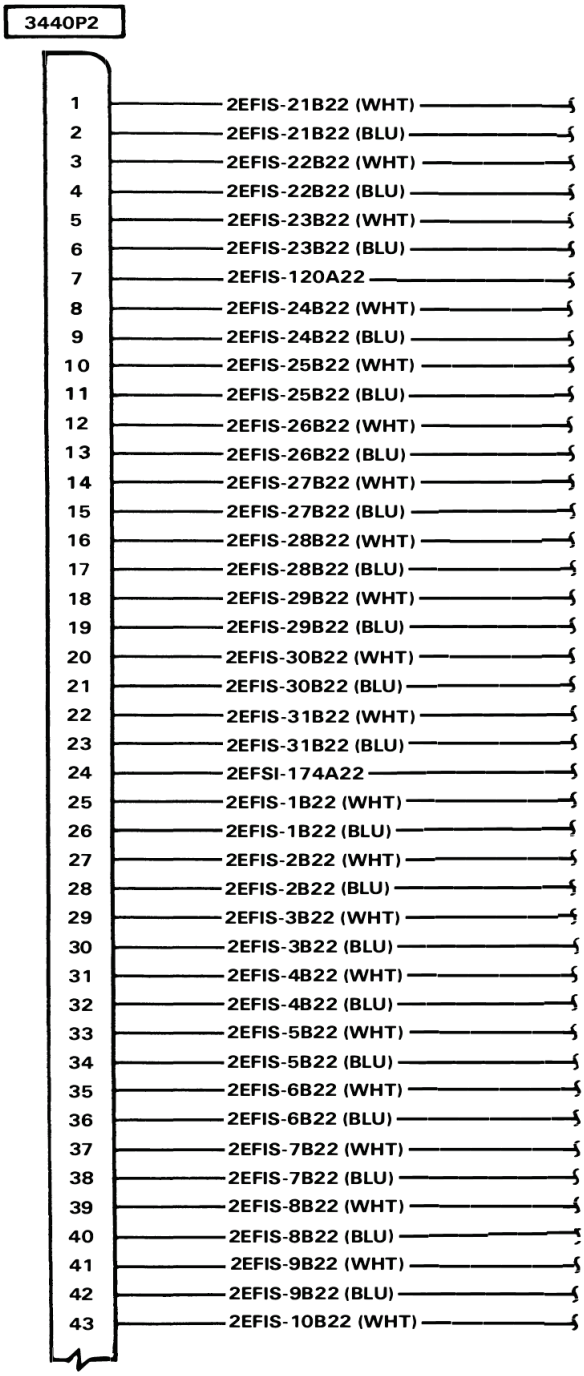
ICN-412-A-980101-B-97499-00072-A-001-01

Figure 1. Component Replacement (Sheet 44 of 59)



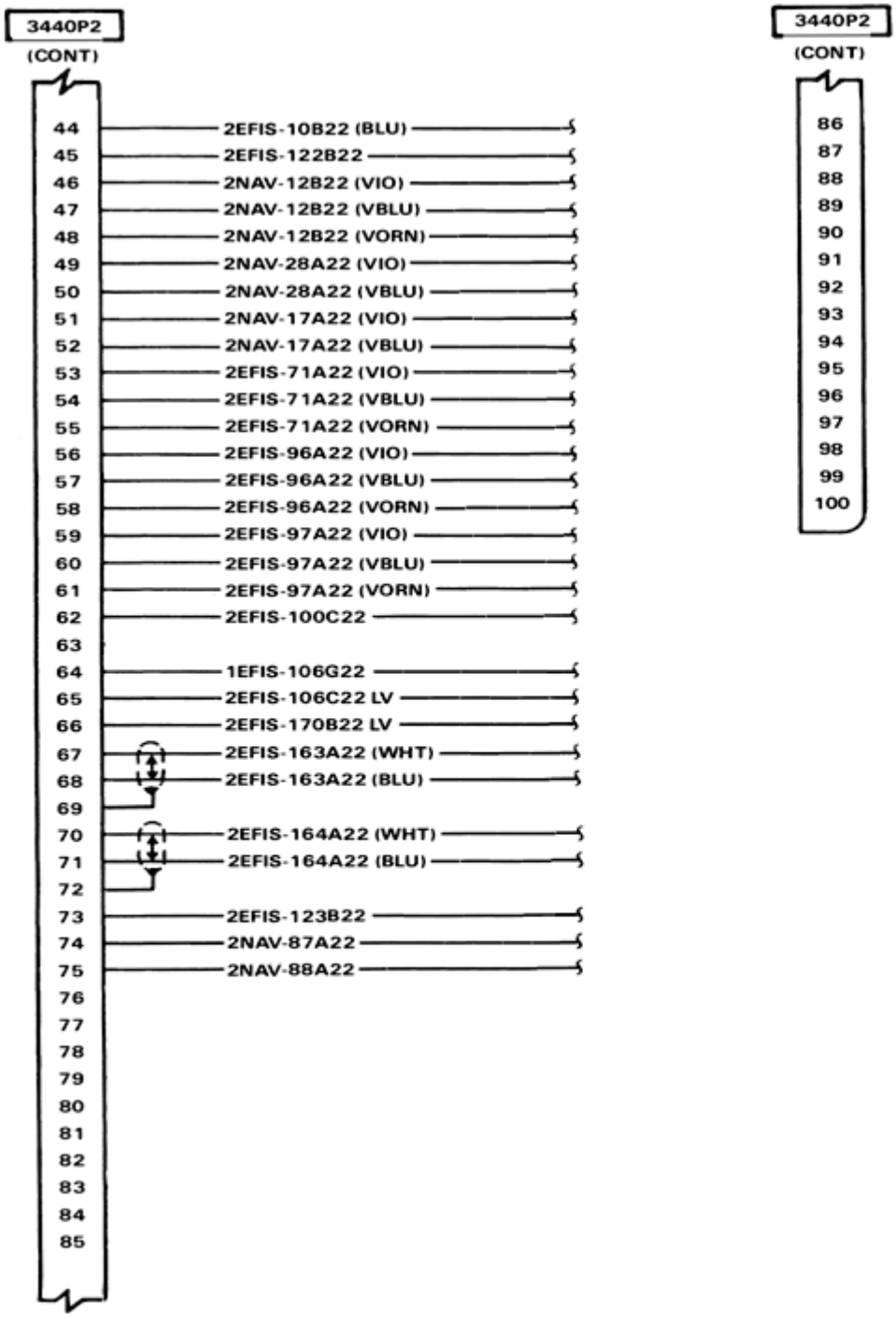
ICN-412-A-980101-B-97499-00073-A-001-01

Figure 1. Component Replacement (Sheet 45 of 59)



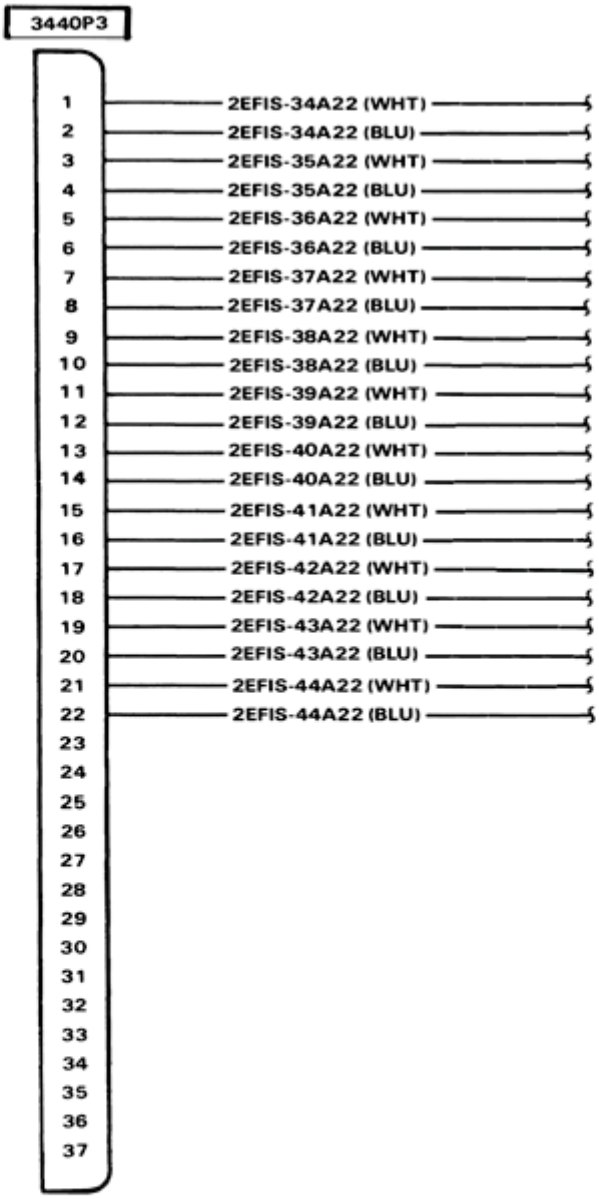
ICN-412-A-980101-B-97499-00074-A-001-01

Figure 1. Component Replacement (Sheet 46 of 59)



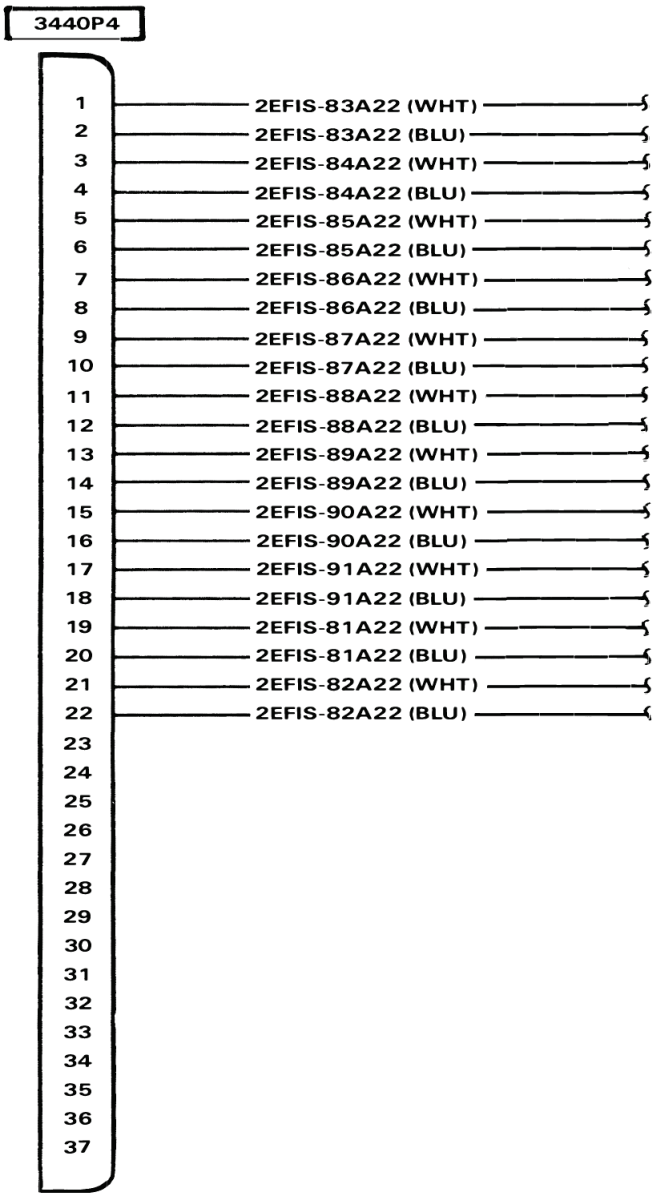
ICN-412-A-980101-B-97499-00075-A-001-01

Figure 1. Component Replacement (Sheet 47 of 59)



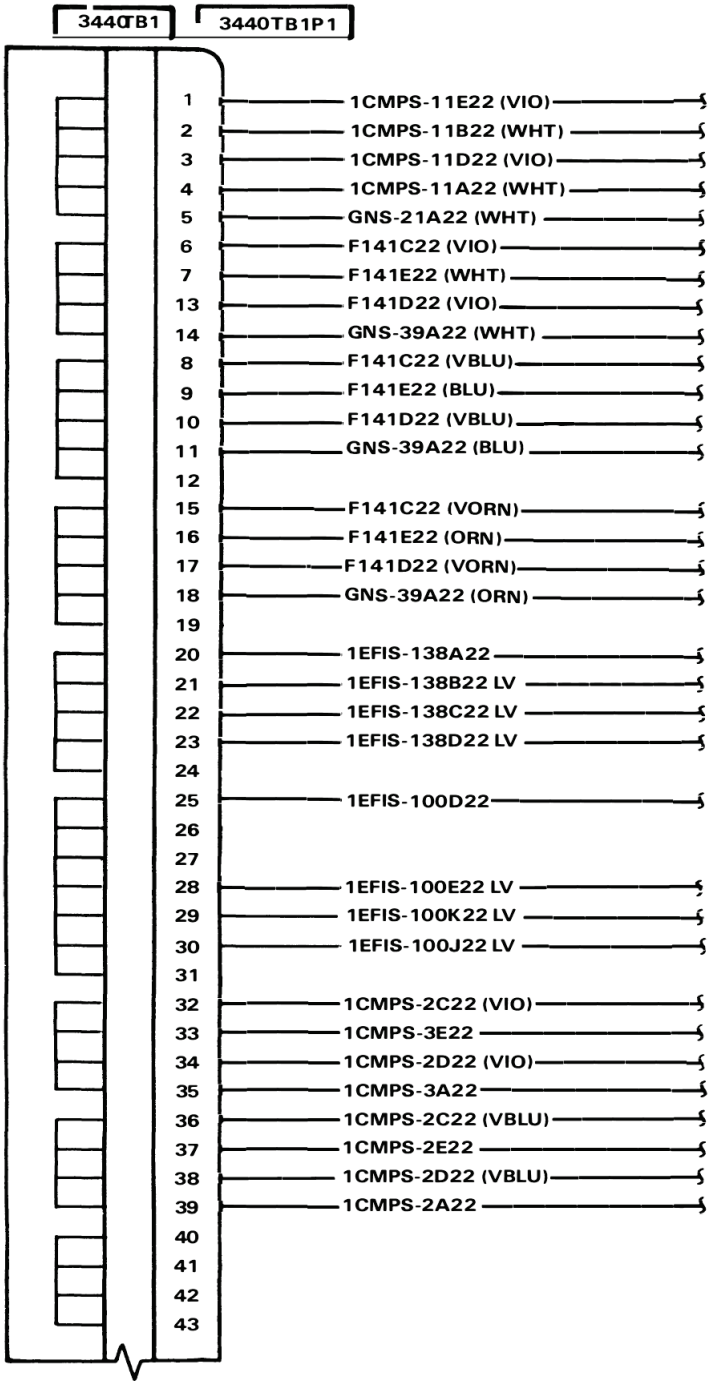
ICN-412-A-980101-B-97499-00076-A-001-01

Figure 1. Component Replacement (Sheet 48 of 59)



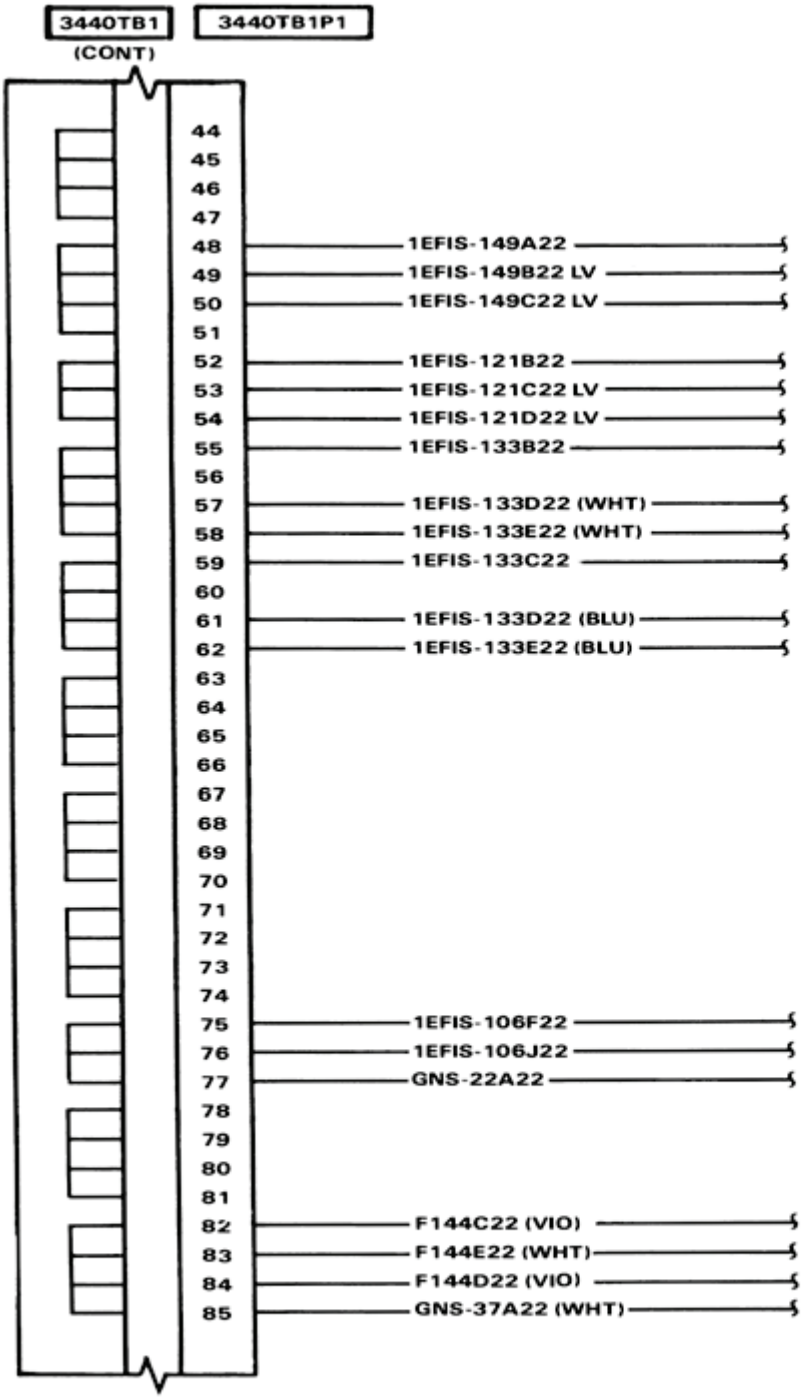
ICN-412-A-980101-B-97499-00077-A-001-01

Figure 1. Component Replacement (Sheet 49 of 59)



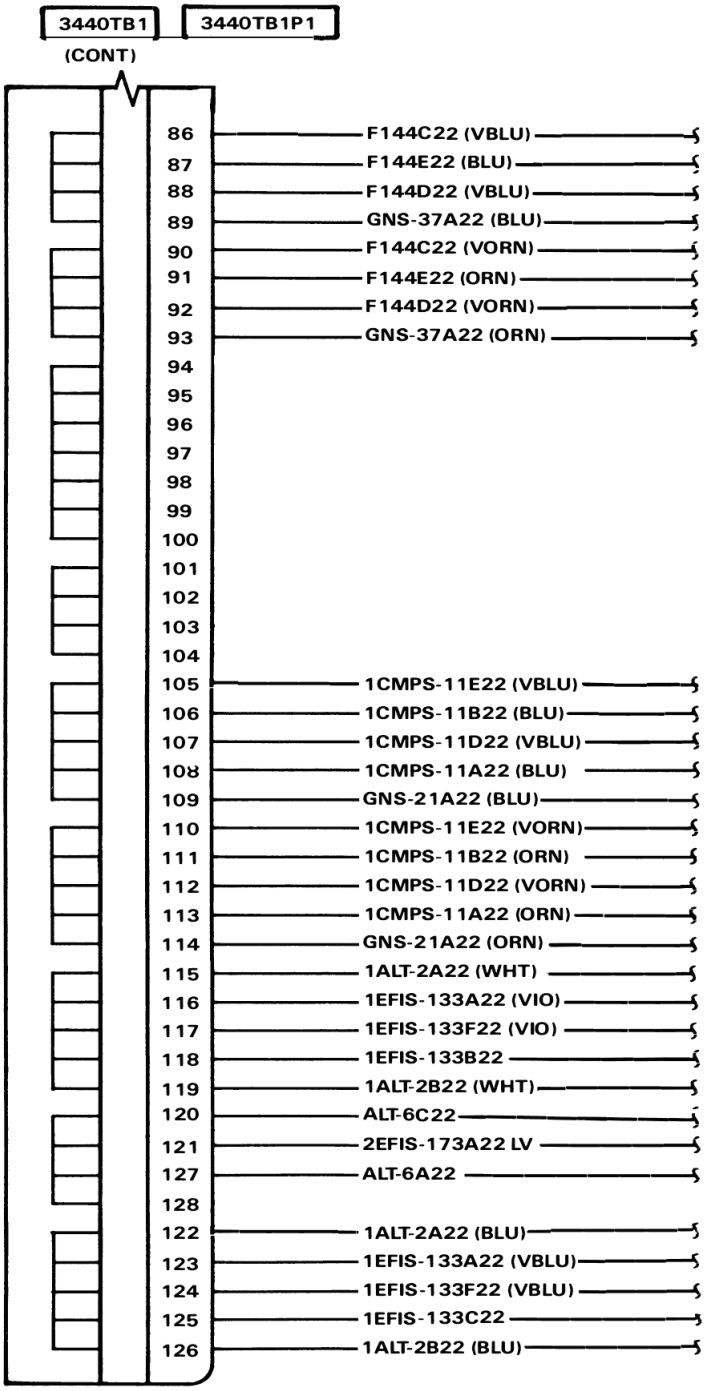
ICN-412-A-980101-B-97499-00078-A-001-01

Figure 1. Component Replacement (Sheet 50 of 59)



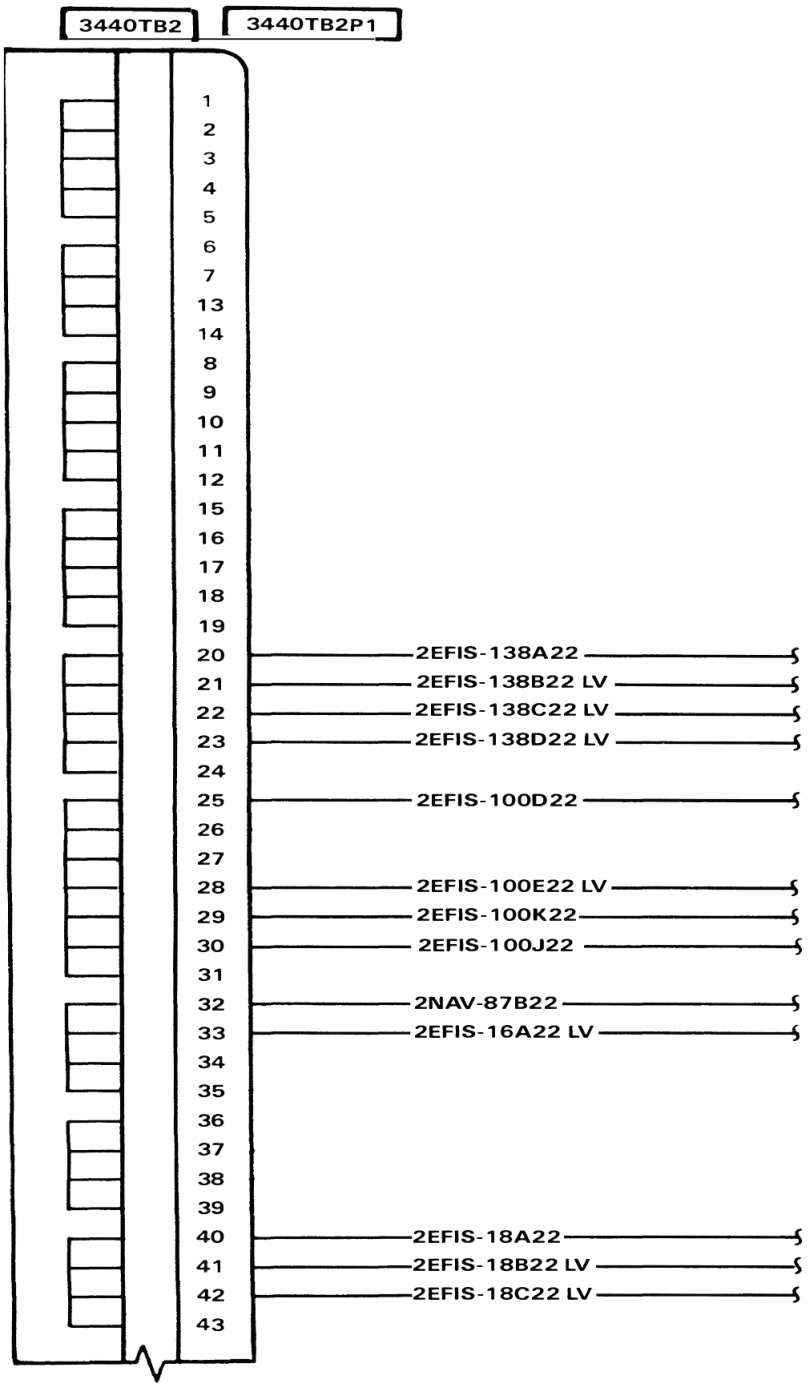
ICN-412-A-980101-B-97499-00079-A-001-01

Figure 1. Component Replacement (Sheet 51 of 59)



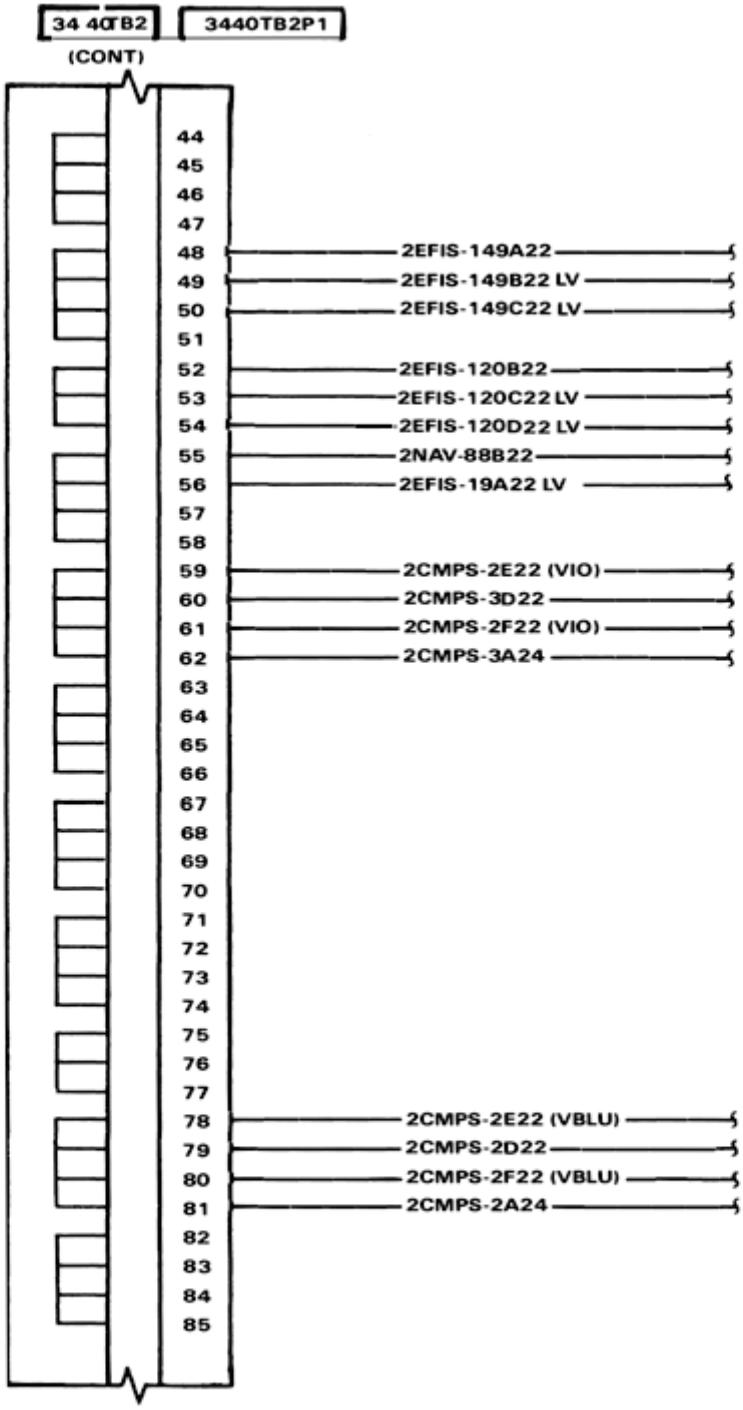
ICN-412-A-980101-B-97499-00080-A-001-01

Figure 1. Component Replacement (Sheet 52 of 59)



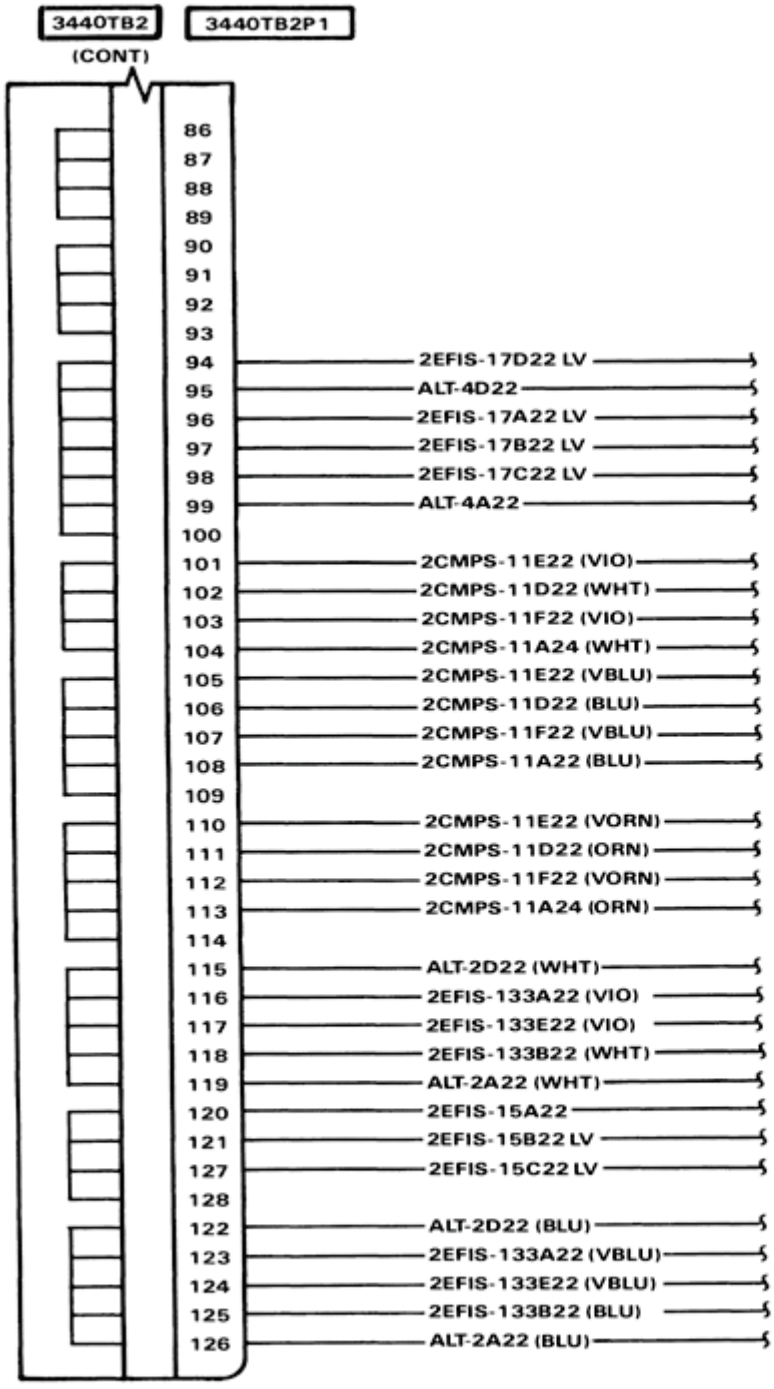
ICN-412-A-980101-B-97499-00081-A-001-01

Figure 1. Component Replacement (Sheet 53 of 59)



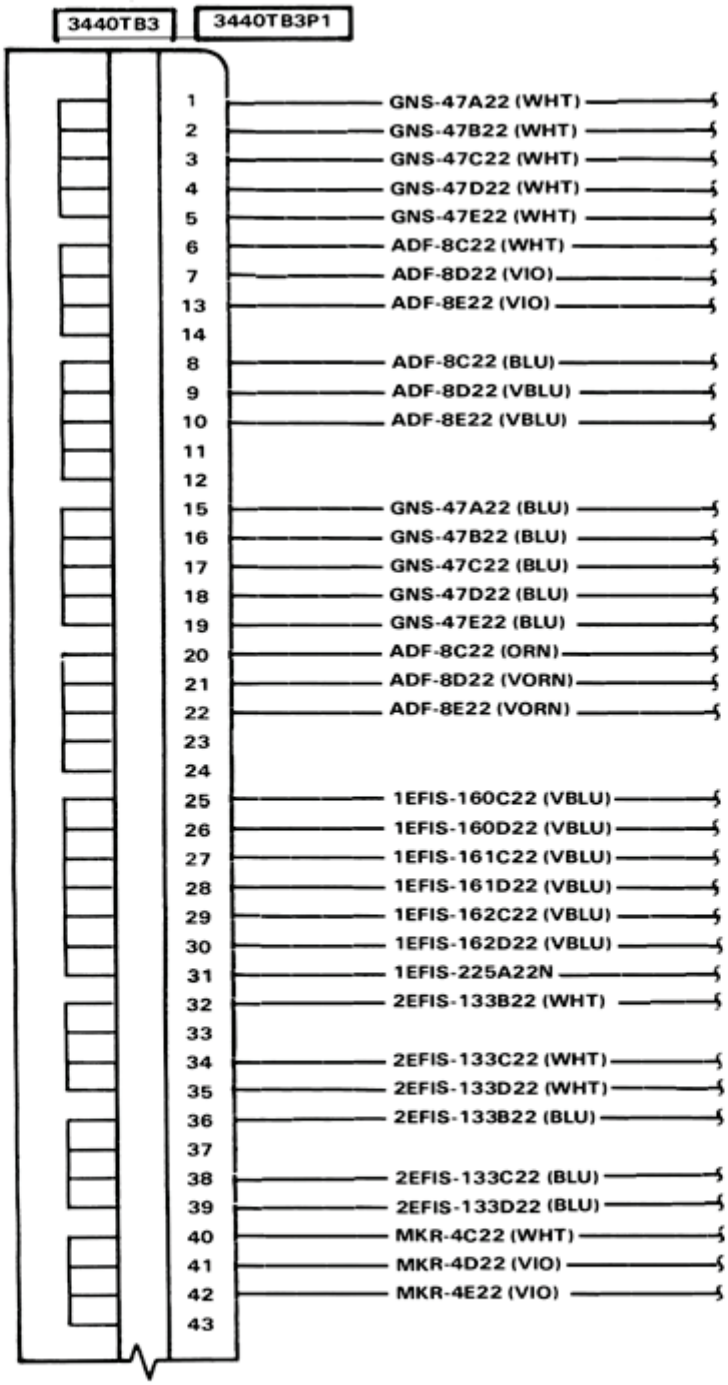
ICN-412-A-980101-B-97499-00082-A-001-01

Figure 1. Component Replacement (Sheet 54 of 59)



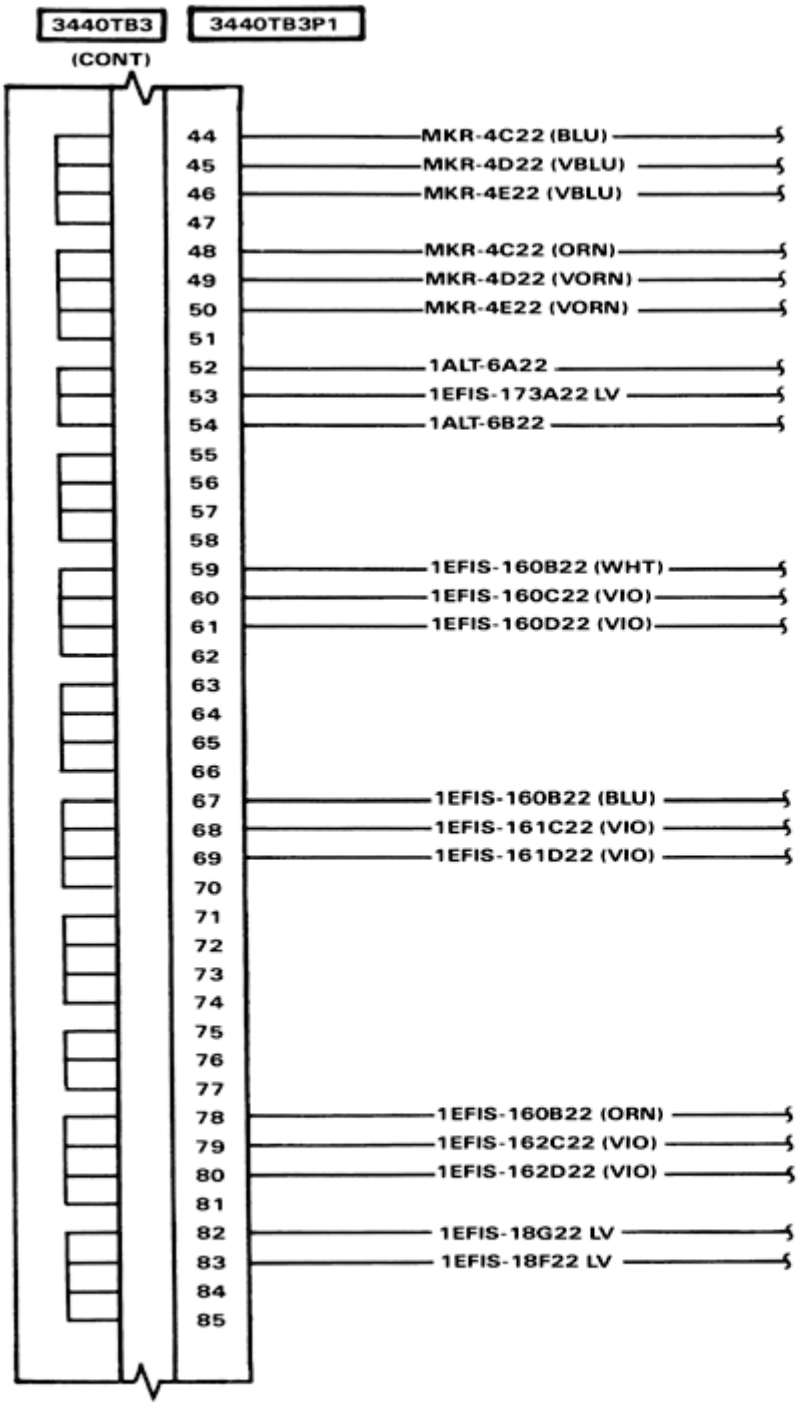
ICN-412-A-980101-B-97499-00083-A-001-01

Figure 1. Component Replacement (Sheet 55 of 59)



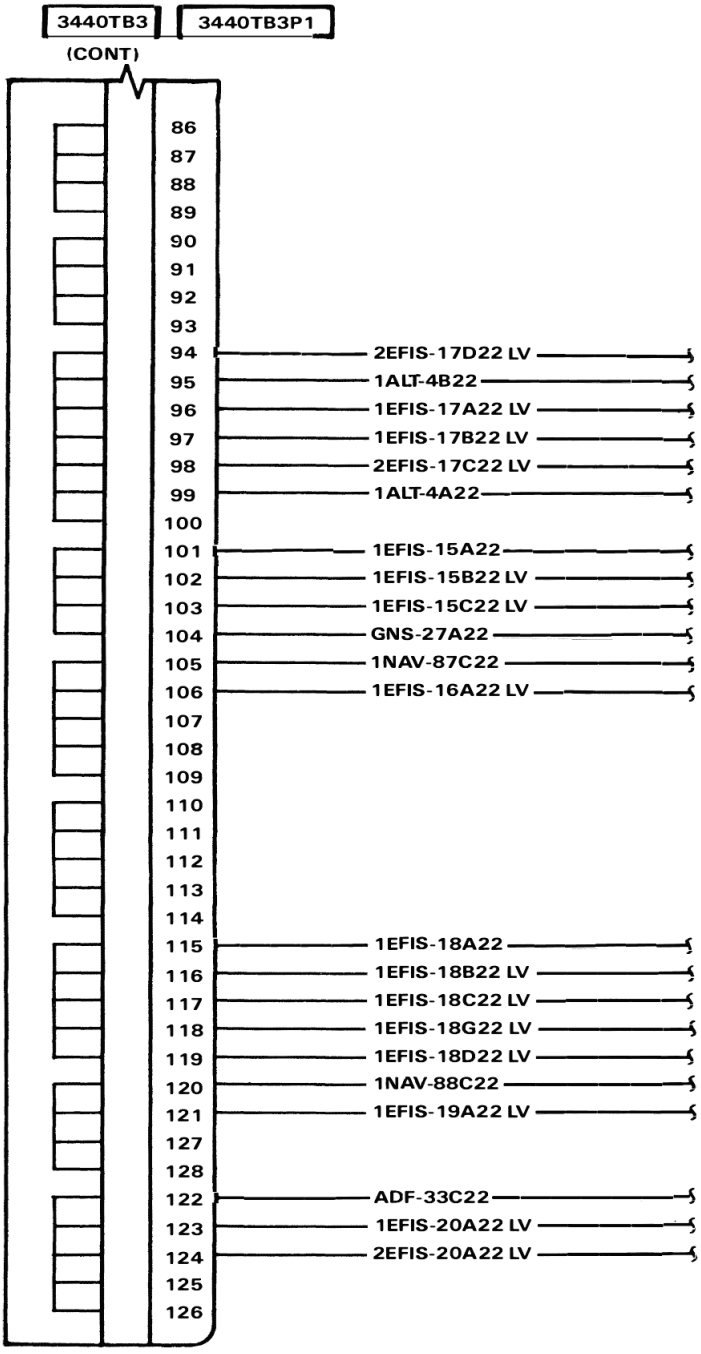
ICN-412-A-980101-B-97499-00084-A-001-01

Figure 1. Component Replacement (Sheet 56 of 59)



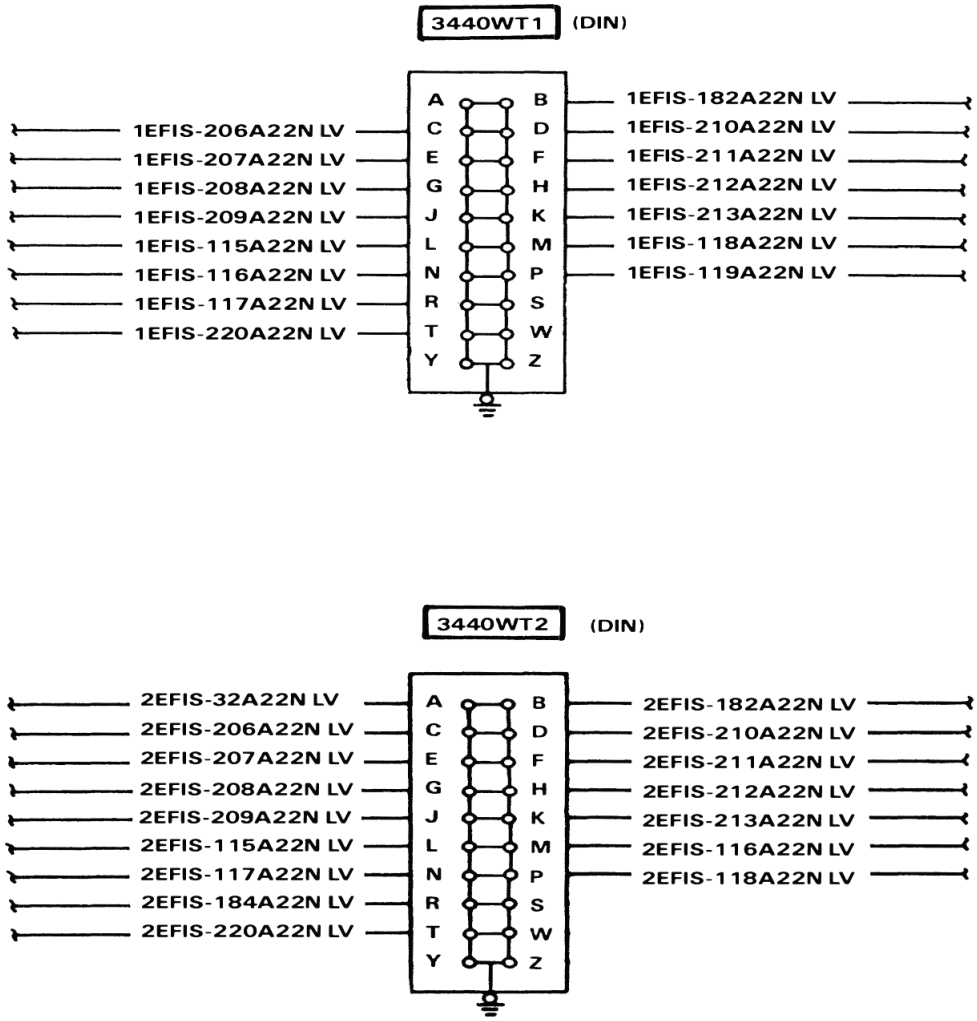
ICN-412-A-980101-B-97499-00085-A-001-01

Figure 1. Component Replacement (Sheet 57 of 59)



ICN-412-A-980101-B-97499-00086-A-001-01

Figure 1. Component Replacement (Sheet 58 of 59)



ICN-412-A-980101-B-97499-00087-A-001-01

Figure 1. Component Replacement (Sheet 59 of 59)

Intentionally blank page